

Old (ZbAnalysis_boosted) vs new (QCD_sf) control plots, 2018

Old: /uscms/home/psrivast/nobackup/ZbExercise/CMSSW_14_0_6/src/Zb/CMSSW_14_0_6/src/ZbAnalysis_boosted/condor_output_mSD/2018 (*_amcnlo.png / .C, amc@NLO DY)

New: /uscms/home/psrivast/nobackup/BTVNanoCommissioning/hists_run2018_mSD_SF/merged_run2018.coffea

Luminosity 59832 /pb. Same selection, lepton SFs, cross sections and lumi mask in both (see SESSION_NOTES.md reference table).

Each variable page: per channel, left = old framework's plot as produced; middle = new result in the same style (same processes, colours, stacking order, x-range, log scale, hatched MC stat. band); right = new/old per bin for data (black) and total MC (red), with integrals. Old numbers are read from the old canvas macros (.C), so the right-hand panel compares the exact plotted histograms.

Binning: when old and new bin edges coincide on ≥ 8 bins in range, both are summed onto that common grid (exact). Otherwise the new histogram is redistributed onto the old bins assuming a flat distribution inside each new bin; those pages say 'approx.' and the middle panel shows the new native binning.

In the new/old panel, neighbouring bins are merged until each holds ≥ 50 old data events.

Its error bars combine old and new statistical errors as if independent; since both frameworks select mostly the same events, they overstate the real old/new scatter.

Data: Zee from EGamma only, Zmm from SingleMuon only (as in the old framework).

Leading-jet mSD, jet mass, tau21, tau32 exist in the new output only for Z_jet and without a Zee/Zmm split:

they are compared with the SUM of the old Zee and Zmm plots, and their new data sums both primary datasets ($\sim 2\%$ of events are in both), so expect new/old data ~ 1.02 there.

Known differences (not bugs in the new code):

- old applies Rochester muon corrections (-0.2% at the two-muon step); old jet ID reads the AK4 branch with the AK8 index (likely the $\sim +1-1.5\%$ new/old at the jet step, in data and MC alike).
- old swaps subjet eta and phi (src/Plots.cxx:338-339): our eta_sub* is paired with its 'phi_sub*' plot and vice versa.
- Z_bjet: the old framework fills subjet histograms from the leading jet's subjets, ours from the tagged jet's subjets; differences are expected.
- Not filled in the new framework (no comparison): dPhi_Z_jet, deltaR_lep_lep, deltaR_lep_jet, deltaR_ZJet, deltaR_lep_Z.

Yield summary (same layout as old summary_eventCount_amcnlo.txt)

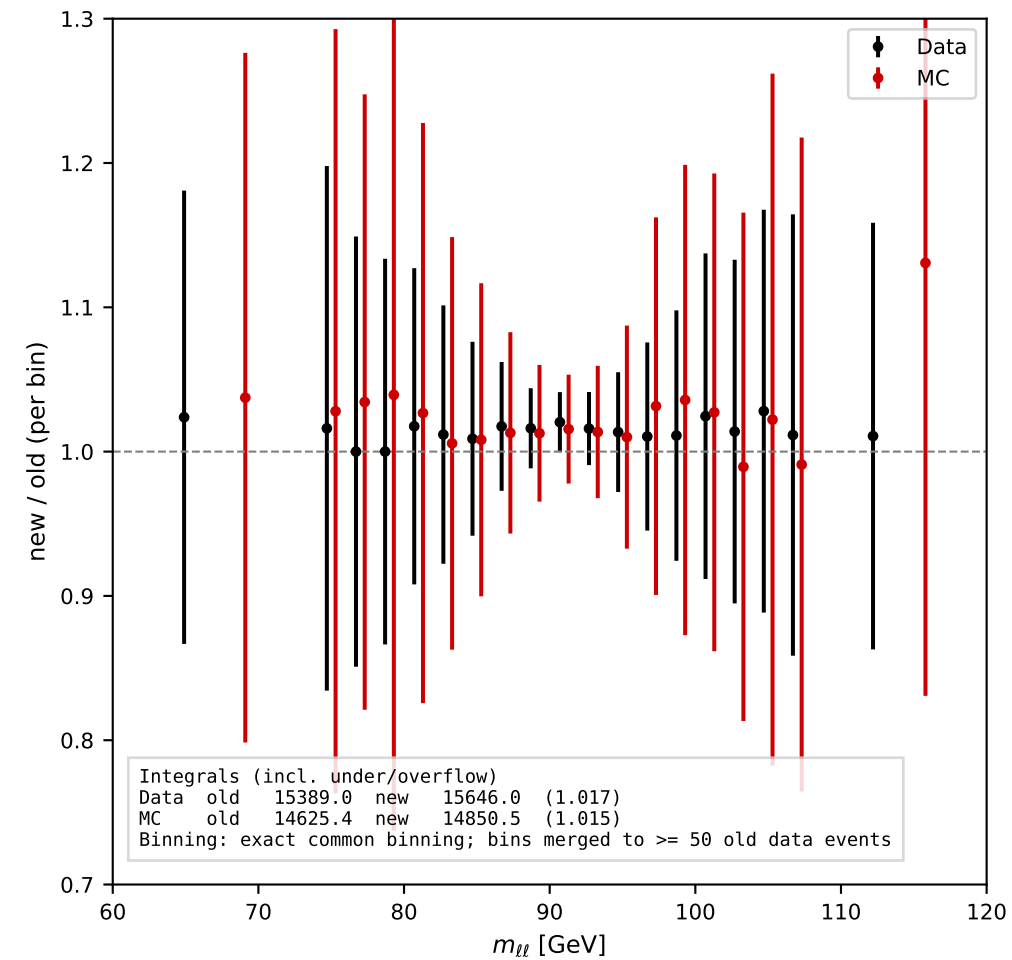
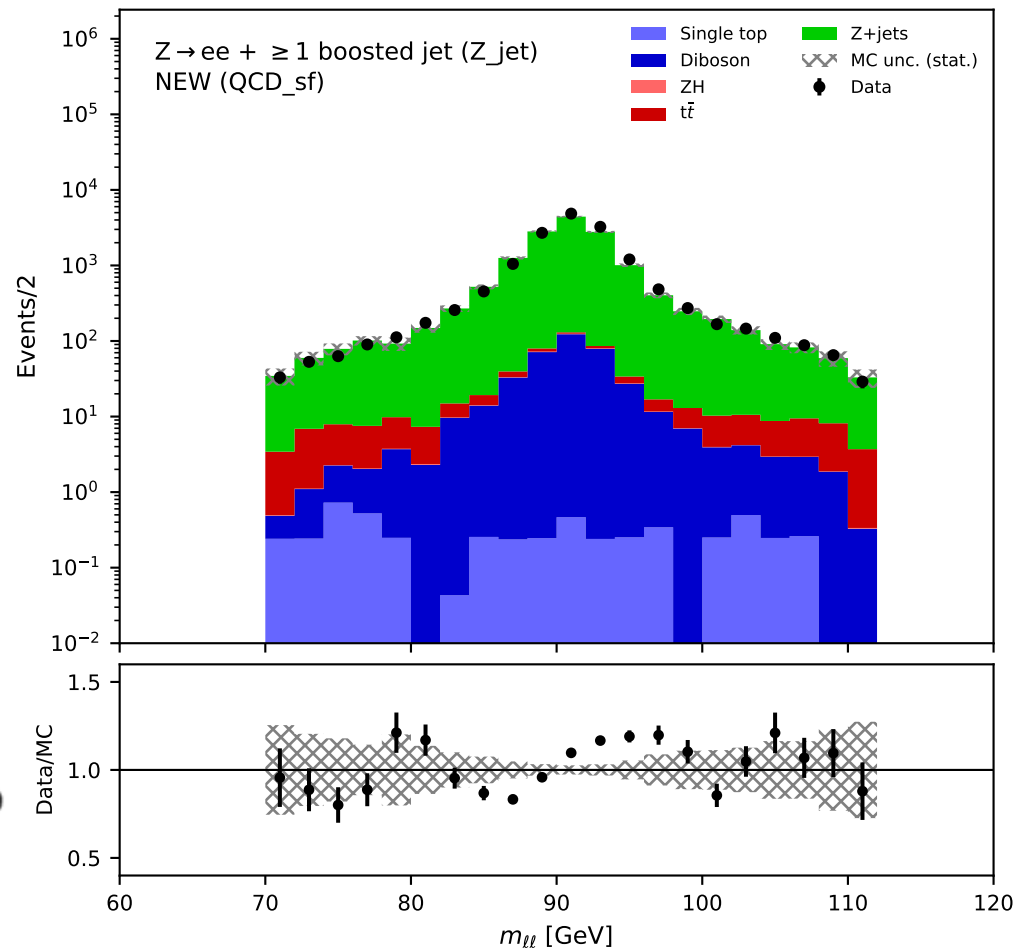
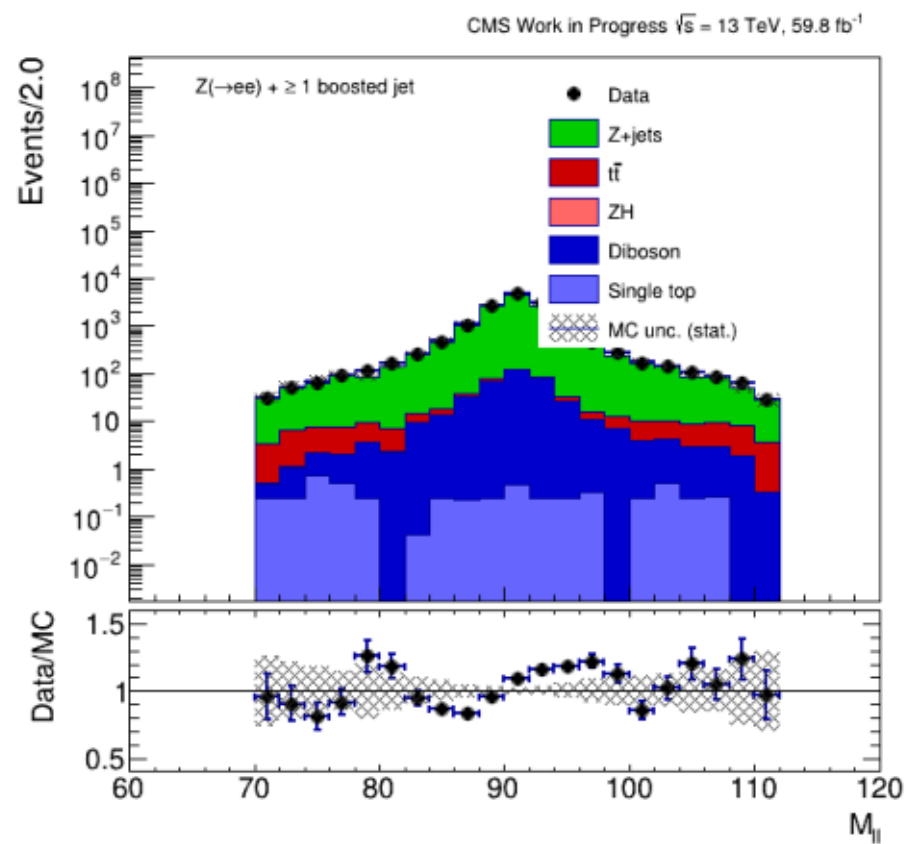
Old (ZbAnalysis_boosted, condor_output_mSD) vs new (QCD_sf) yields, 2018, 59.8 fb⁻¹
Layout of the old summary_eventCount_amcnlo.txt tables.

Z_jet	Electron			Muon		
	old	new	new/old	old	new	new/old
DY	14105.2	14323.2	1.015	25352.3	25672.4	1.013
tt	116.6	118.2	1.014	265.2	268.8	1.014
Single t	5.3	5.3	1.000	14.3	14.3	1.000
VV	392.5	398.0	1.014	633.1	641.7	1.014
ZH	5.7	5.8	1.018	9.5	9.6	1.013
MC	14625.4	14850.5	1.015	26274.3	26606.8	1.013
Data	15389.0	15646.0	1.017	27224.0	27459.0	1.009
Data/MC	1.052	1.054		1.036	1.032	

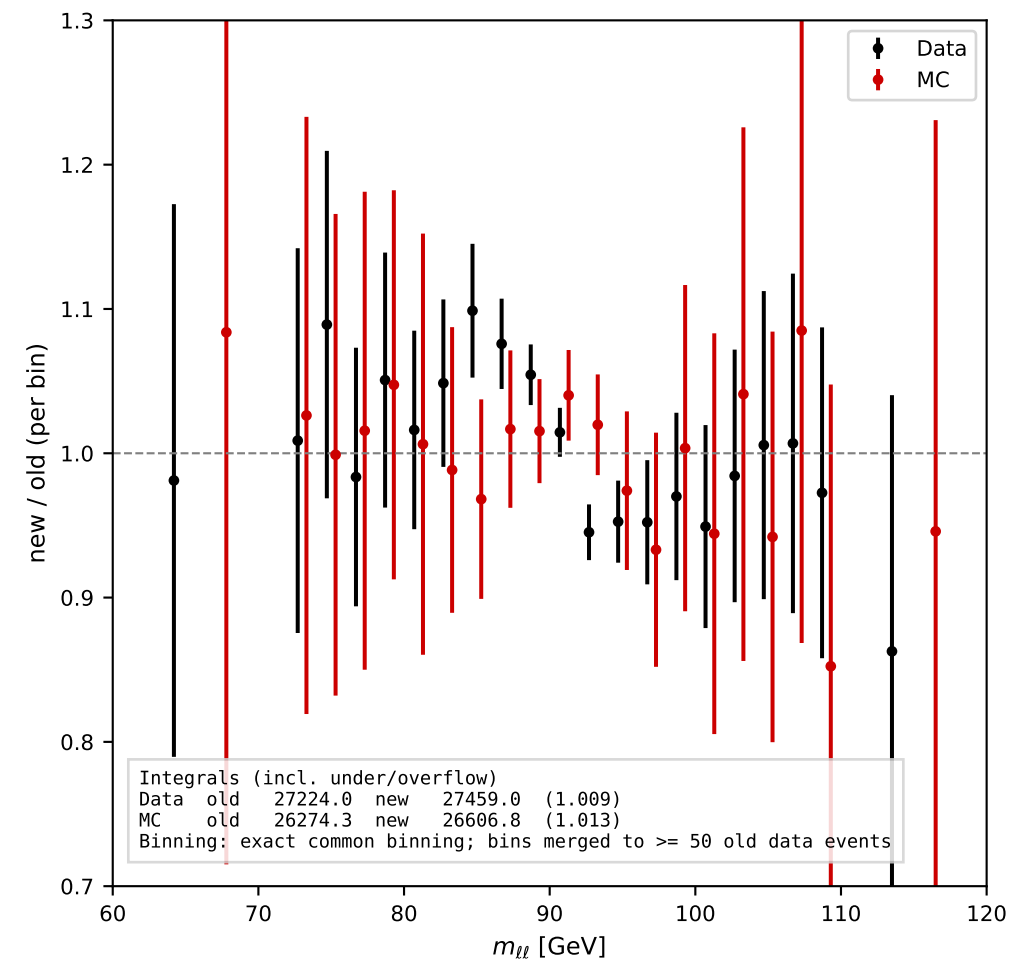
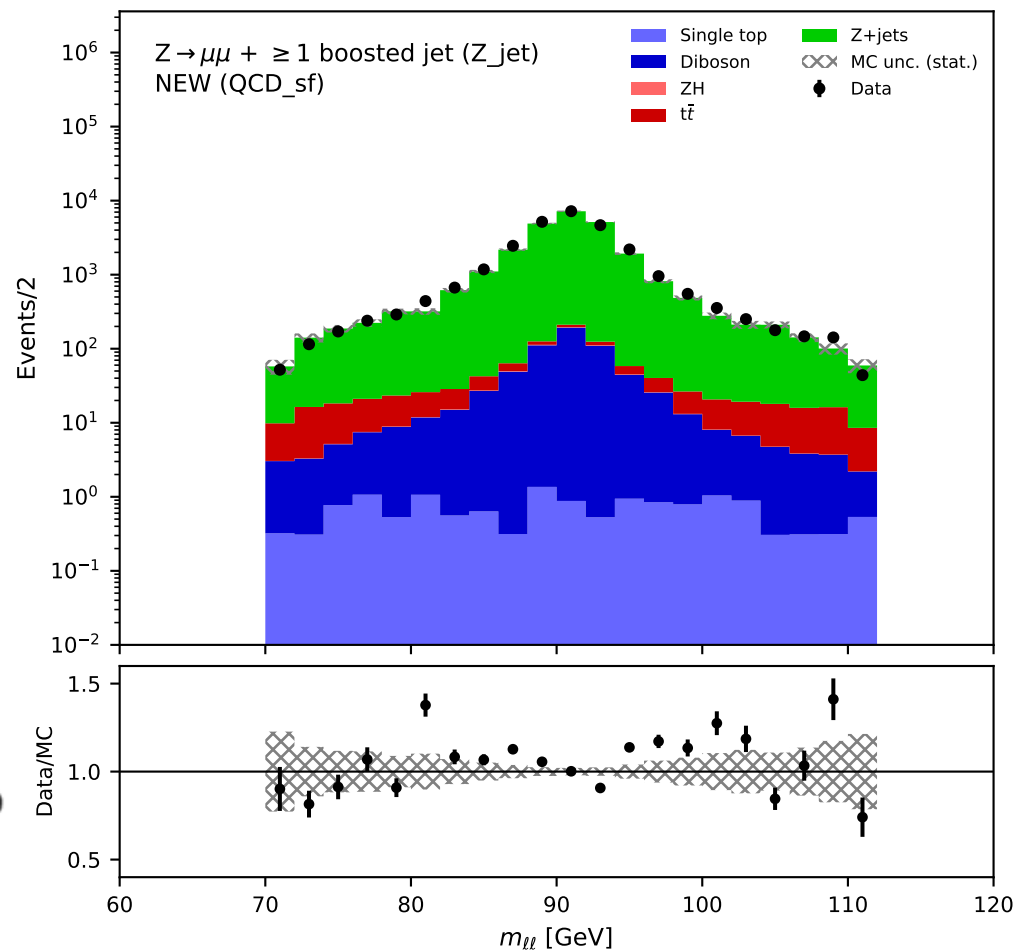
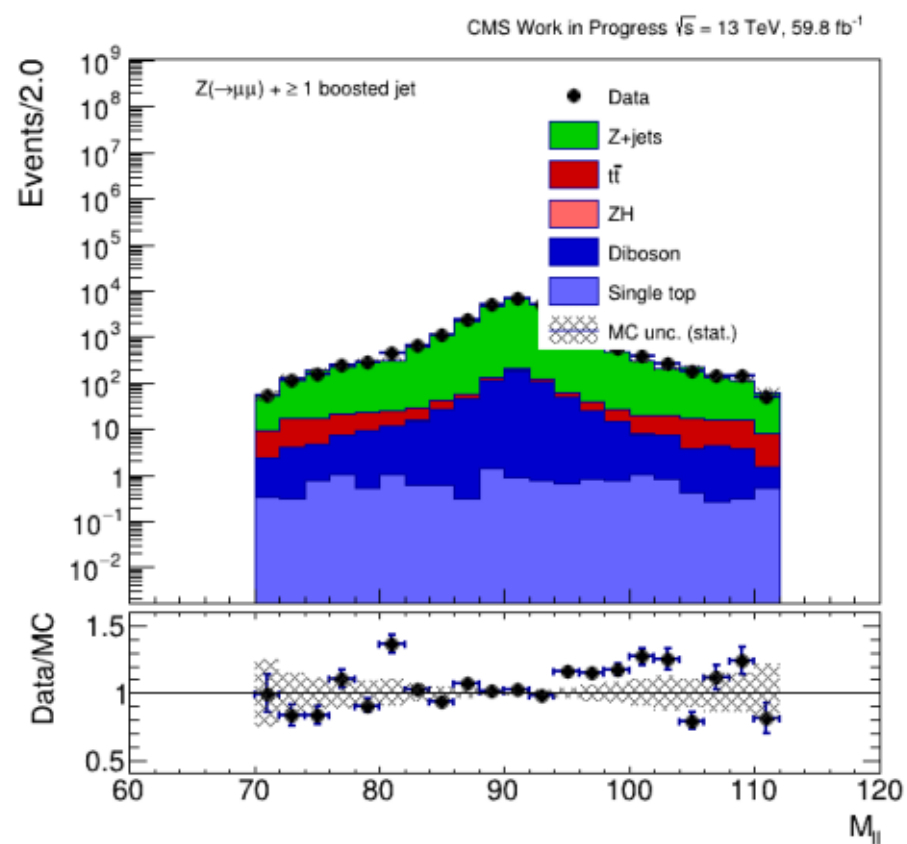
Z_bjet	Electron			Muon		
	old	new	new/old	old	new	new/old
DY	453.2	450.9	0.995	763.4	774.1	1.014
tt	21.9	22.2	1.012	51.6	52.3	1.013
Single t	1.1	1.1	1.000	3.3	3.3	1.000
VV	21.7	22.7	1.042	40.9	42.5	1.040
ZH	3.3	3.4	1.018	5.4	5.5	1.015
MC	501.3	500.2	0.998	864.6	877.7	1.015
Data	585.0	591.0	1.010	1070.0	1076.0	1.006
Data/MC	1.167	1.182		1.238	1.226	

ZMass | Z_jet | 2018

OLD: ZMass_Z_jet_Zee

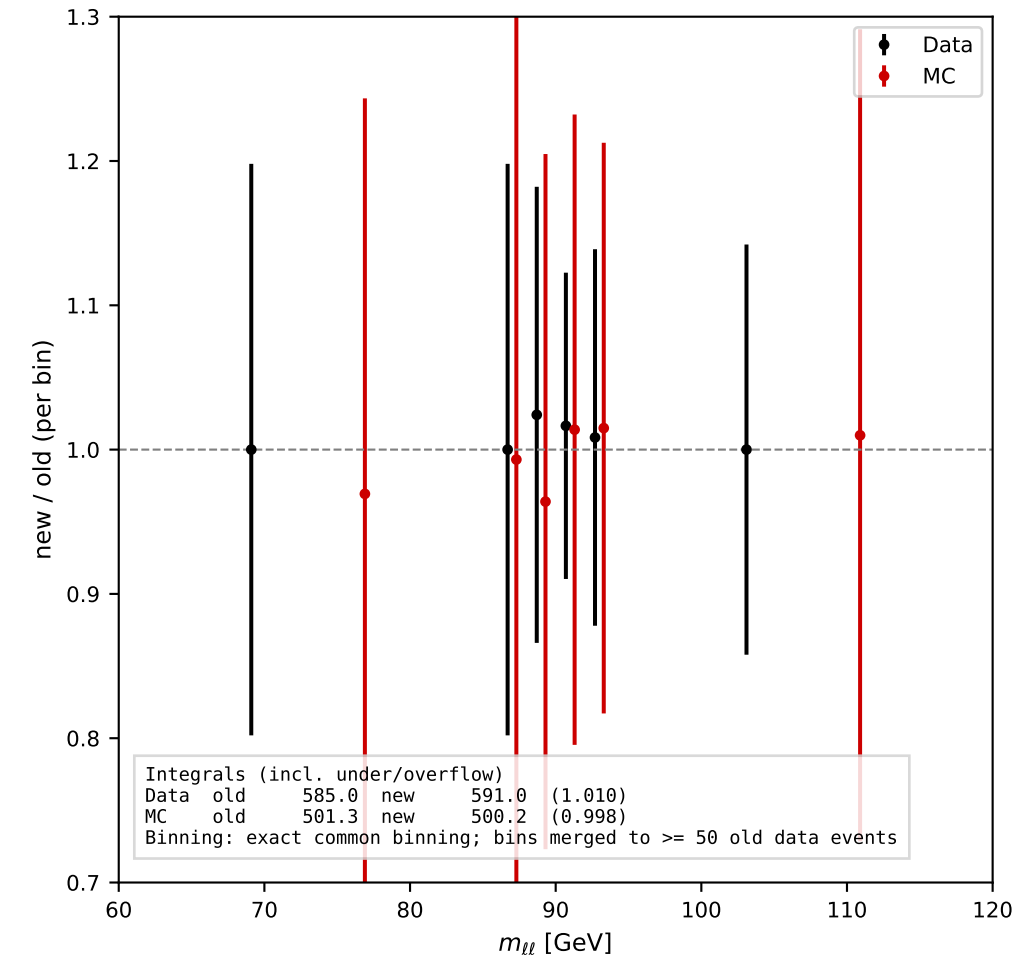
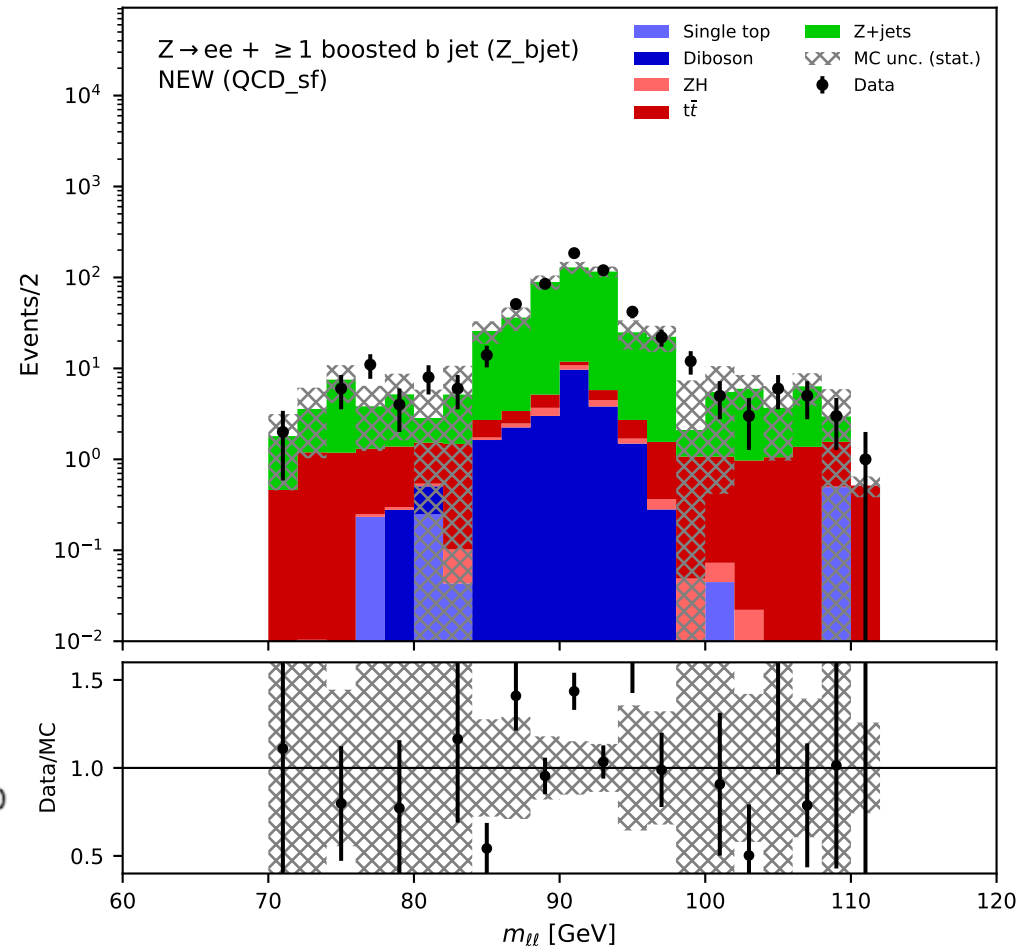
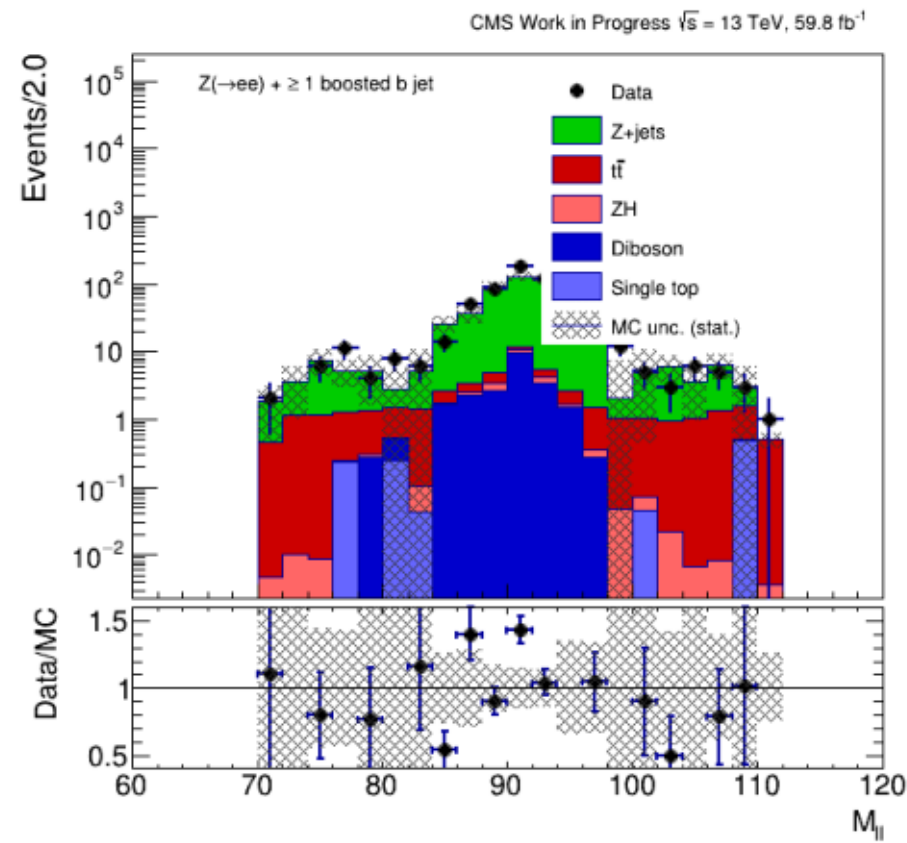


OLD: ZMass_Z_jet_Zmm

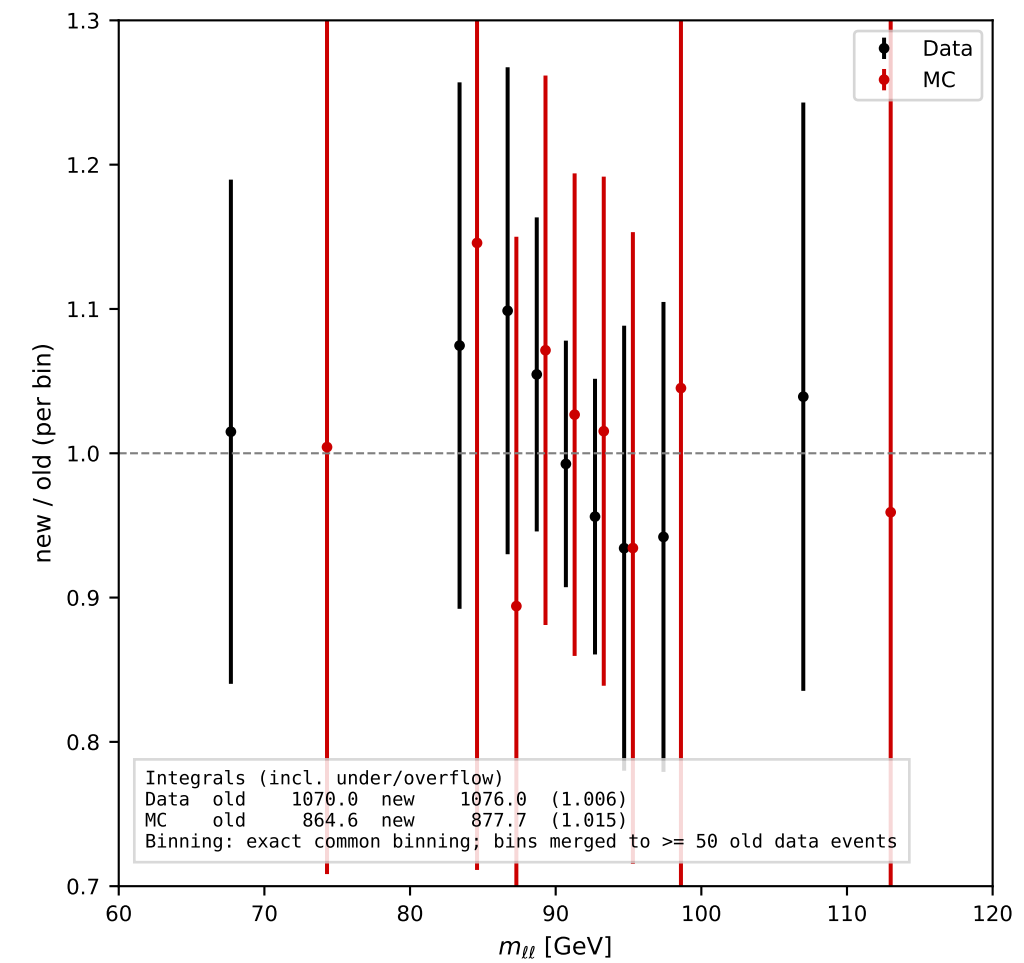
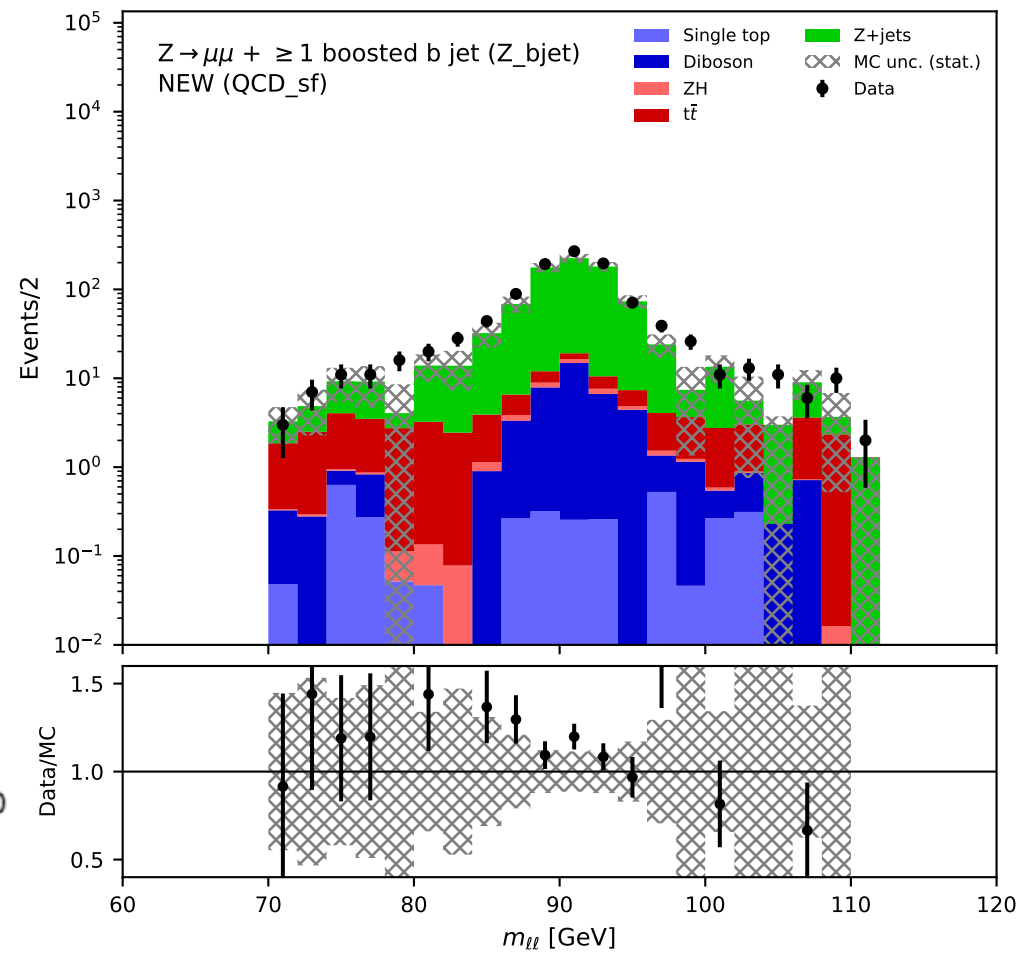
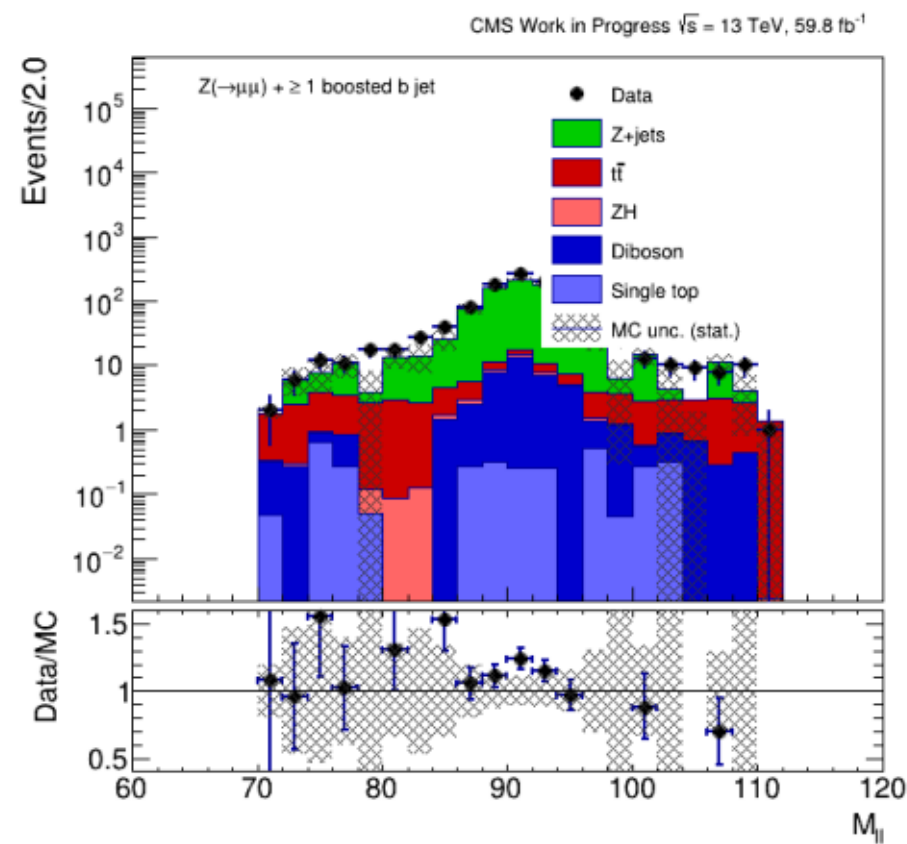


ZMass | Z_bjet | 2018

OLD: ZMass_Z_bjet_Zee

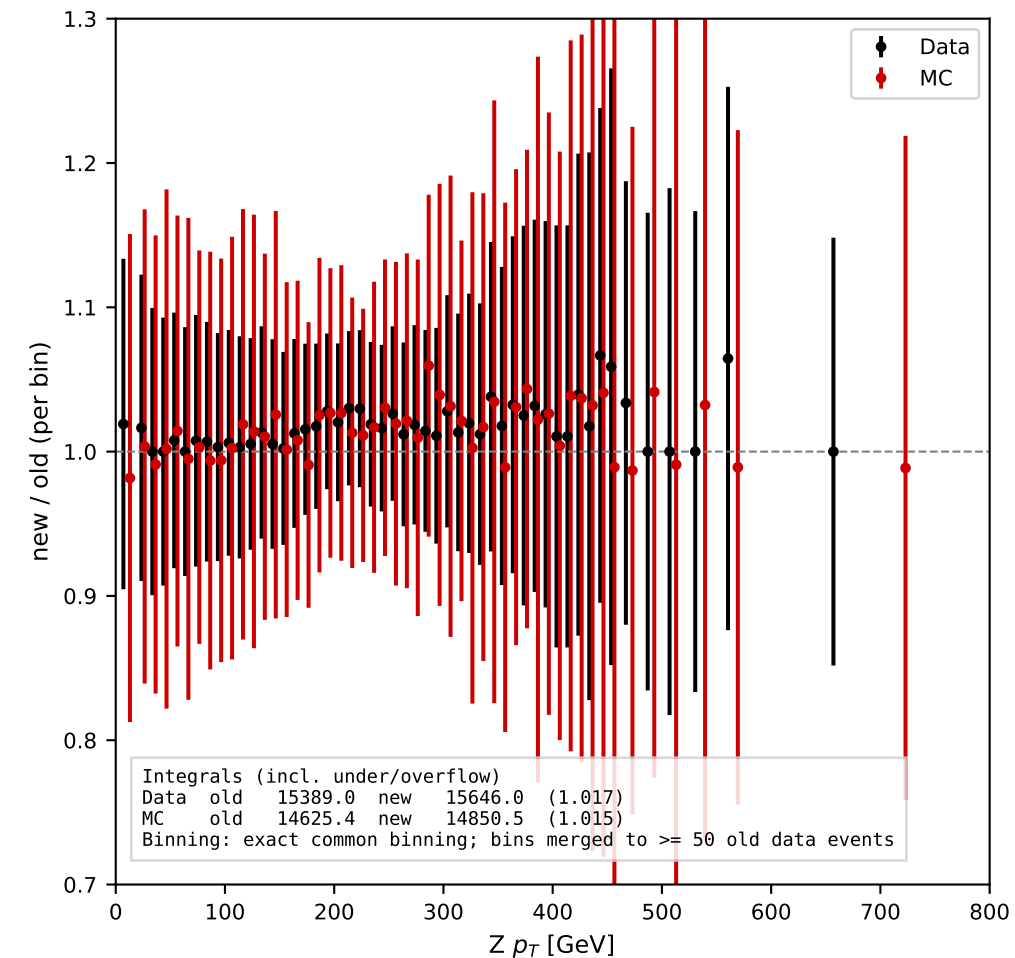
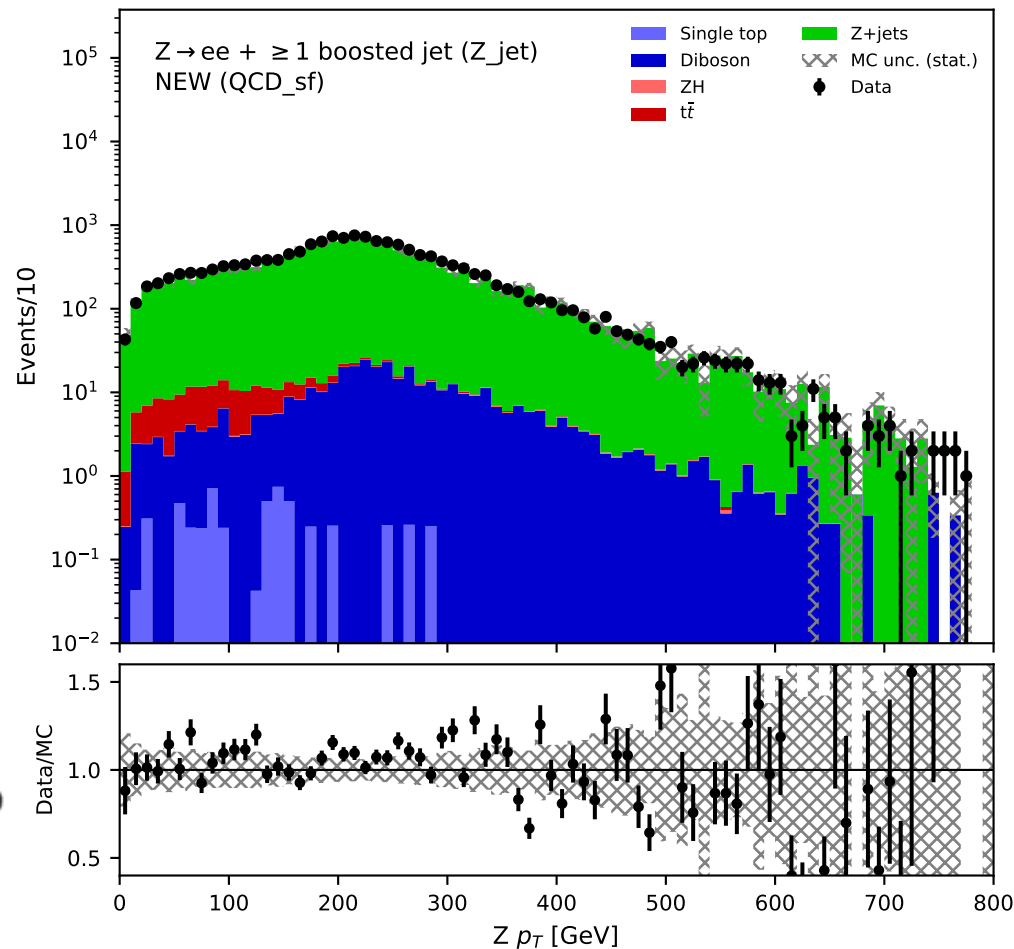
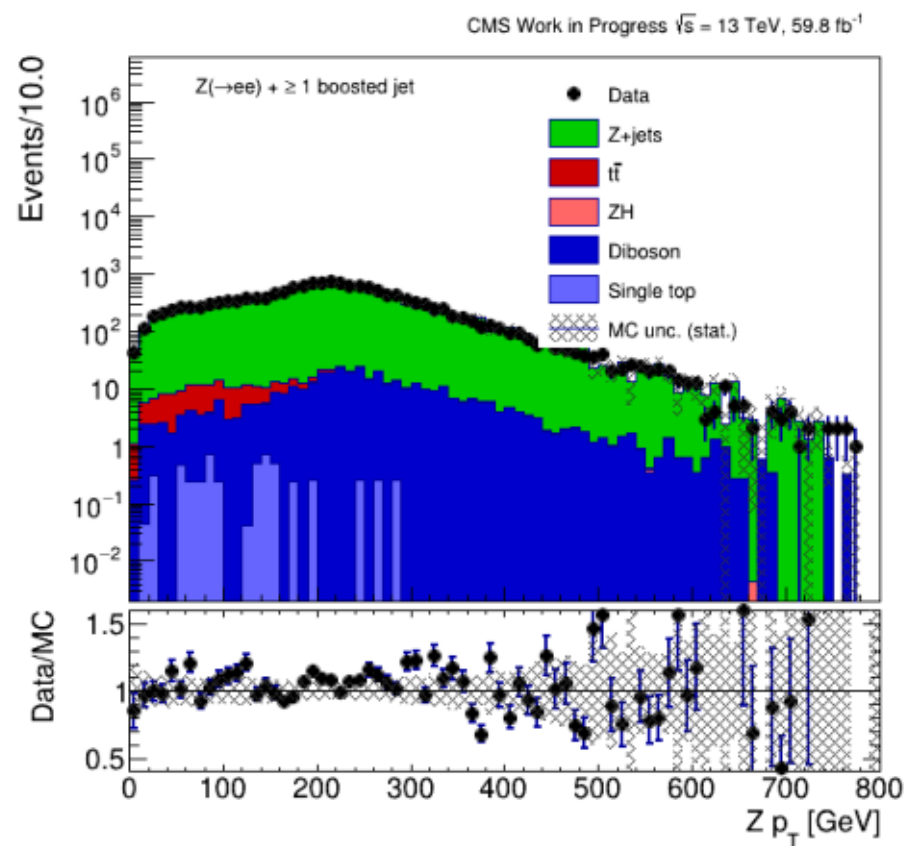


OLD: ZMass_Z_bjet_Zmm

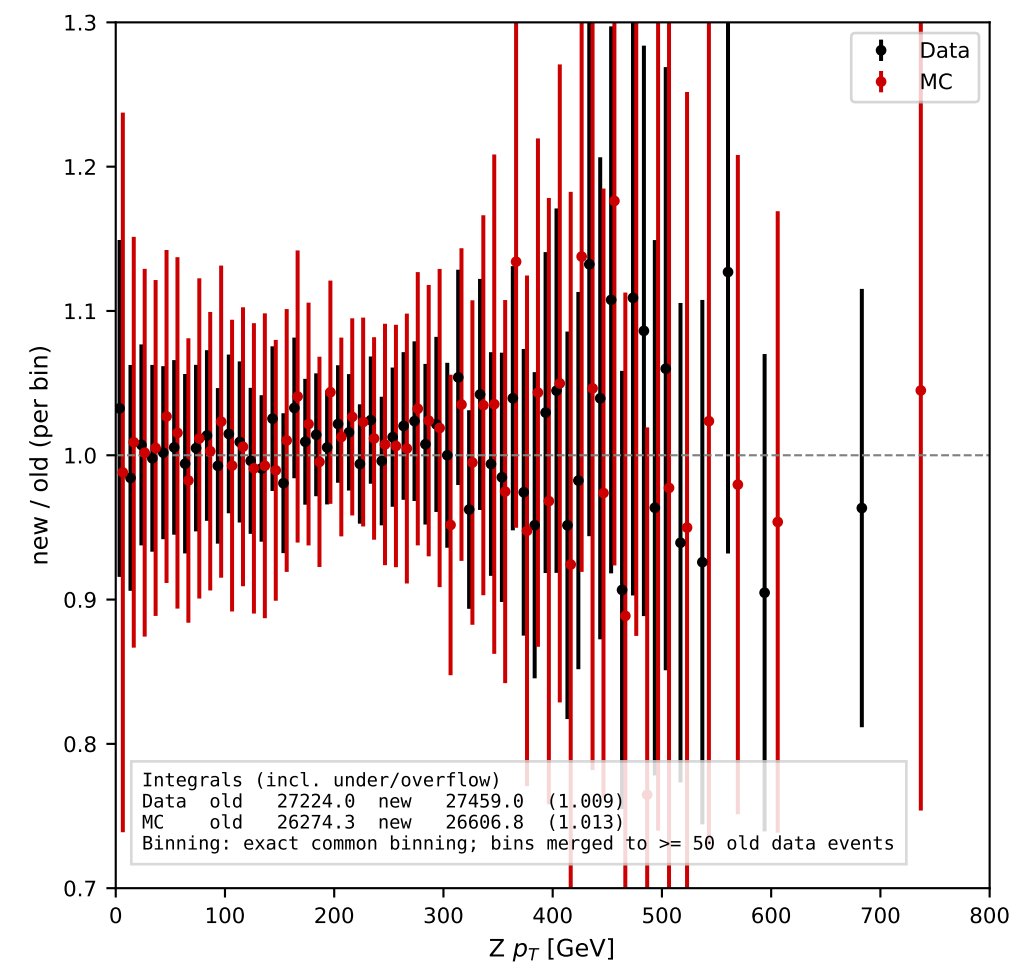
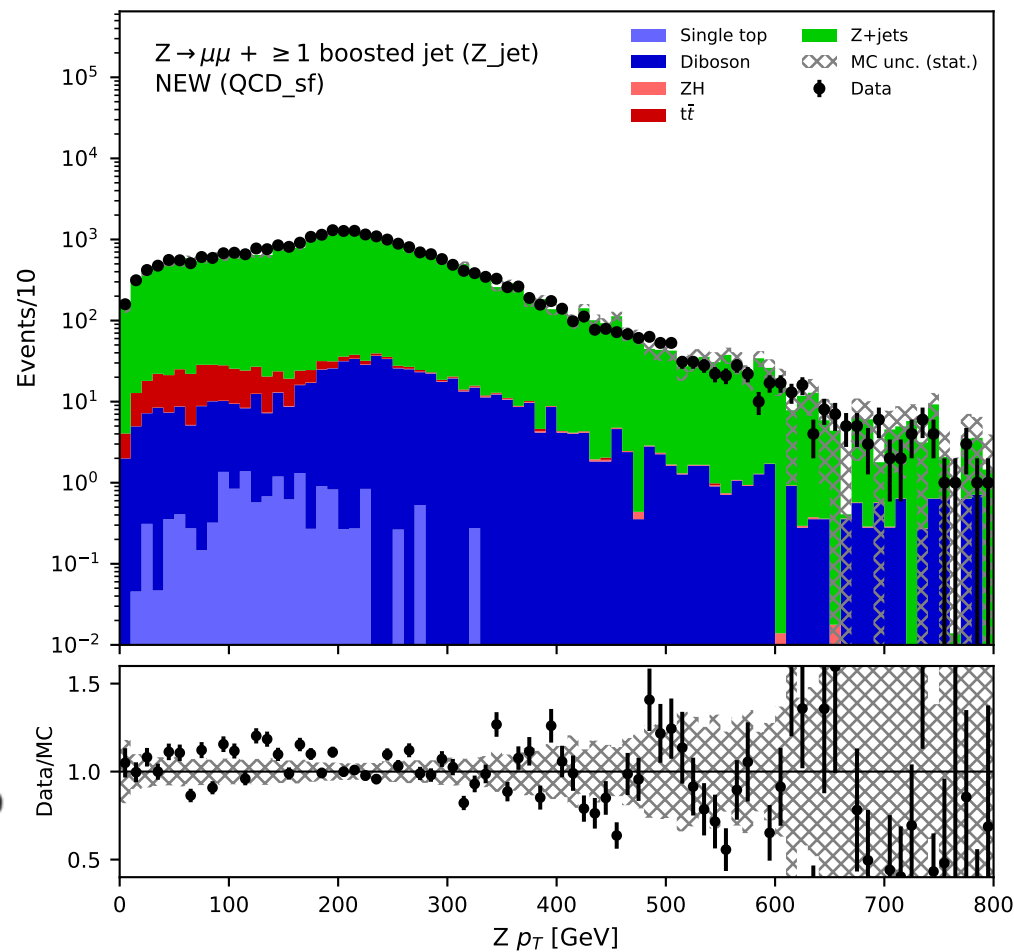
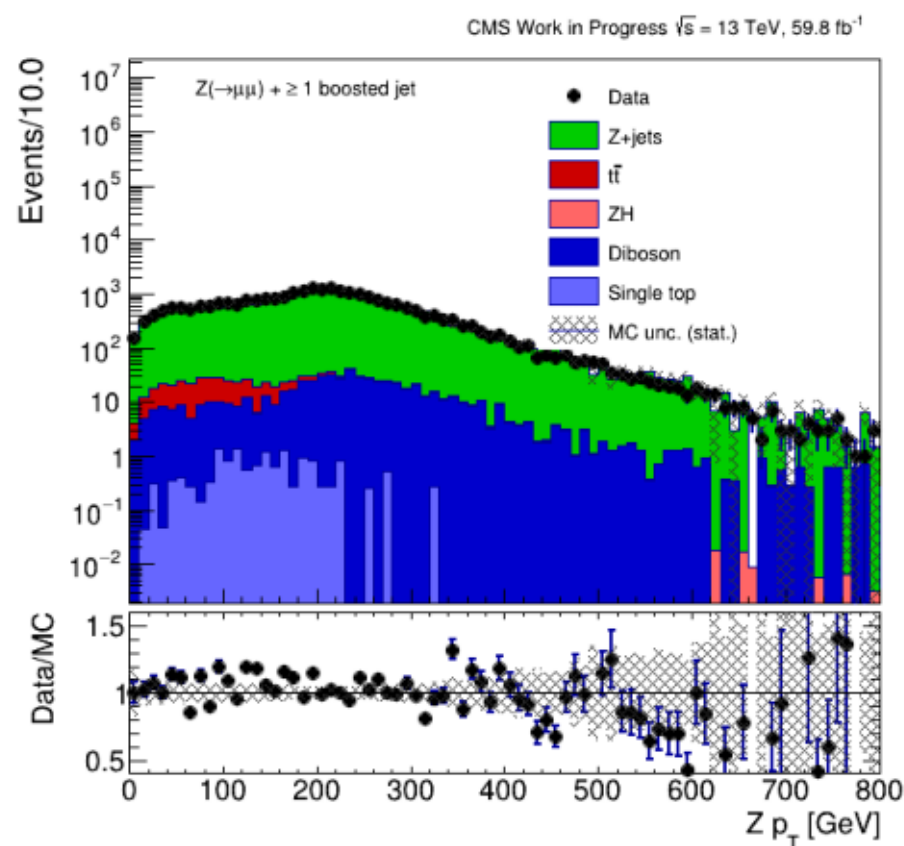


pt_dilepton | Z_jet | 2018

OLD: pt_dilepton_Z_jet_Zee

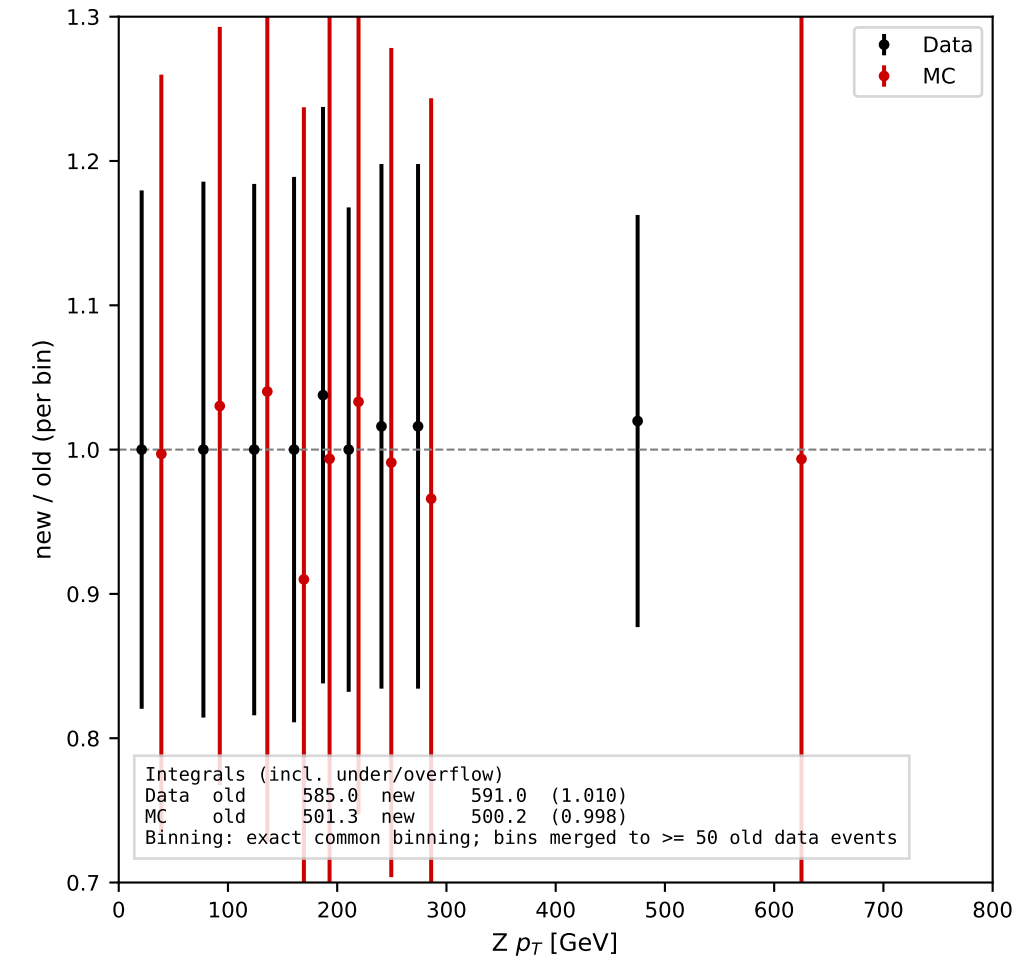
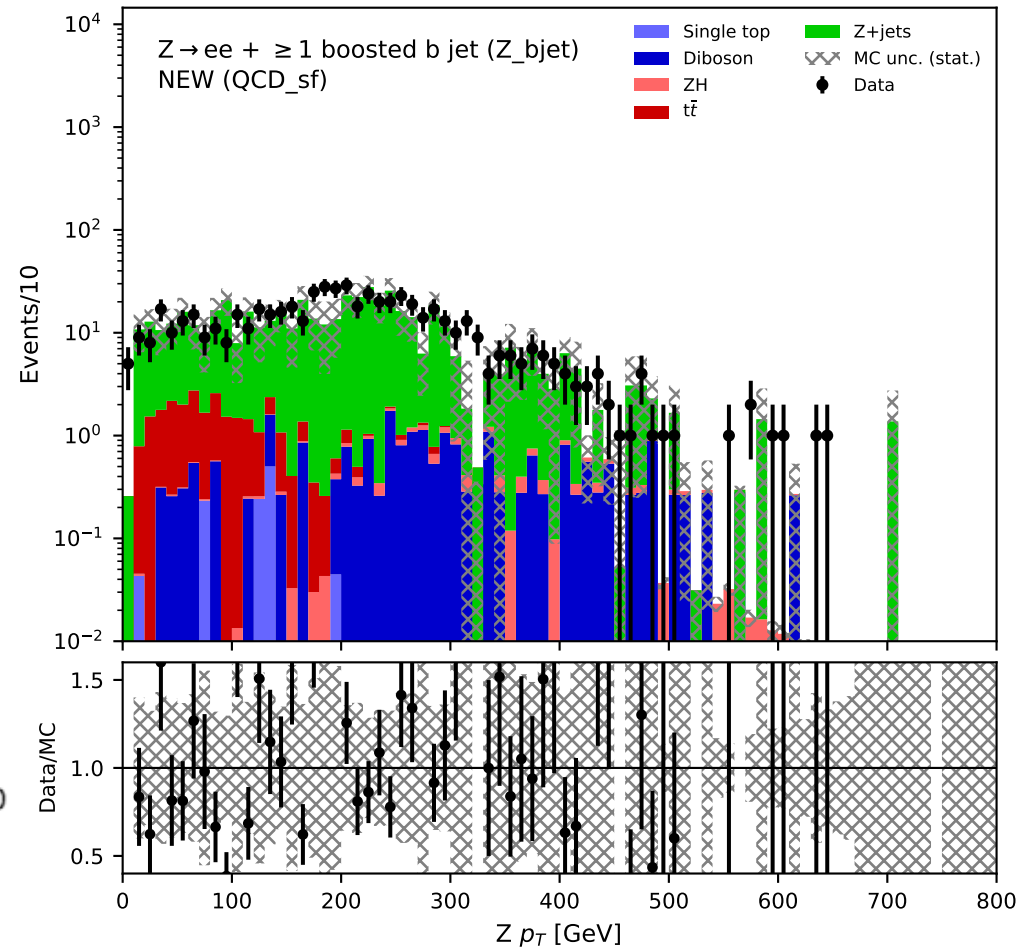
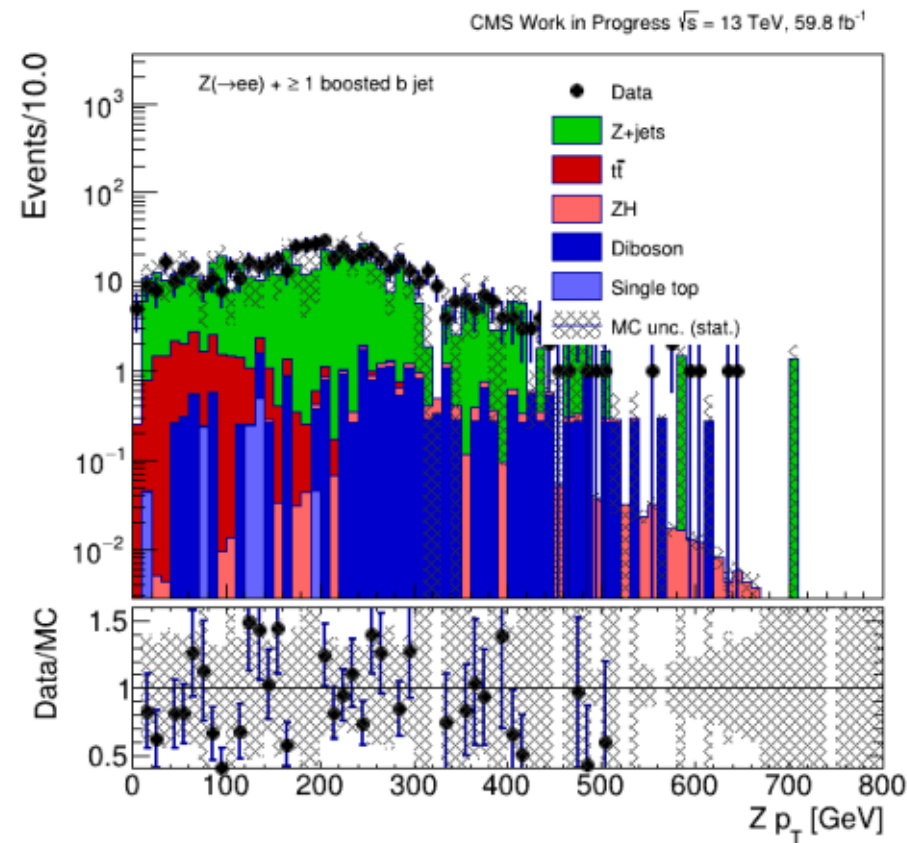


OLD: pt_dilepton_Z_jet_Zmm

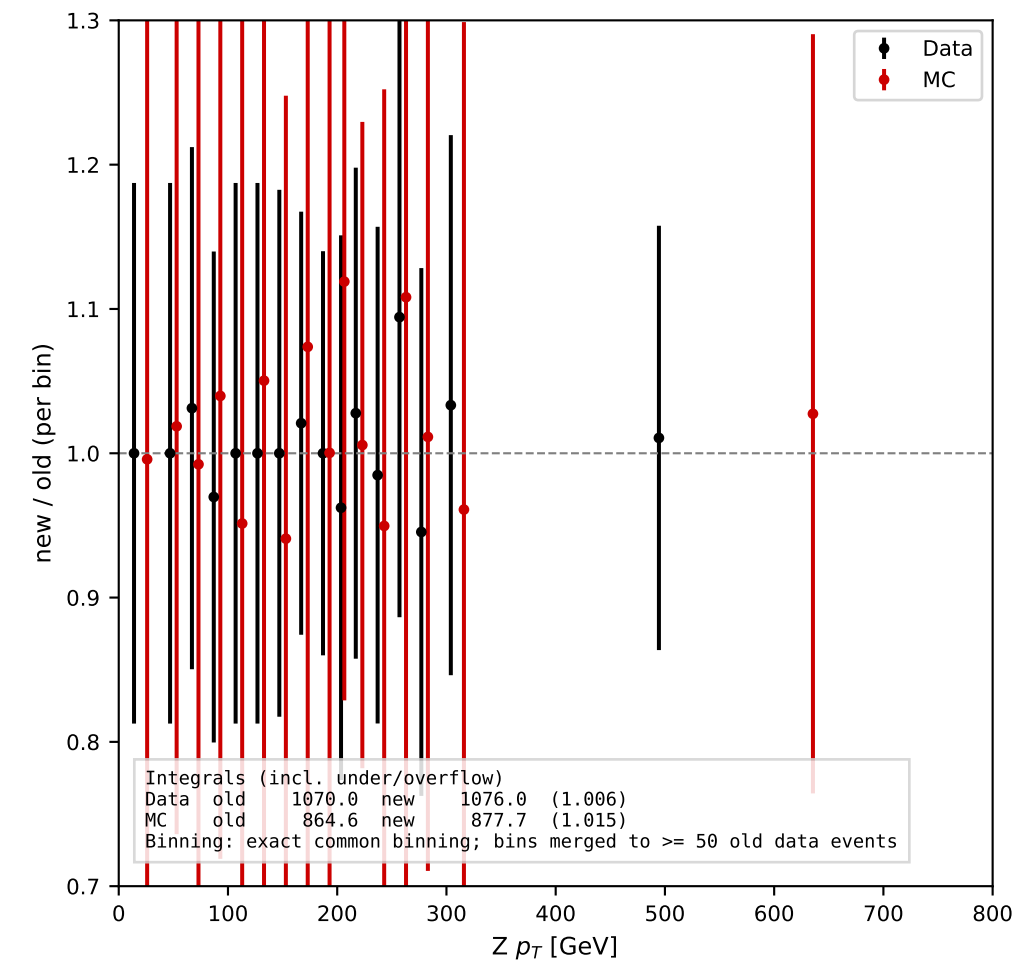
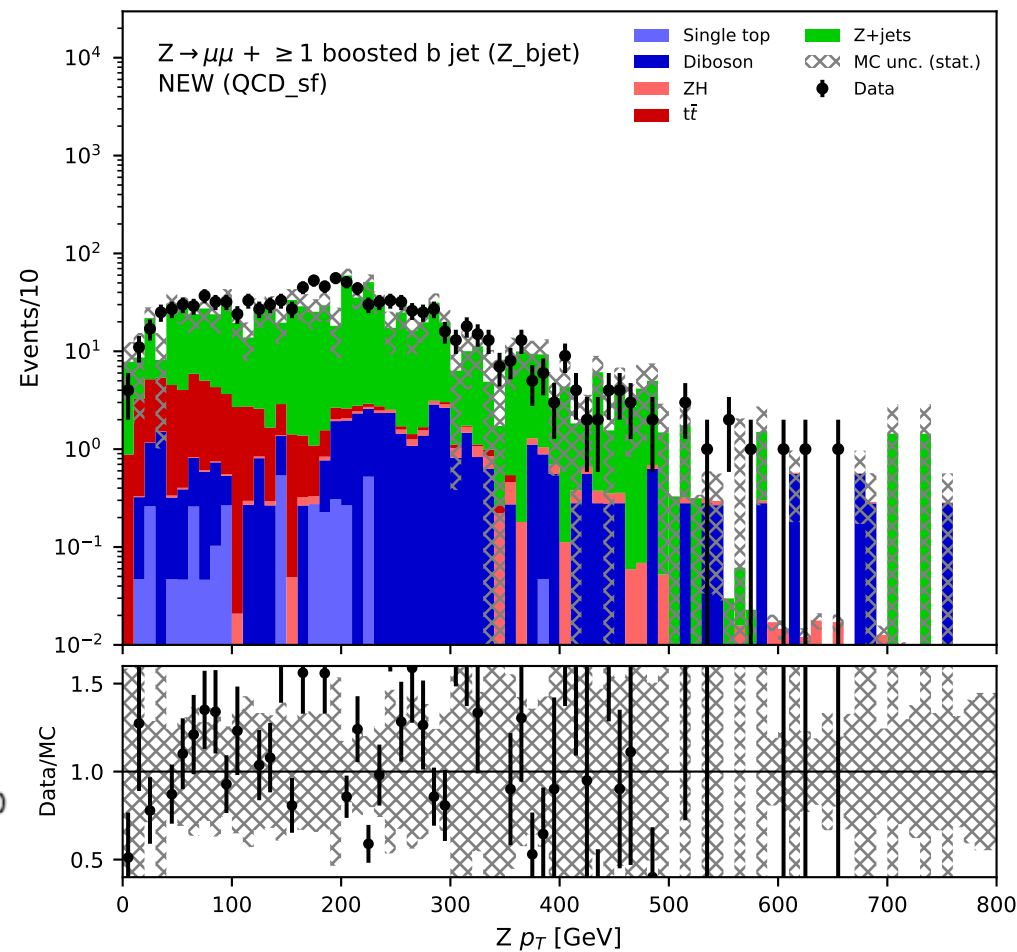
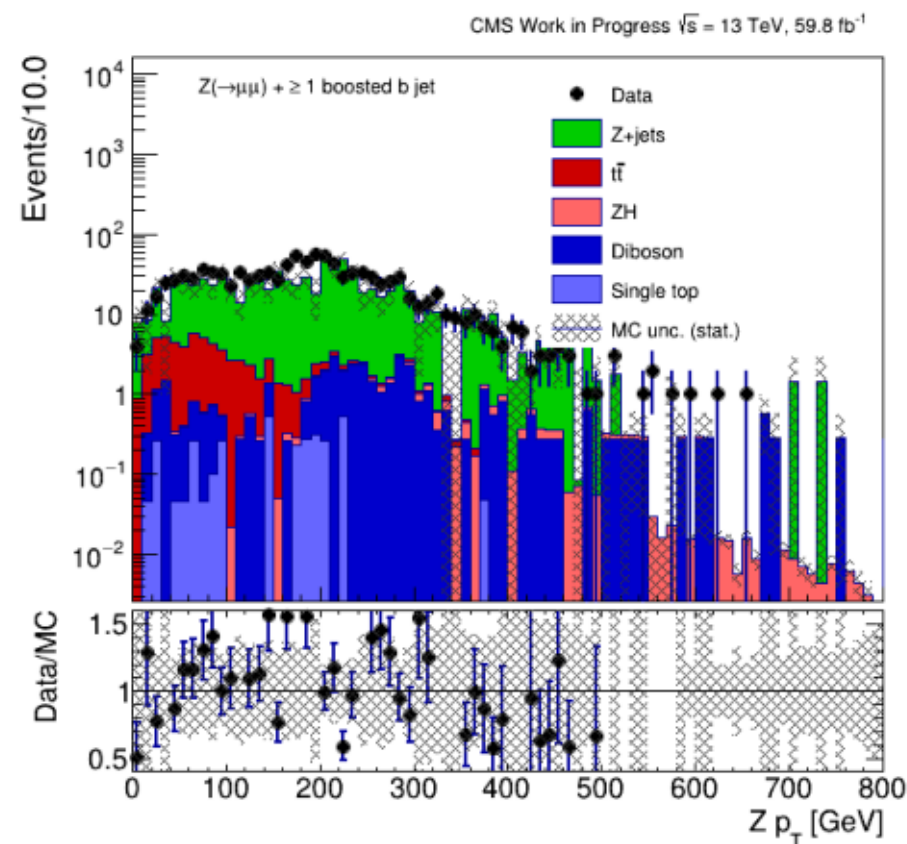


pt_dilepton | Z_bjet | 2018

OLD: pt_dilepton_Z_bjet_Zee

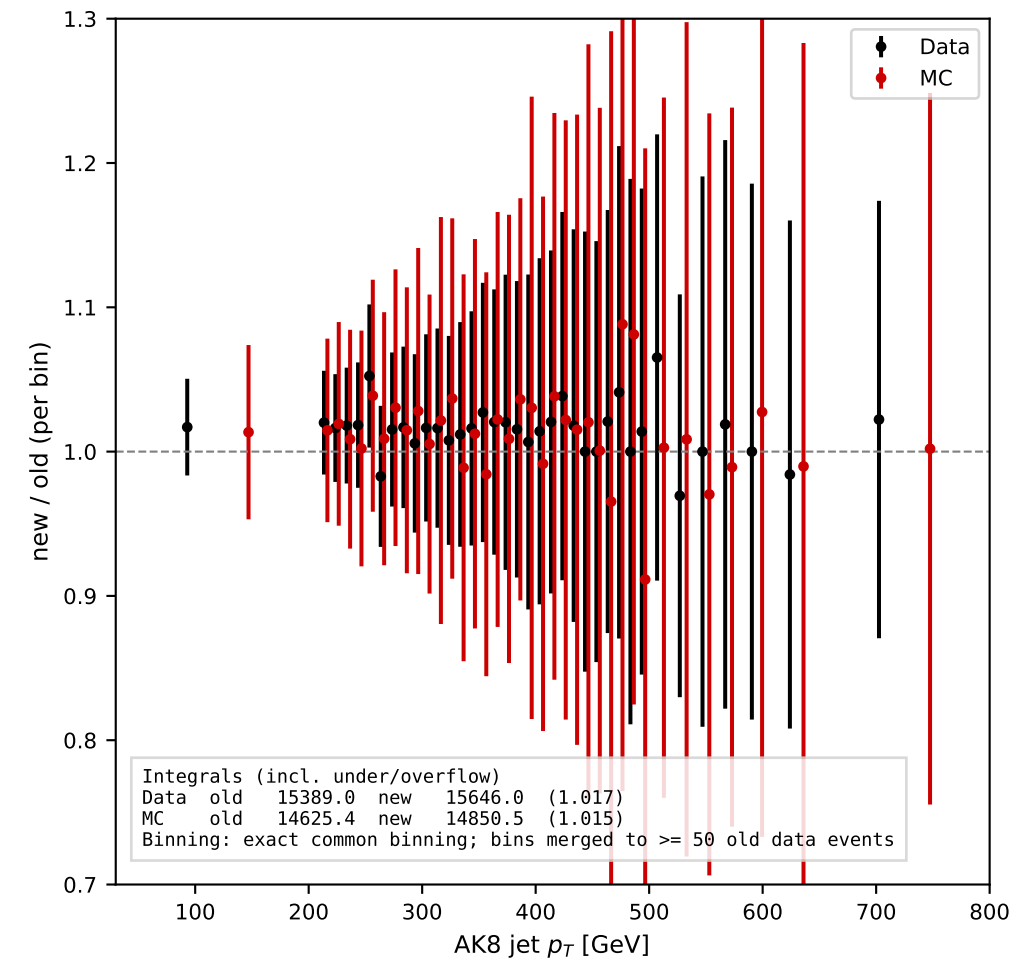
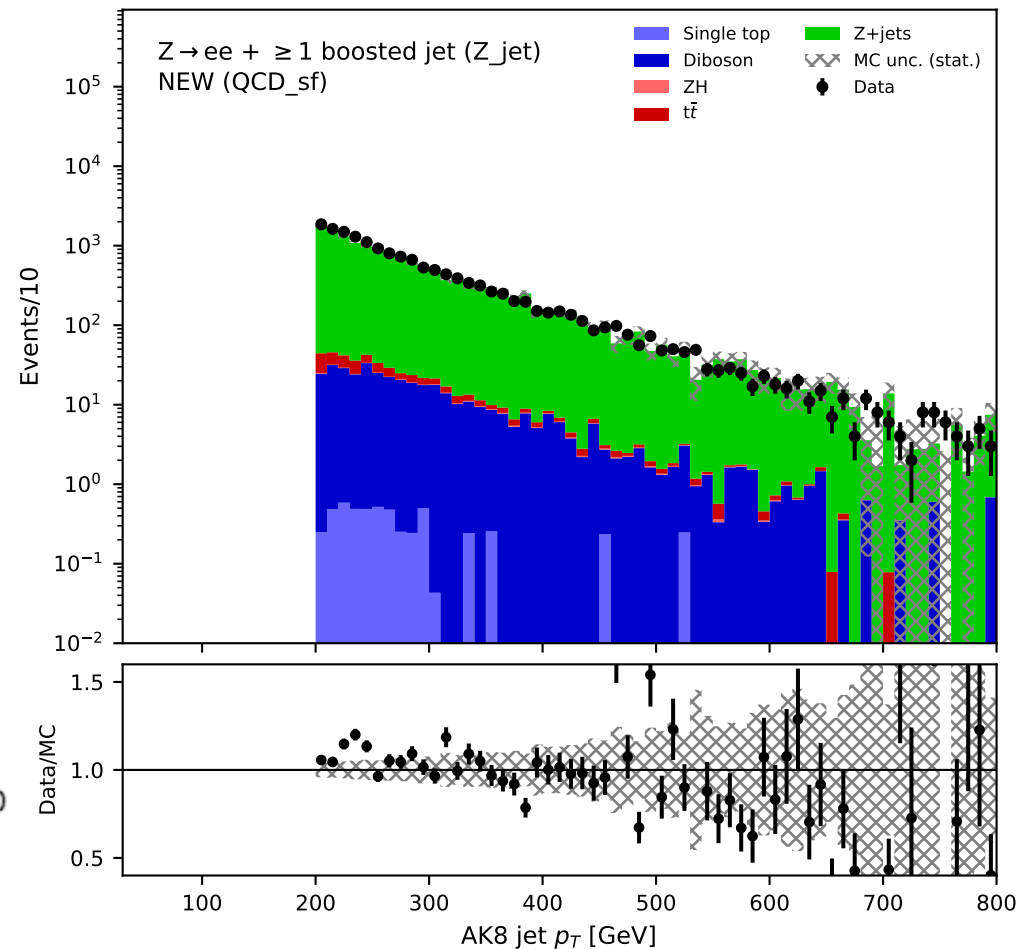
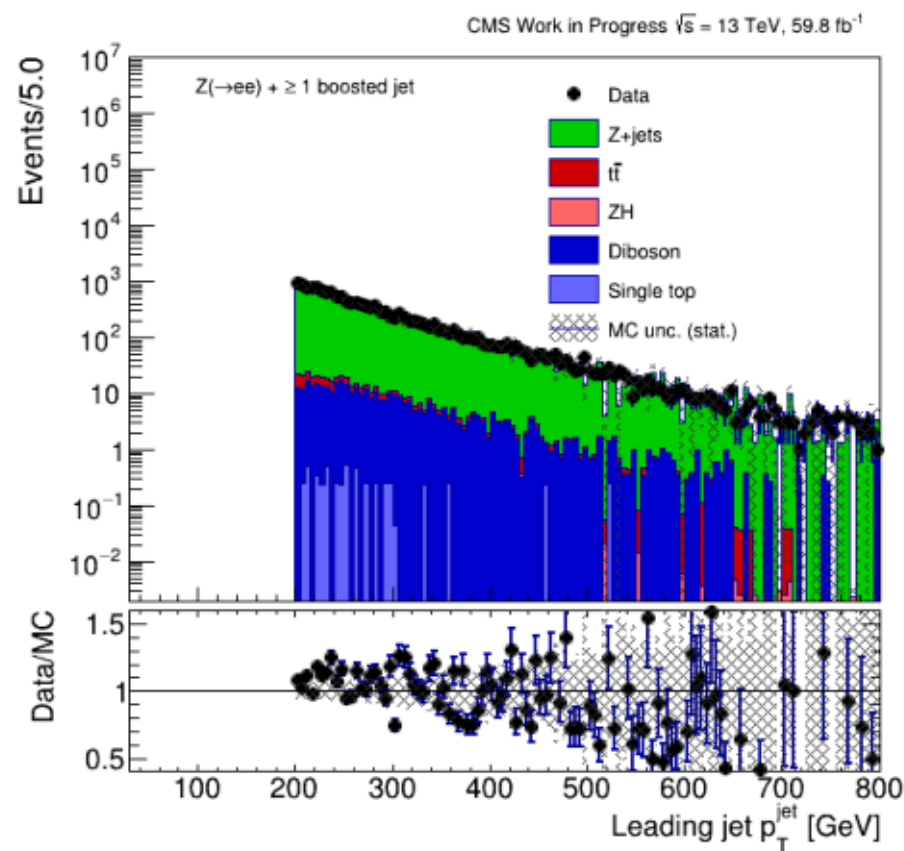


OLD: pt_dilepton_Z_bjet_Zmm

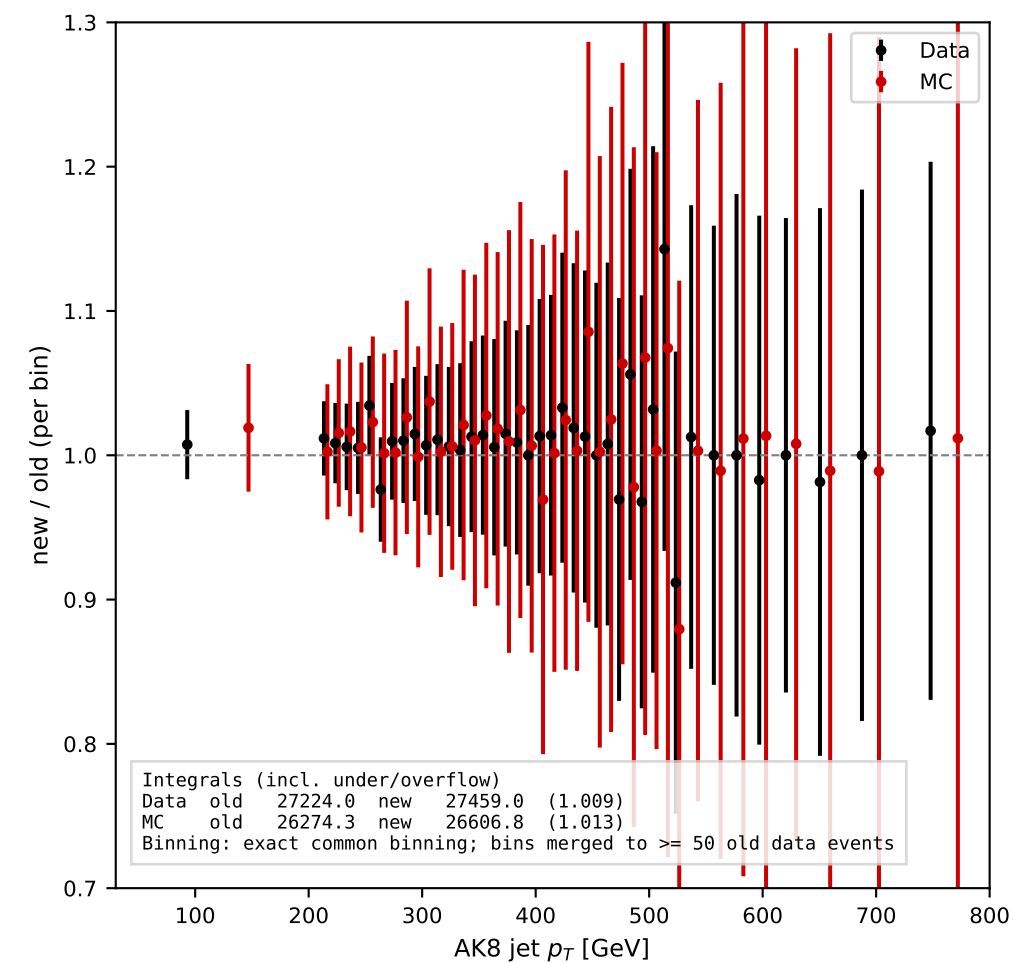
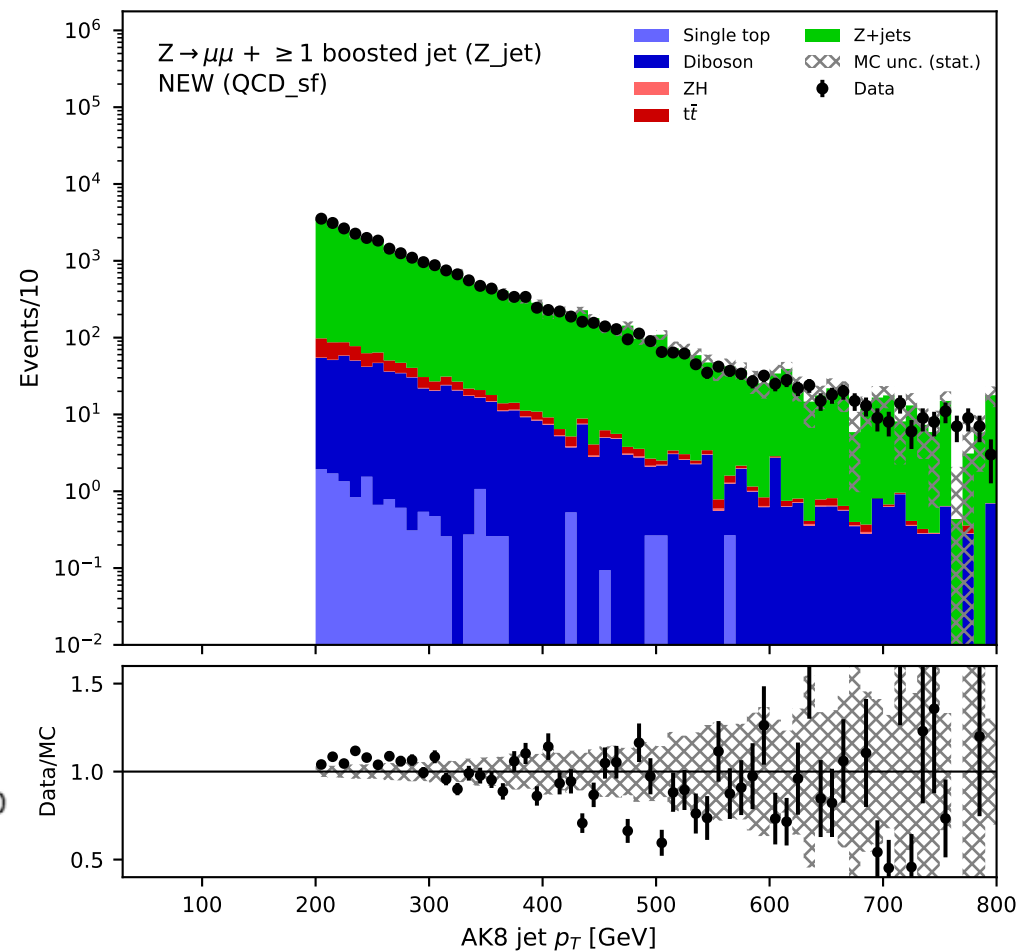
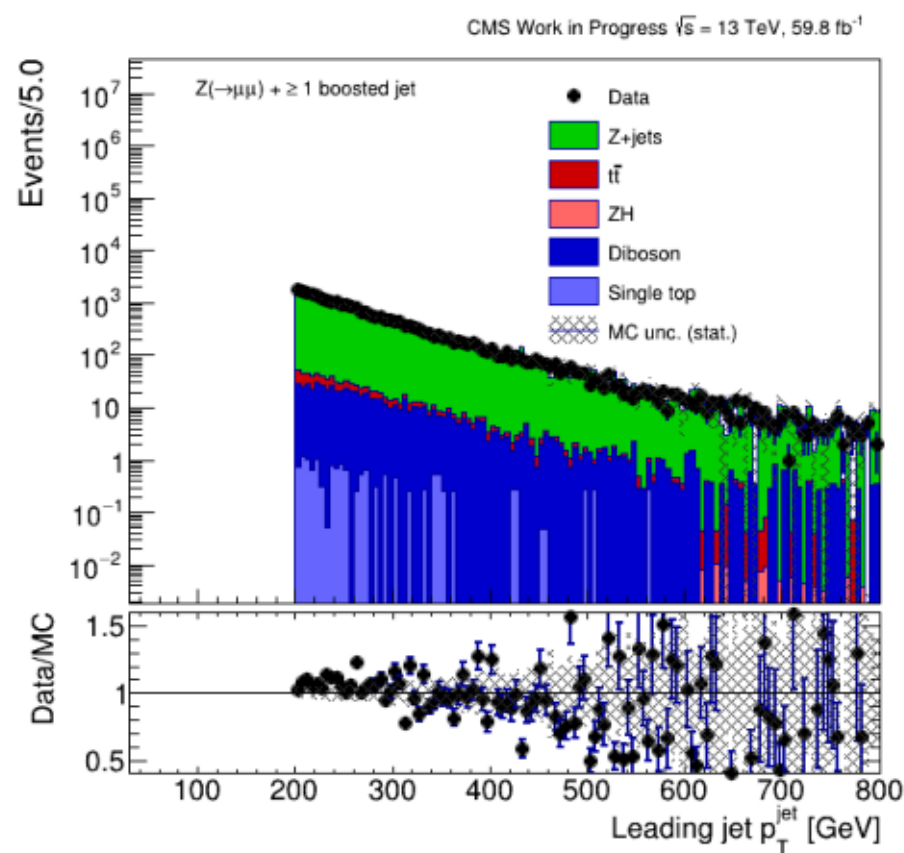


pt_jet | Z_jet | 2018

OLD: pt_jet_Z_jet_Zee

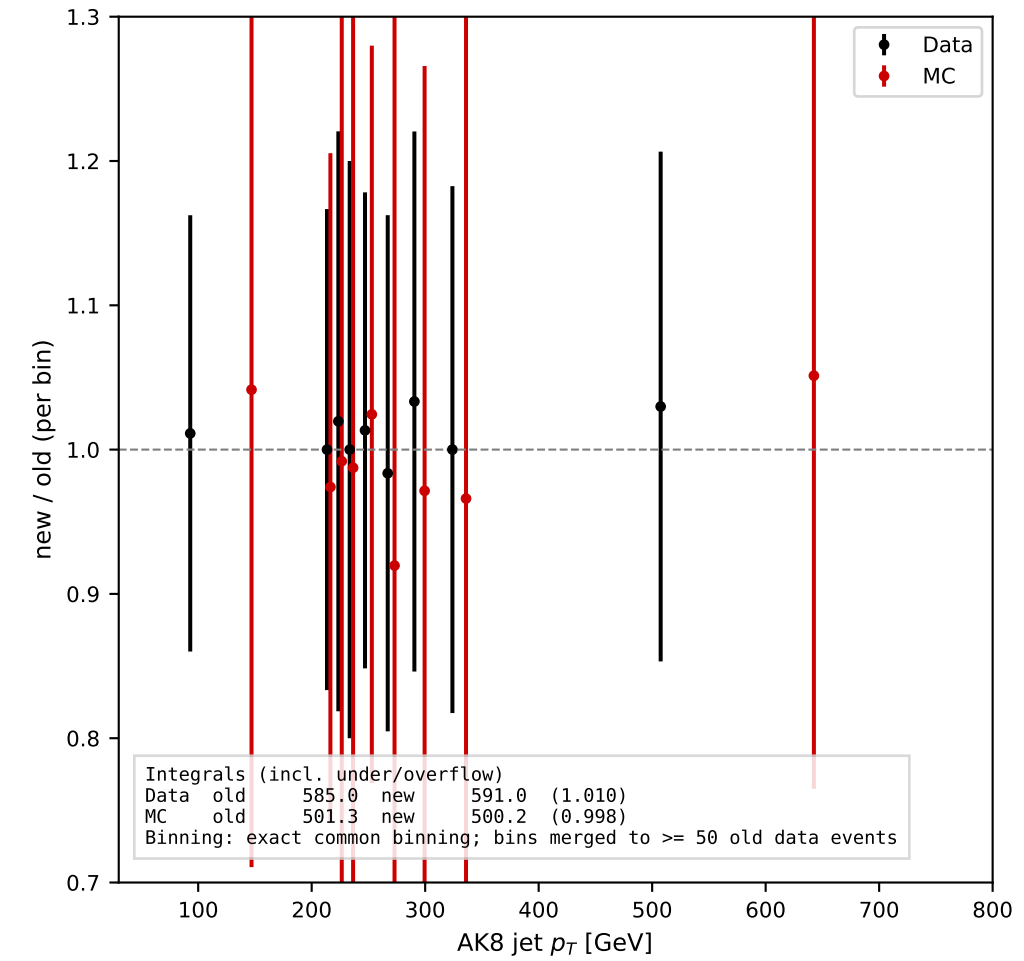
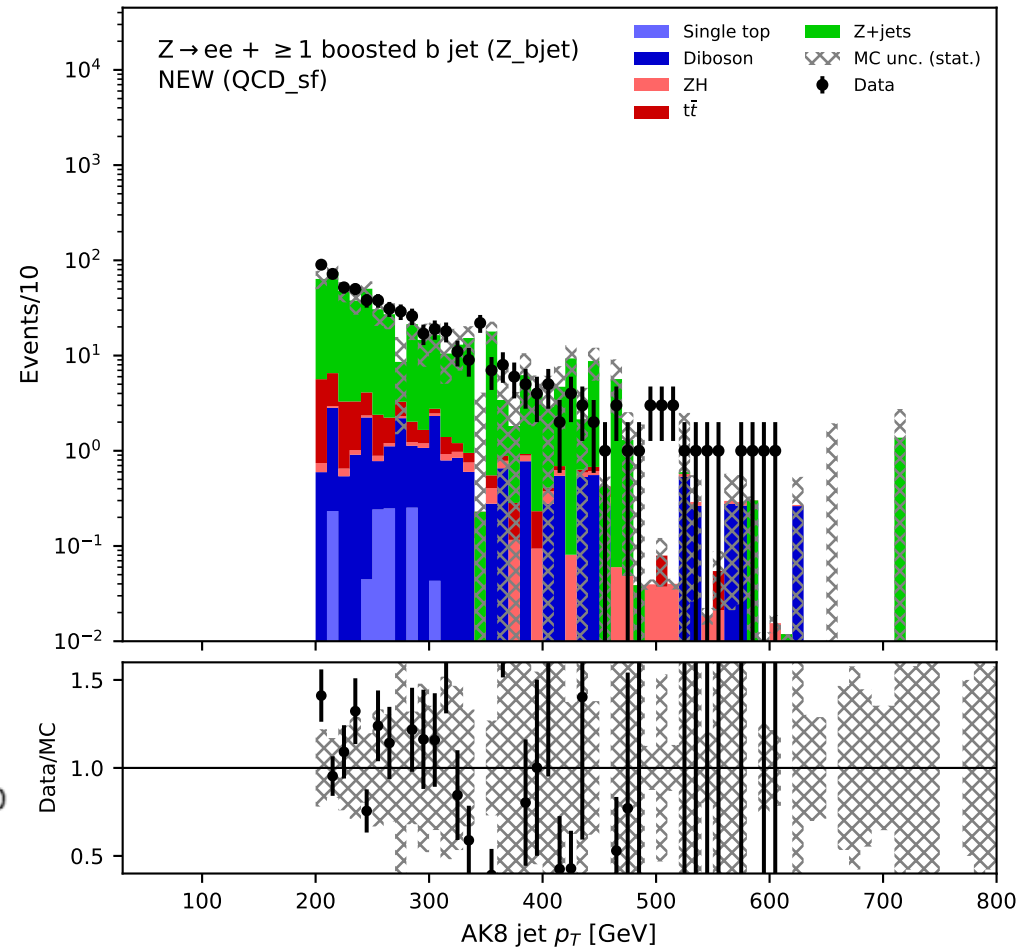
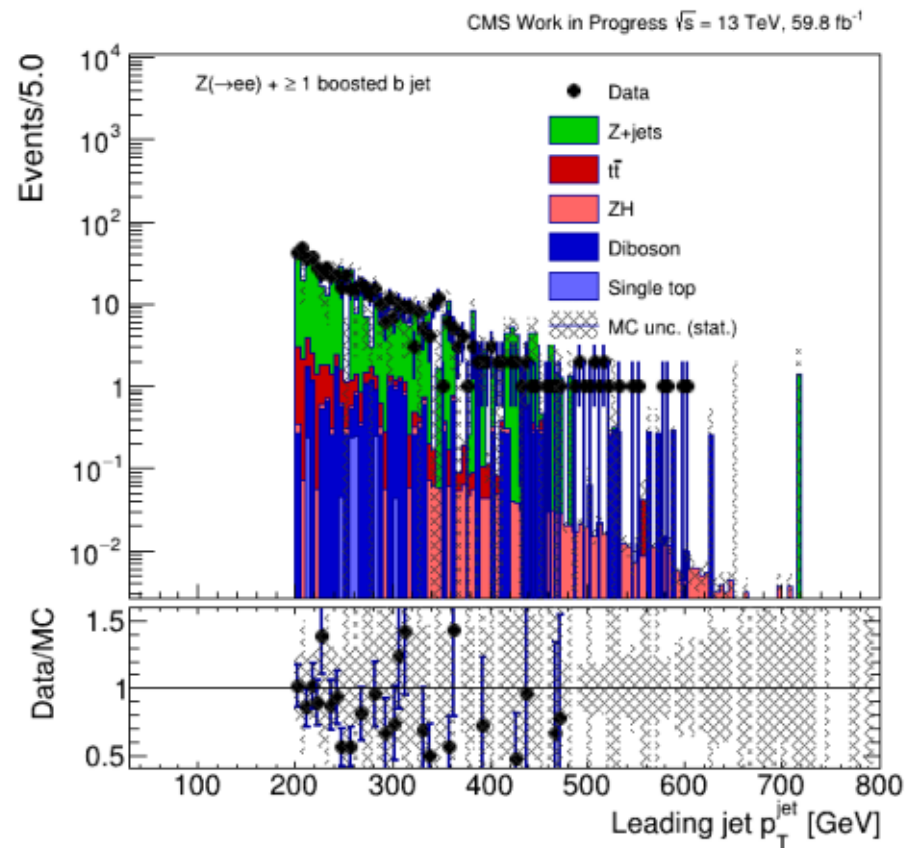


OLD: pt_jet_Z_jet_Zmm

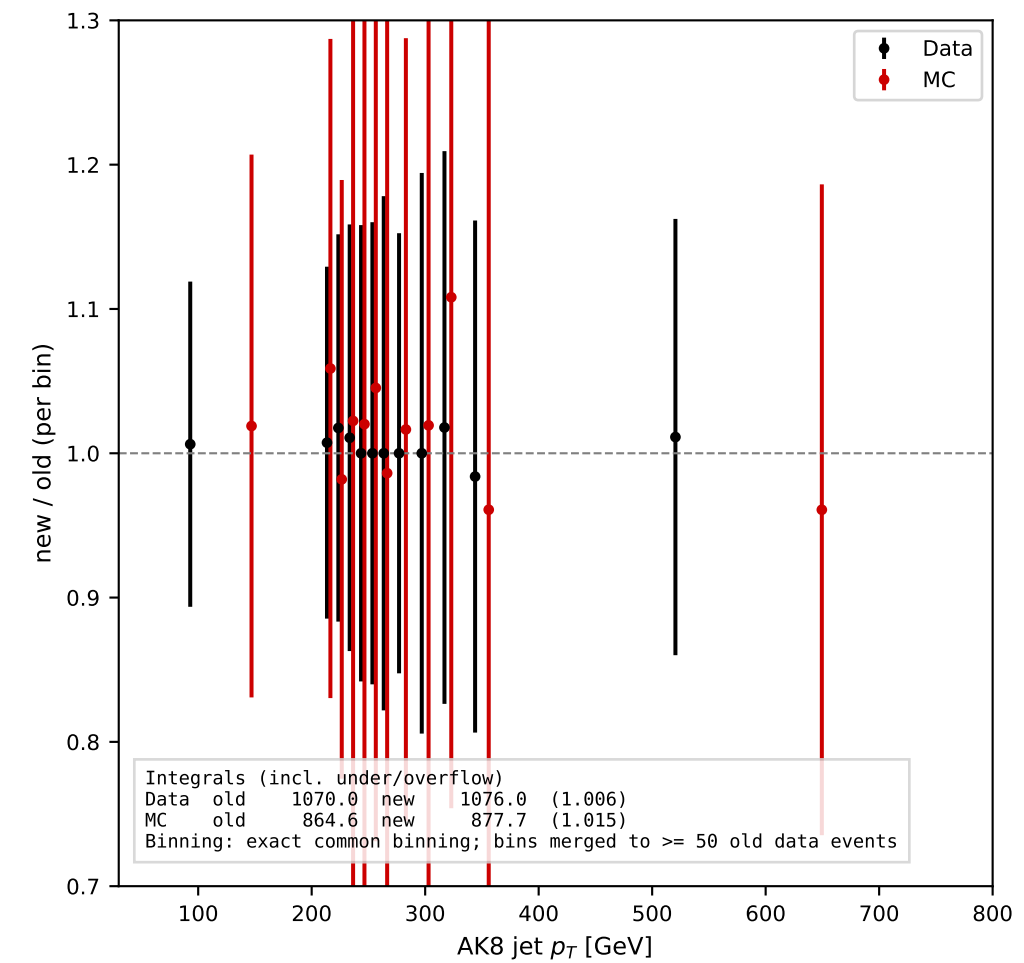
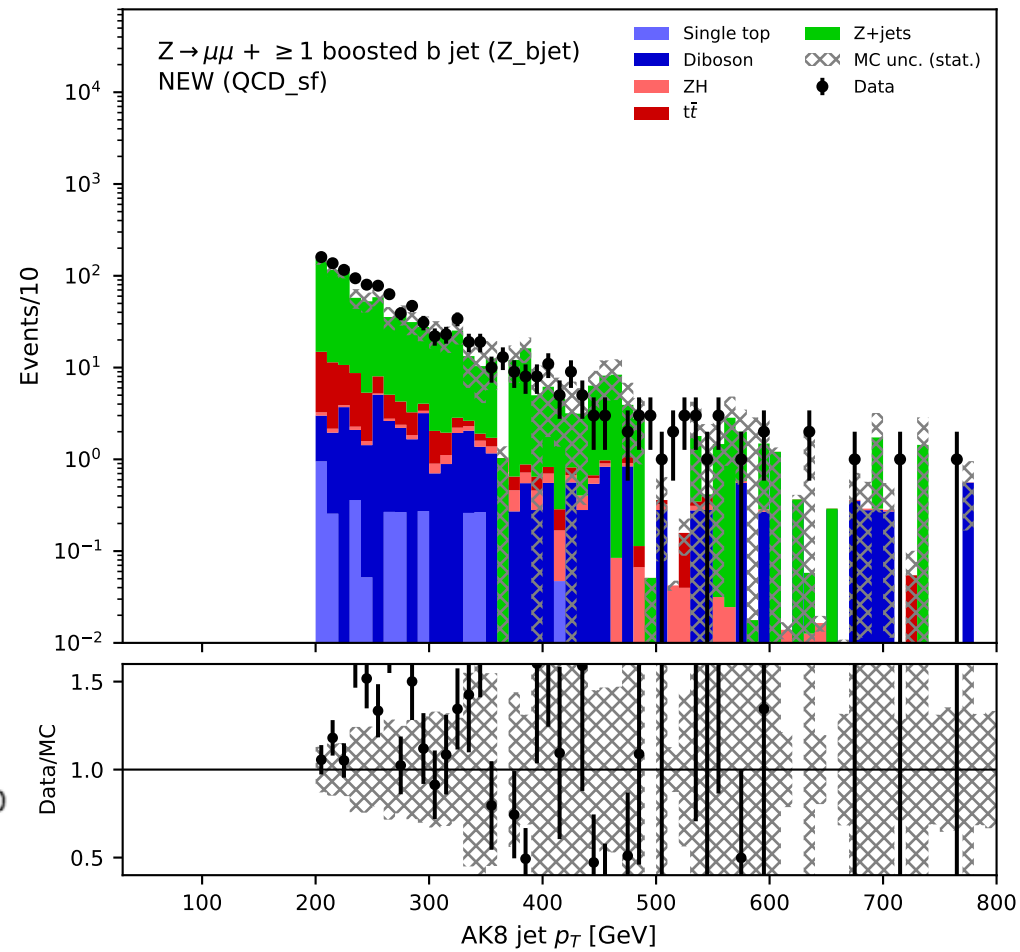
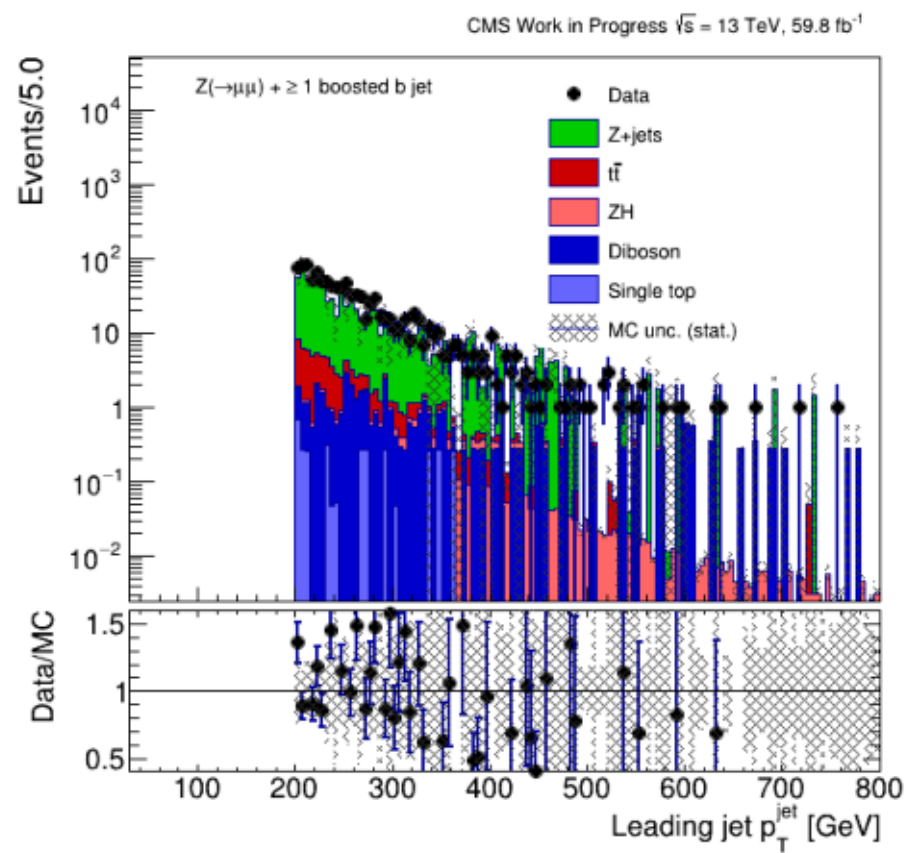


pt_jet | Z_bjet | 2018

OLD: pt_jet_Z_bjet_Zee

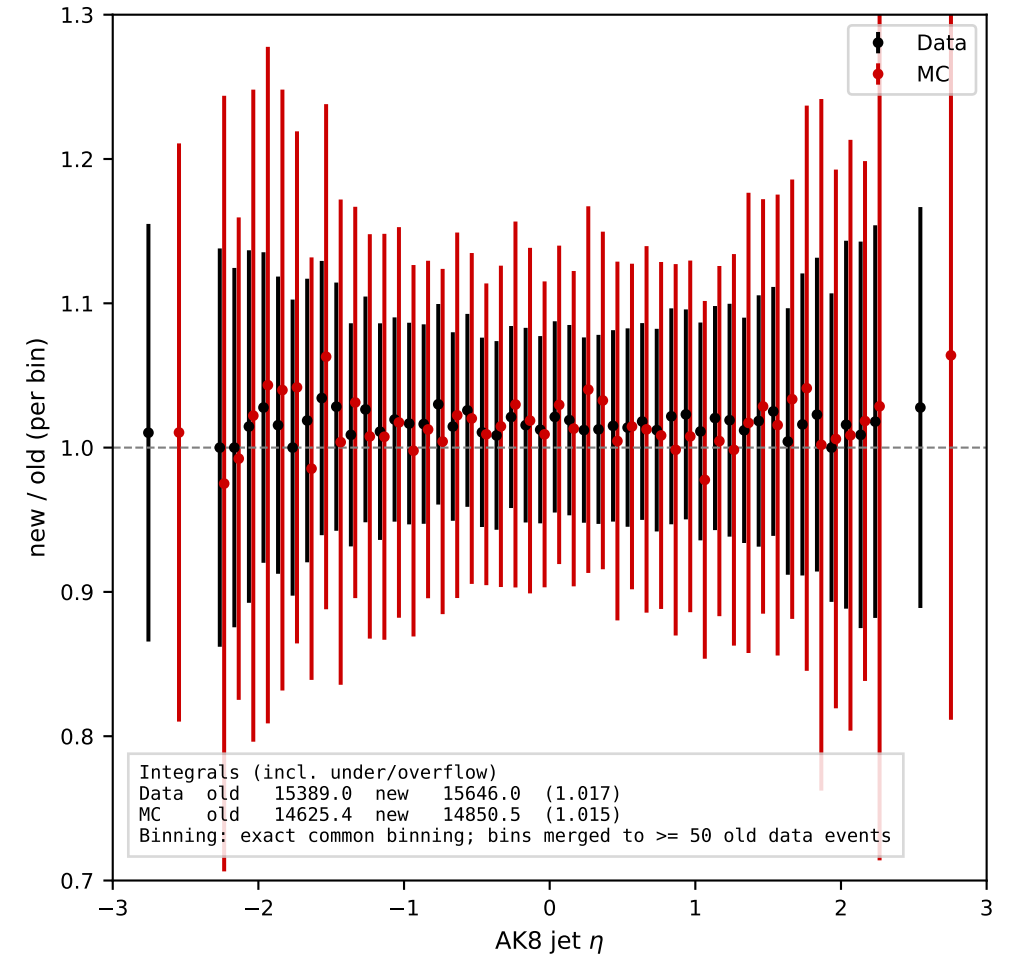
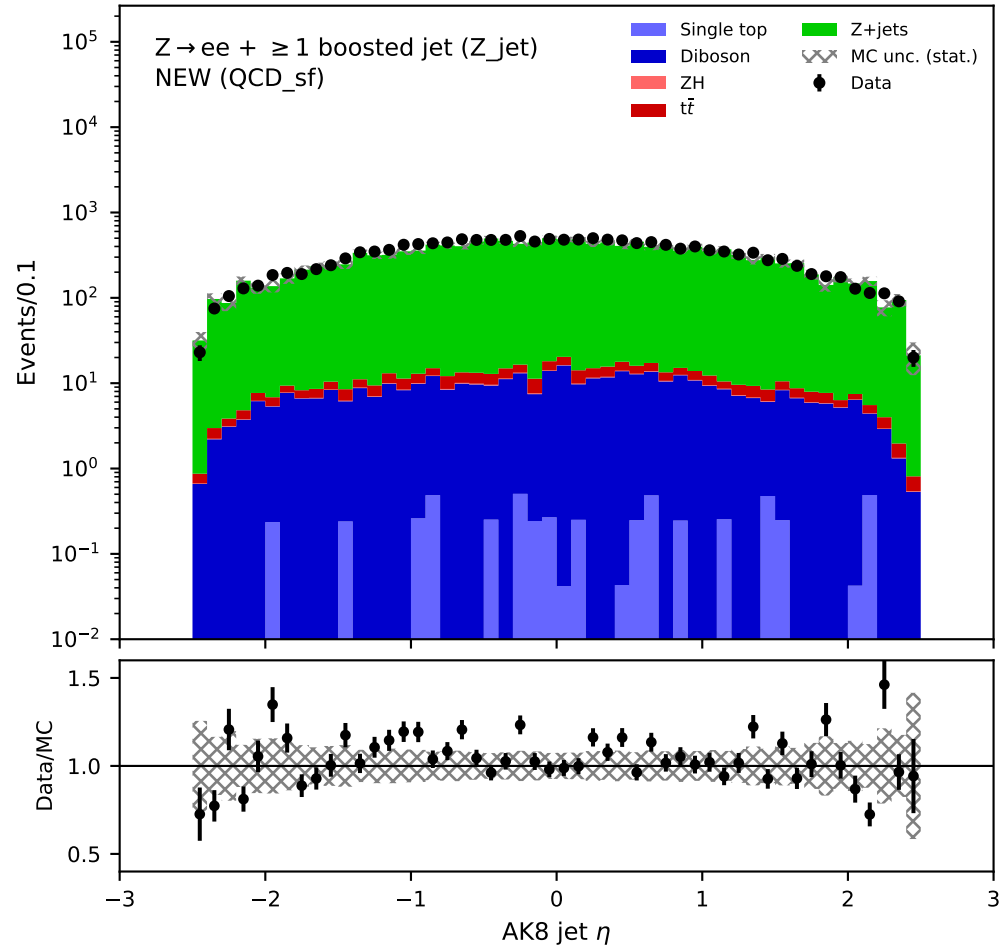
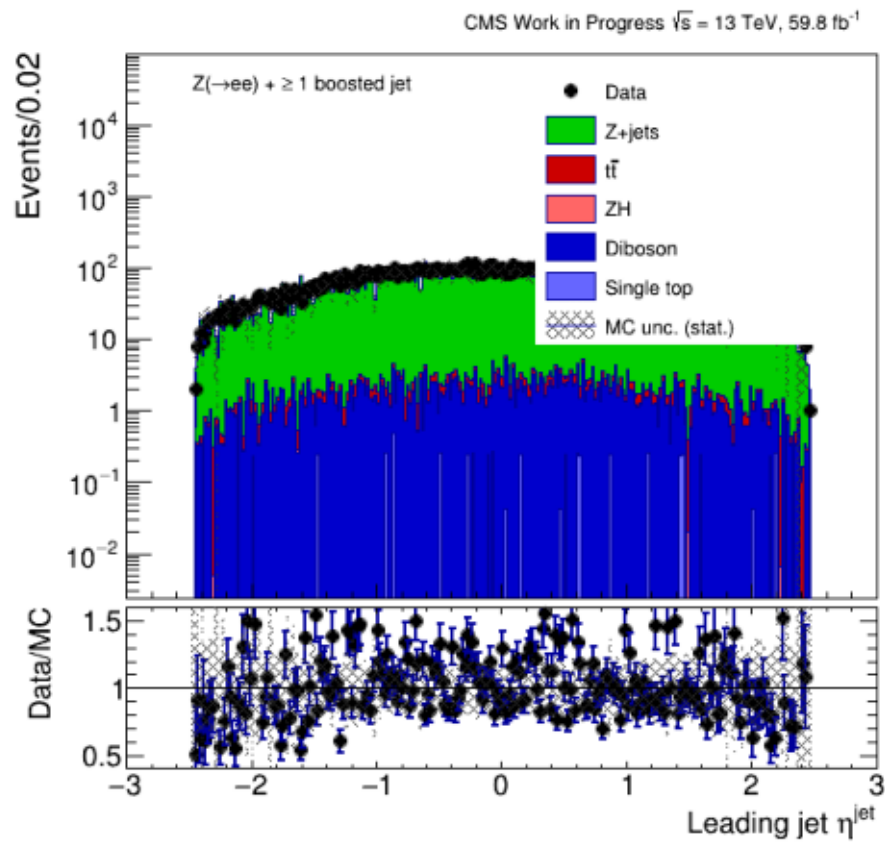


OLD: pt_jet_Z_bjet_Zmm

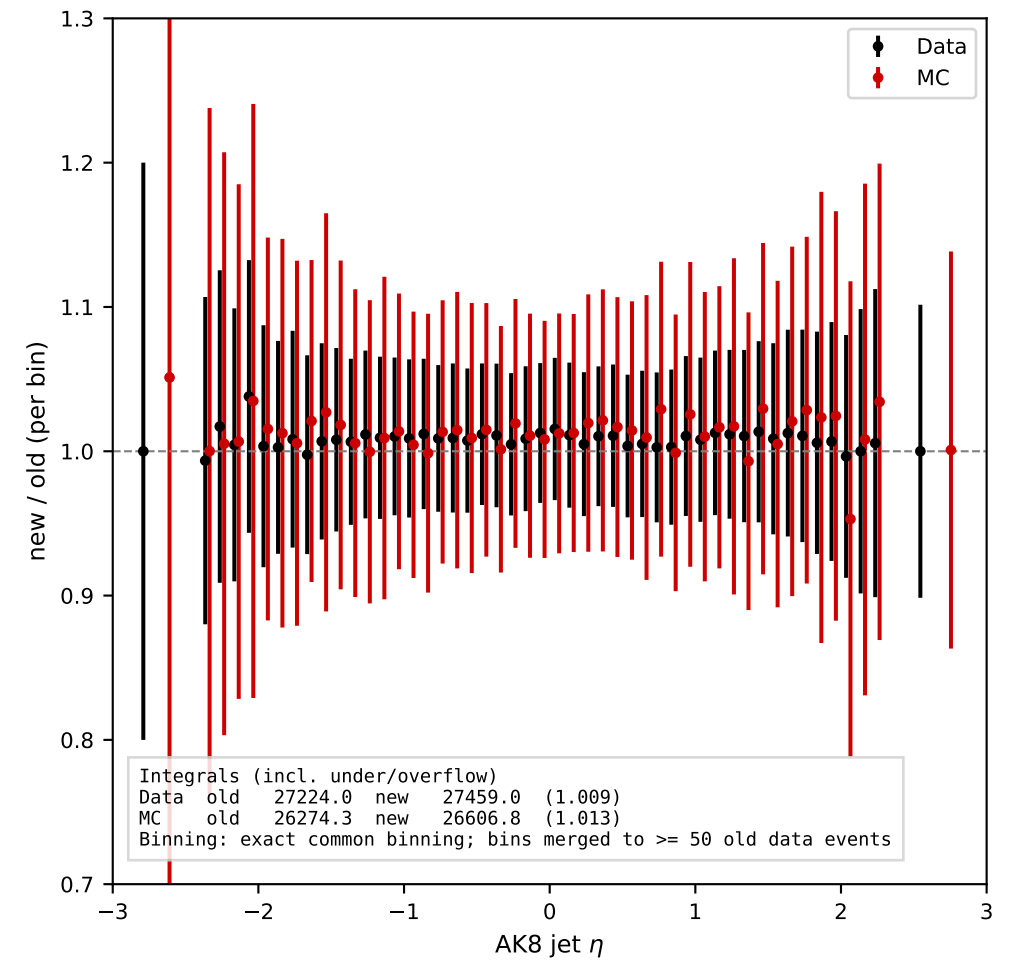
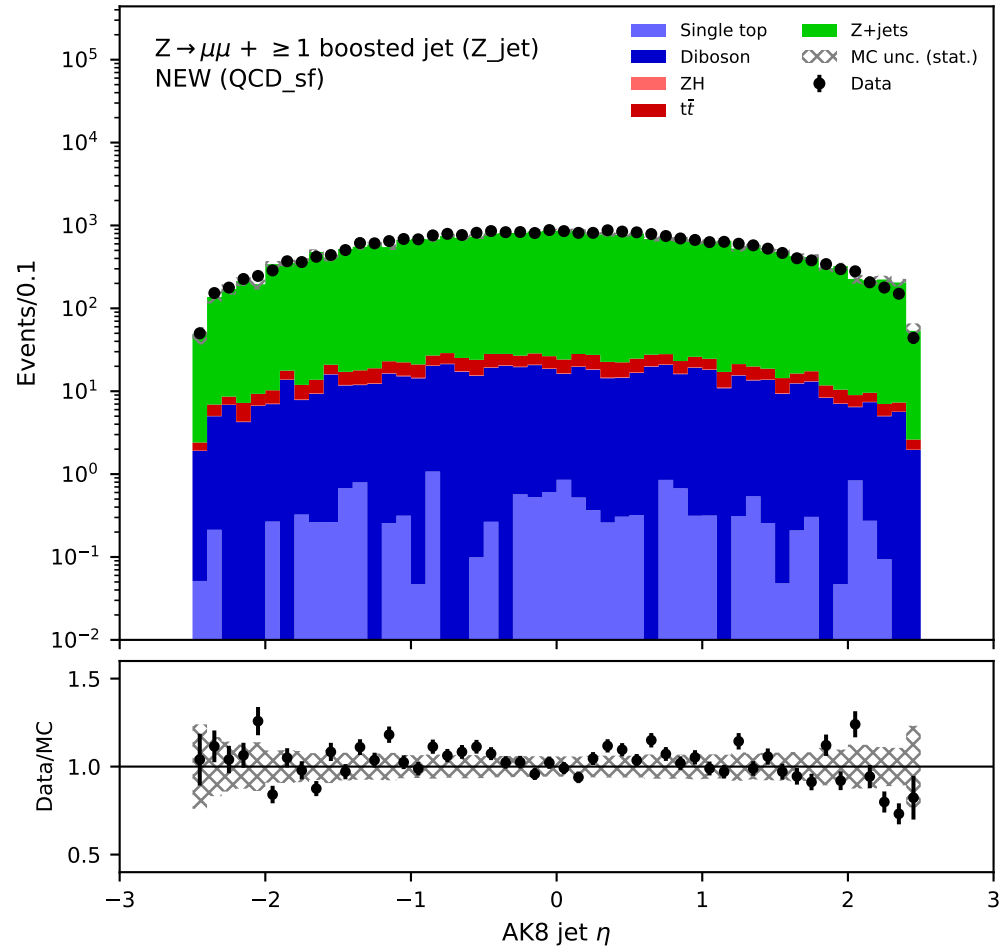
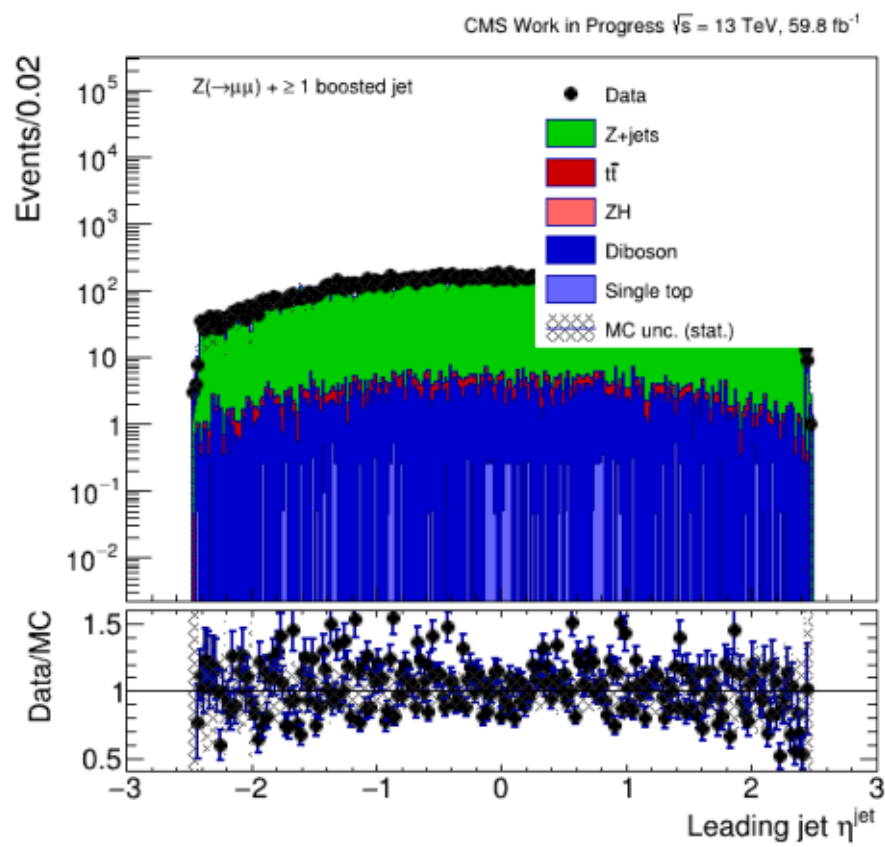


eta_jet | Z_jet | 2018

OLD: eta_jet_Z_jet_Zee

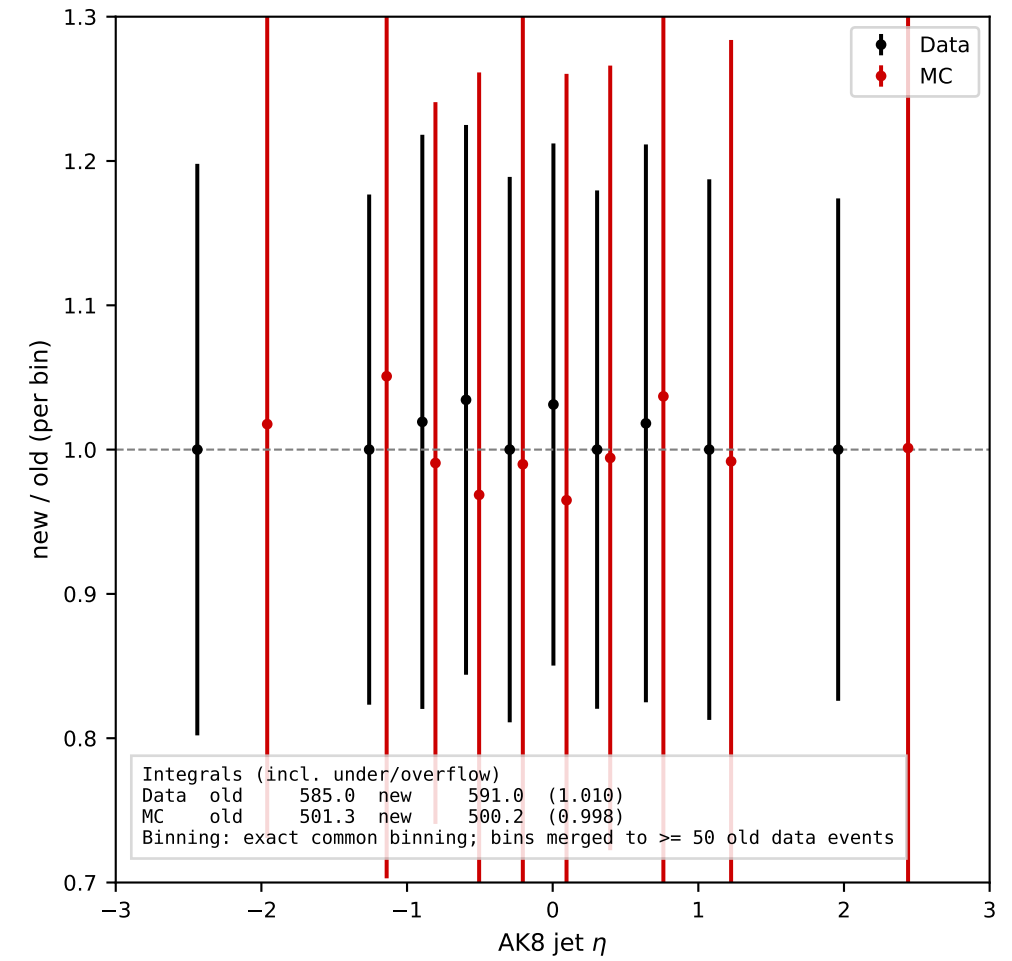
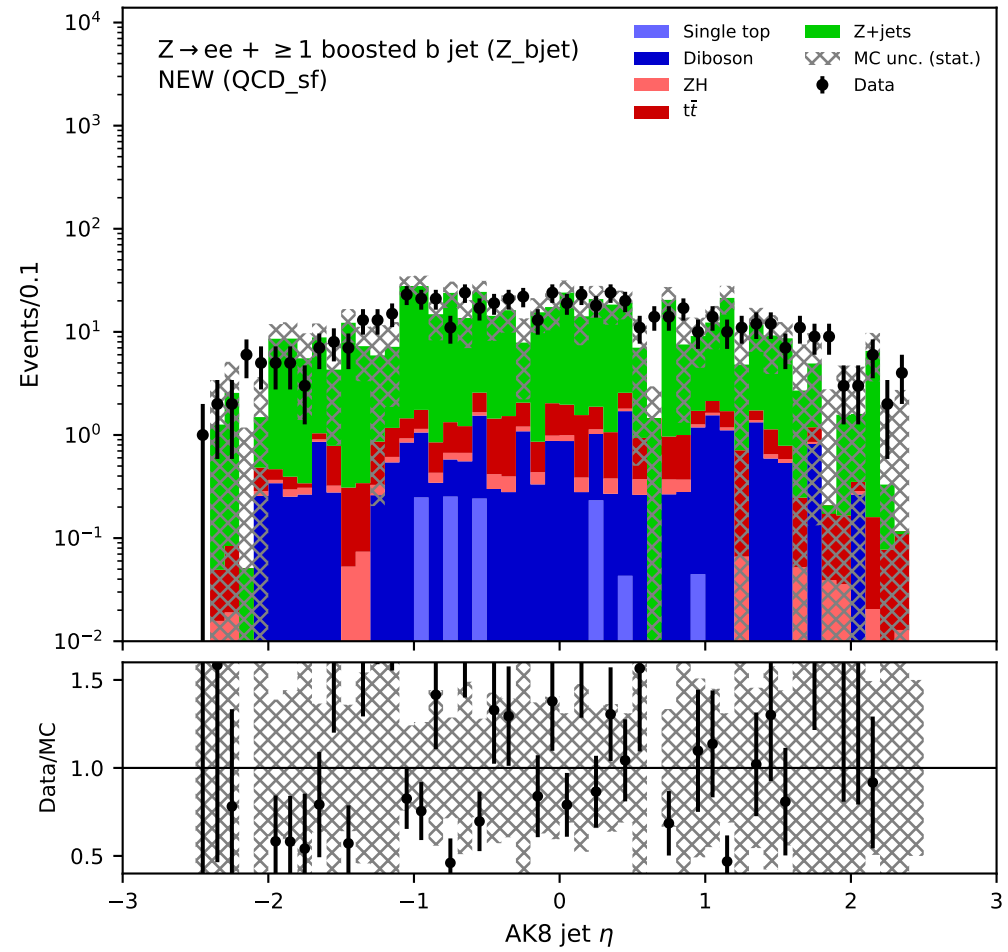
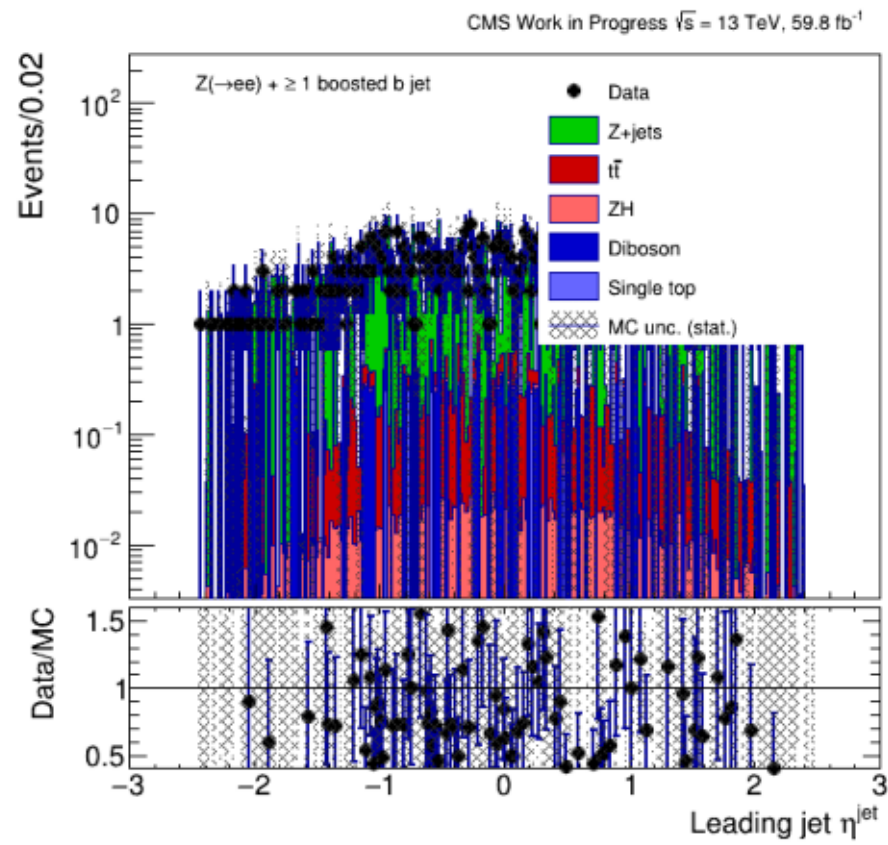


OLD: eta_jet_Z_jet_Zmm

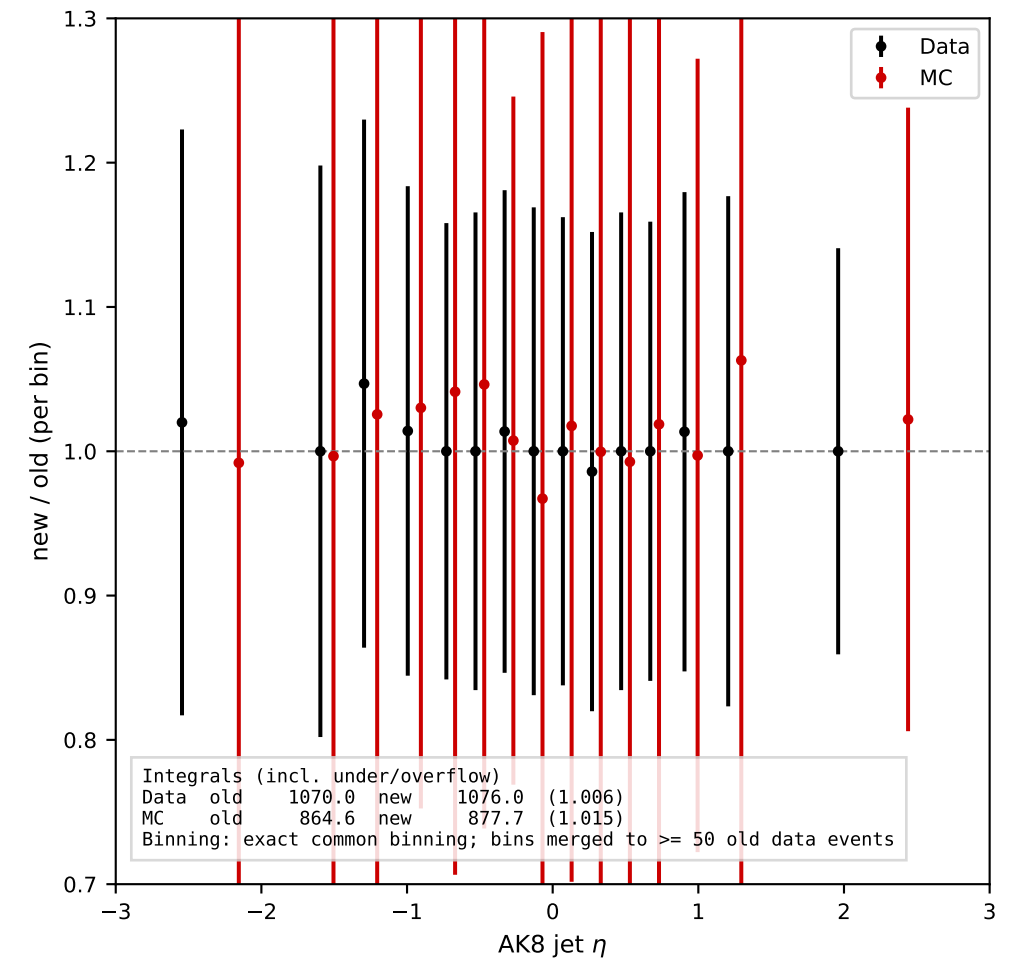
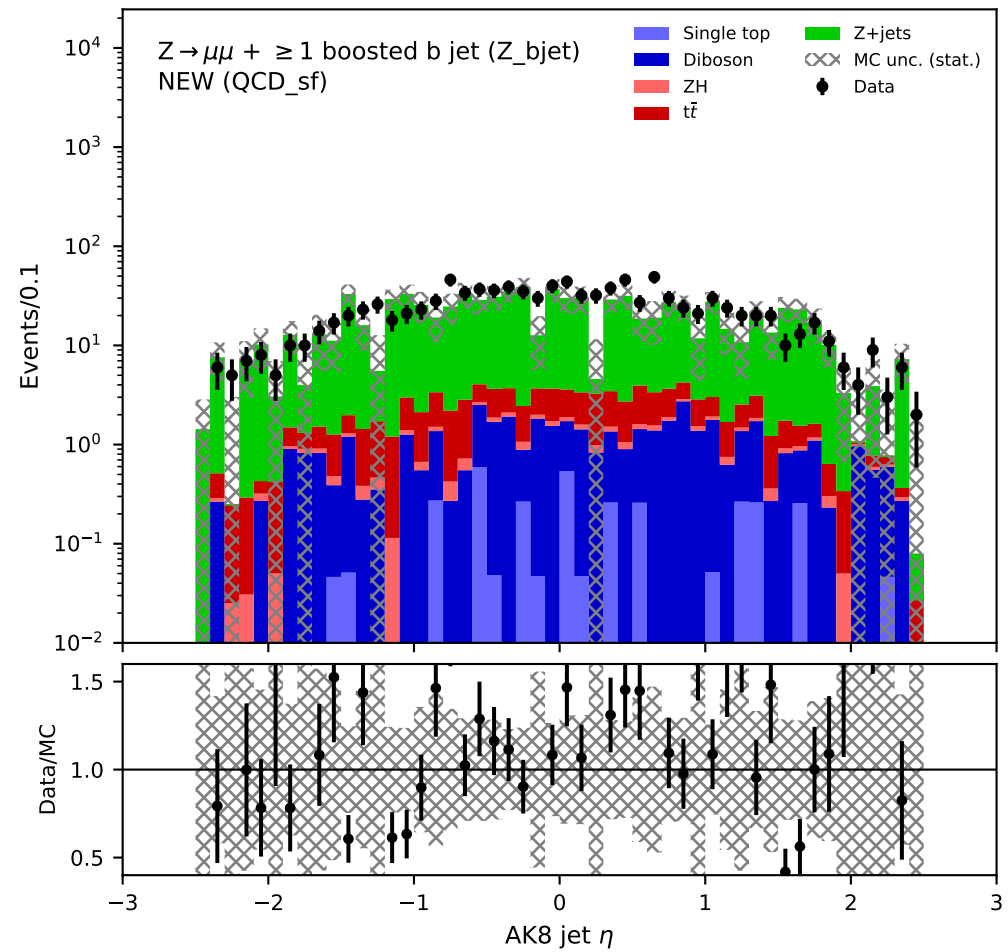
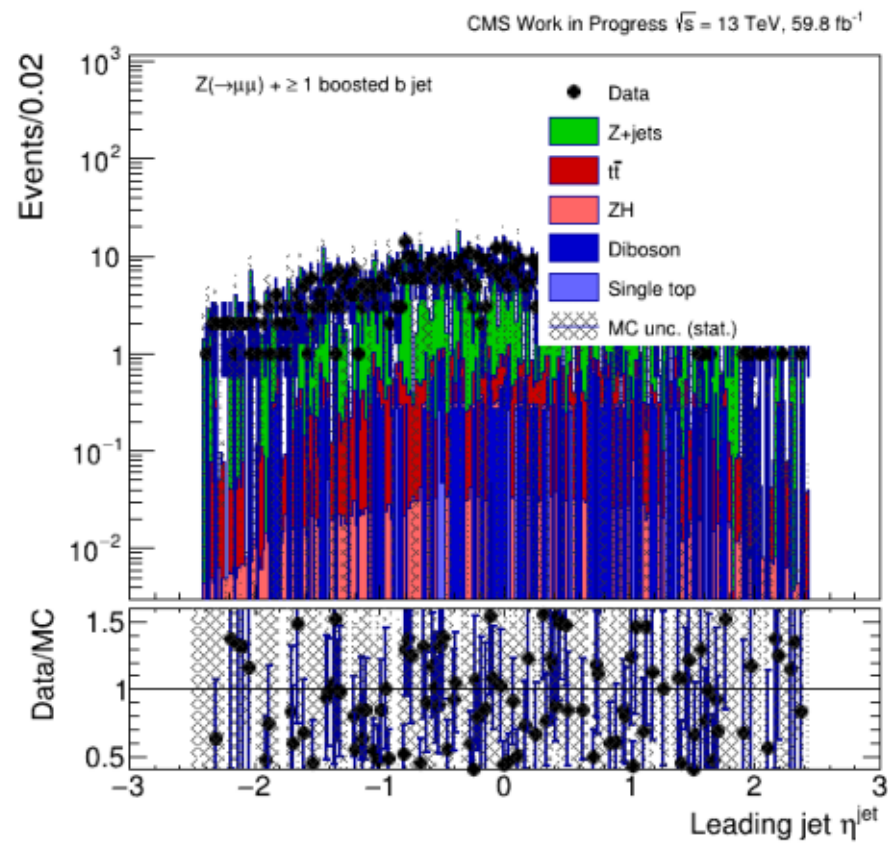


eta_jet | Z_bjet | 2018

OLD: eta_jet_Z_bjet_Zee



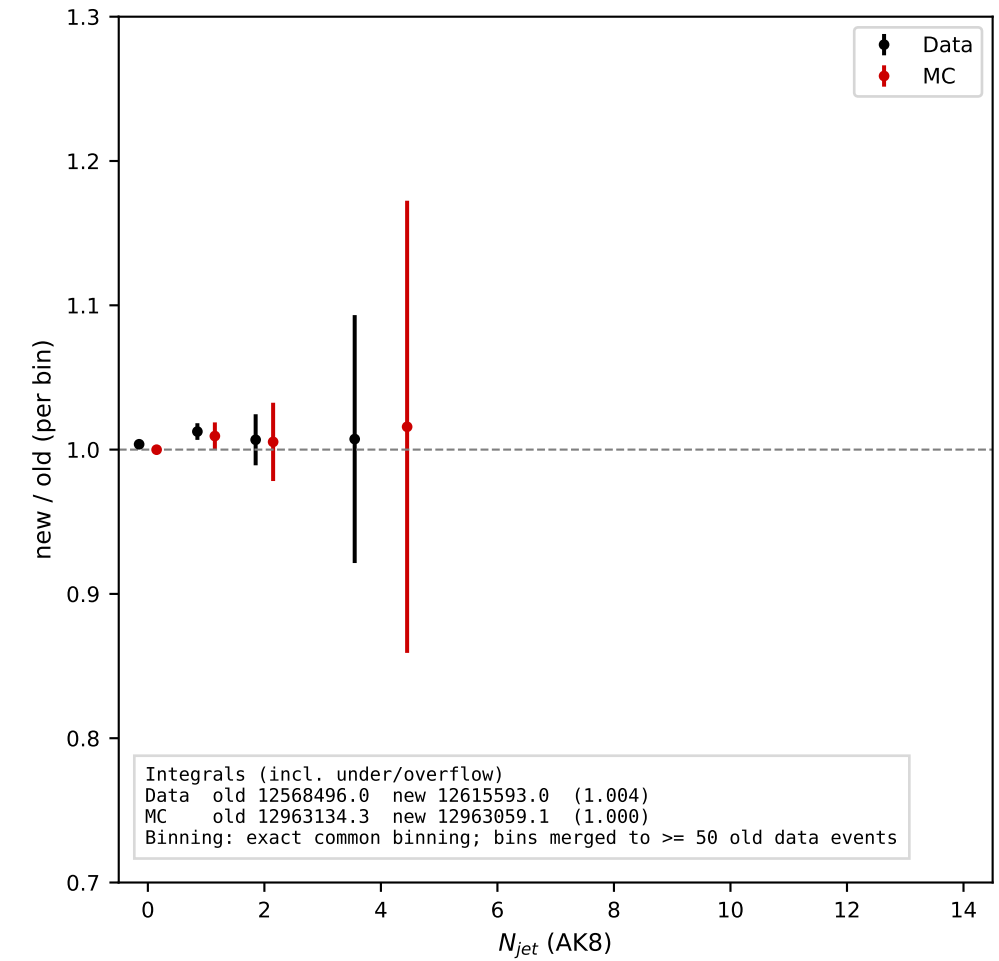
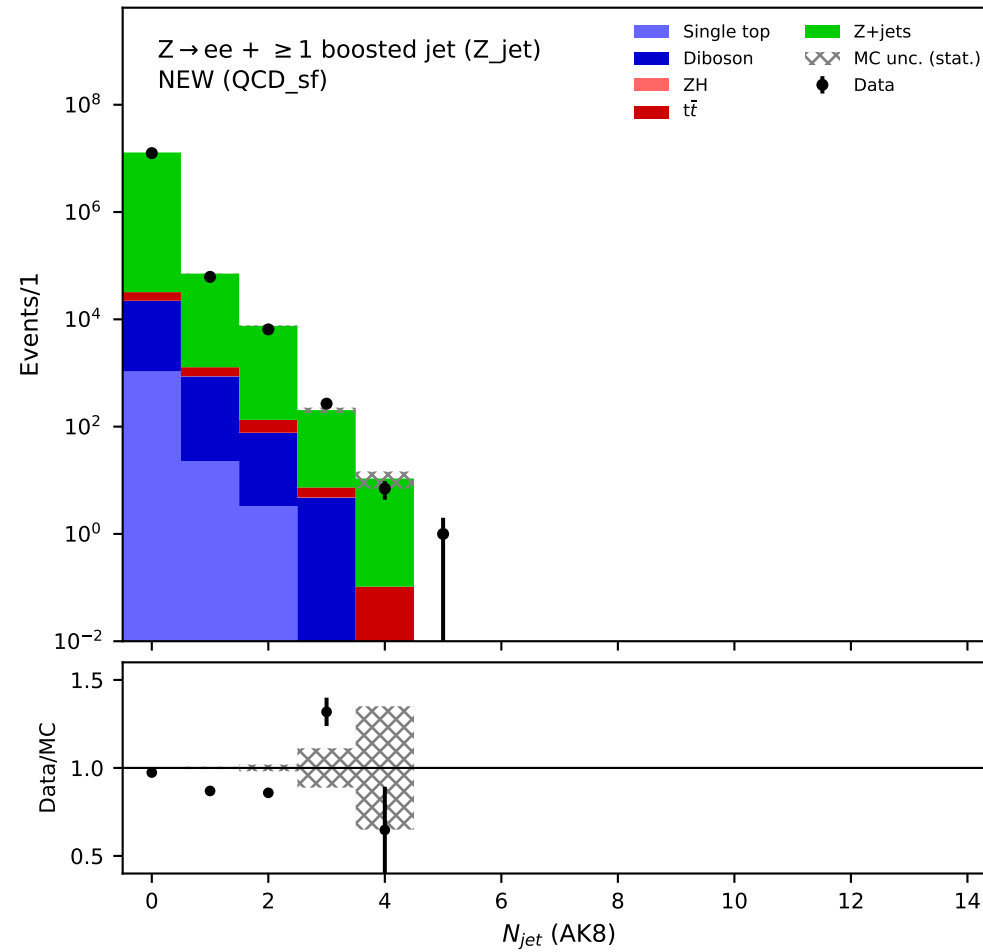
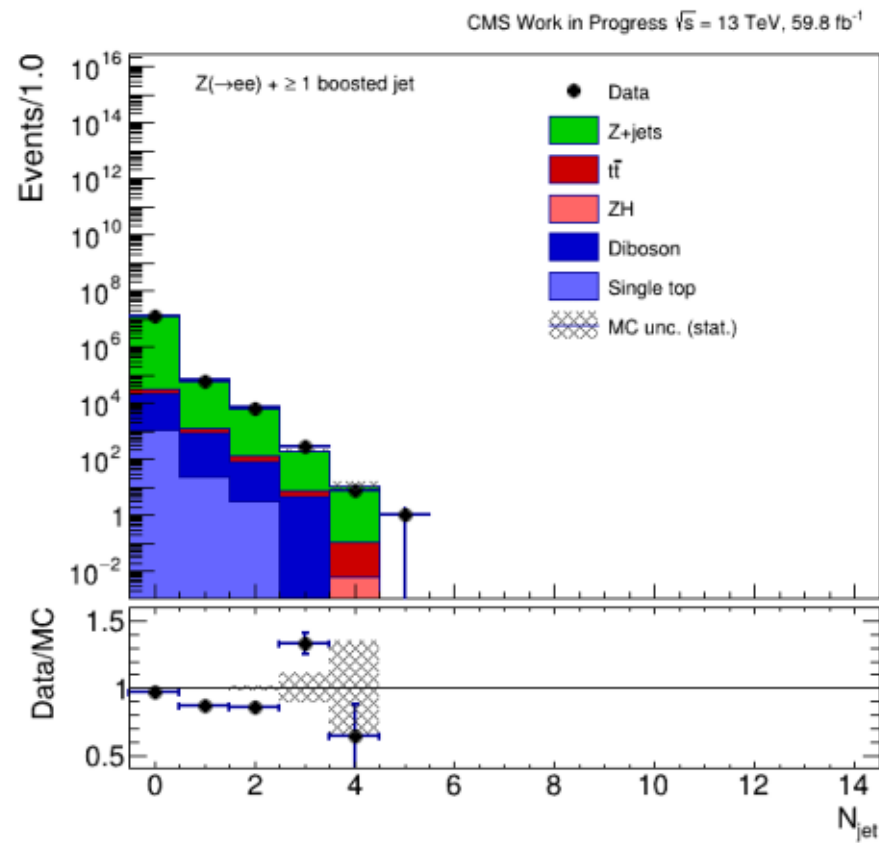
OLD: eta_jet_Z_bjet_Zmm



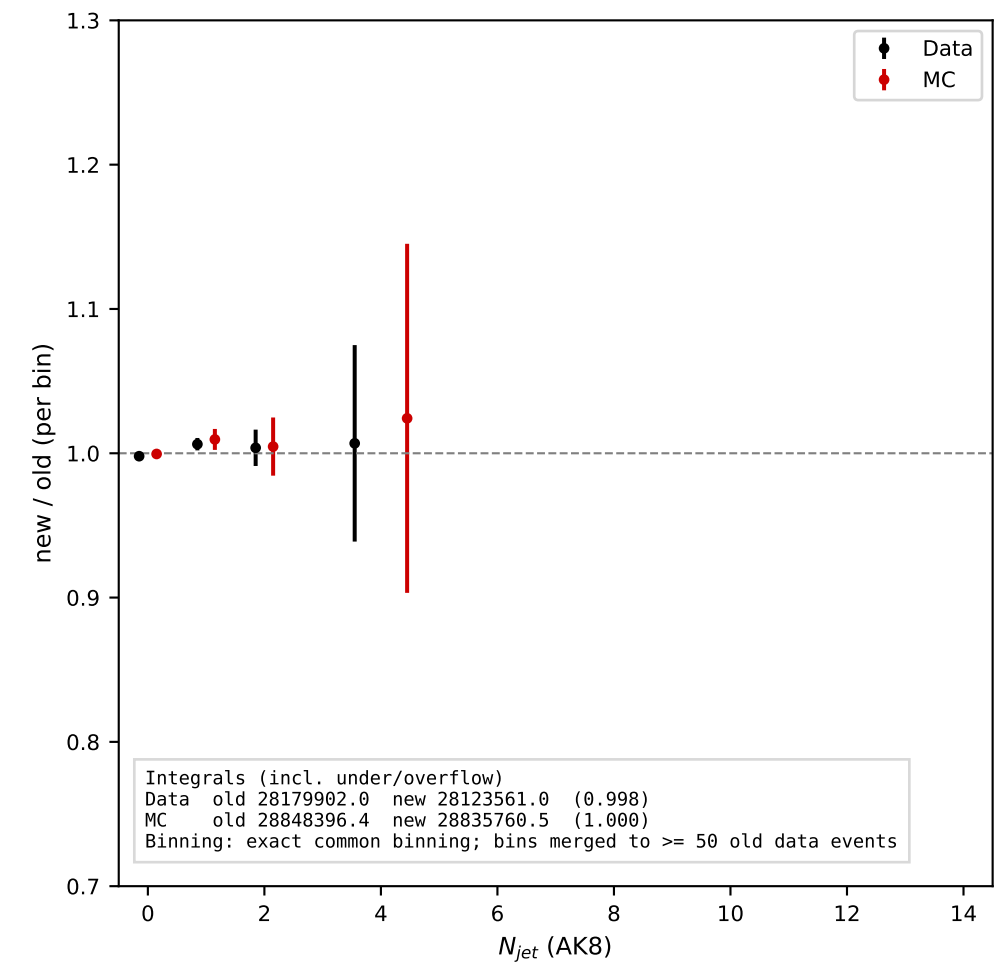
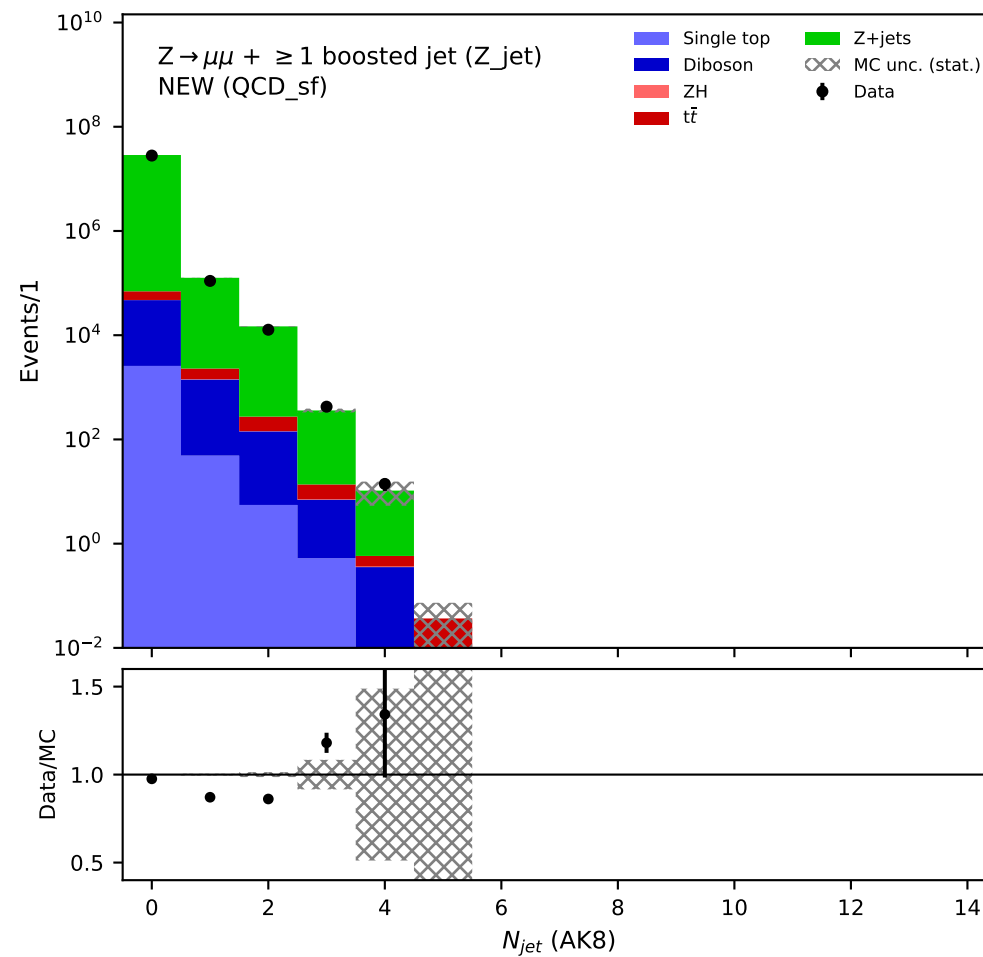
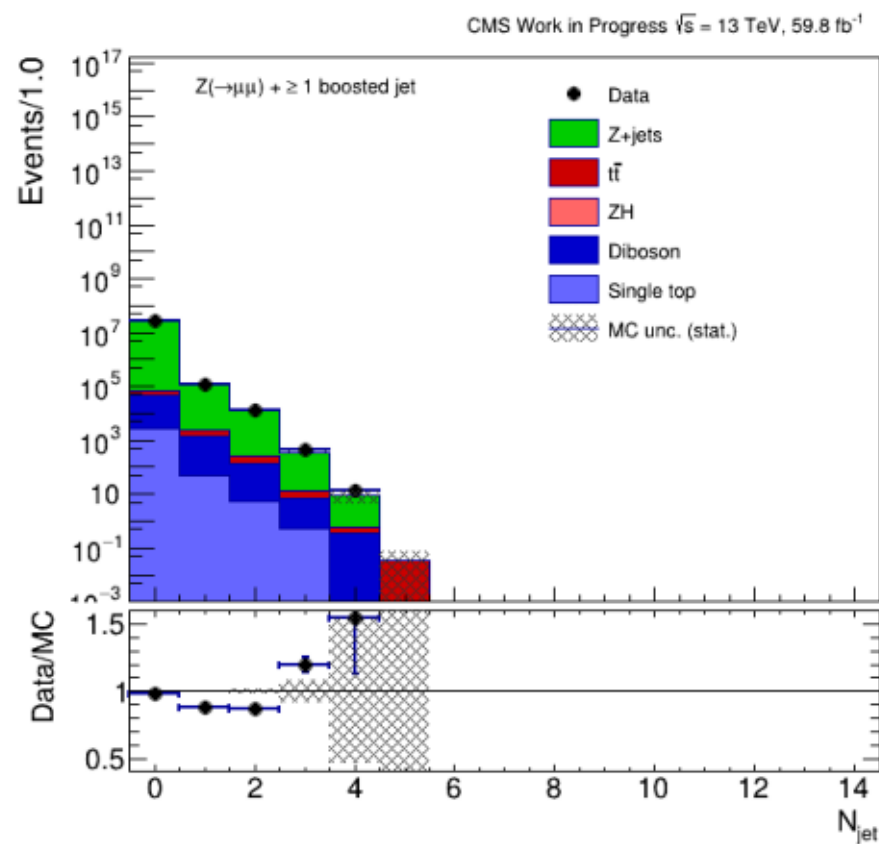
Njet | Z_jet | 2018

Filled before the jet requirement, as in the old FillNjet.

OLD: Njet_Z_jet_Zee



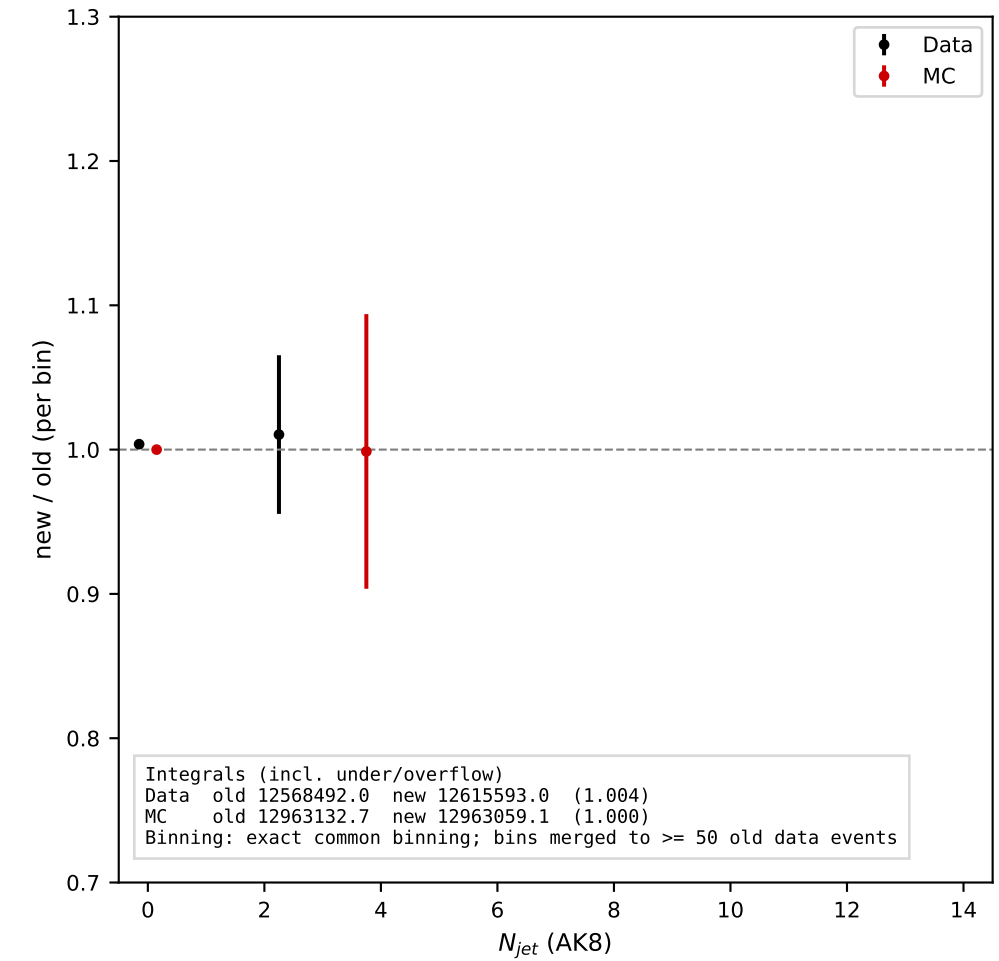
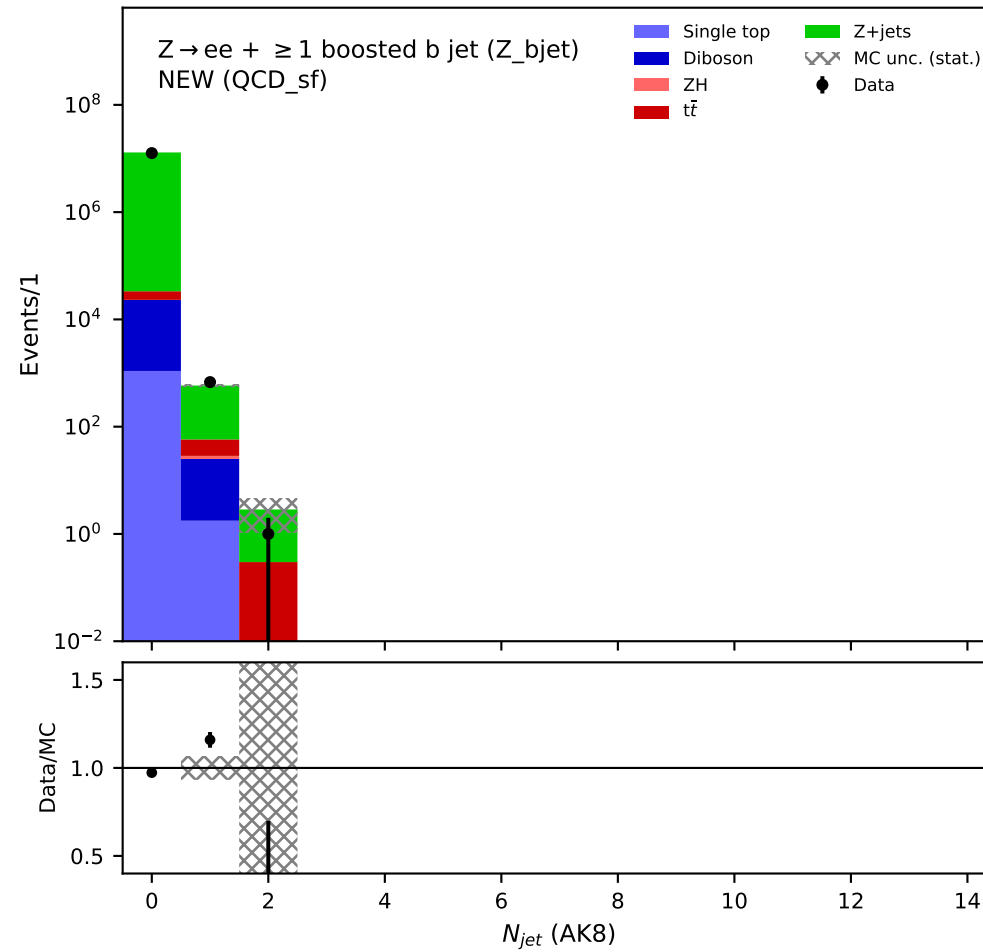
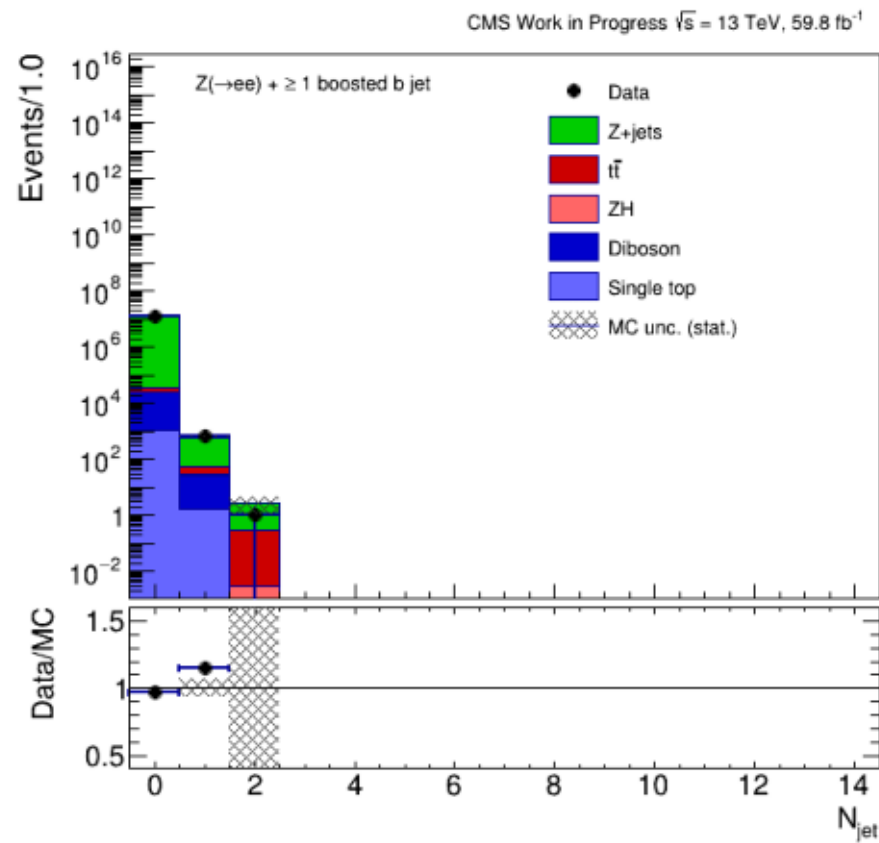
OLD: Njet_Z_jet_Zmm



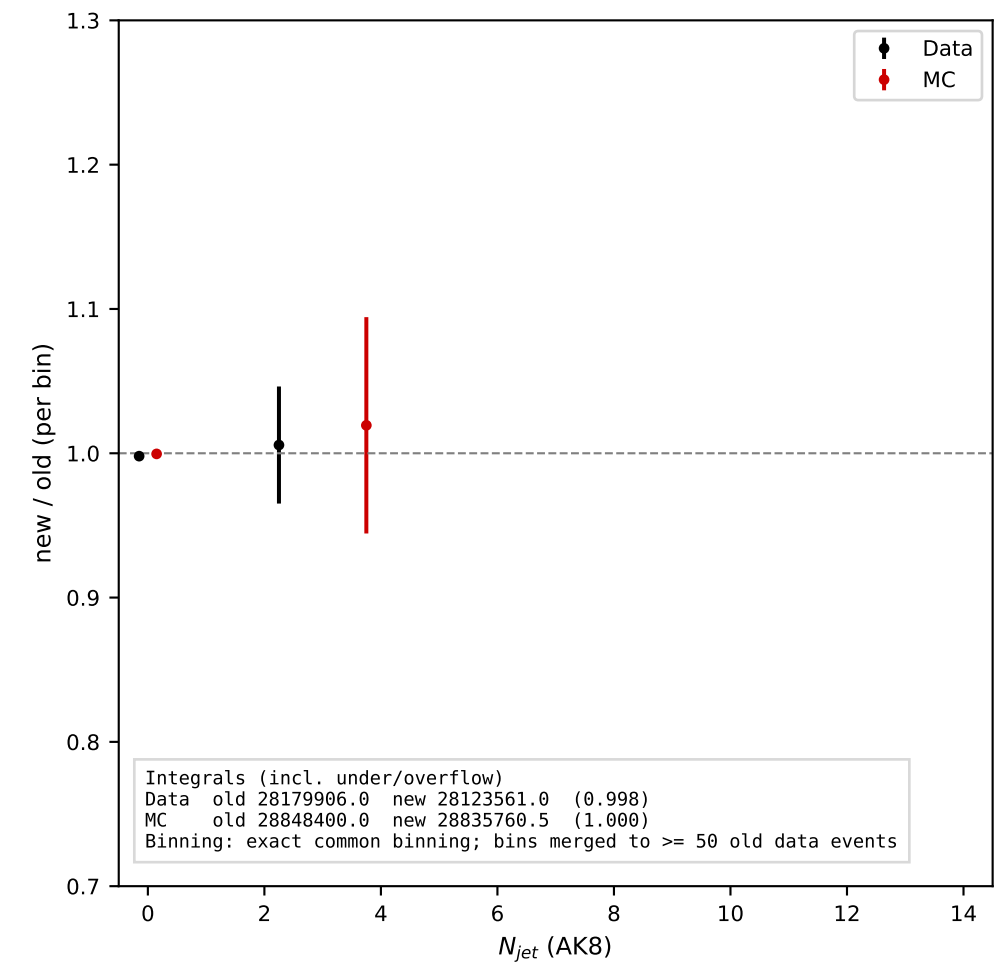
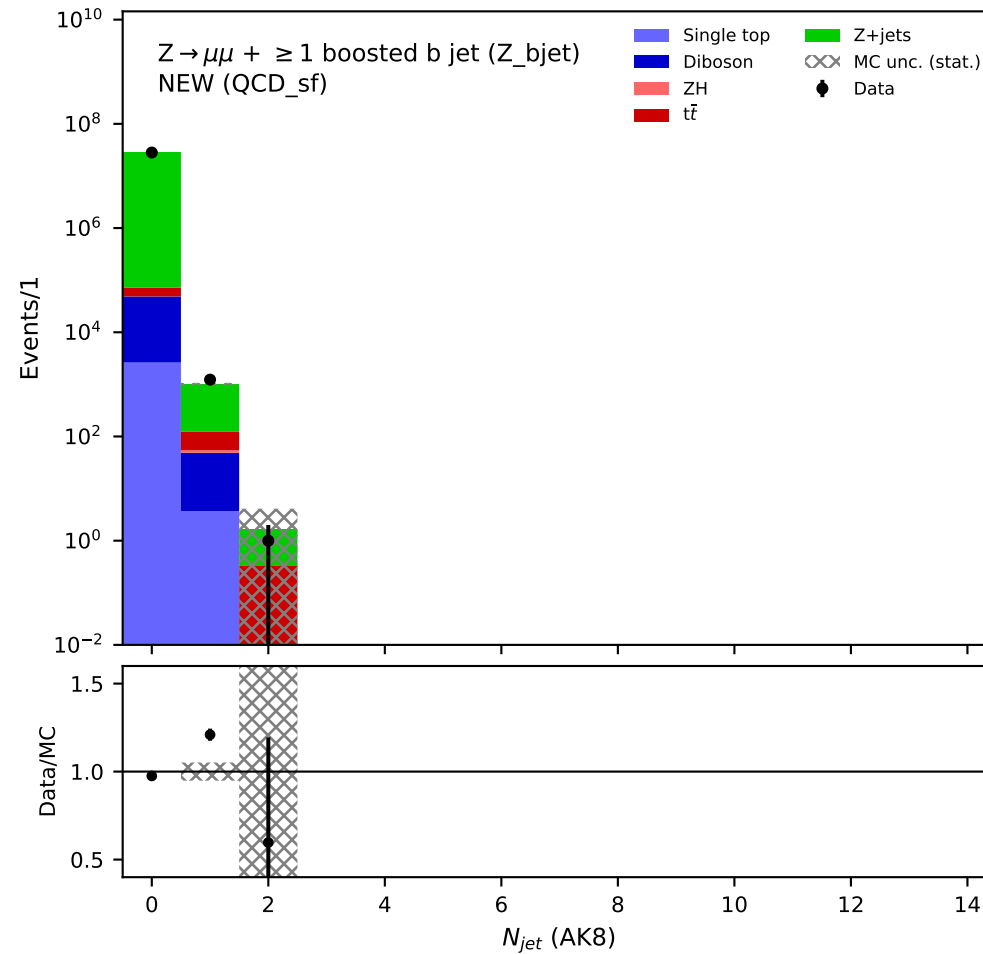
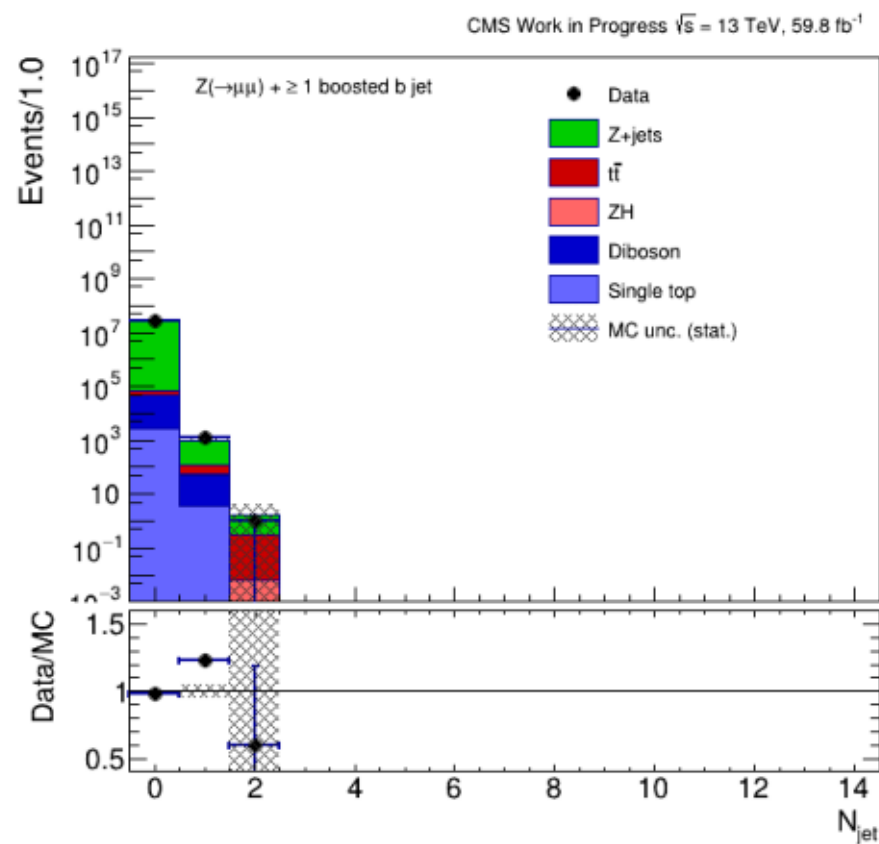
Njet | Z_bjet | 2018

Filled before the jet requirement, as in the old FillNjet.

OLD: Njet_Z_bjet_Zee

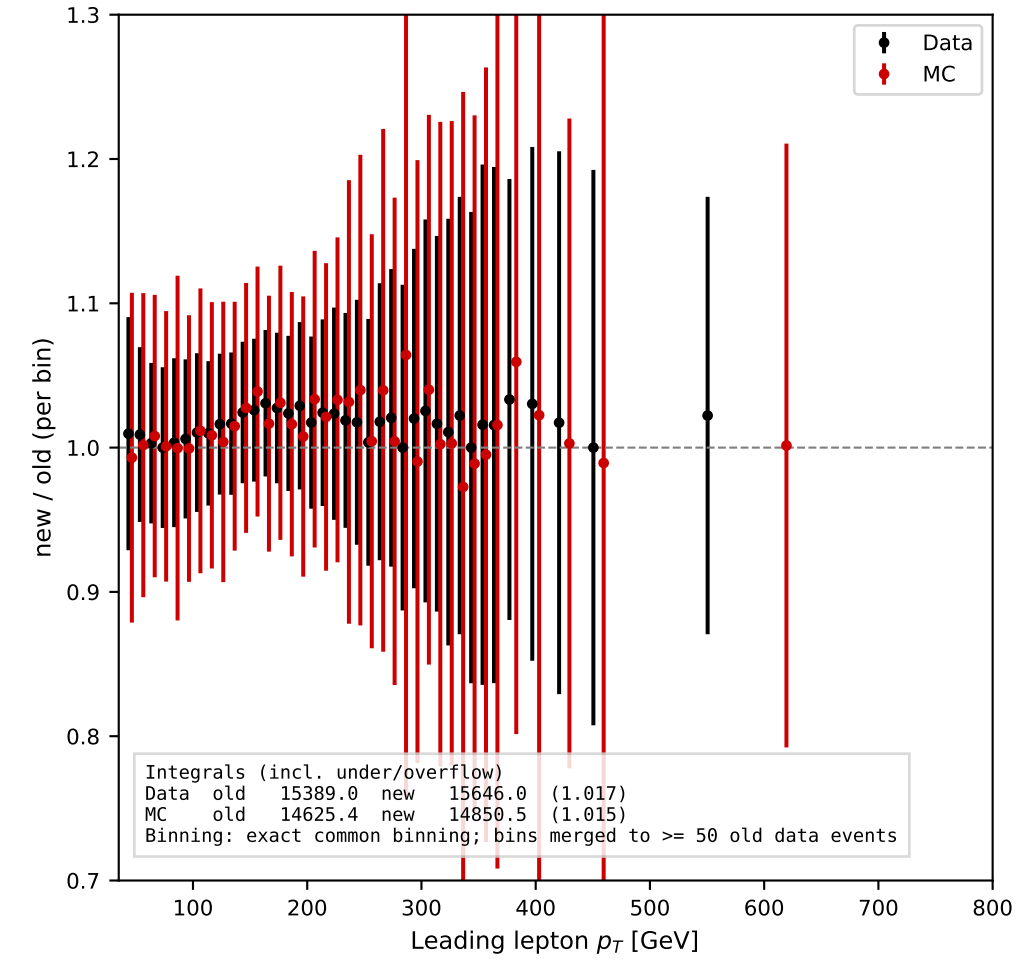
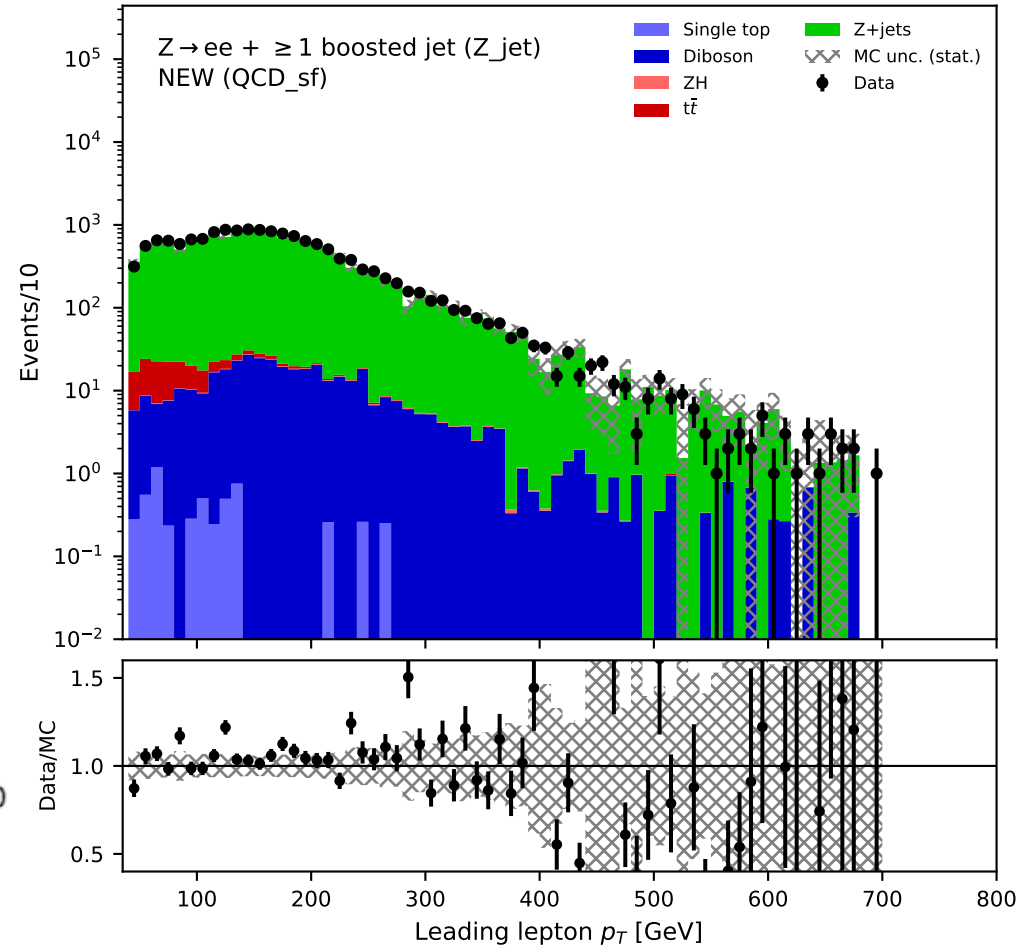
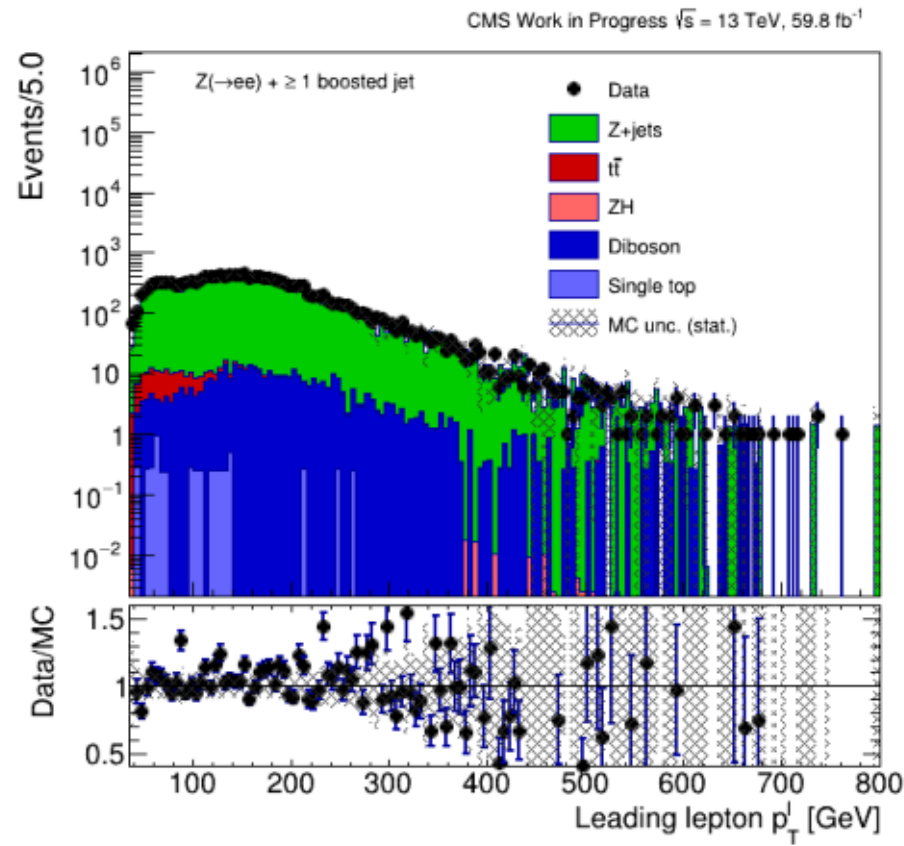


OLD: Njet_Z_bjet_Zmm

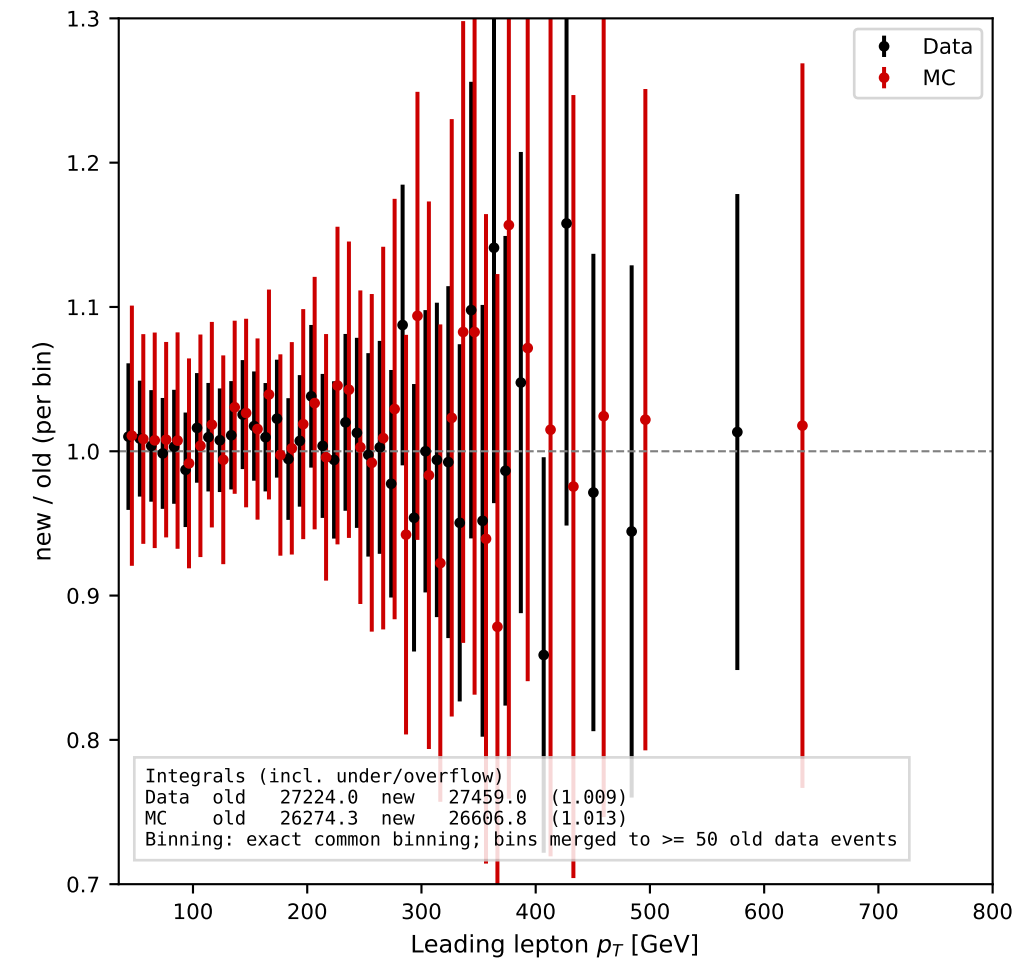
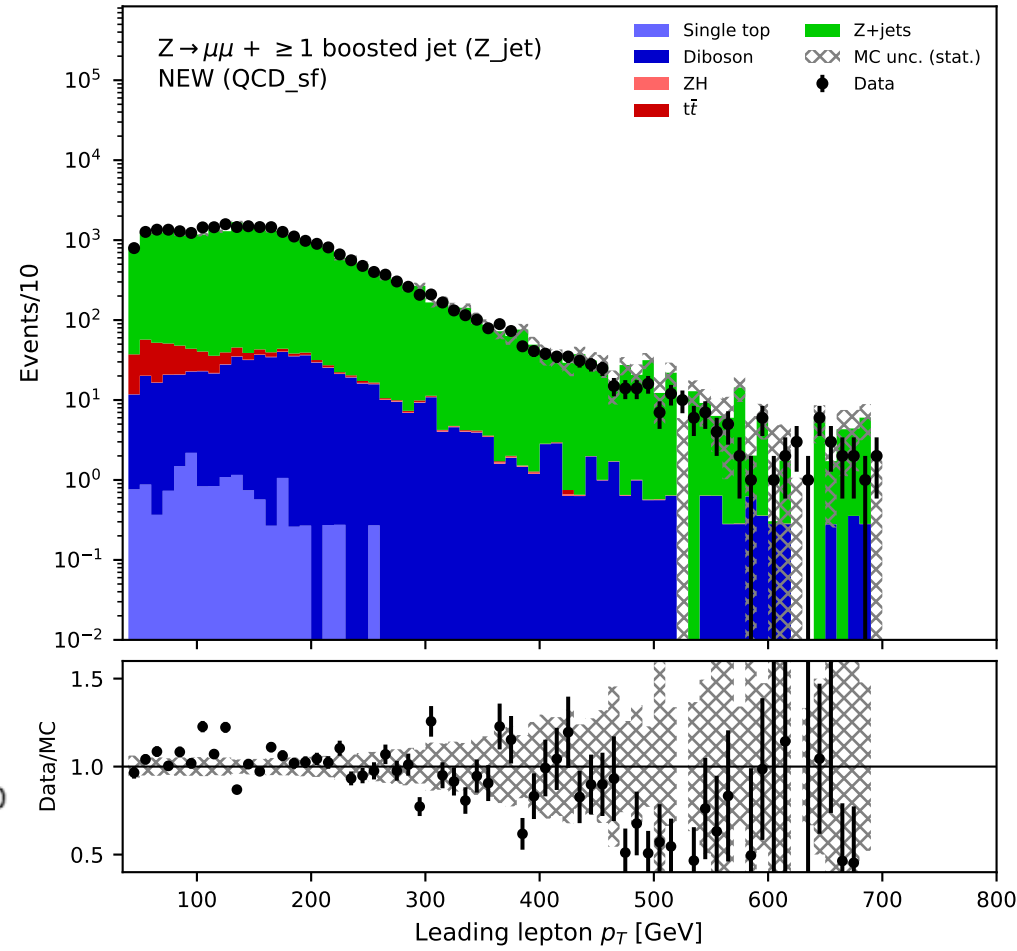
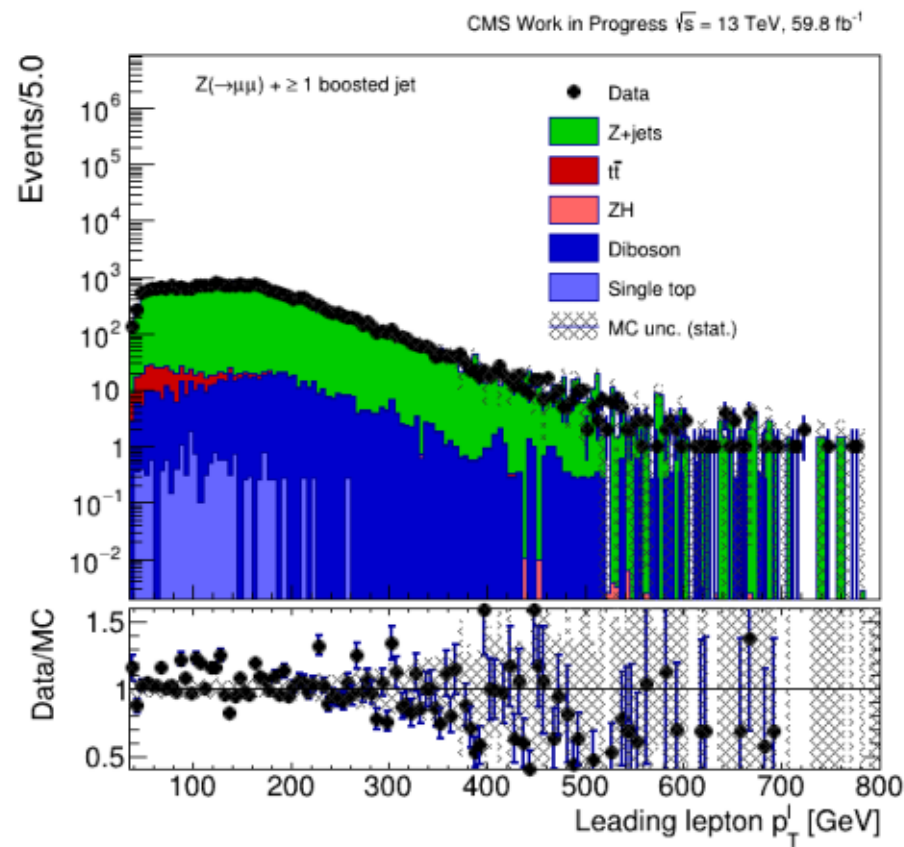


pt_lep0 | Z_jet | 2018

OLD: pt_lep0_Z_jet_Zee

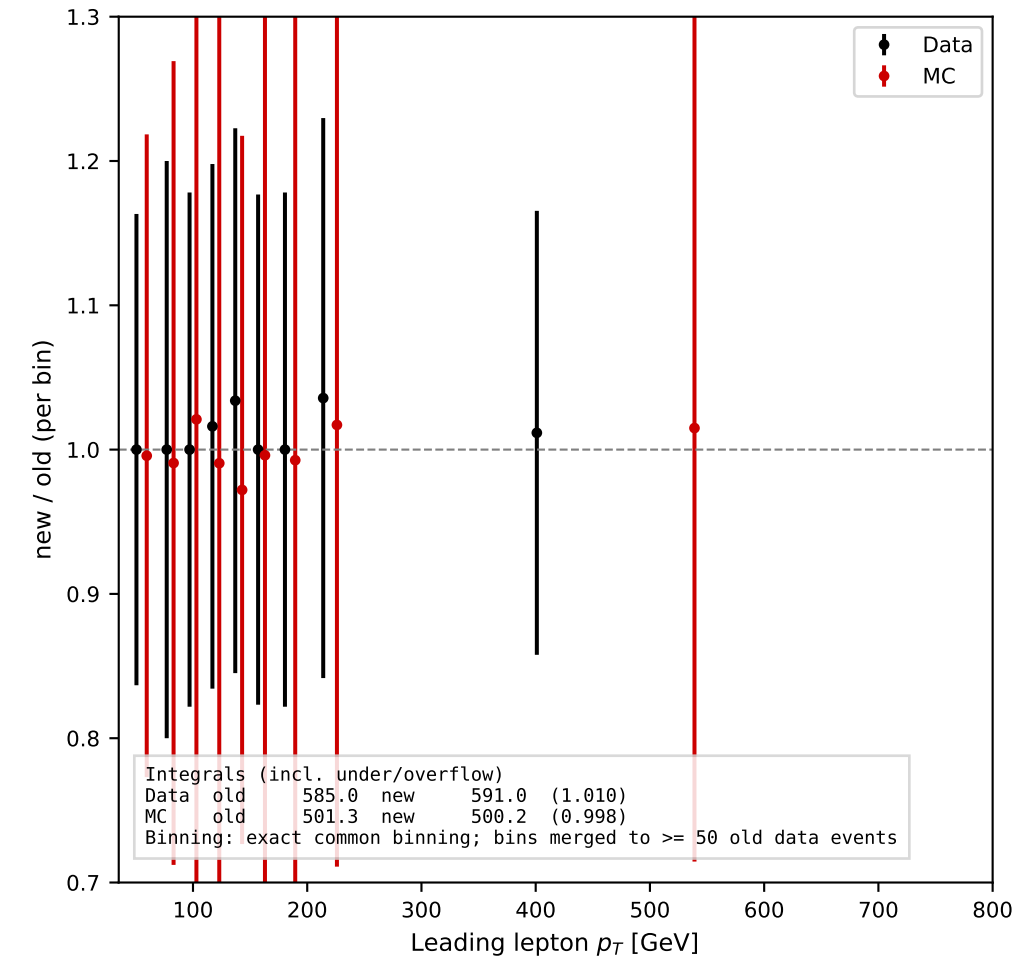
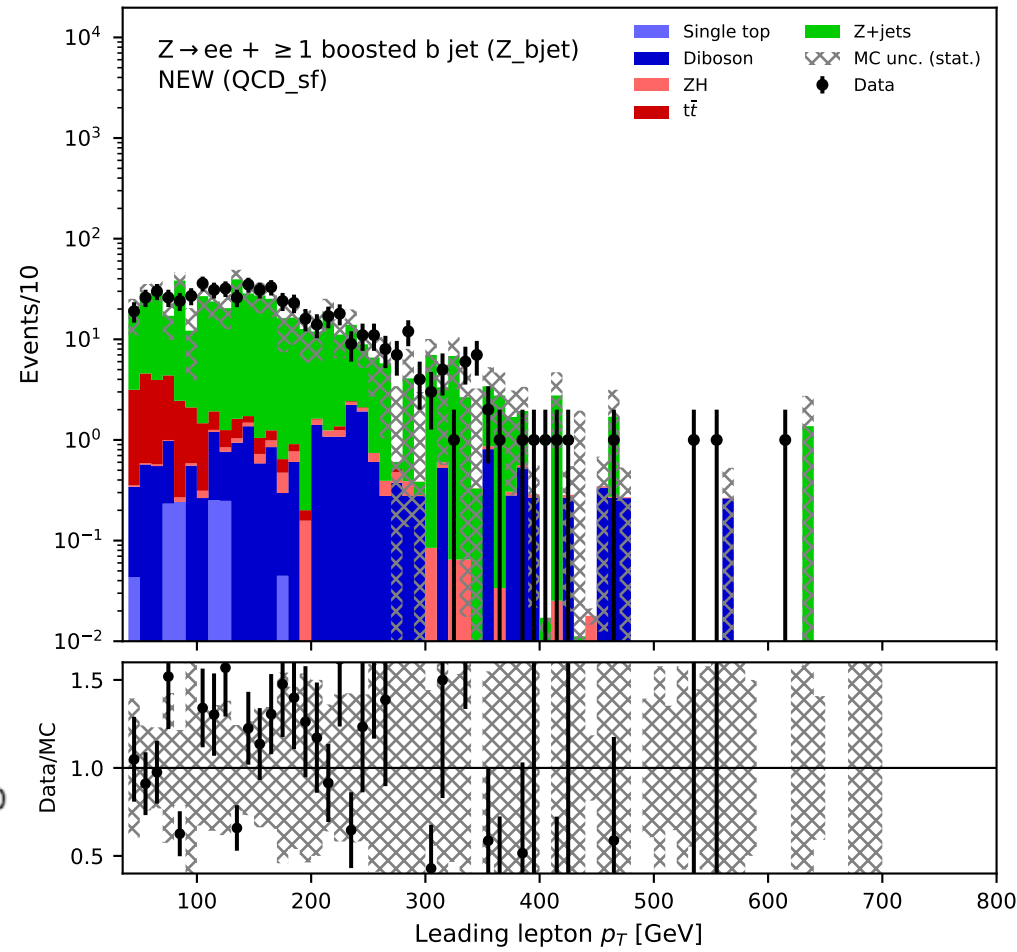
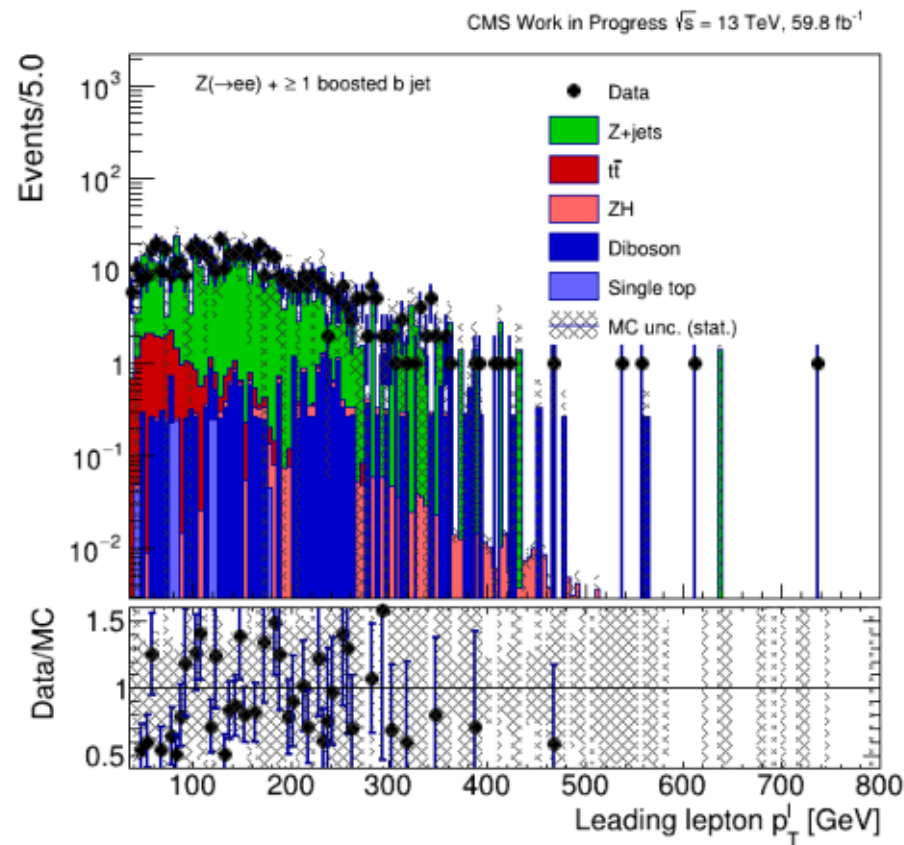


OLD: pt_lep0_Z_jet_Zmm

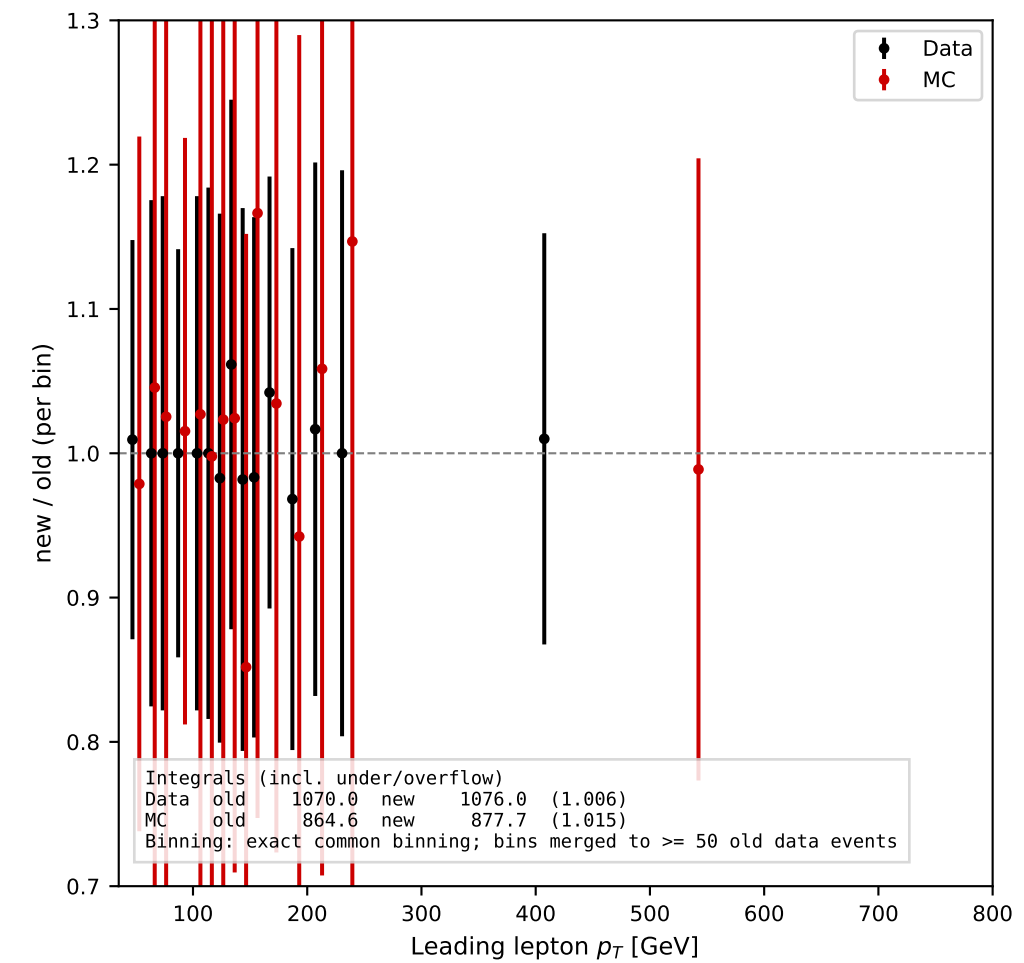
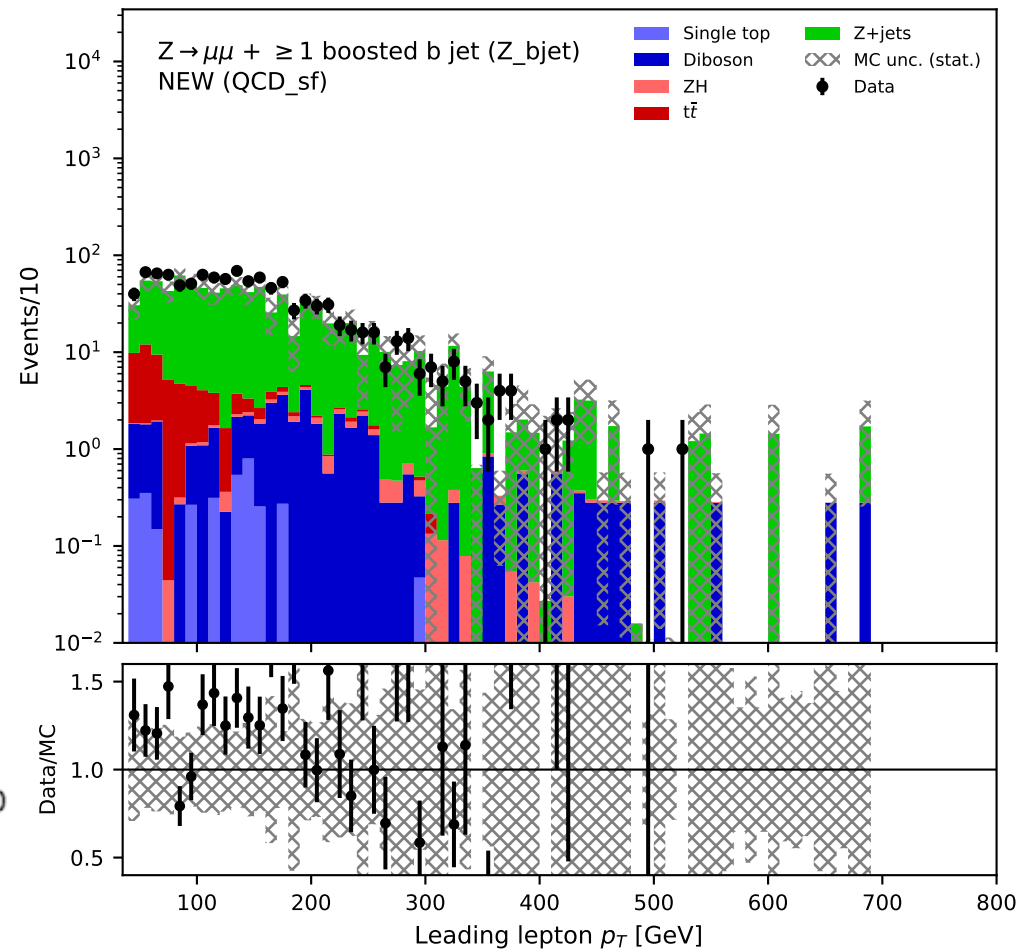
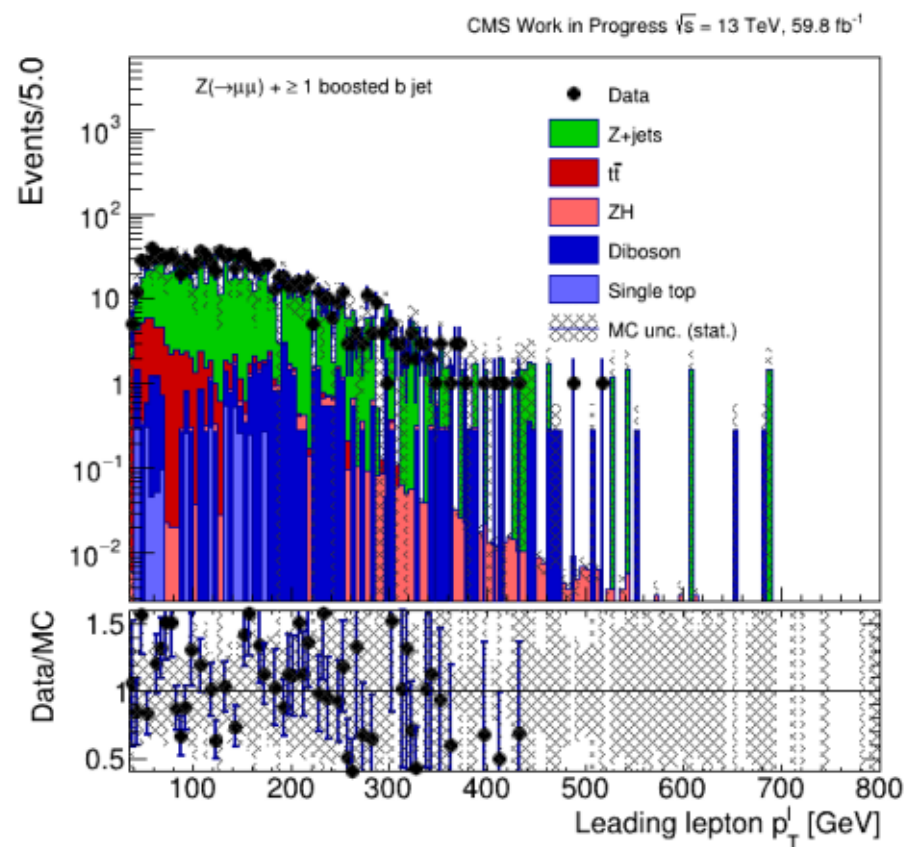


pt_lep0 | Z_bjet | 2018

OLD: pt_lep0_Z_bjet_Zee

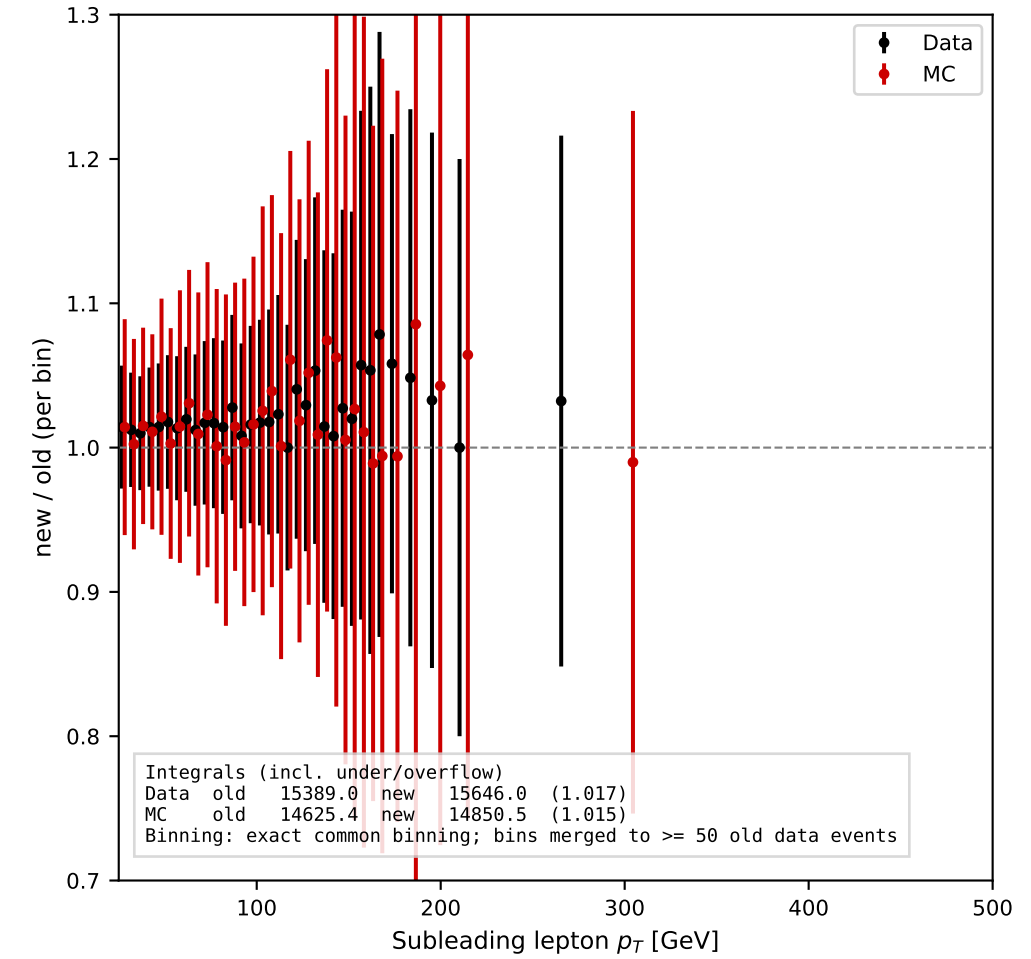
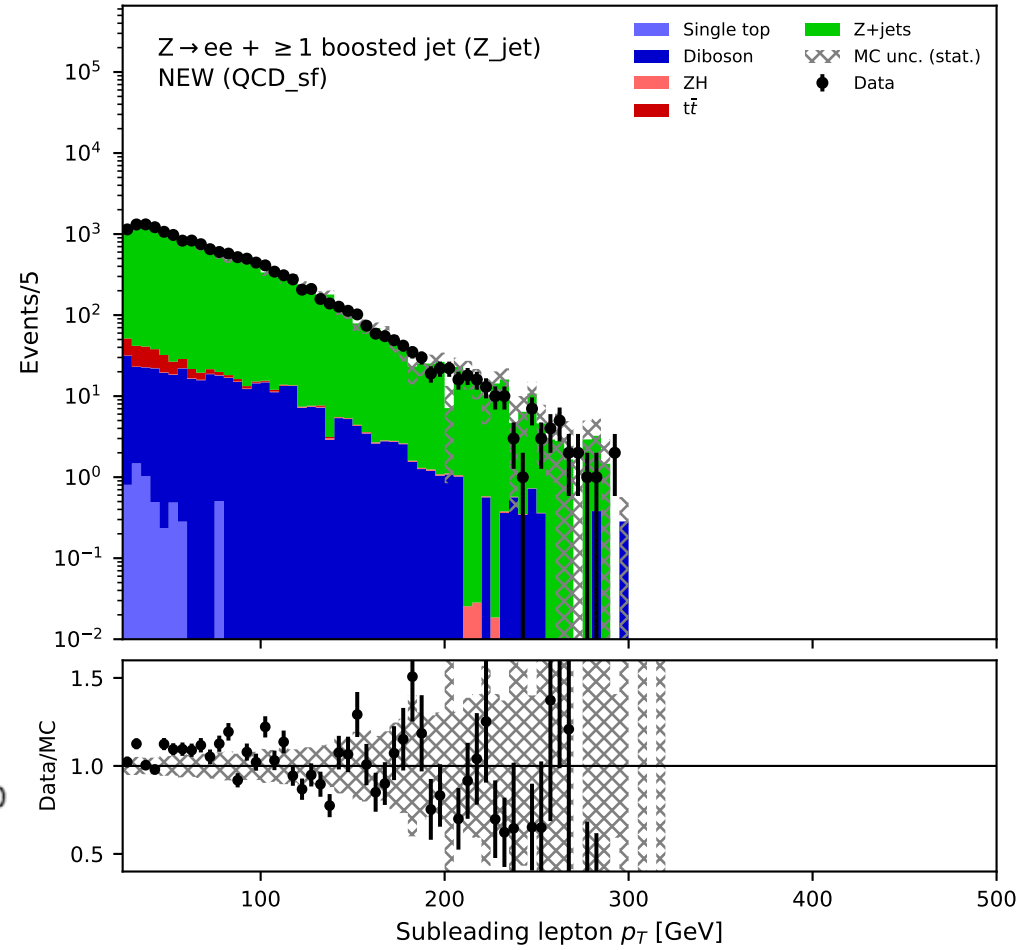
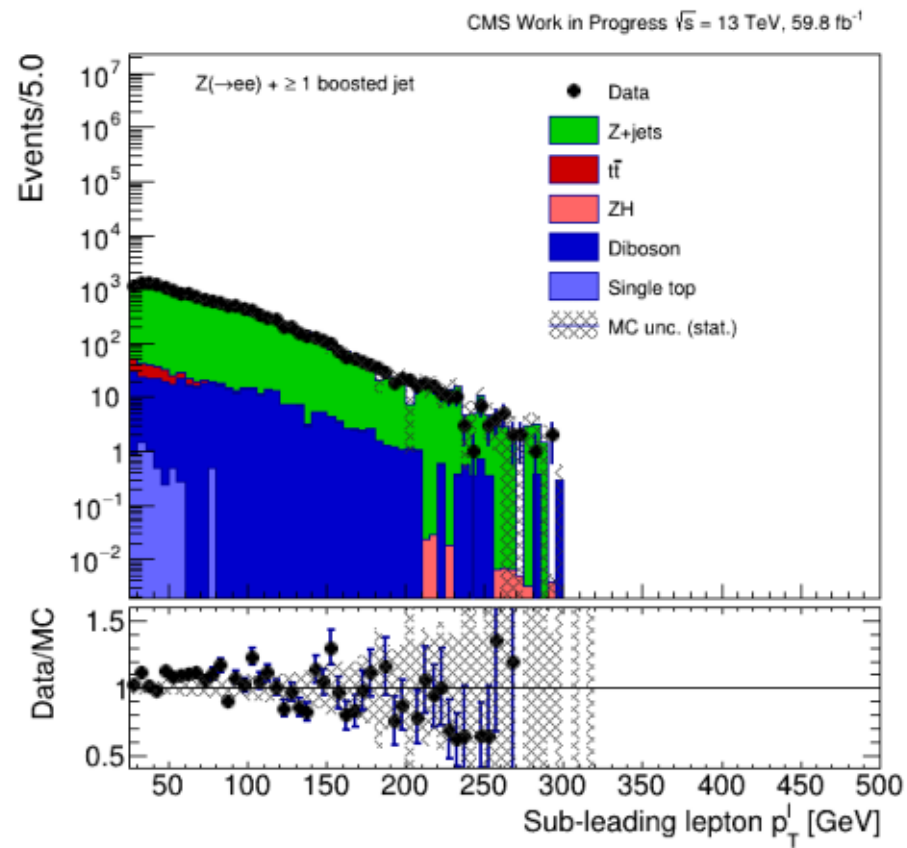


OLD: pt_lep0_Z_bjet_Zmm

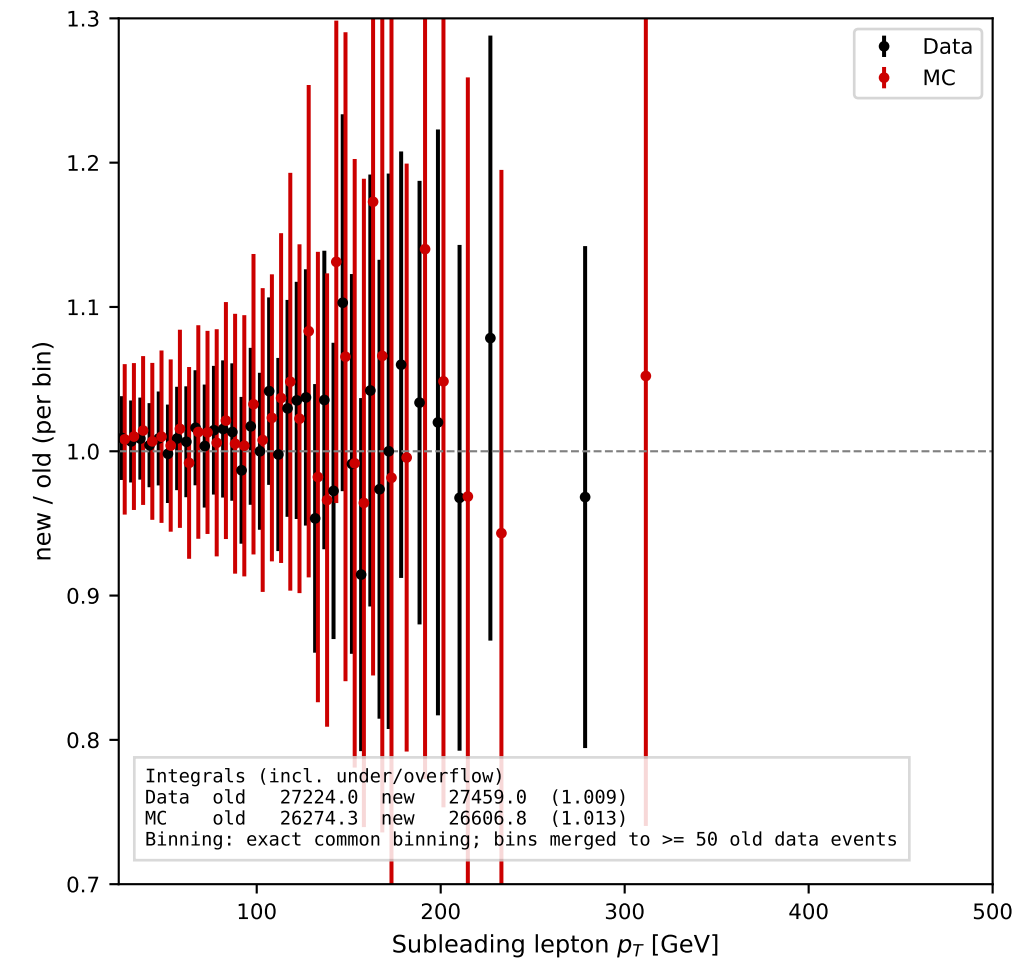
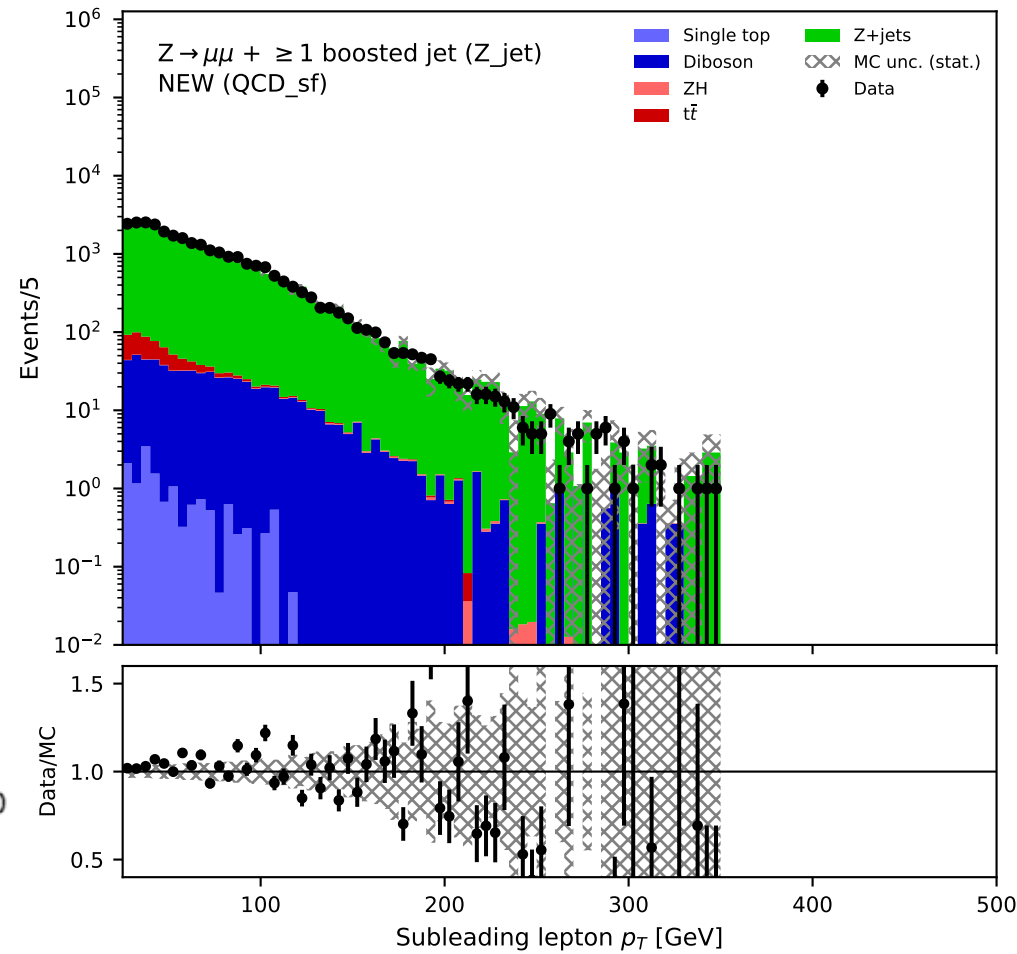
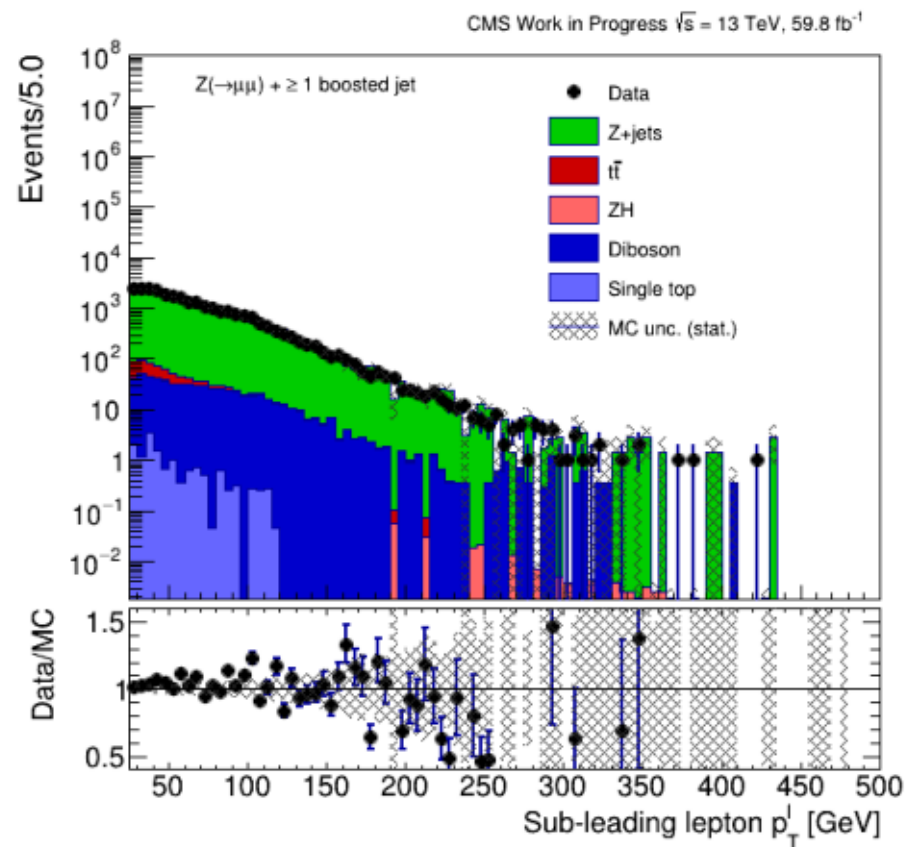


pt_lep1 | Z_jet | 2018

OLD: pt_lep1_Z_jet_Zee

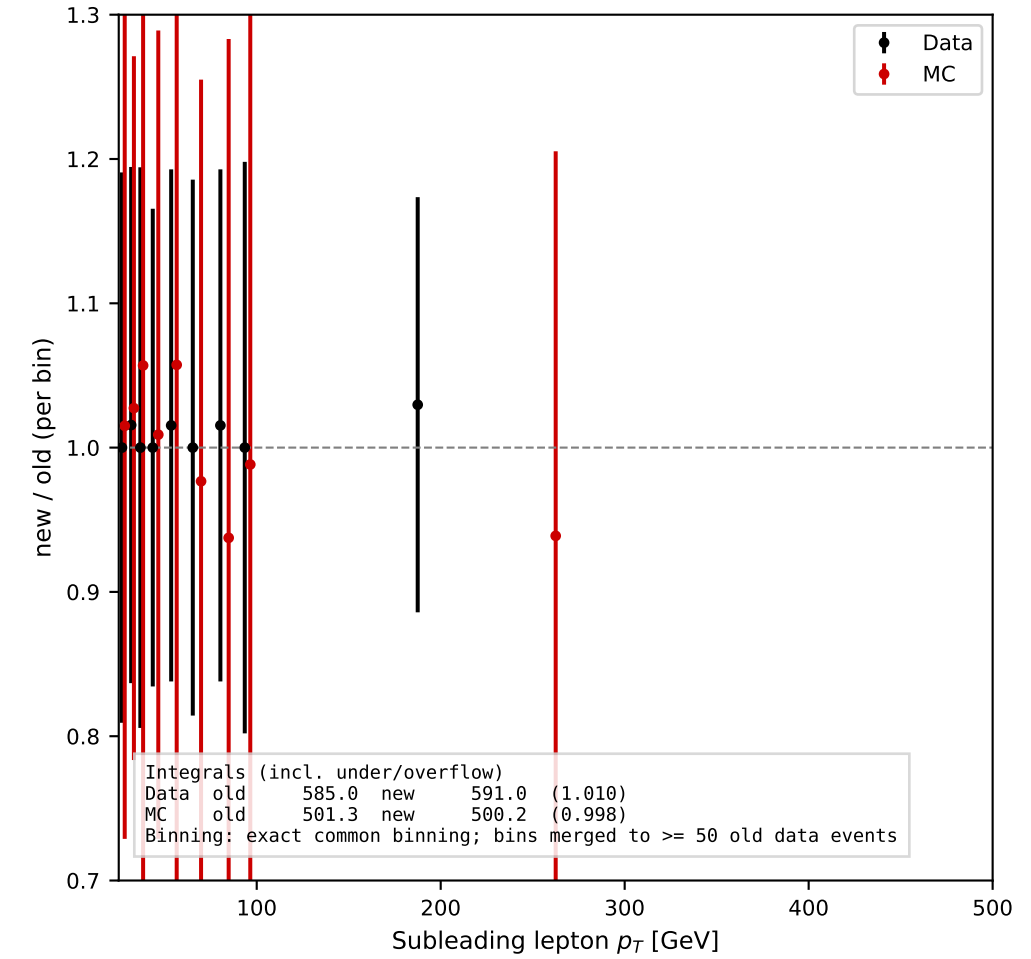
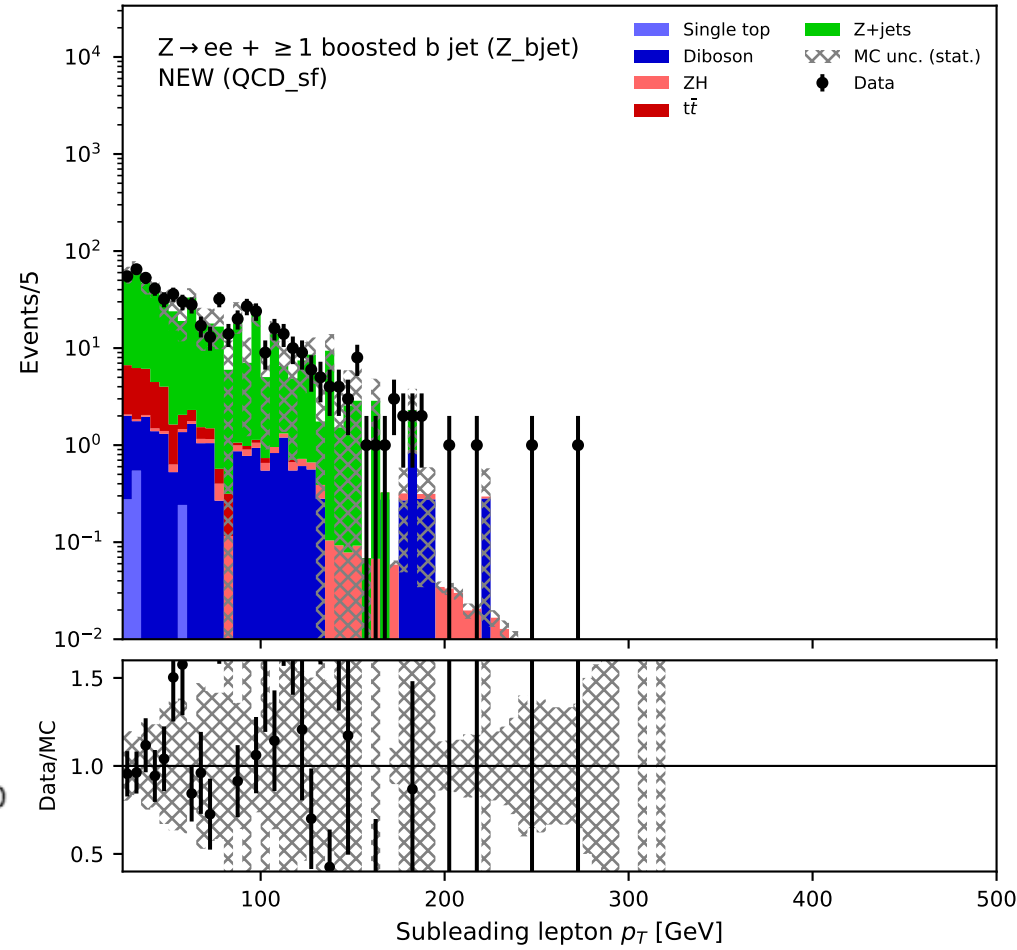
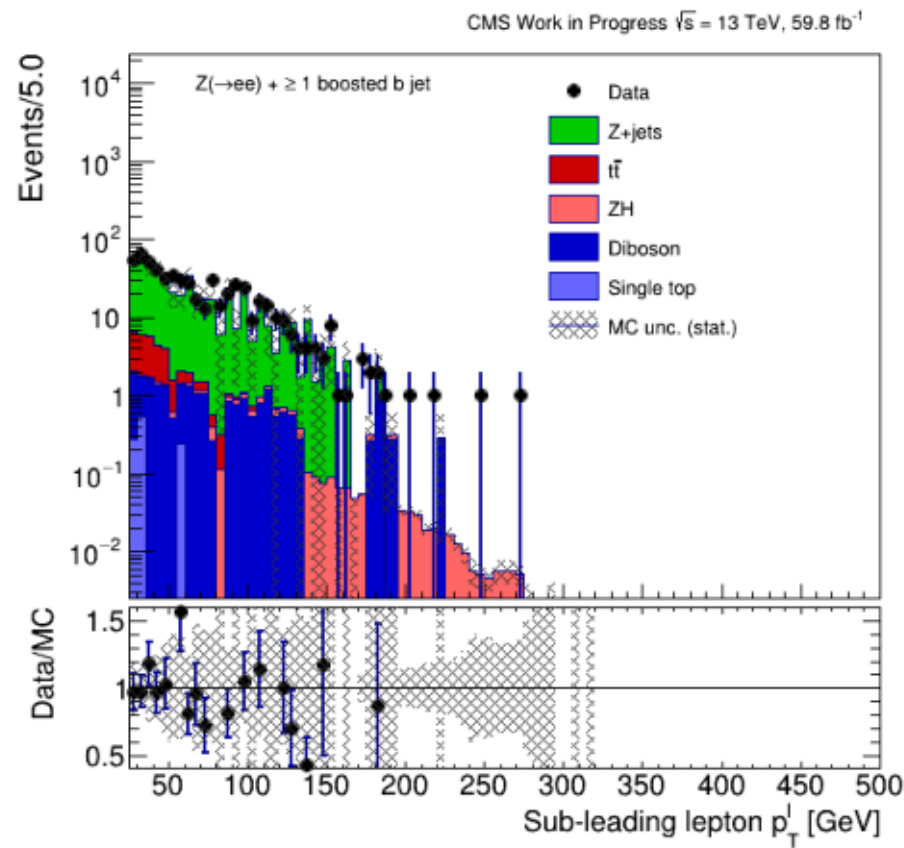


OLD: pt_lep1_Z_jet_Zmm

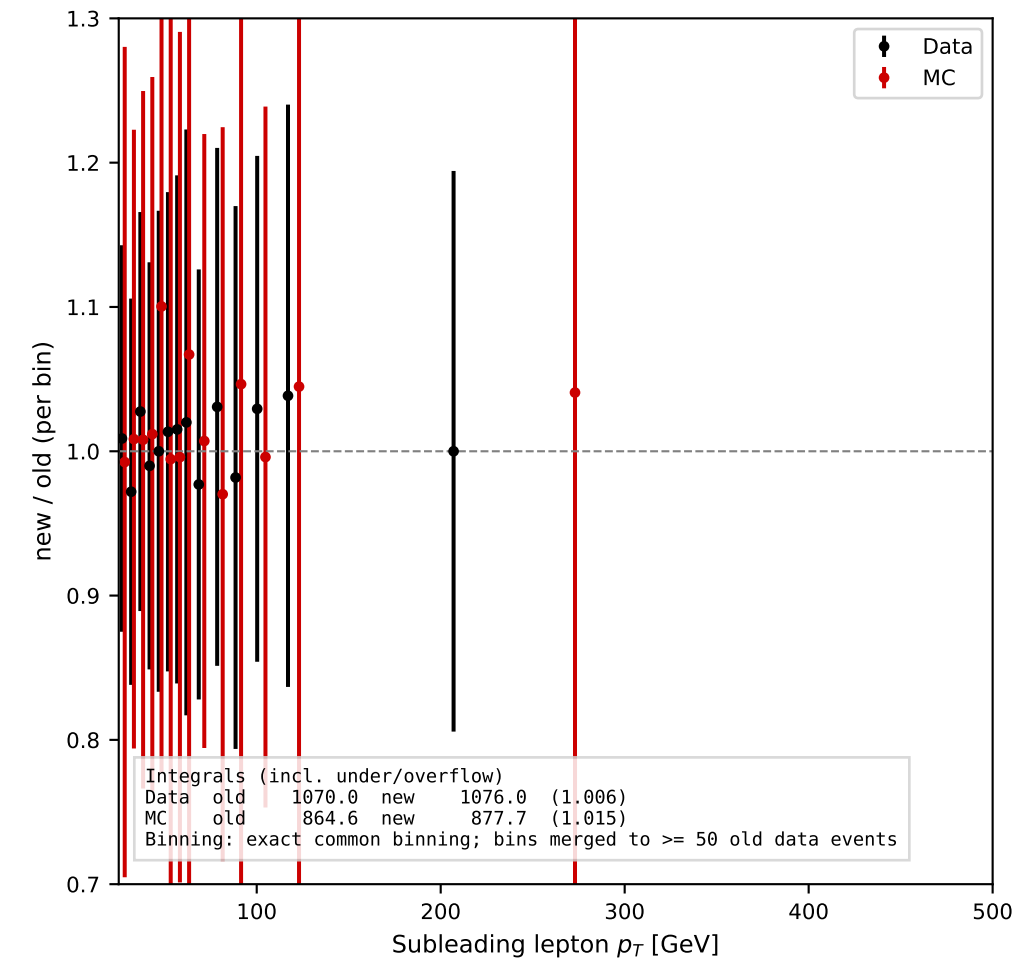
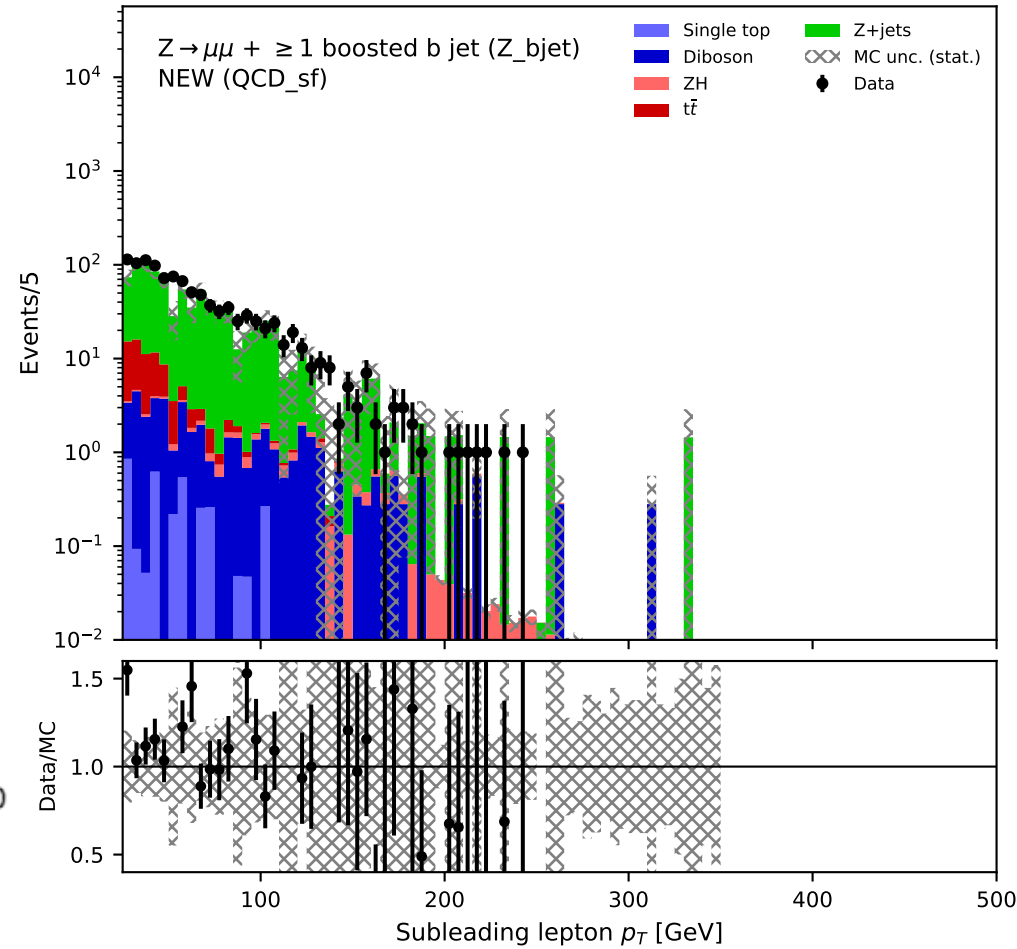
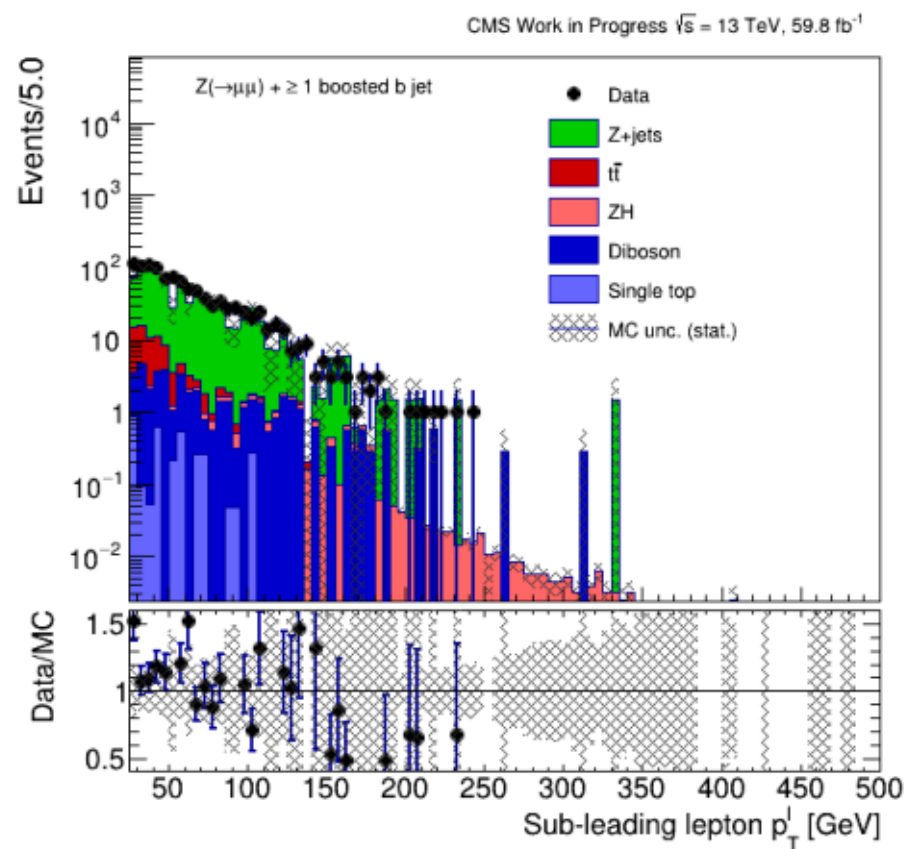


pt_lep1 | Z_bjet | 2018

OLD: pt_lep1_Z_bjet_Zee

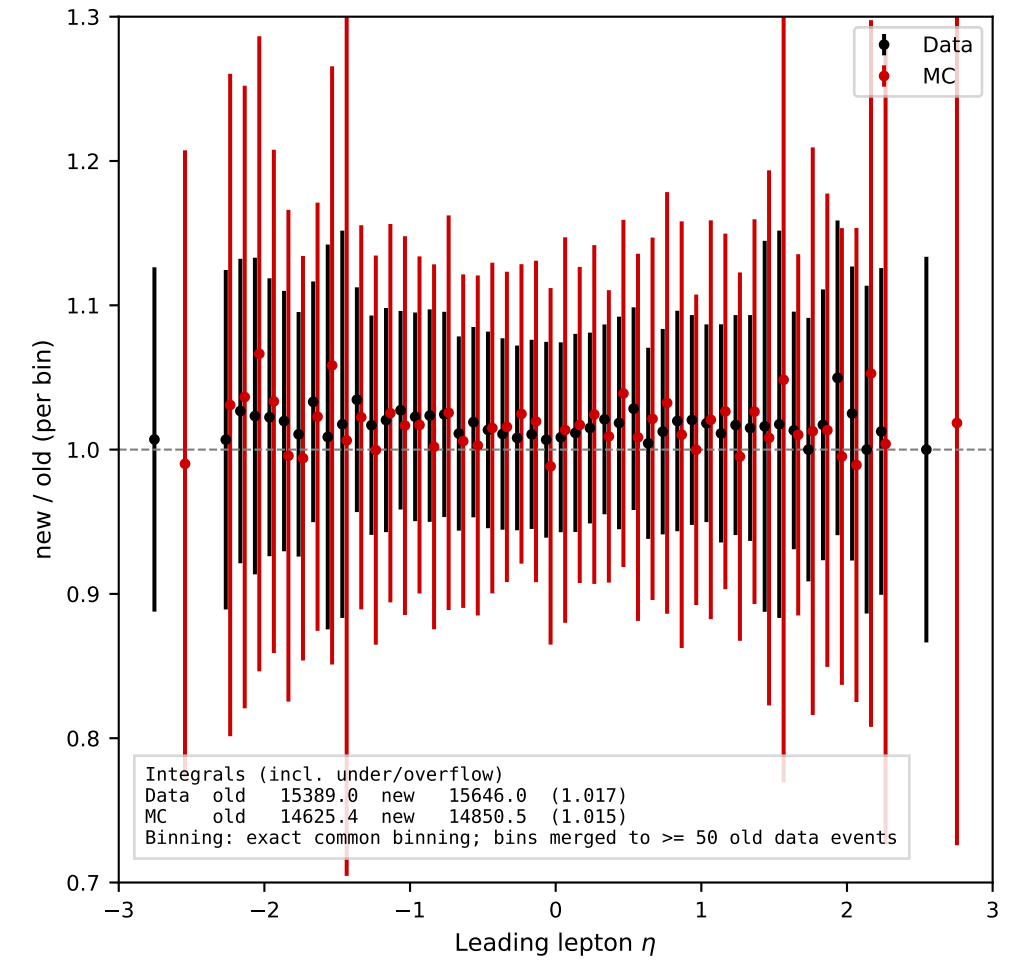
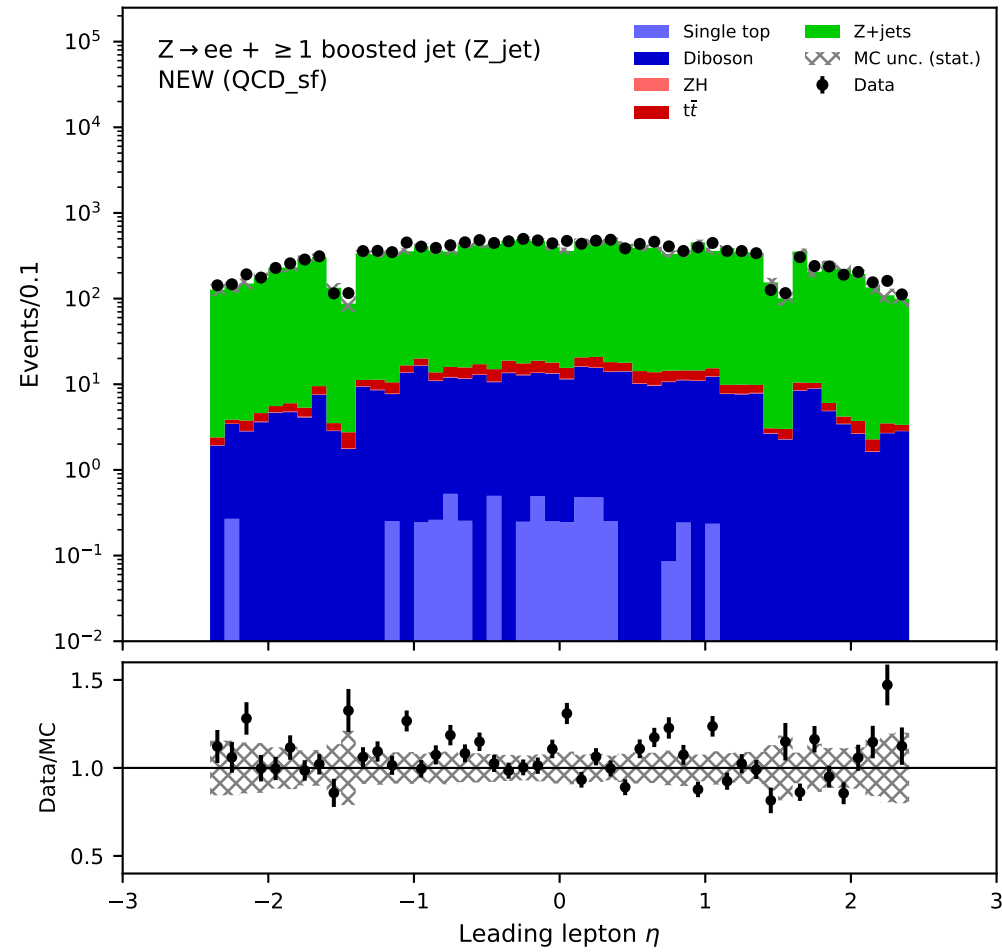
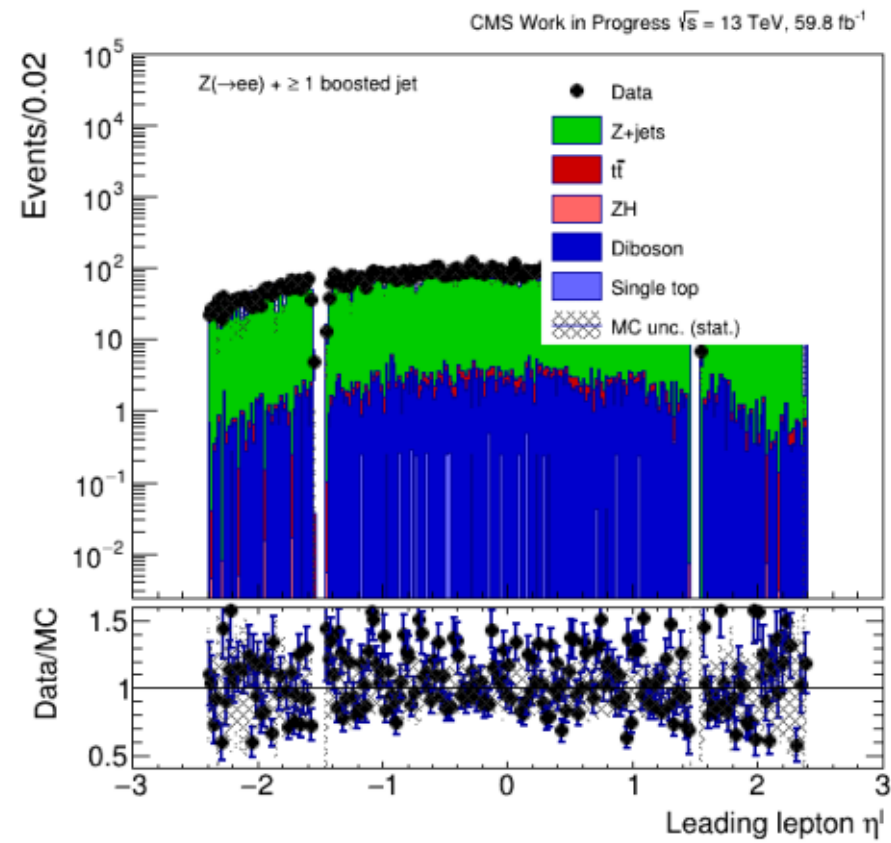


OLD: pt_lep1_Z_bjet_Zmm

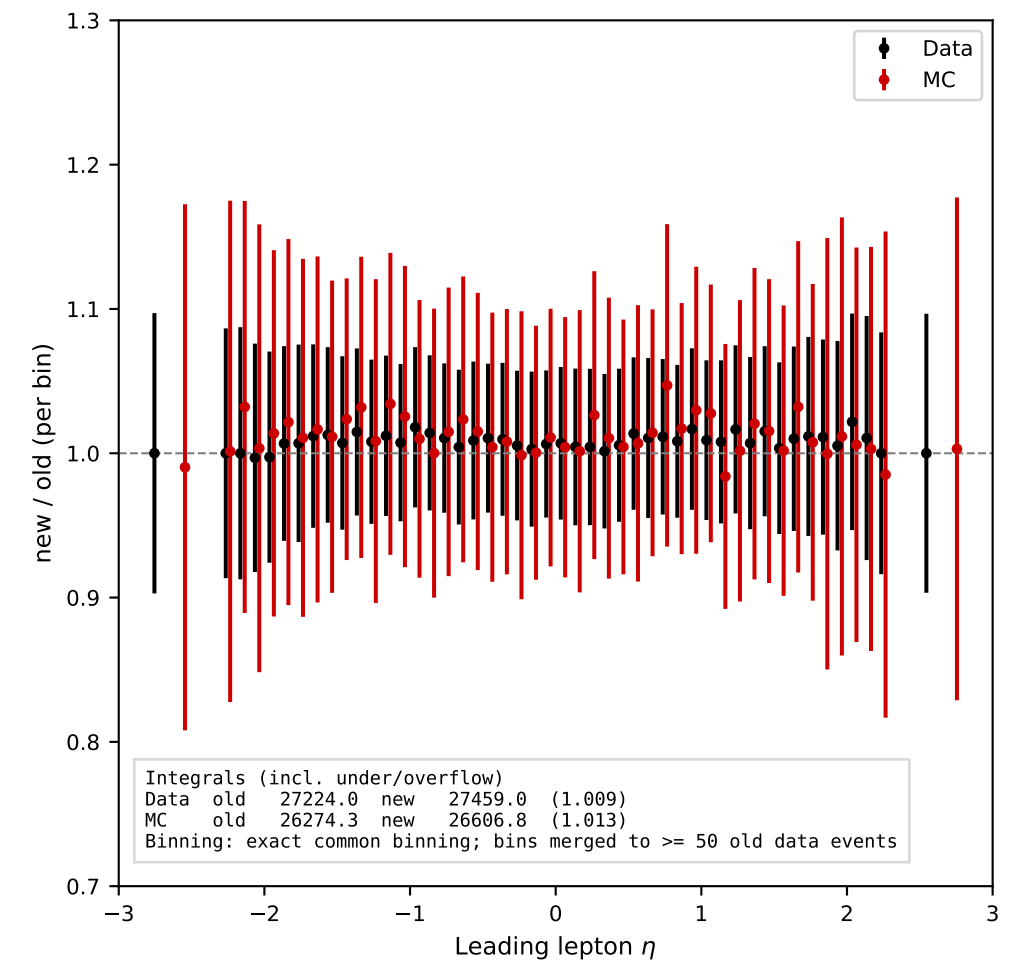
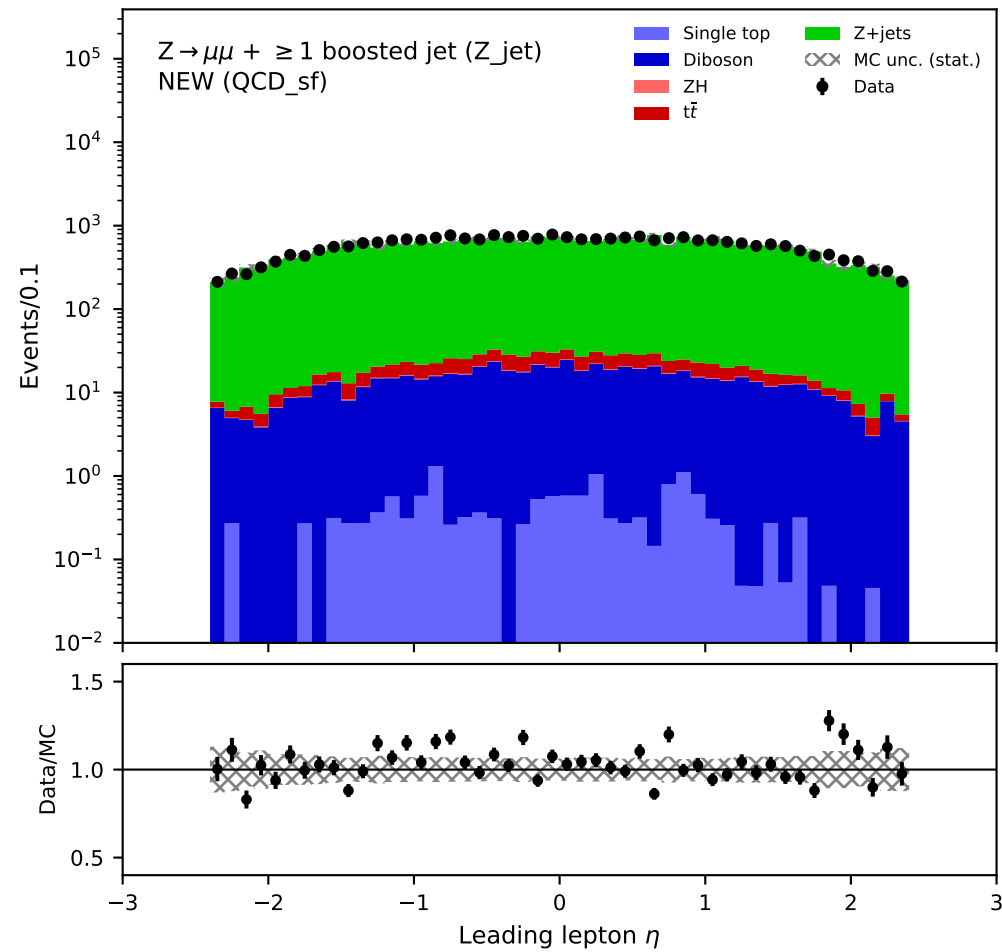
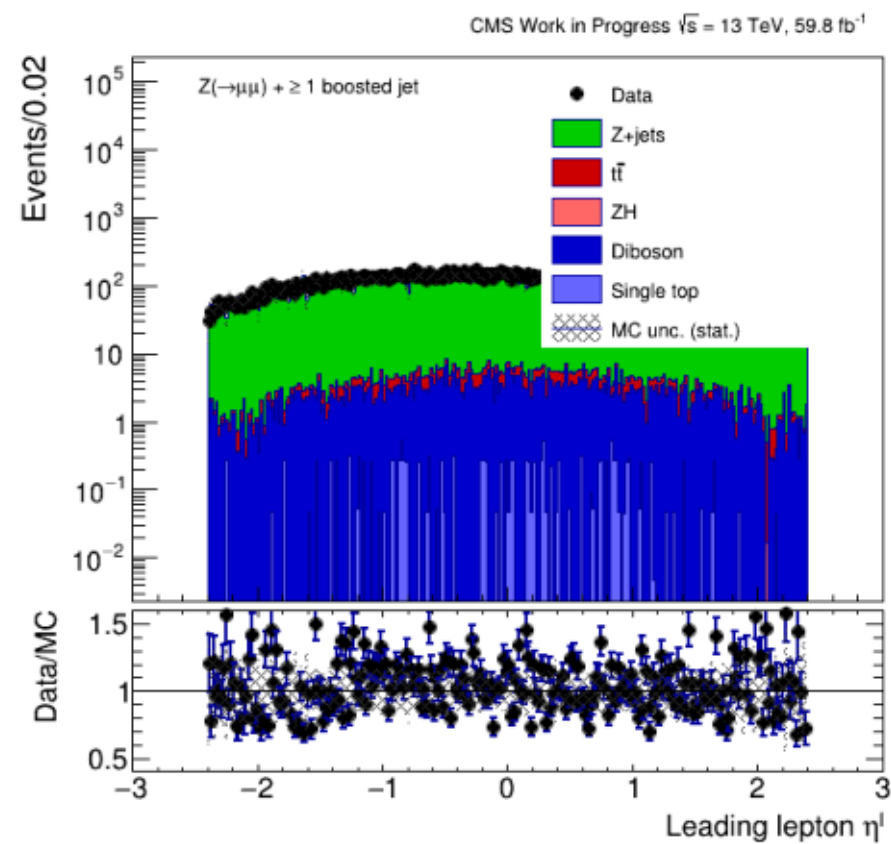


eta_lep0 | Z_jet | 2018

OLD: eta_lep0_Z_jet_Zee

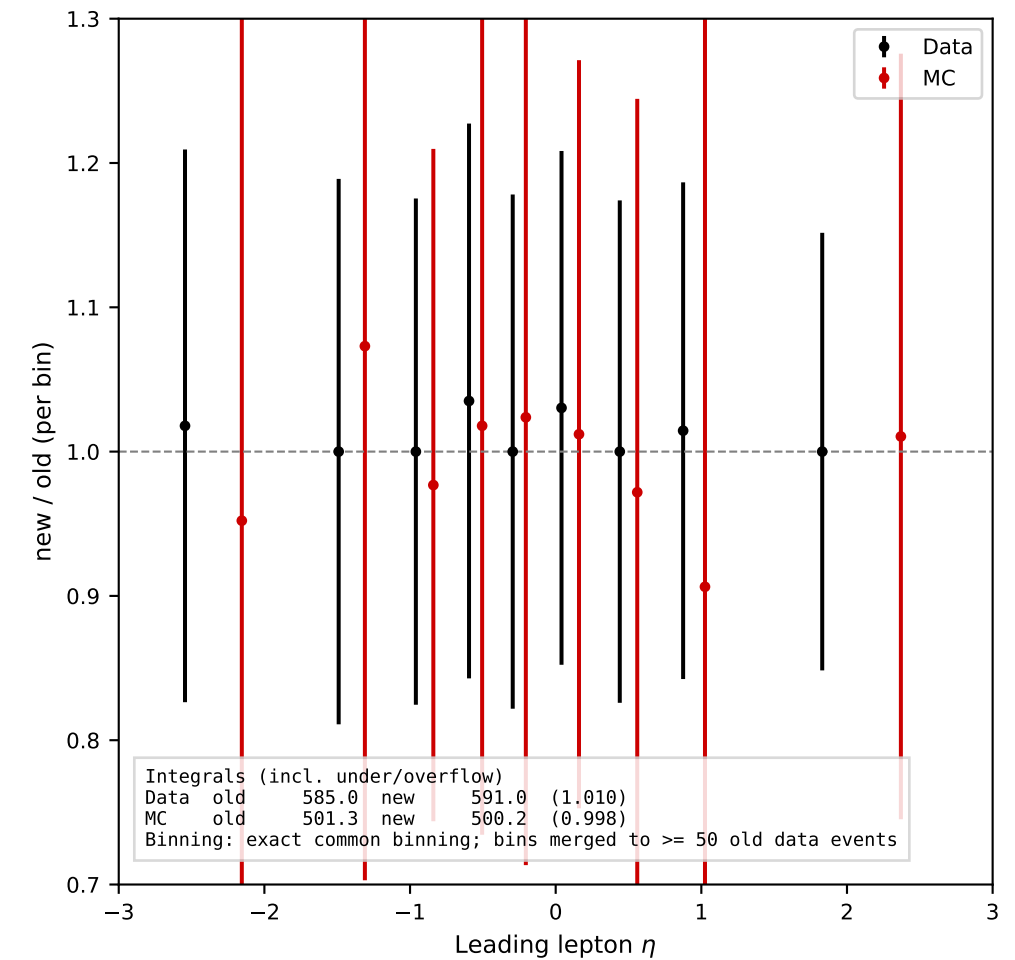
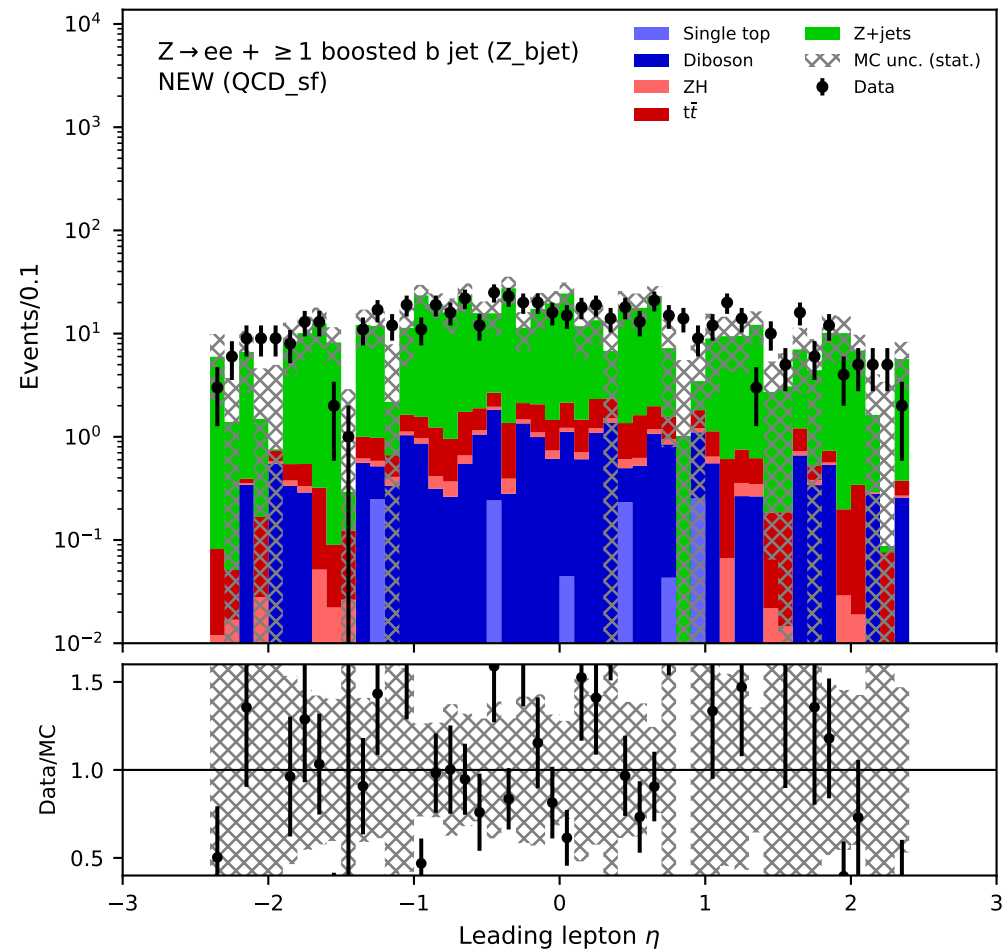
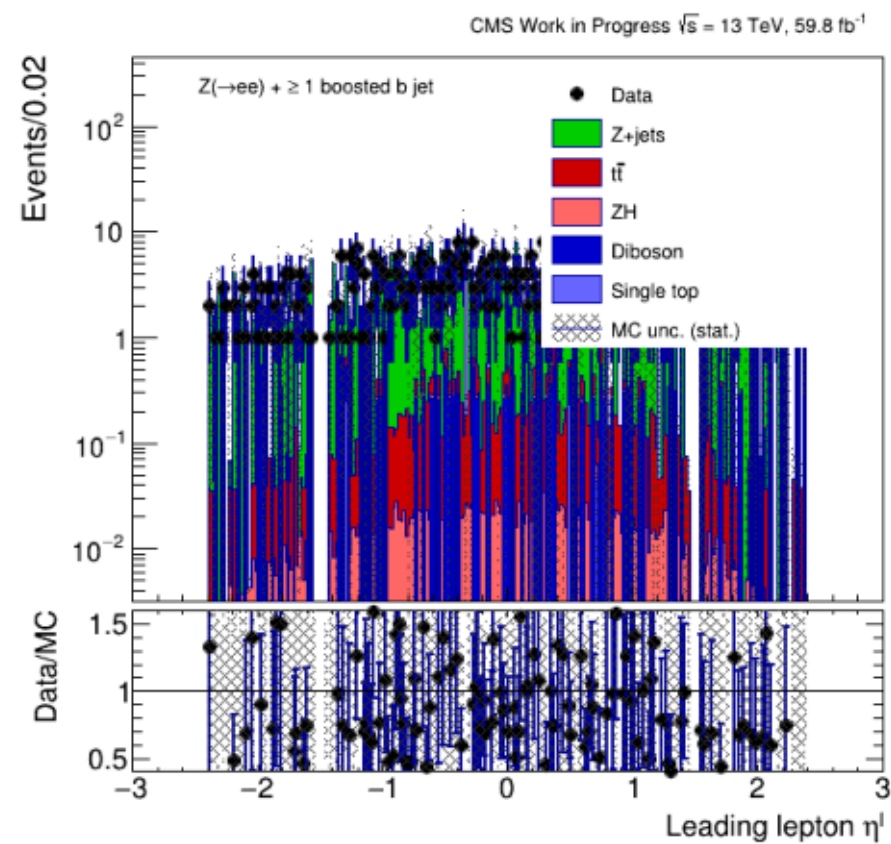


OLD: eta_lep0_Z_jet_Zmm

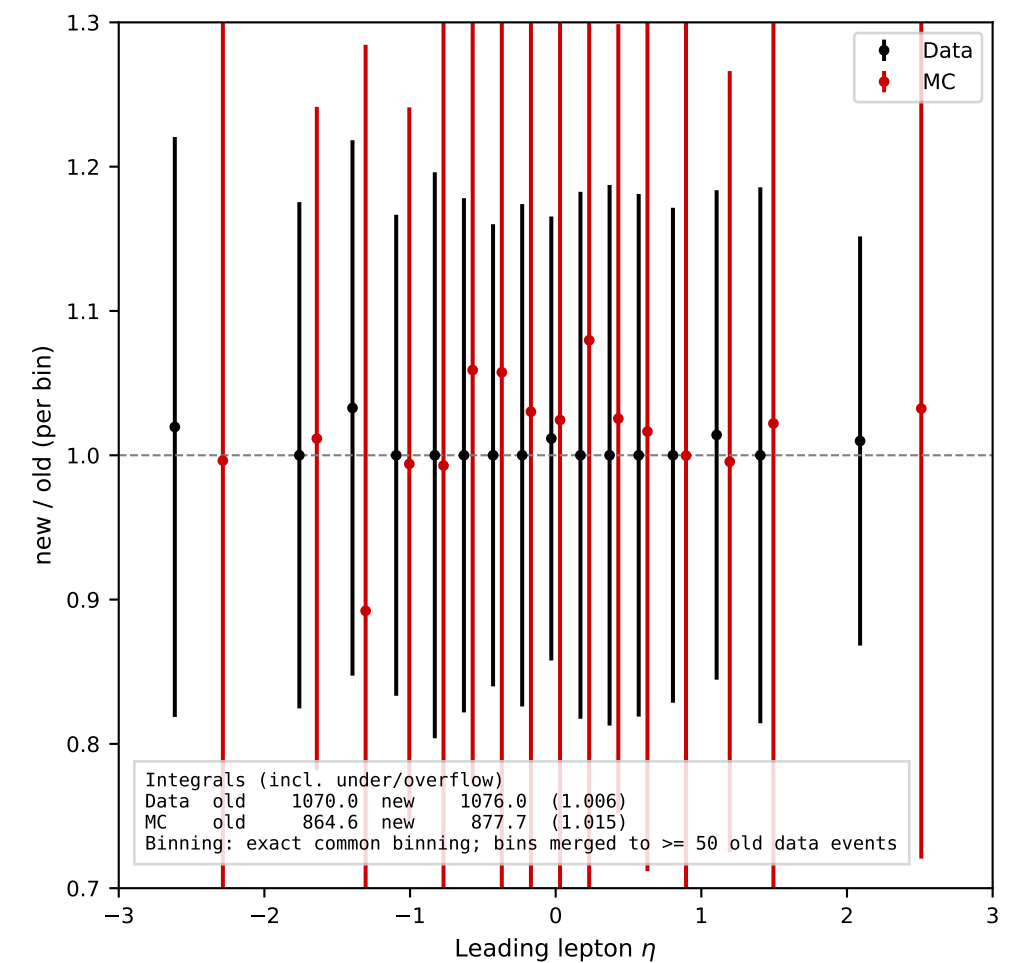
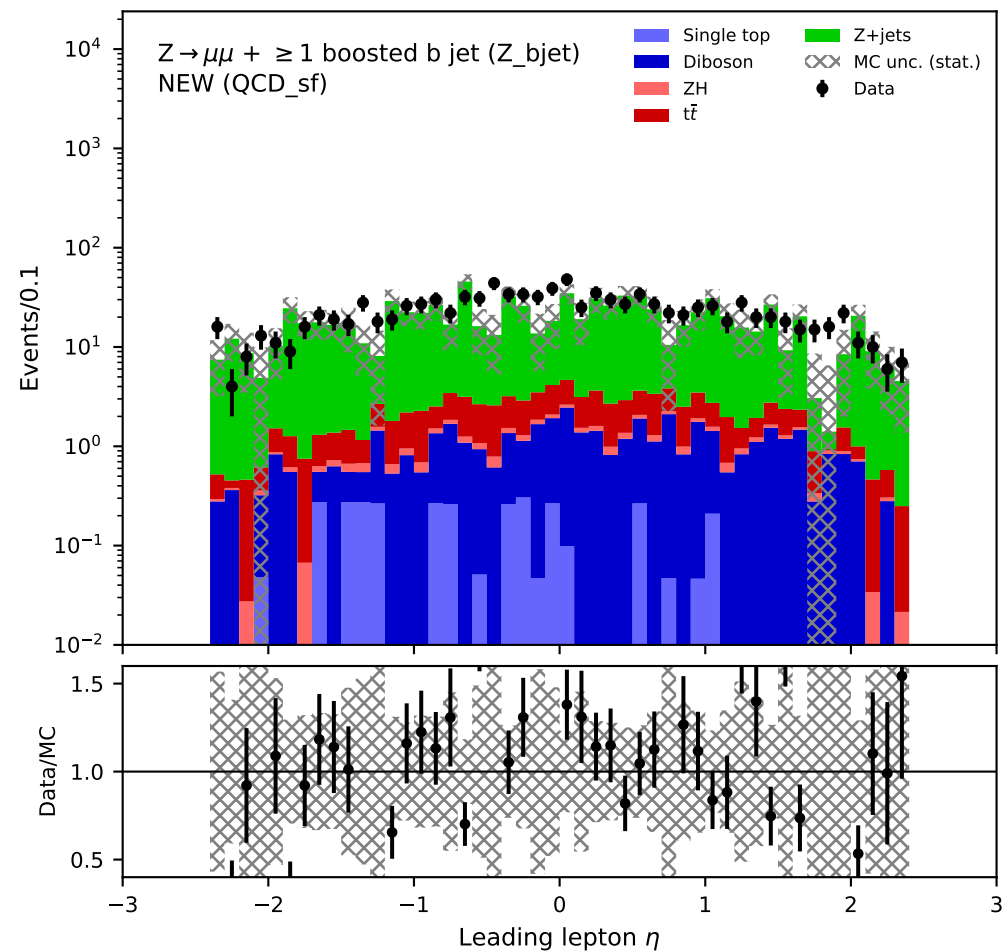
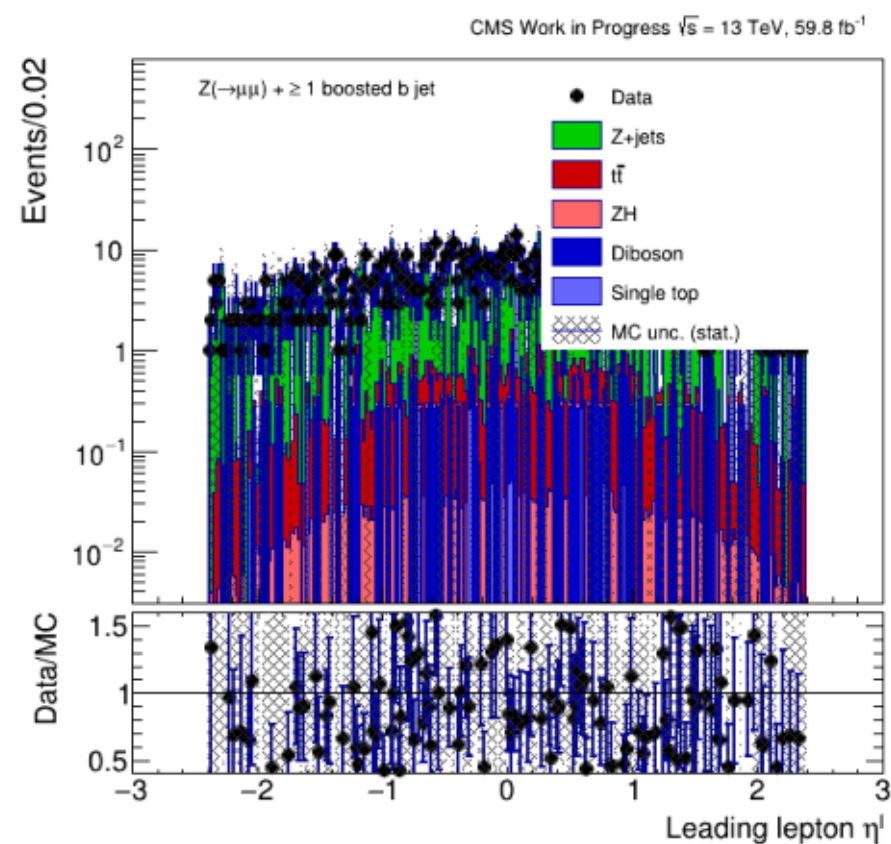


eta_lep0 | Z_bjet | 2018

OLD: eta_lep0_Z_bjet_Zee

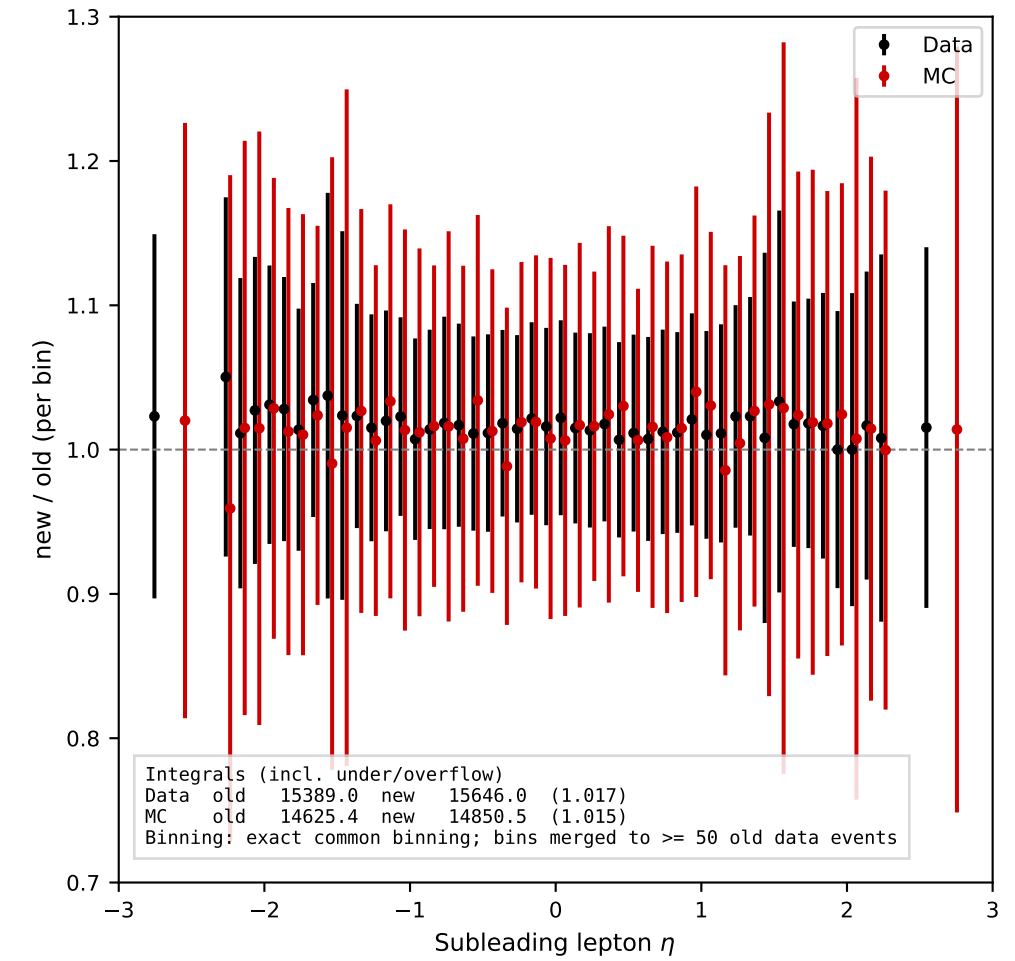
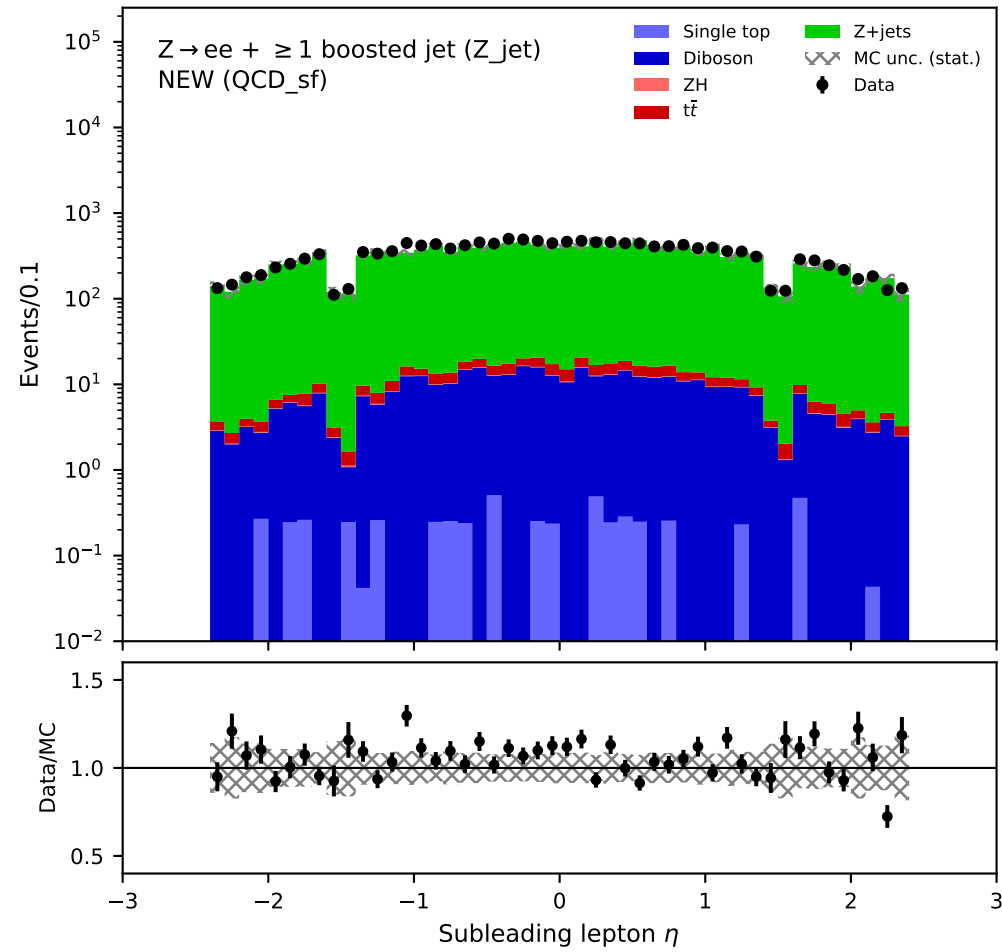
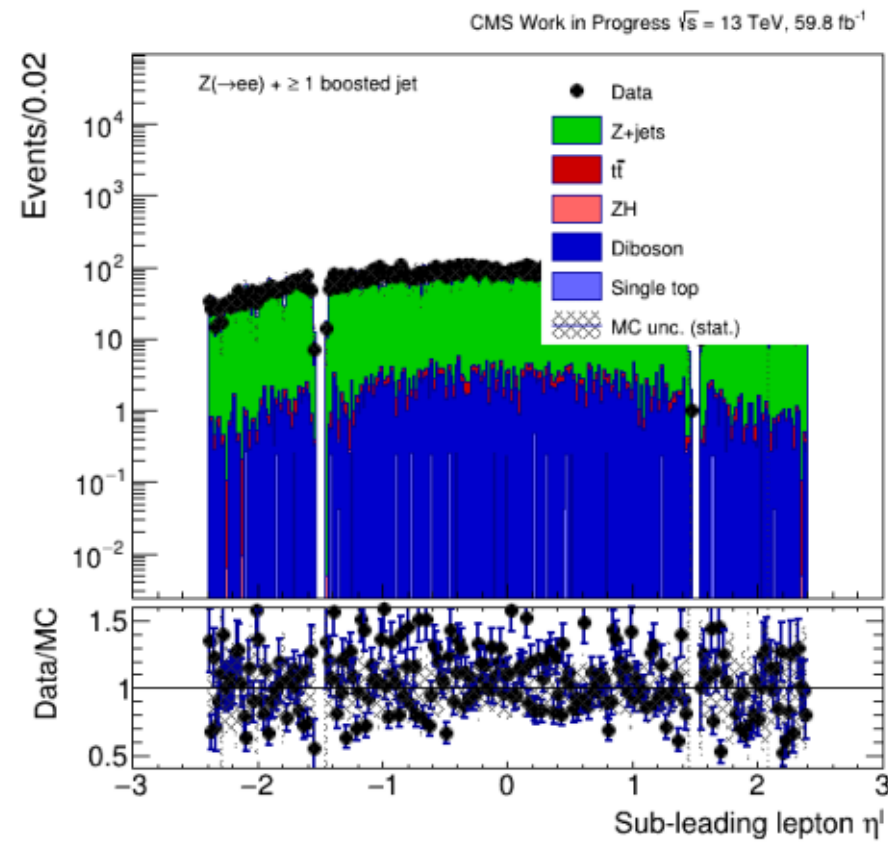


OLD: eta_lep0_Z_bjet_Zmm

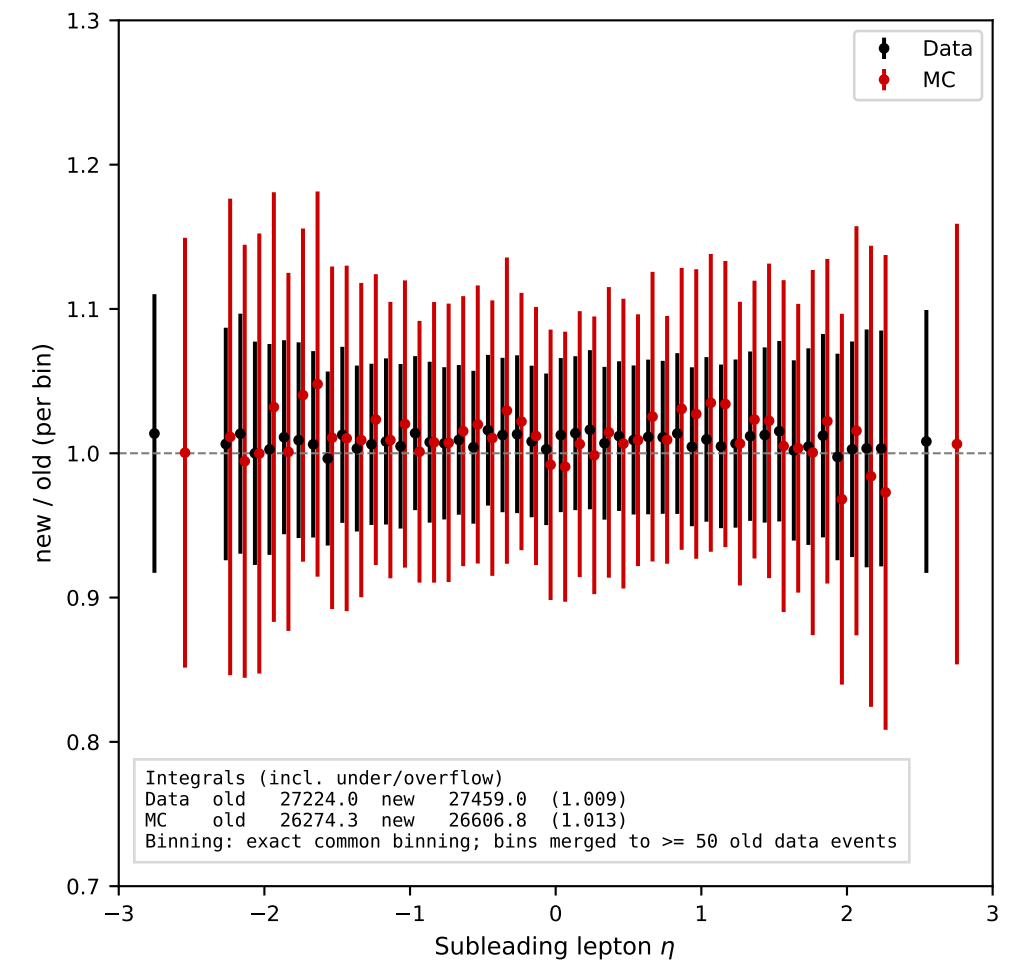
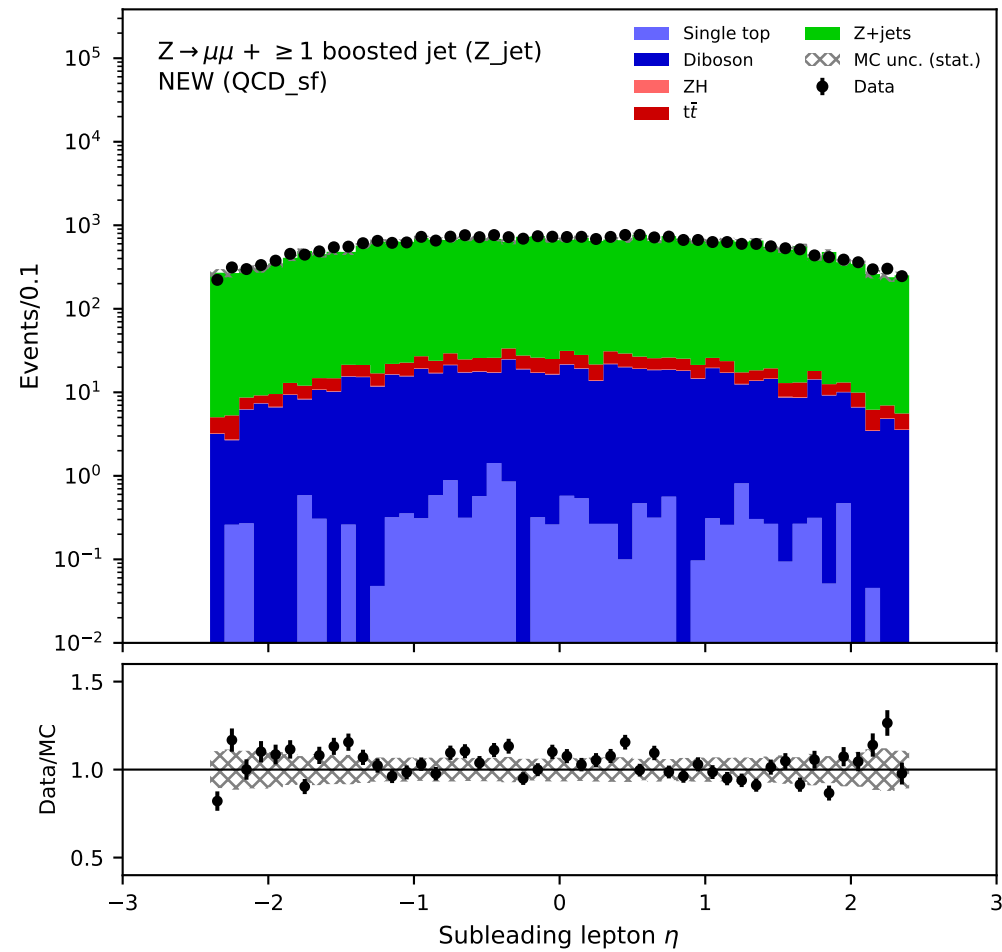
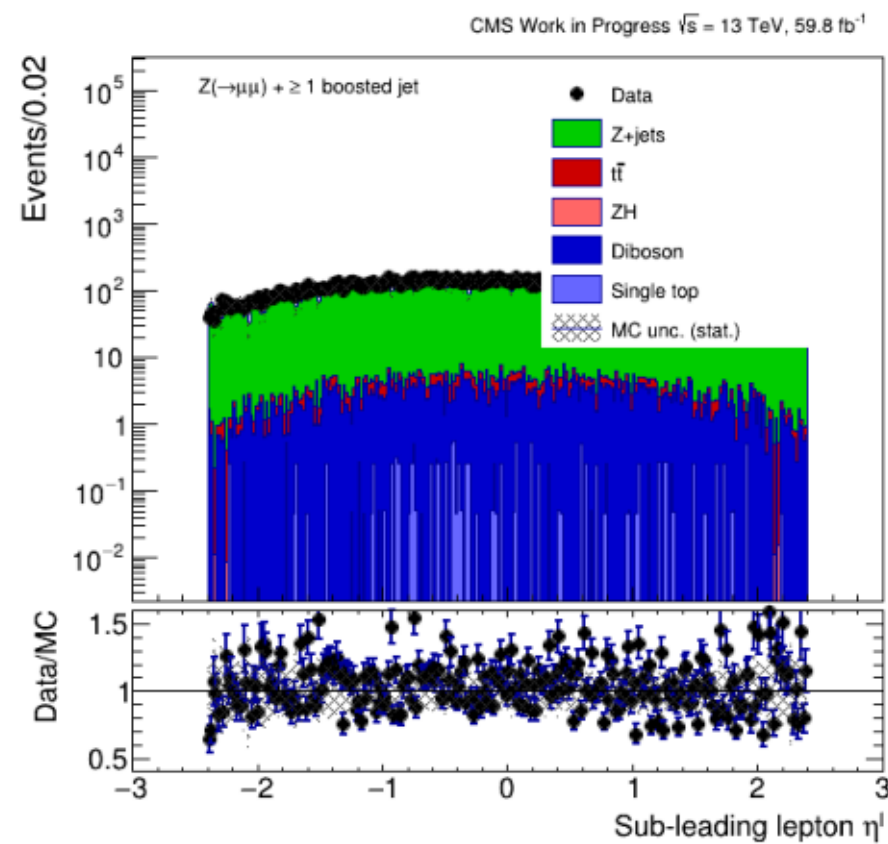


eta_lep1 | Z_jet | 2018

OLD: eta_lep1_Z_jet_Zee

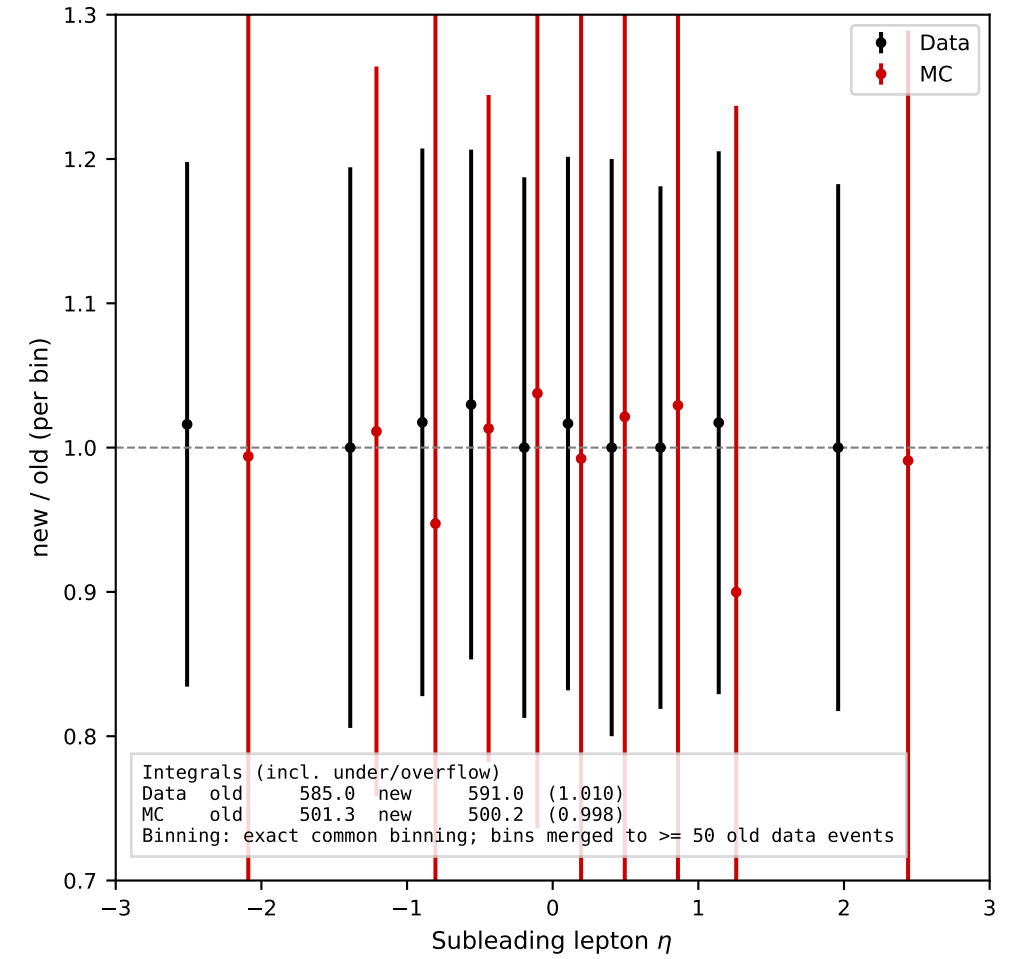
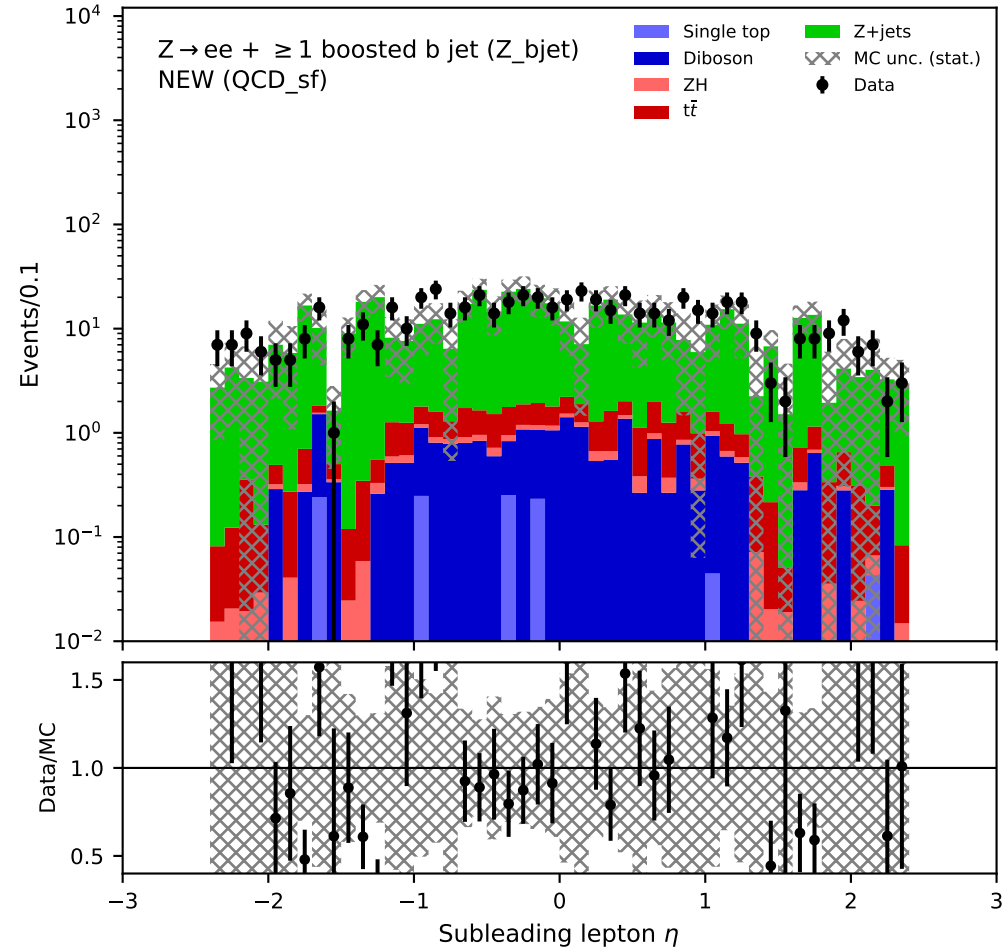
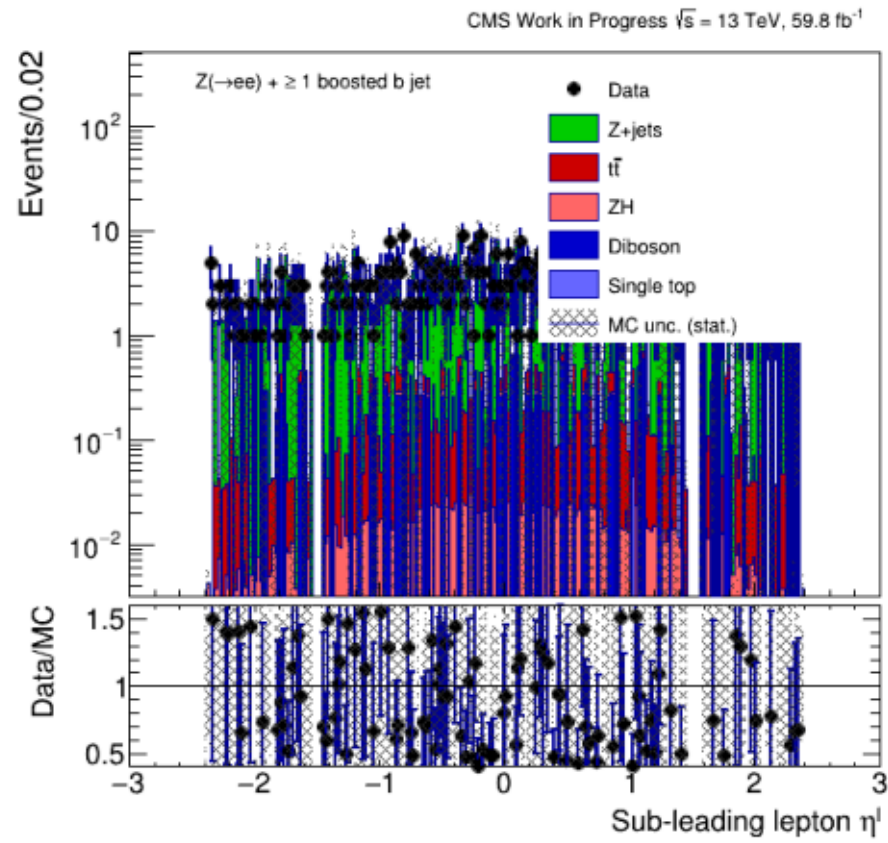


OLD: eta_lep1_Z_jet_Zmm

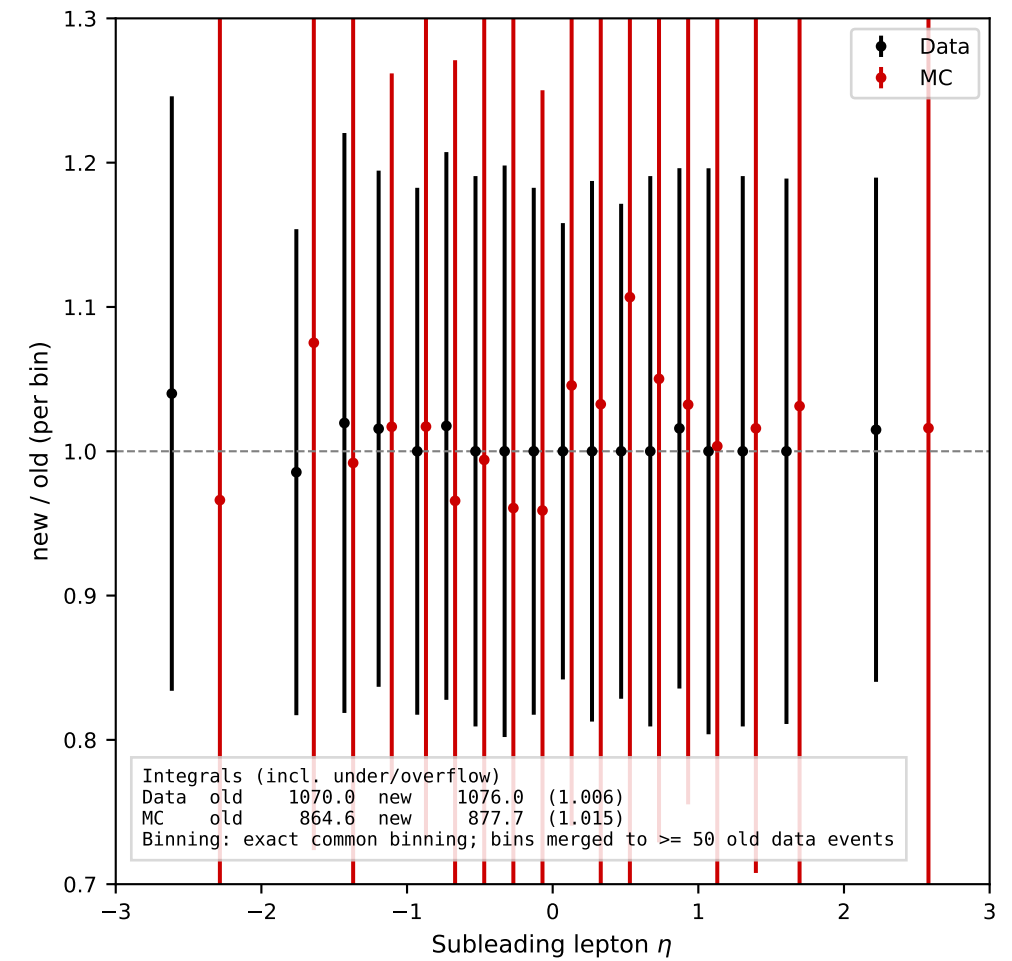
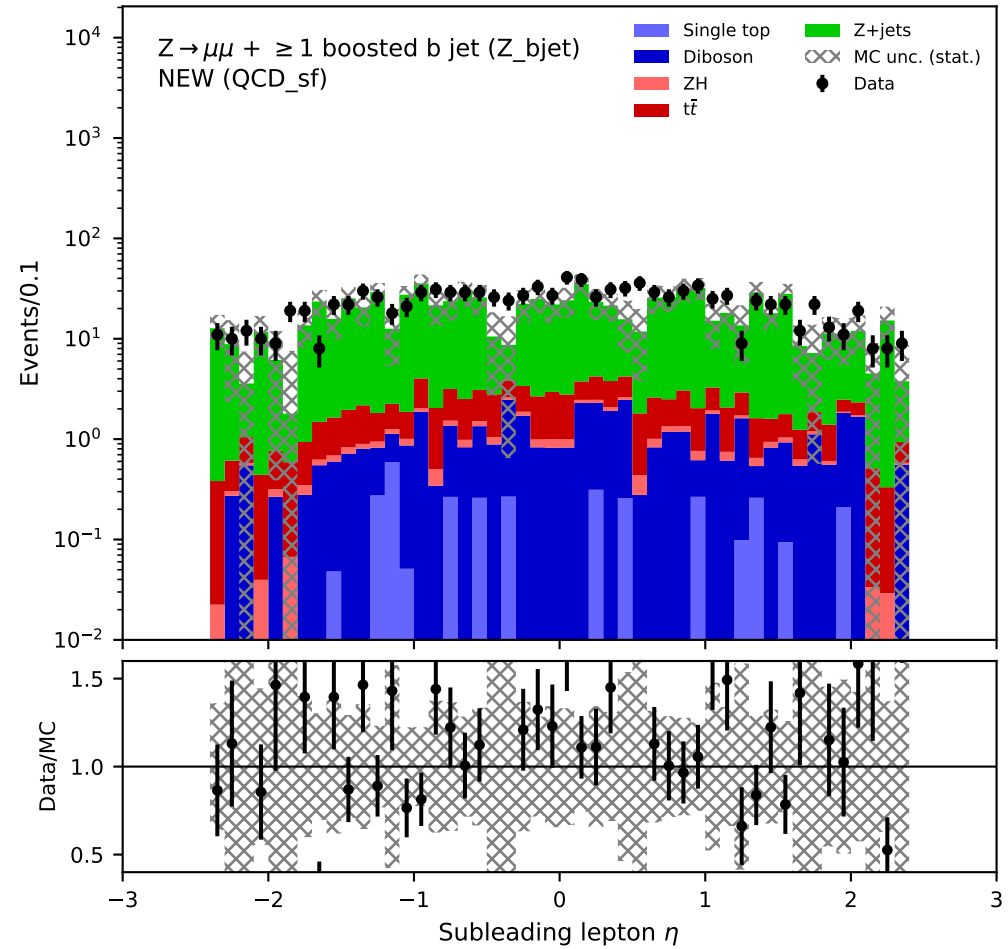
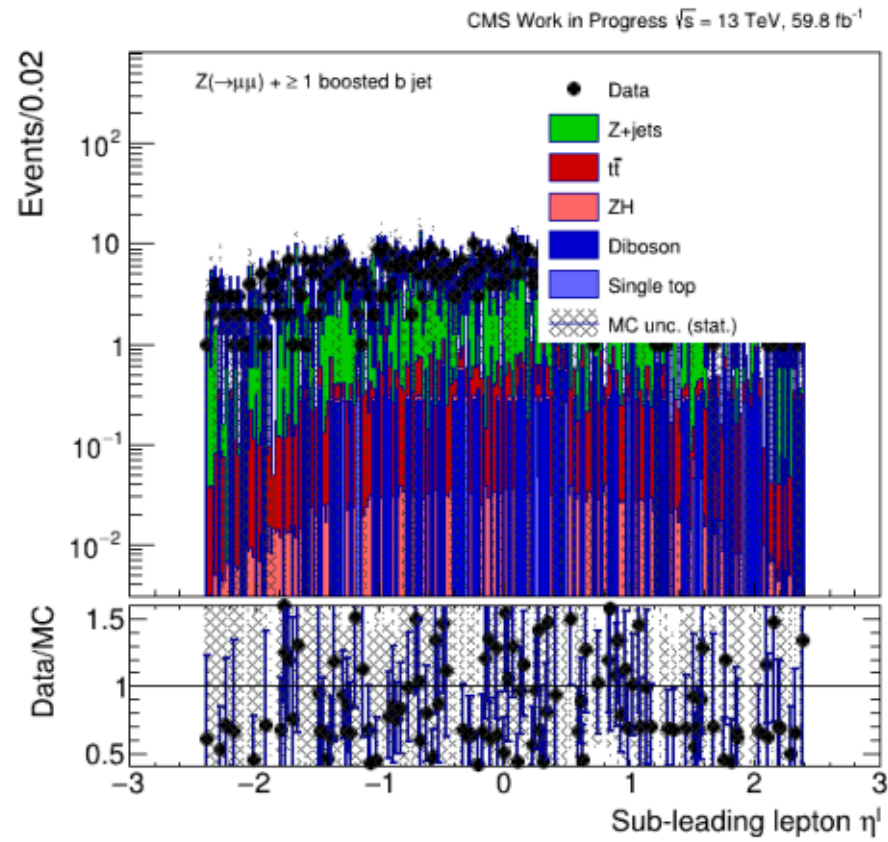


eta_lep1 | Z_bjet | 2018

OLD: eta_lep1_Z_bjet_Zee

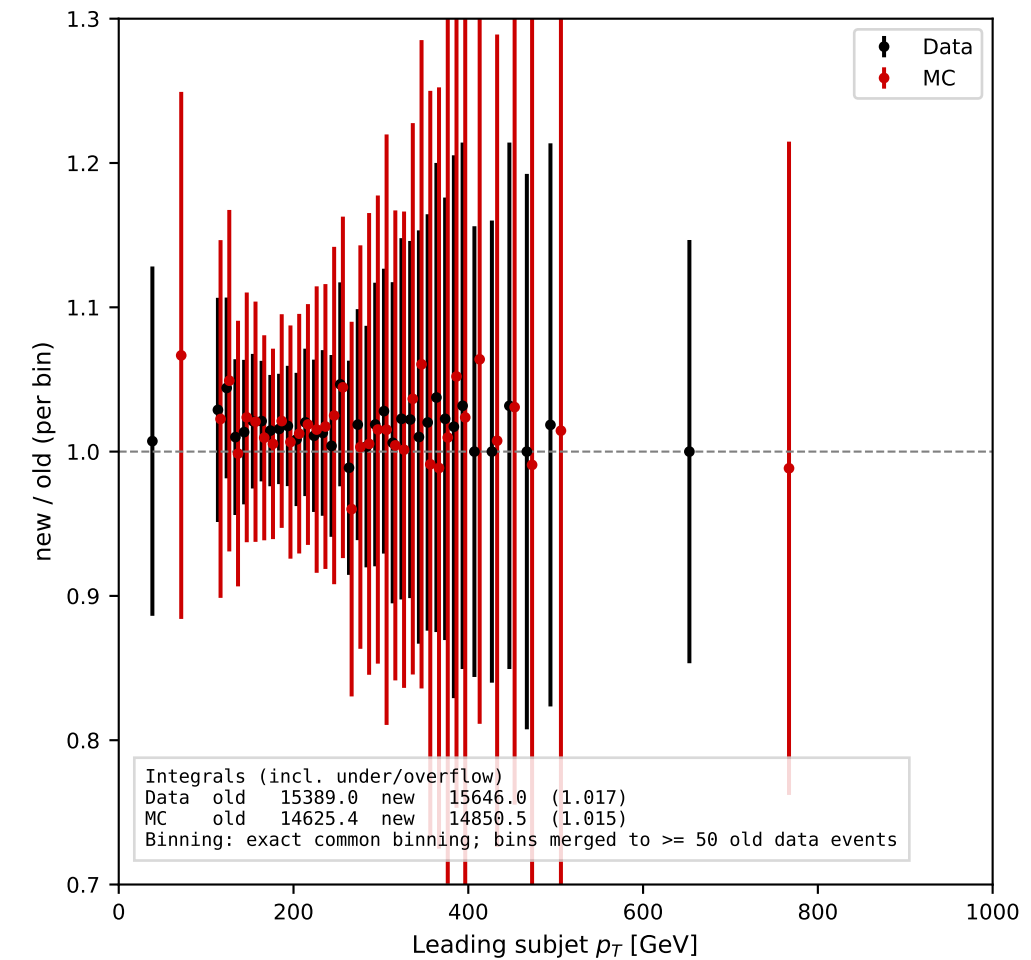
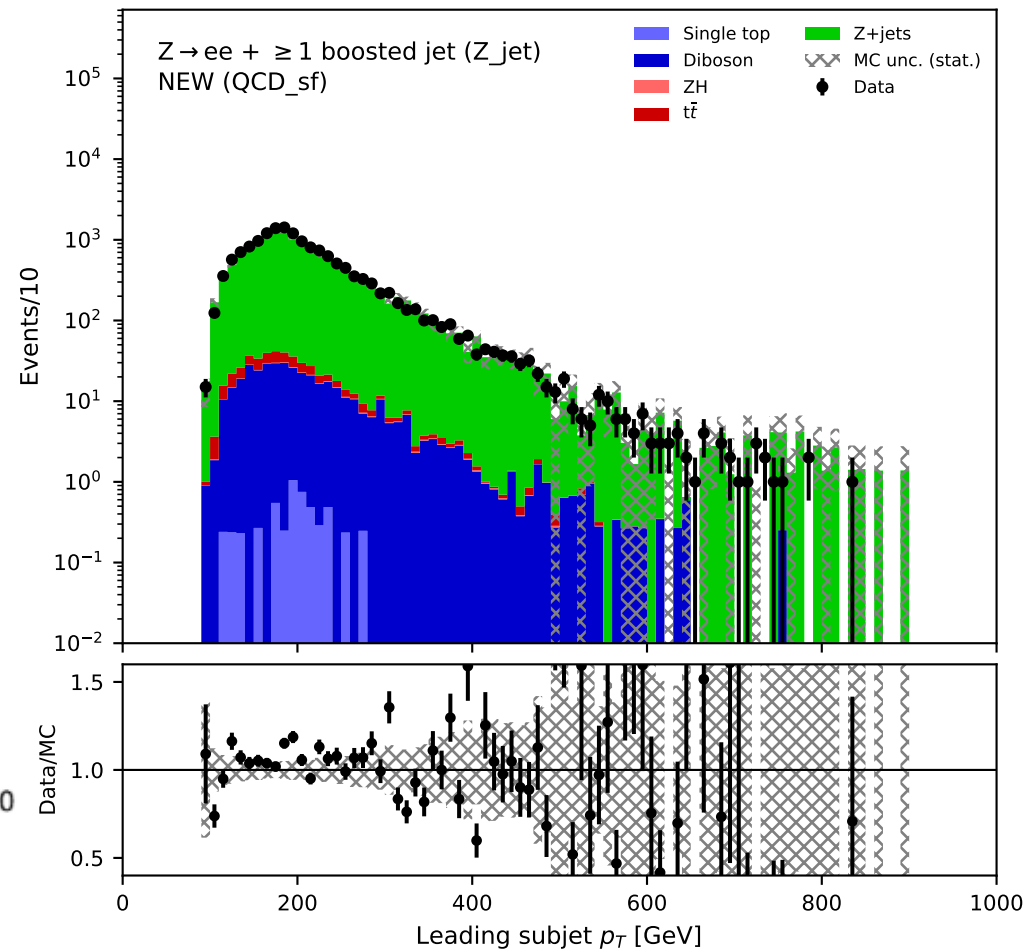
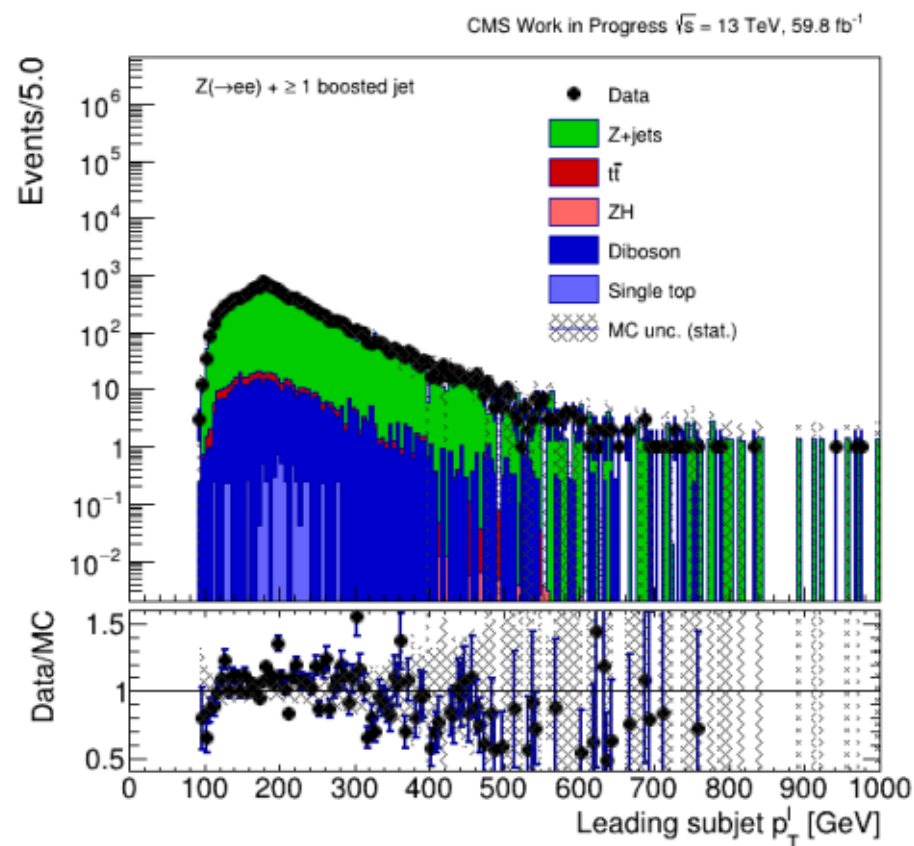


OLD: eta_lep1_Z_bjet_Zmm

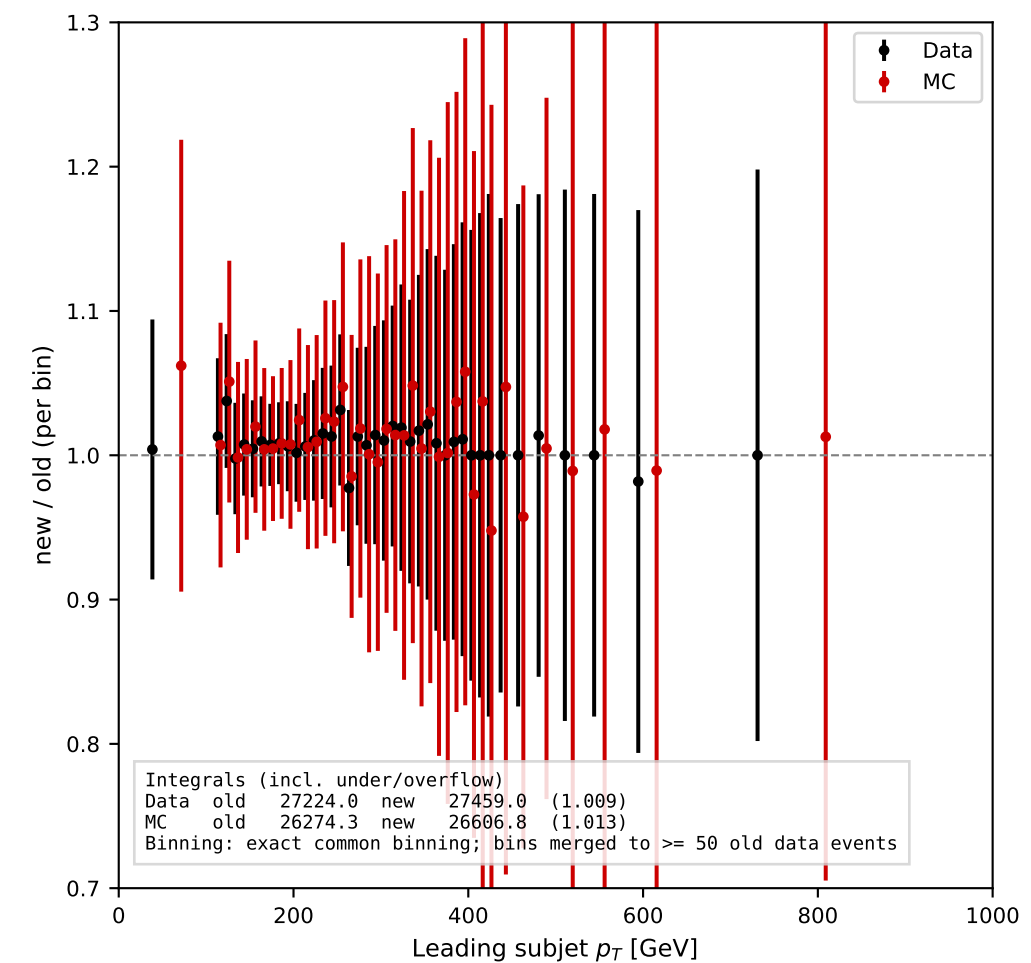
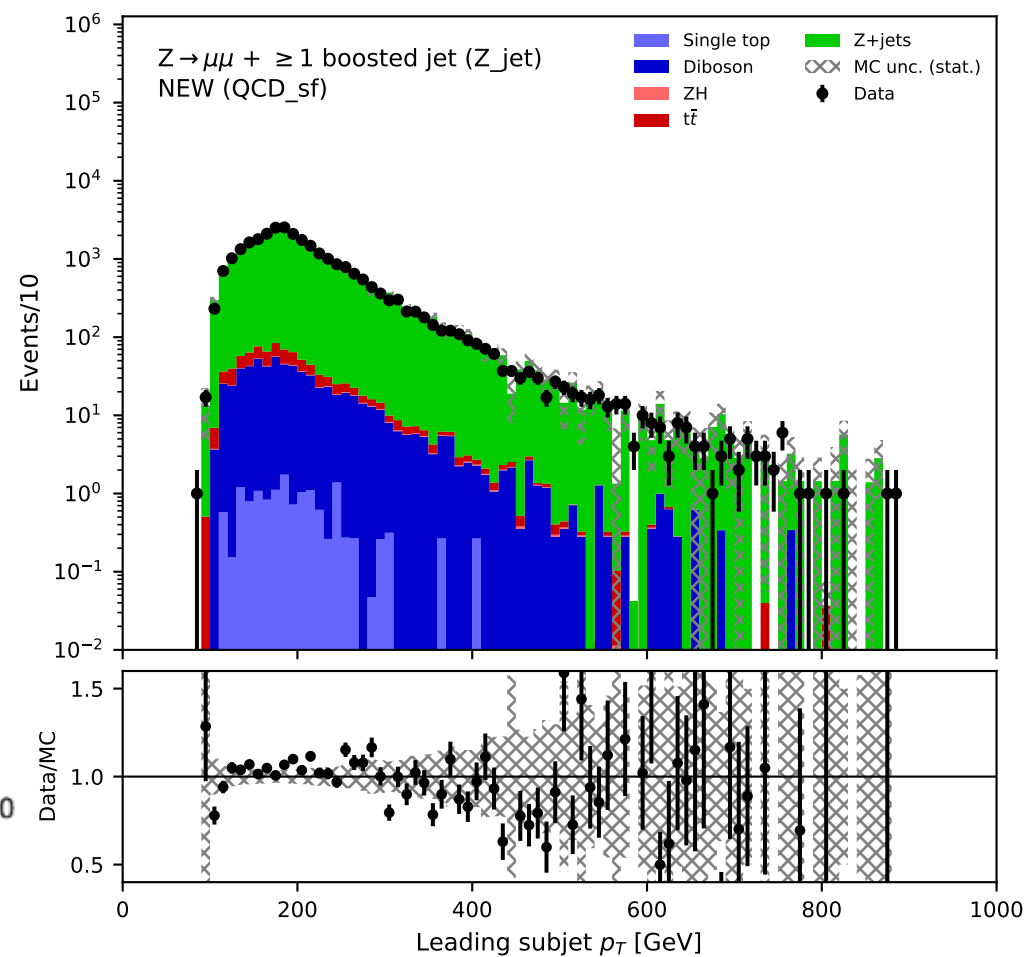
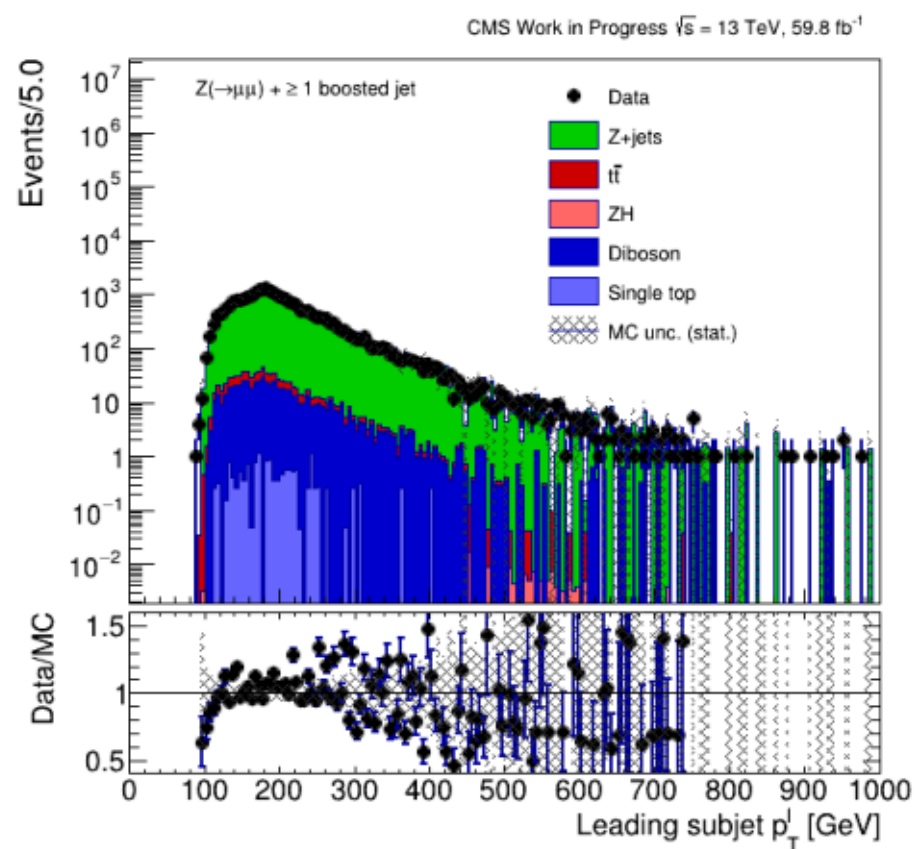


pt_sub0 | Z_jet | 2018

OLD: pt_sub0_Z_jet_Zee



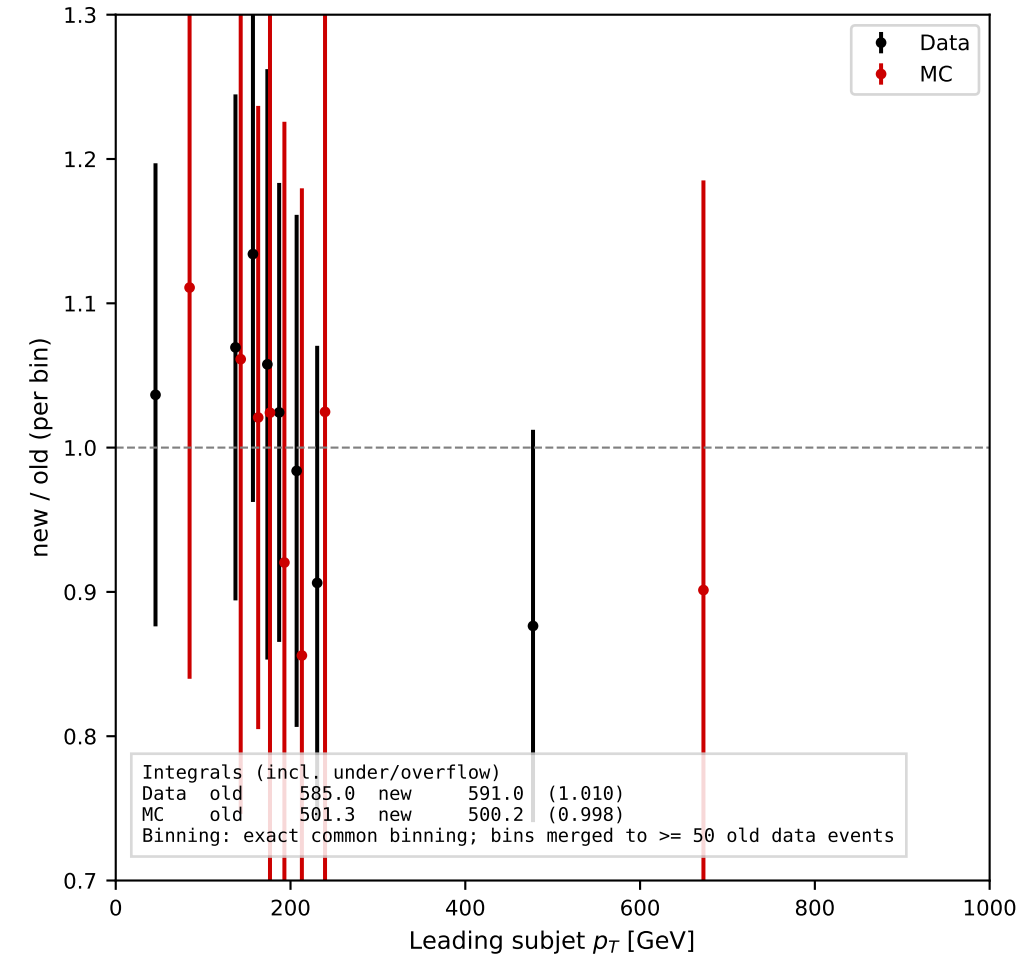
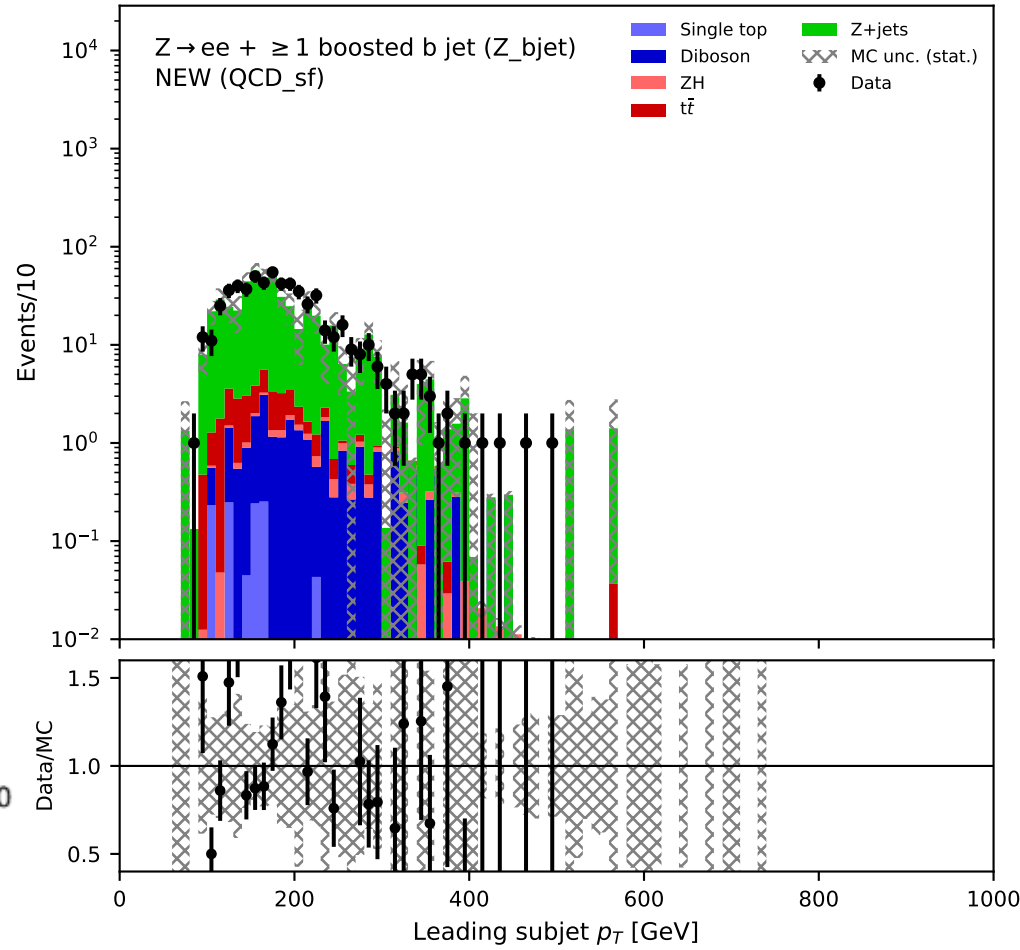
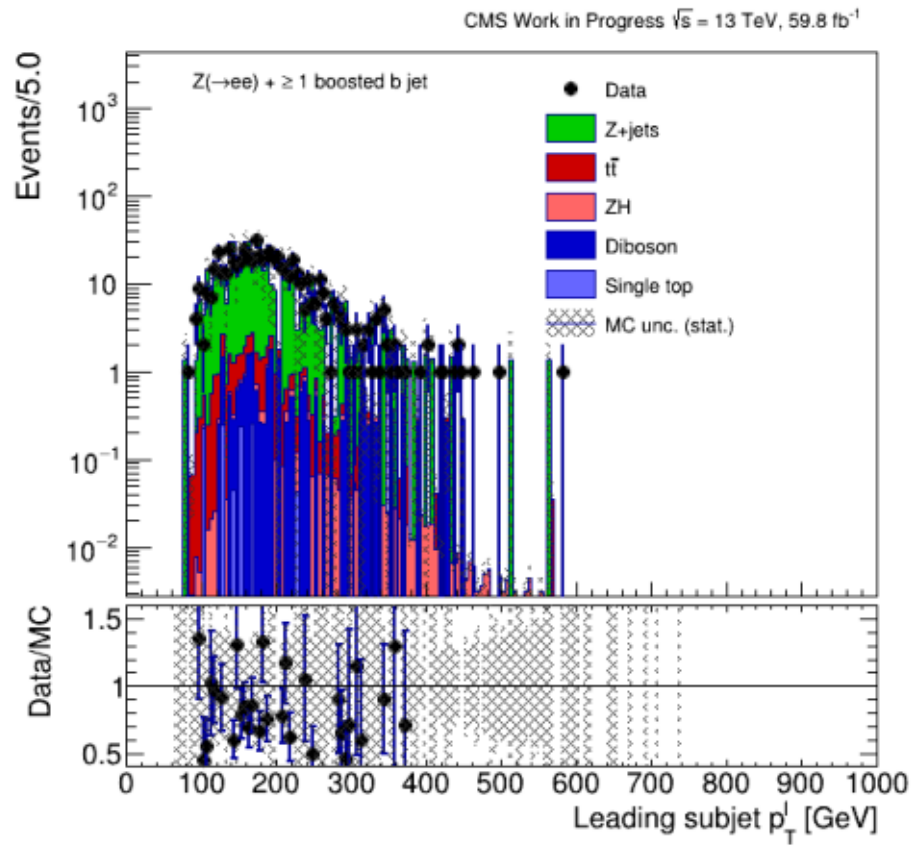
OLD: pt_sub0_Z_jet_Zmm



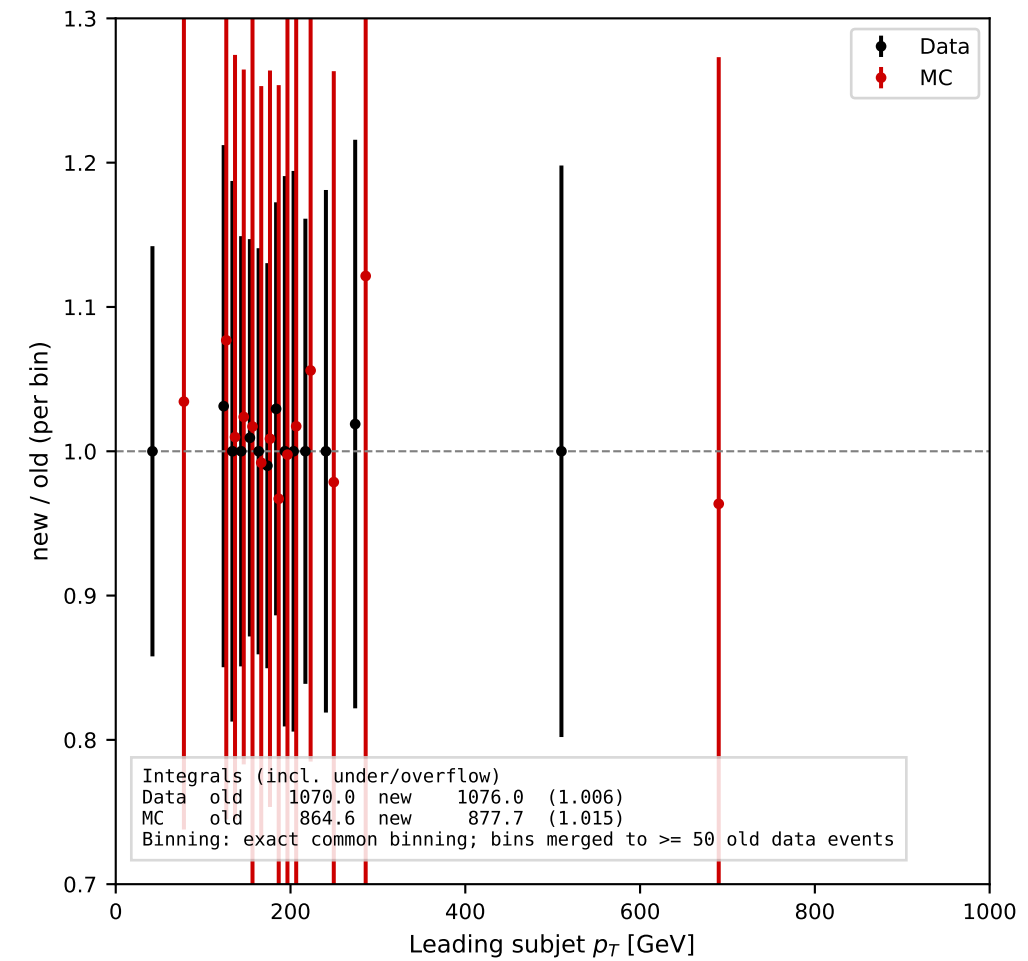
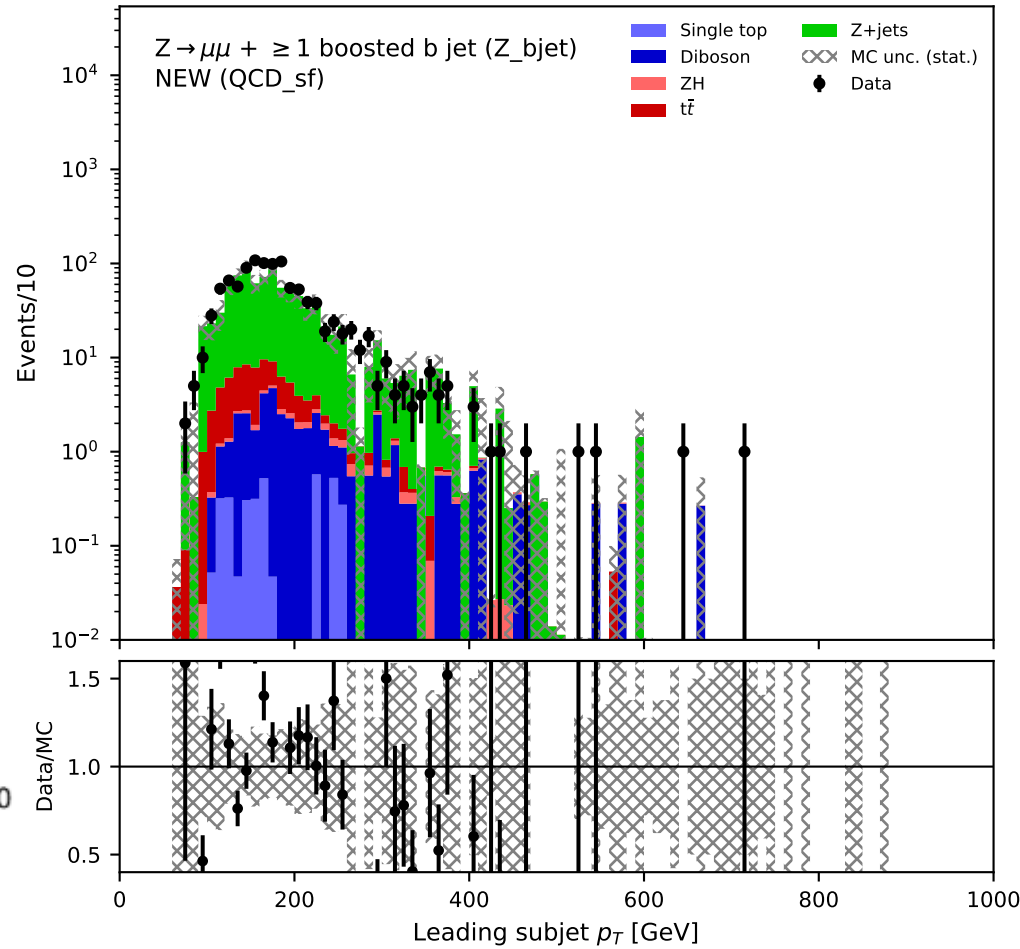
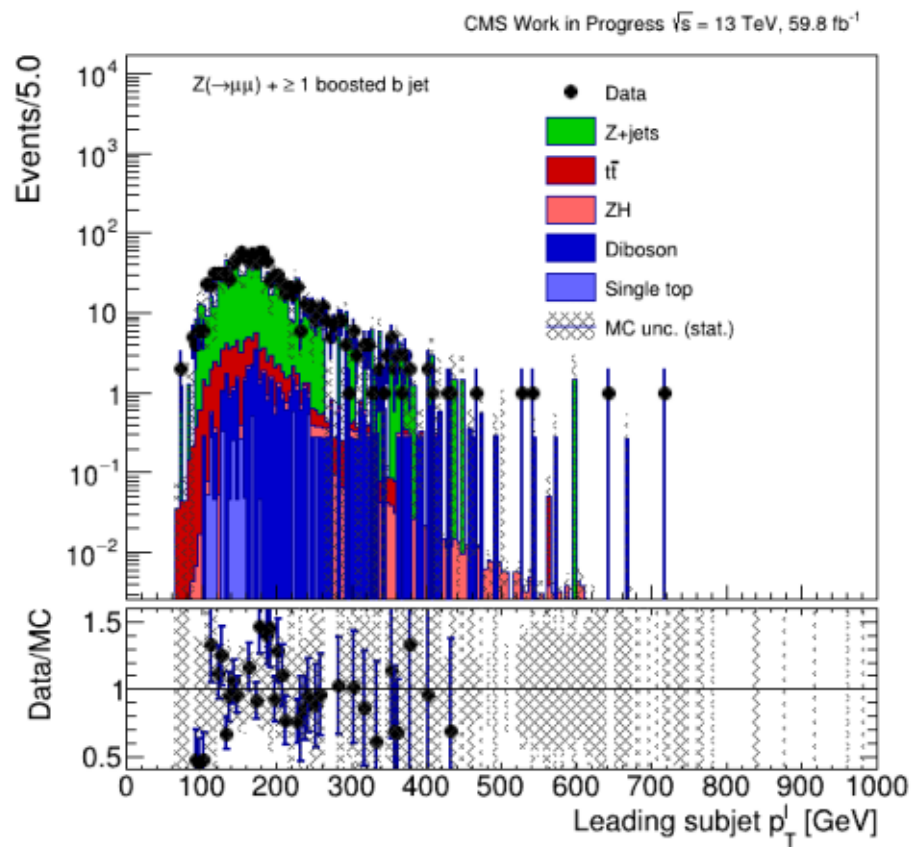
pt_sub0 | Z_bjet | 2018

Z_bjet: the old framework fills subjet histograms from the leading jet's subjects, ours from the tagged jet's subjects; differences are expected.

OLD: pt_sub0_Z_bjet_Zee

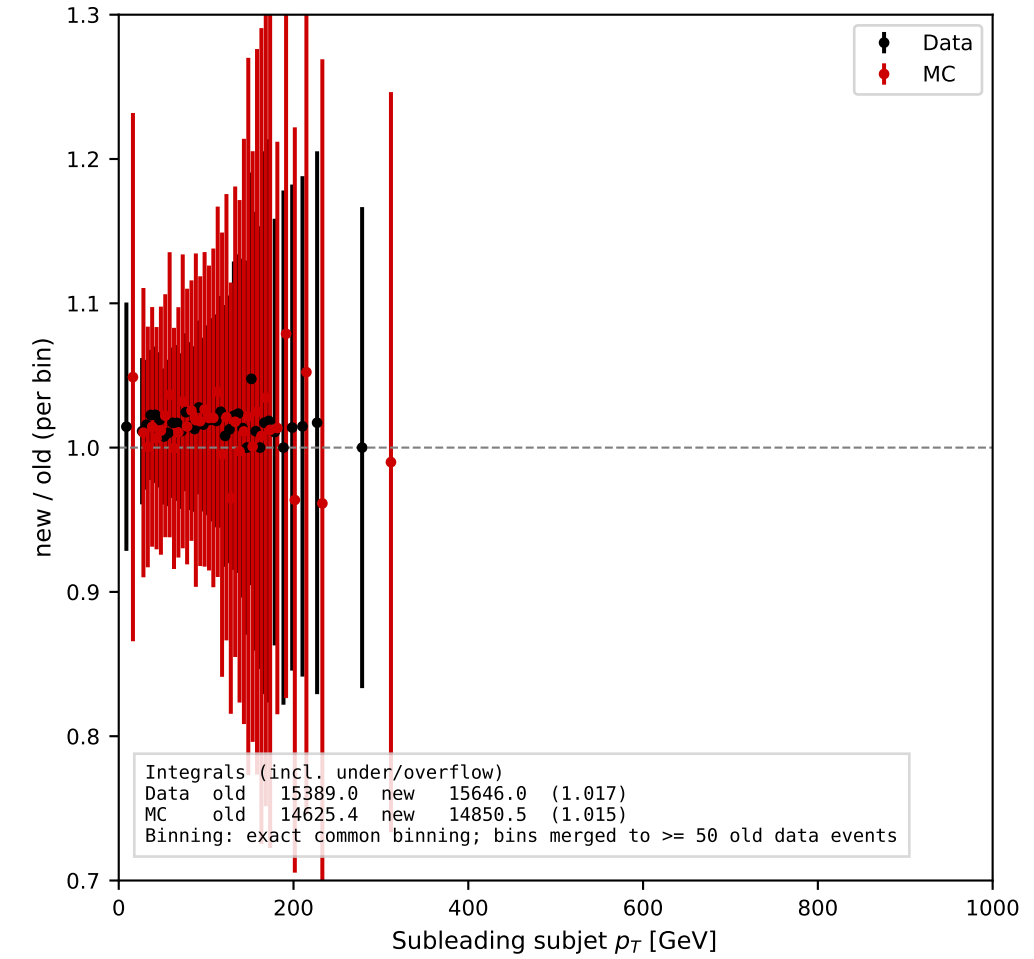
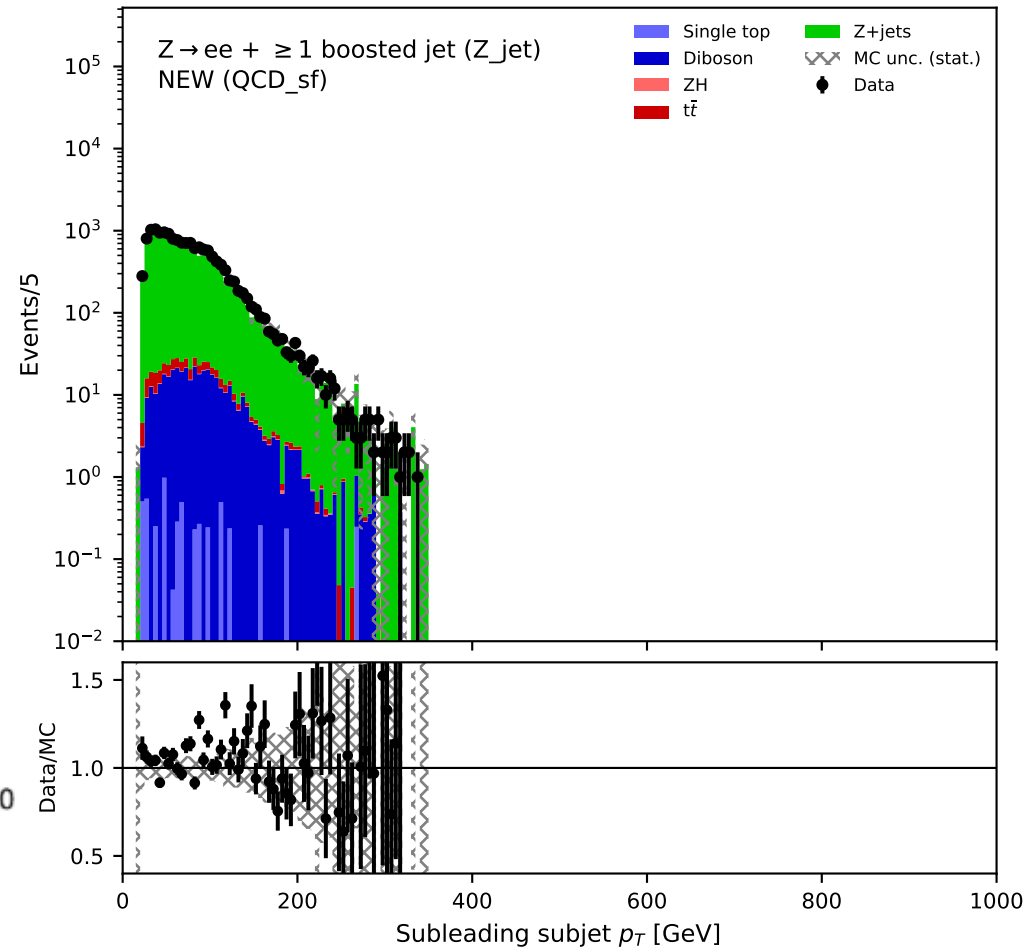
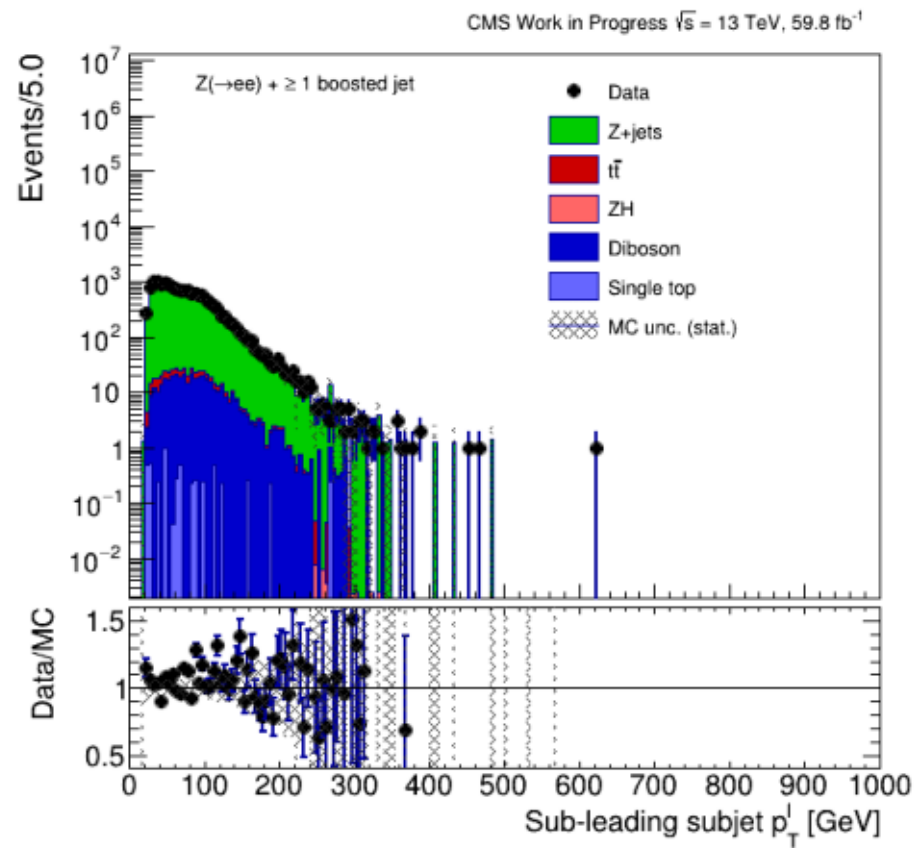


OLD: pt_sub0_Z_bjet_Zmm

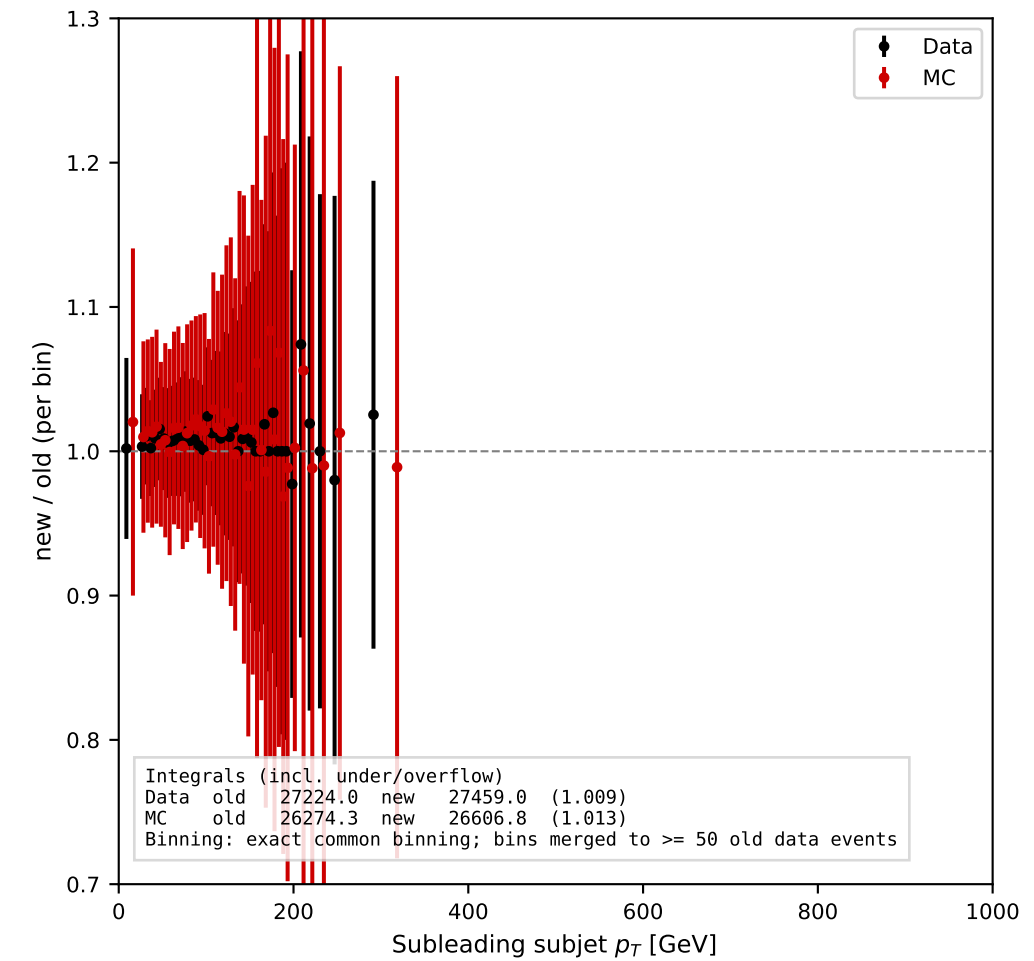
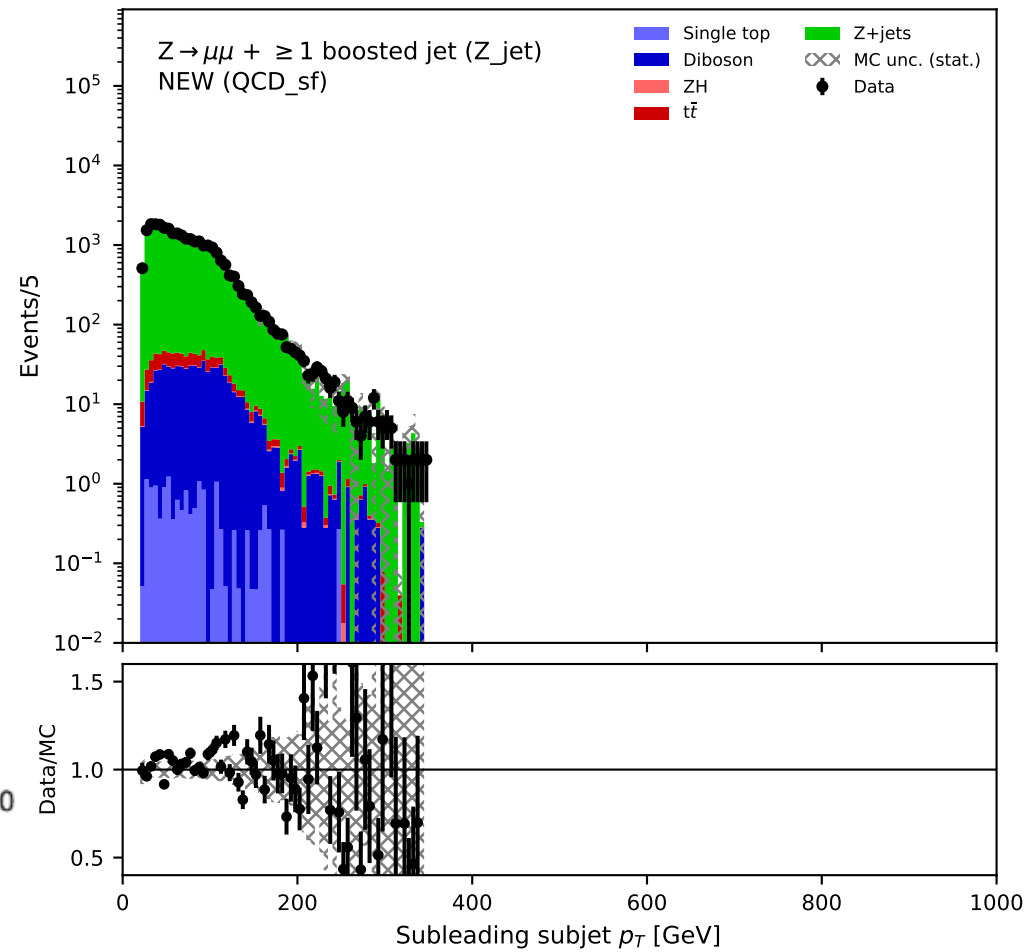
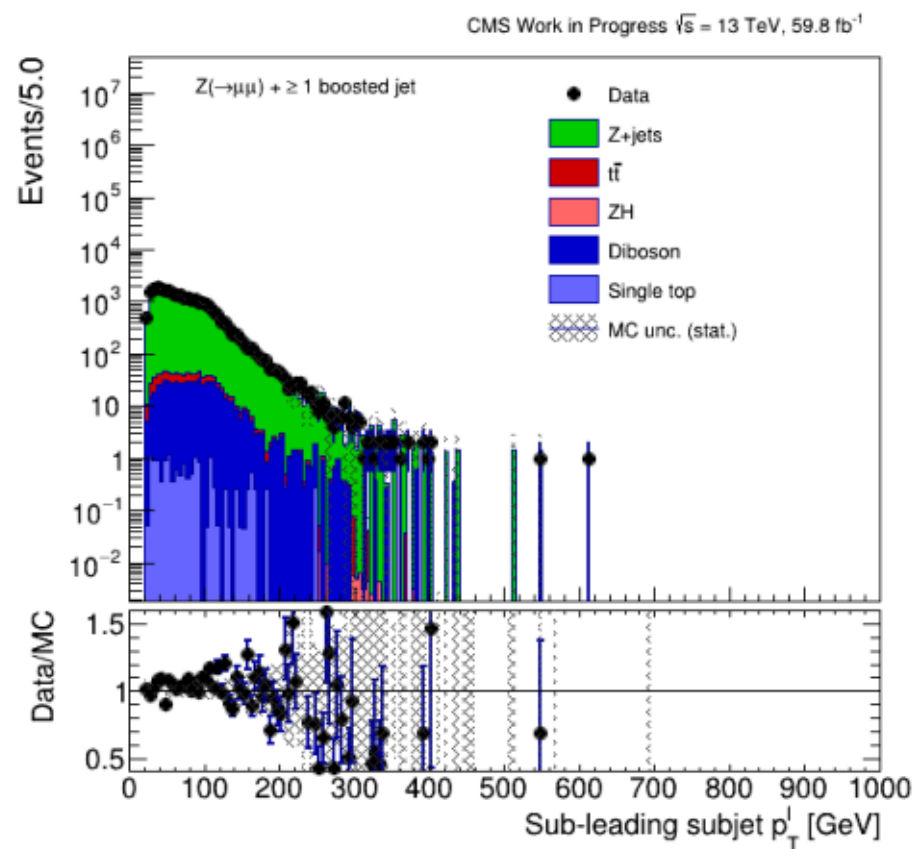


pt_sub1 | Z_jet | 2018

OLD: pt_sub1_Z_jet_Zee



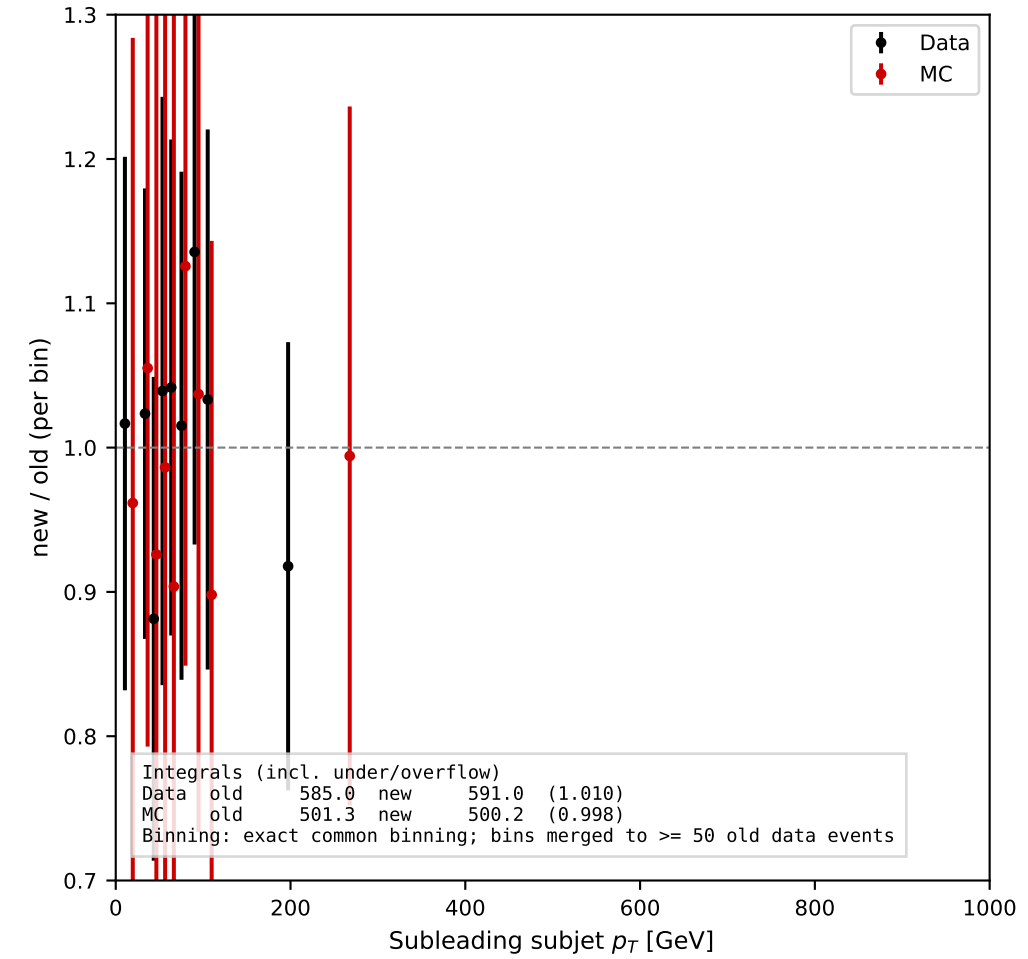
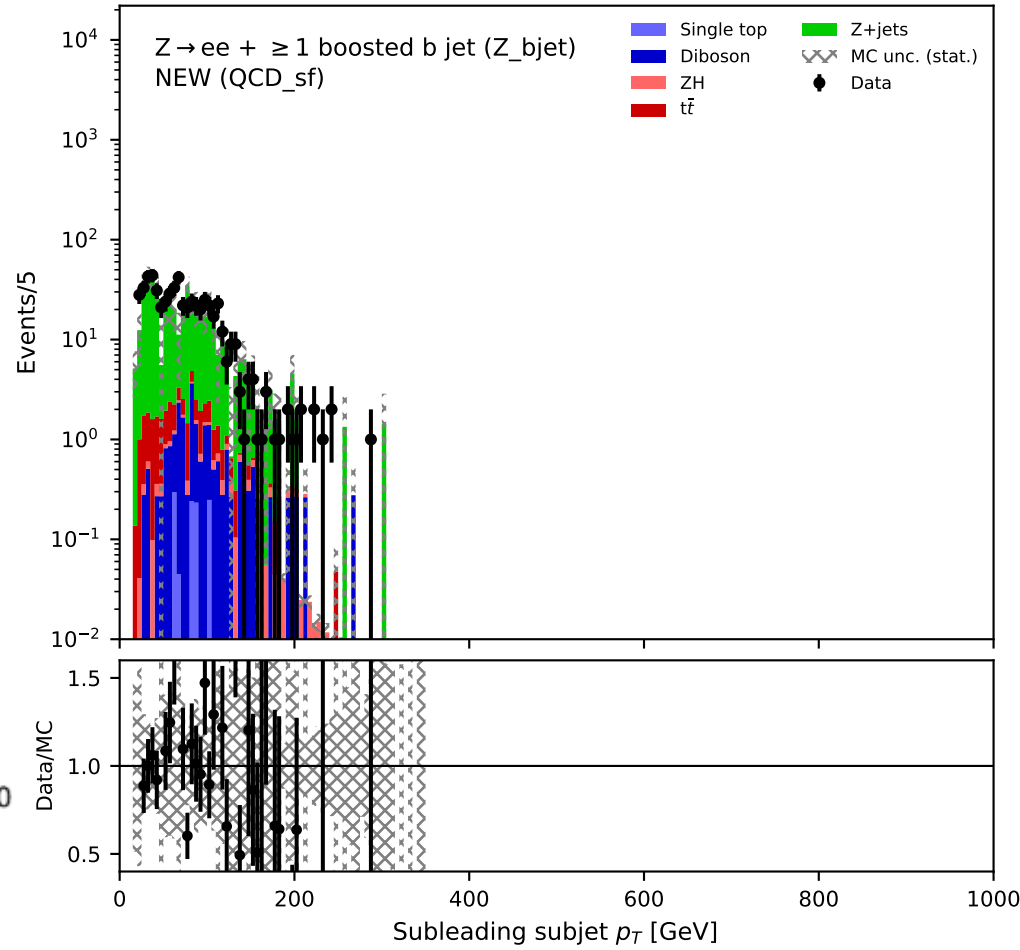
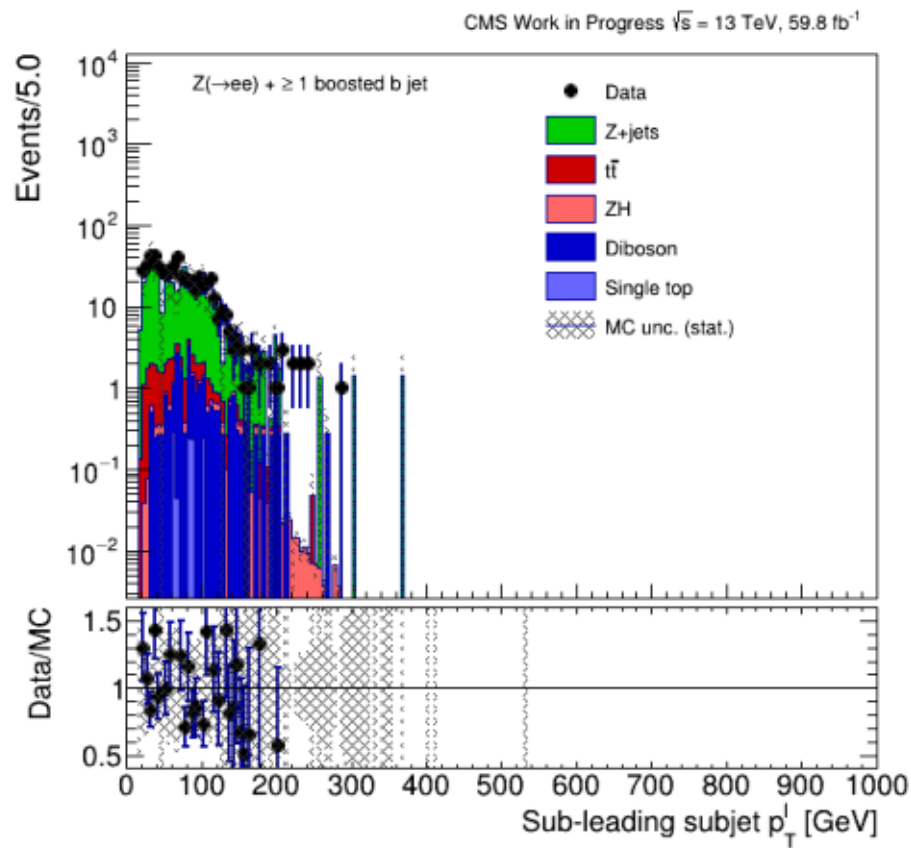
OLD: pt_sub1_Z_jet_Zmm



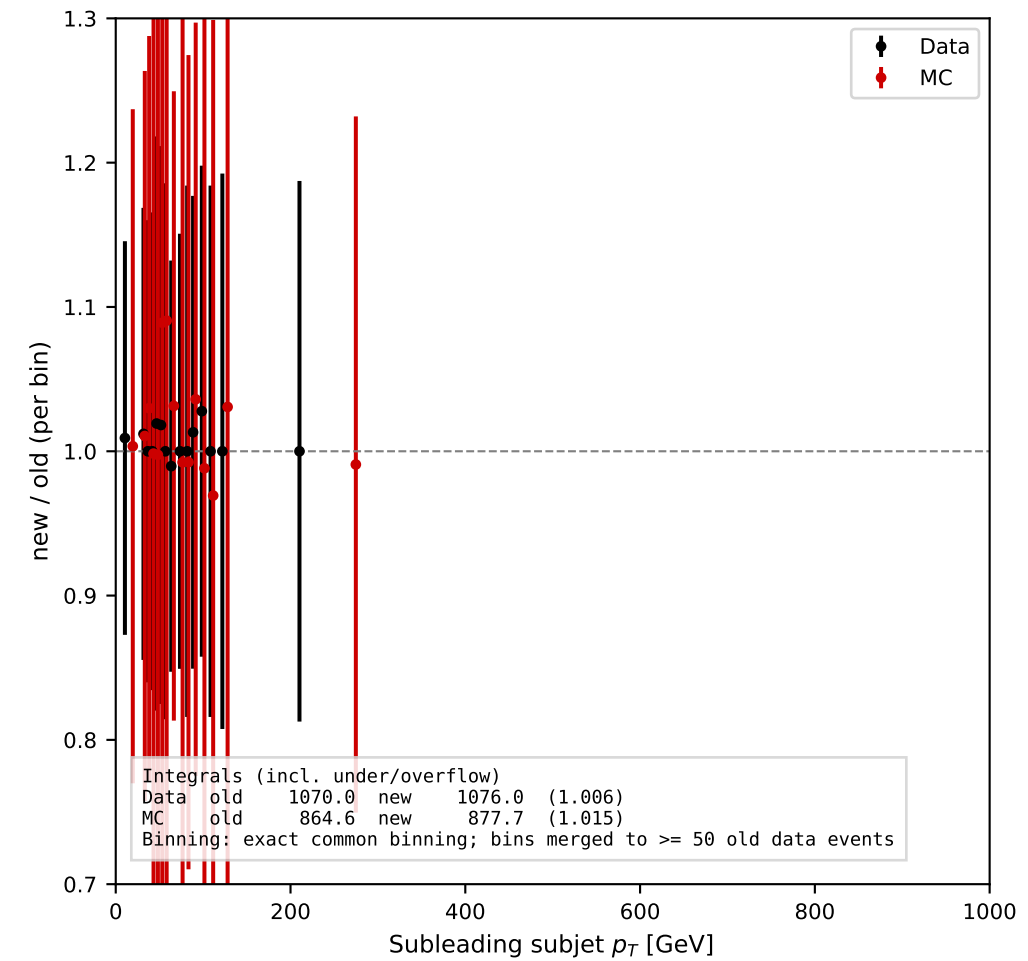
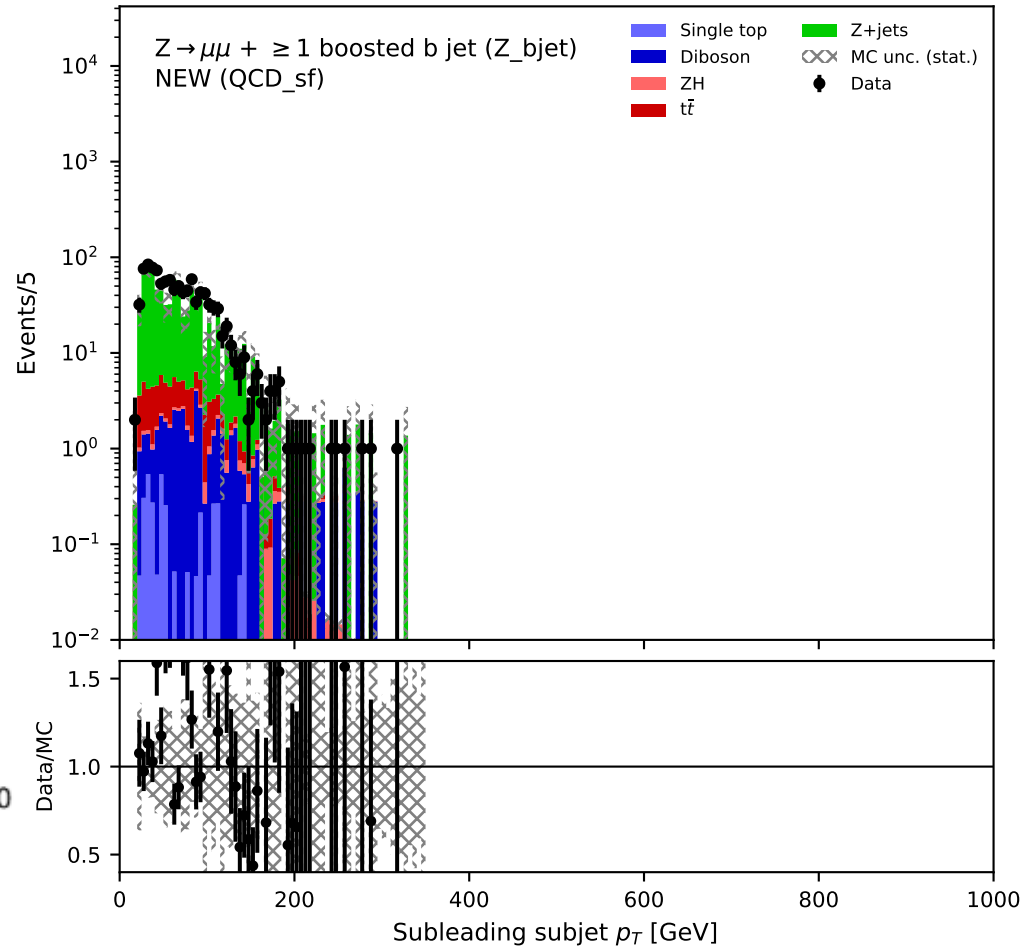
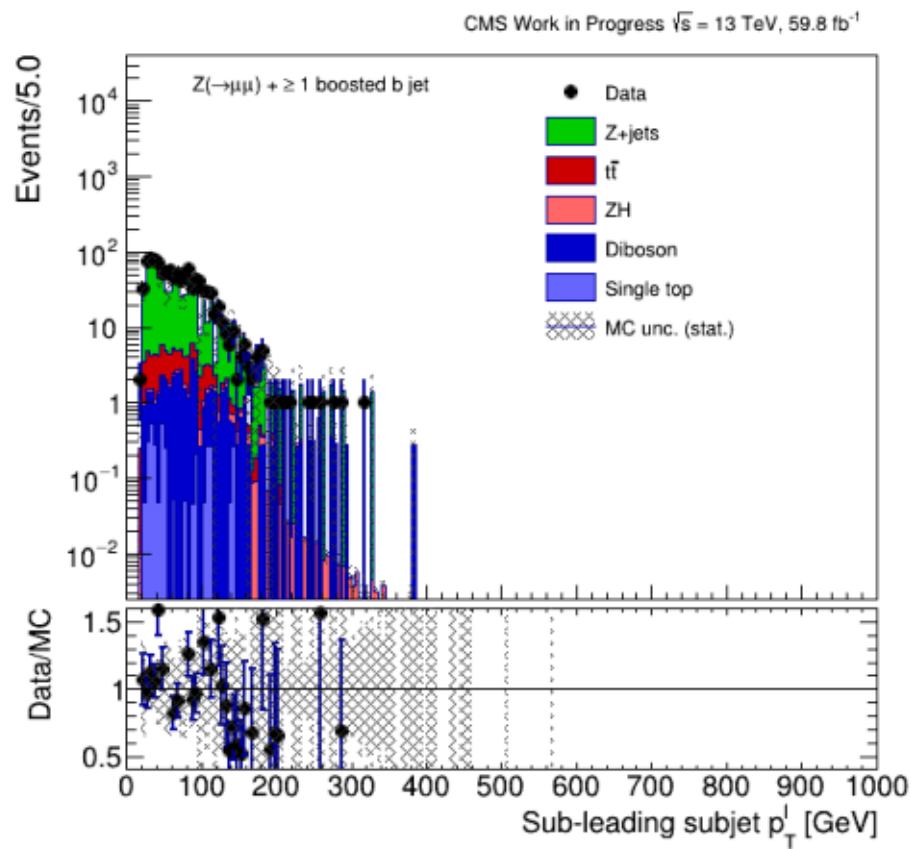
pt_sub1 | Z_bjet | 2018

Z_bjet: the old framework fills subjet histograms from the leading jet's subjects, ours from the tagged jet's subjects; differences are expected.

OLD: pt_sub1_Z_bjet_Zee



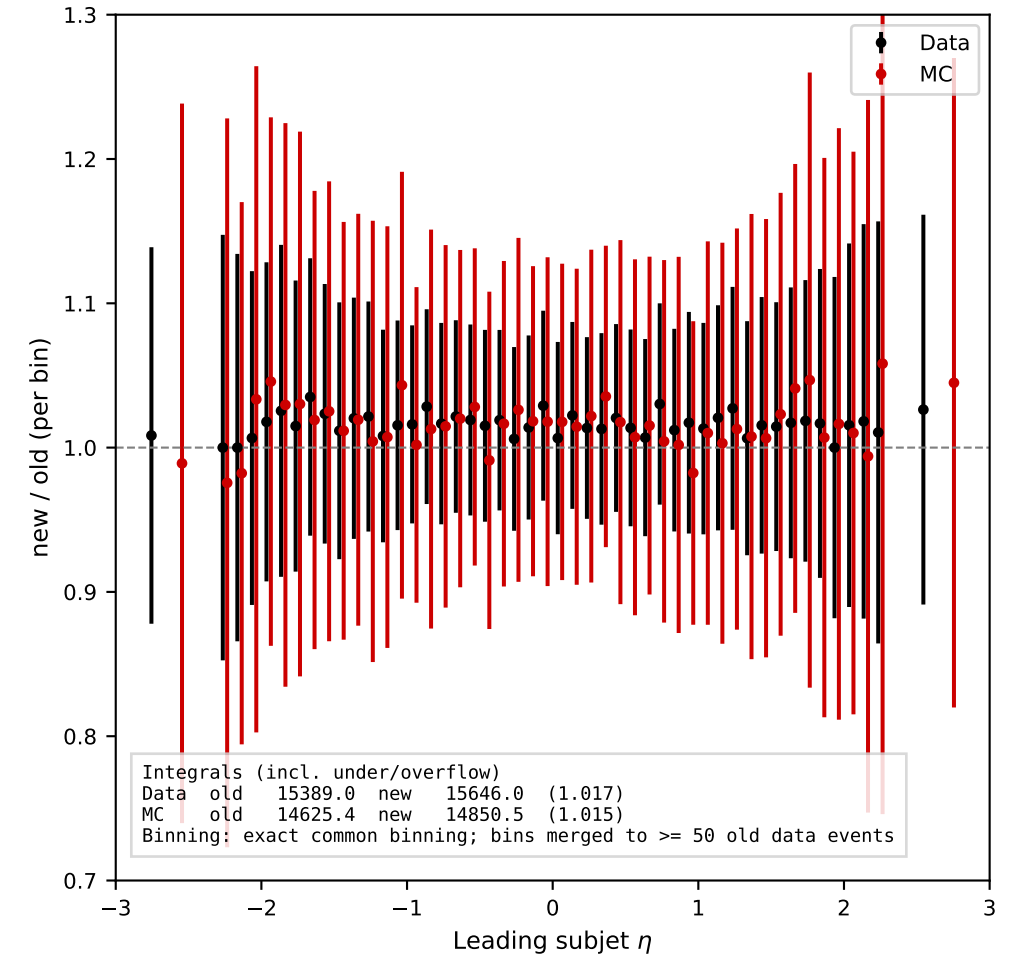
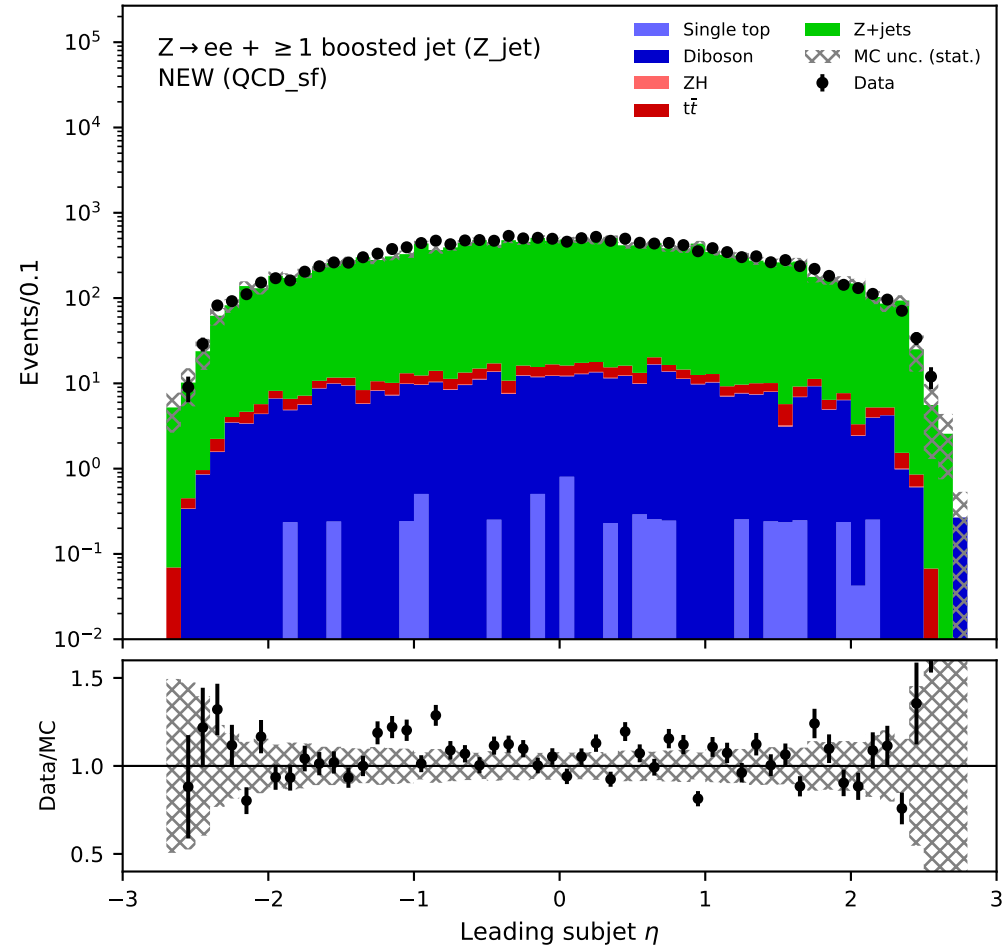
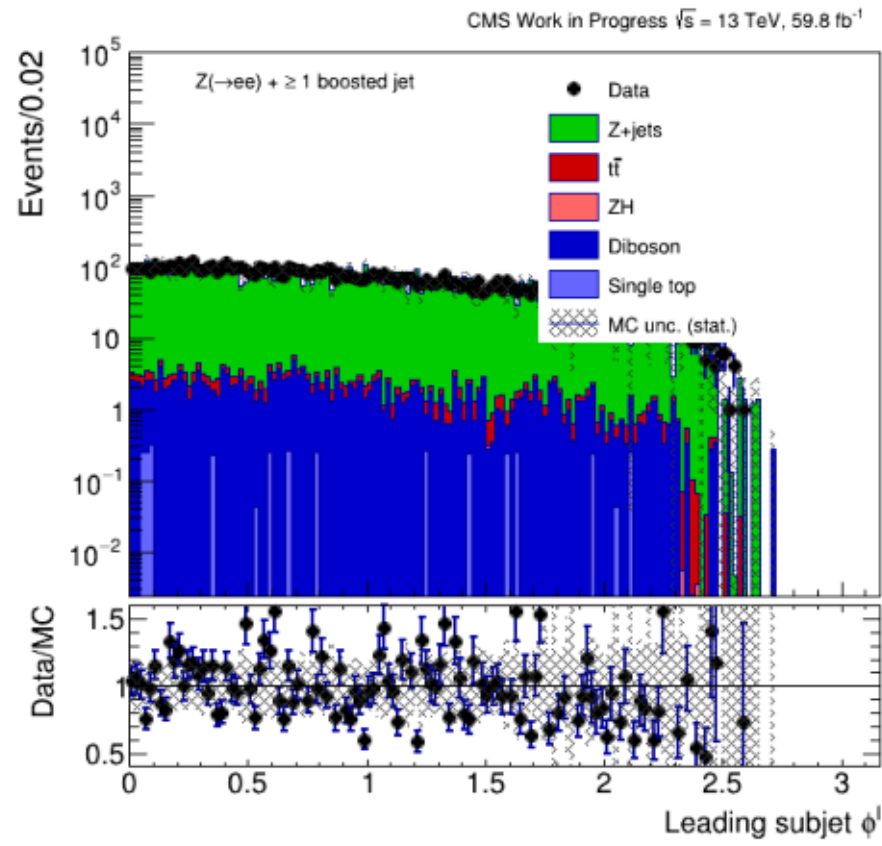
OLD: pt_sub1_Z_bjet_Zmm



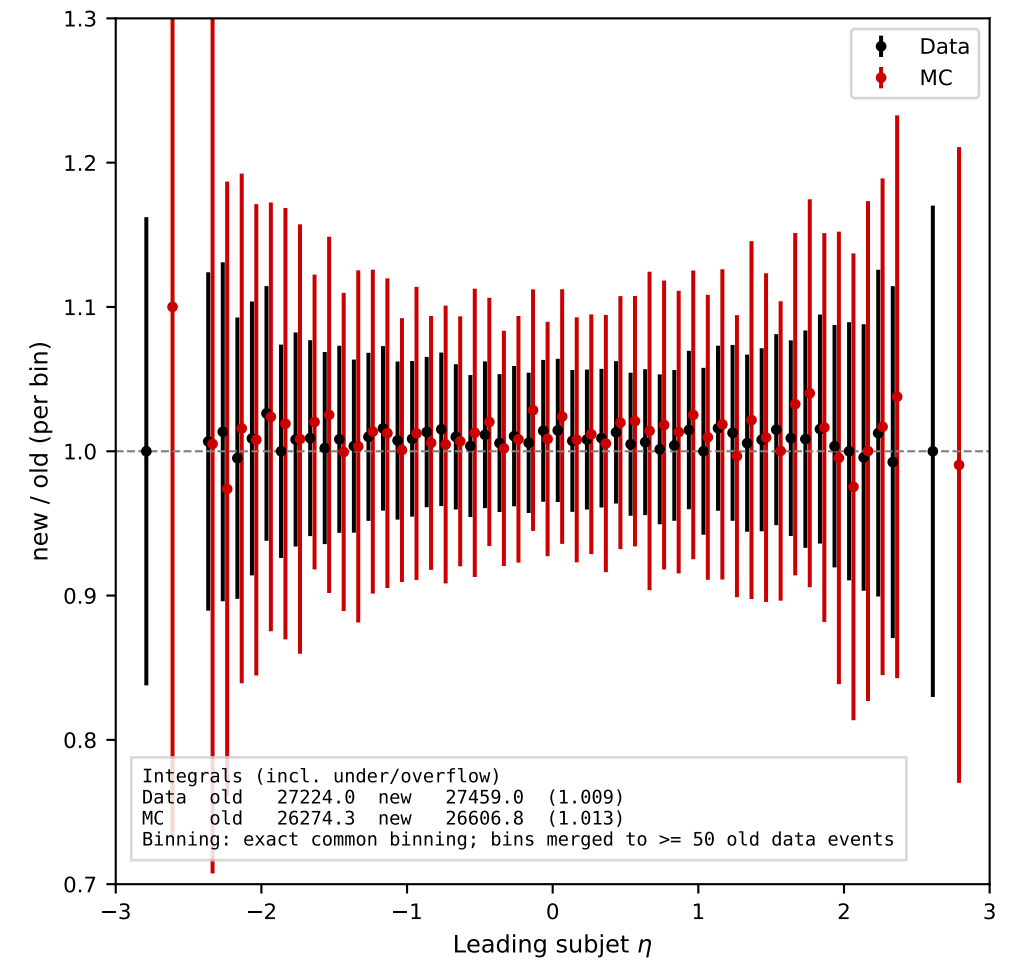
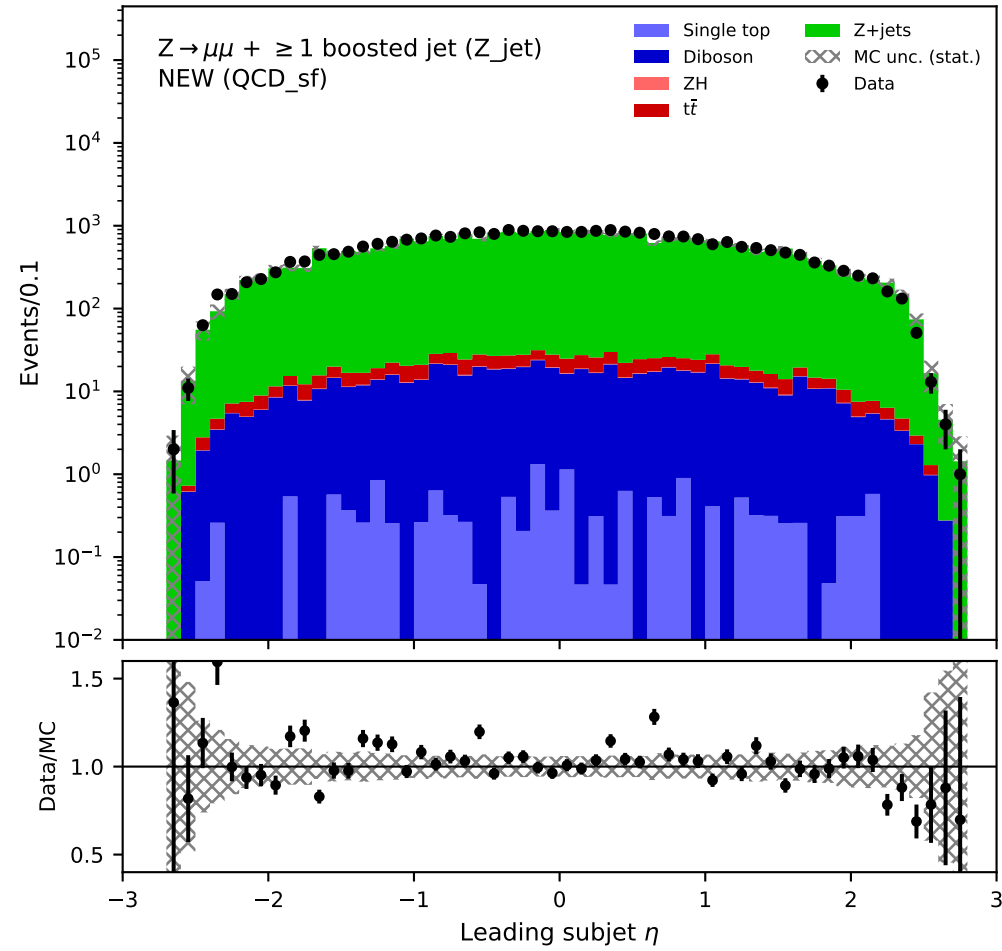
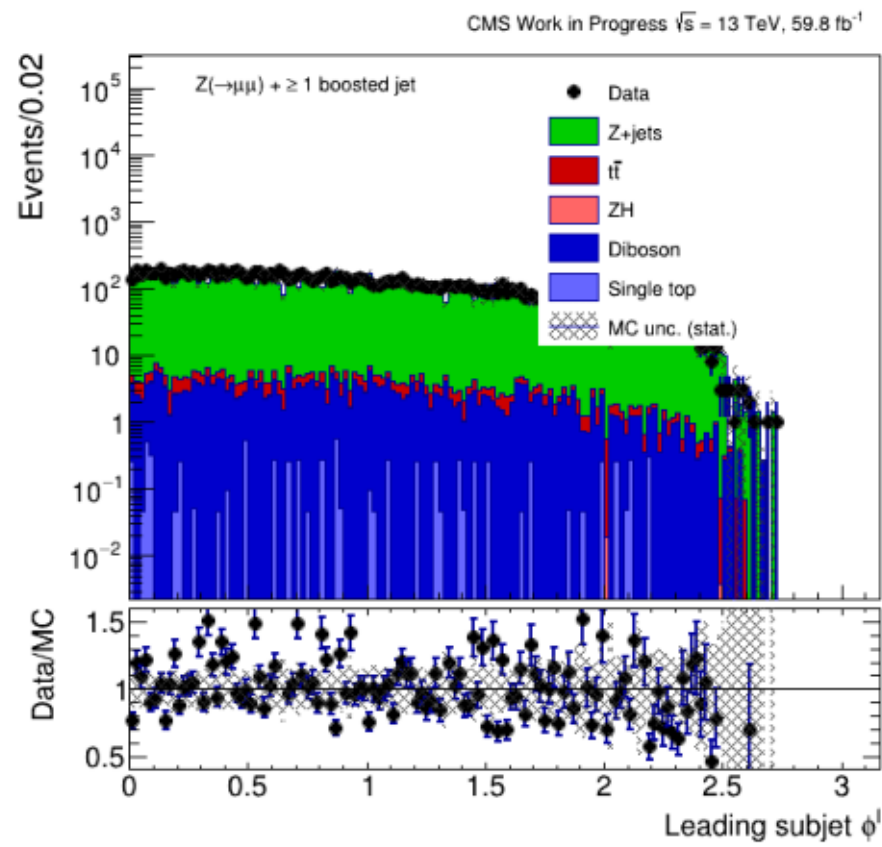
eta_sub0 | Z_jet | 2018

Old plot shown is its 'phi_sub0', which holds subjet eta (old eta/phi swap).

OLD: phi_sub0_Z_jet_Zee



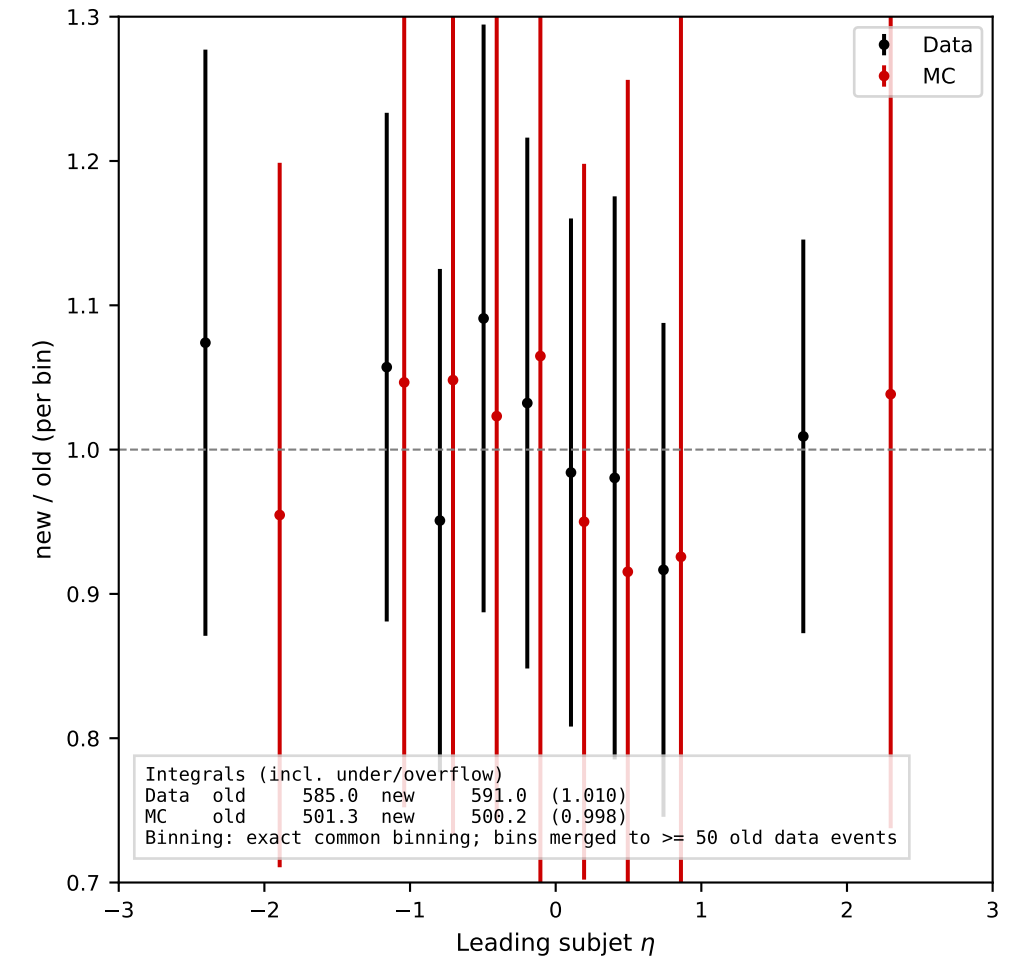
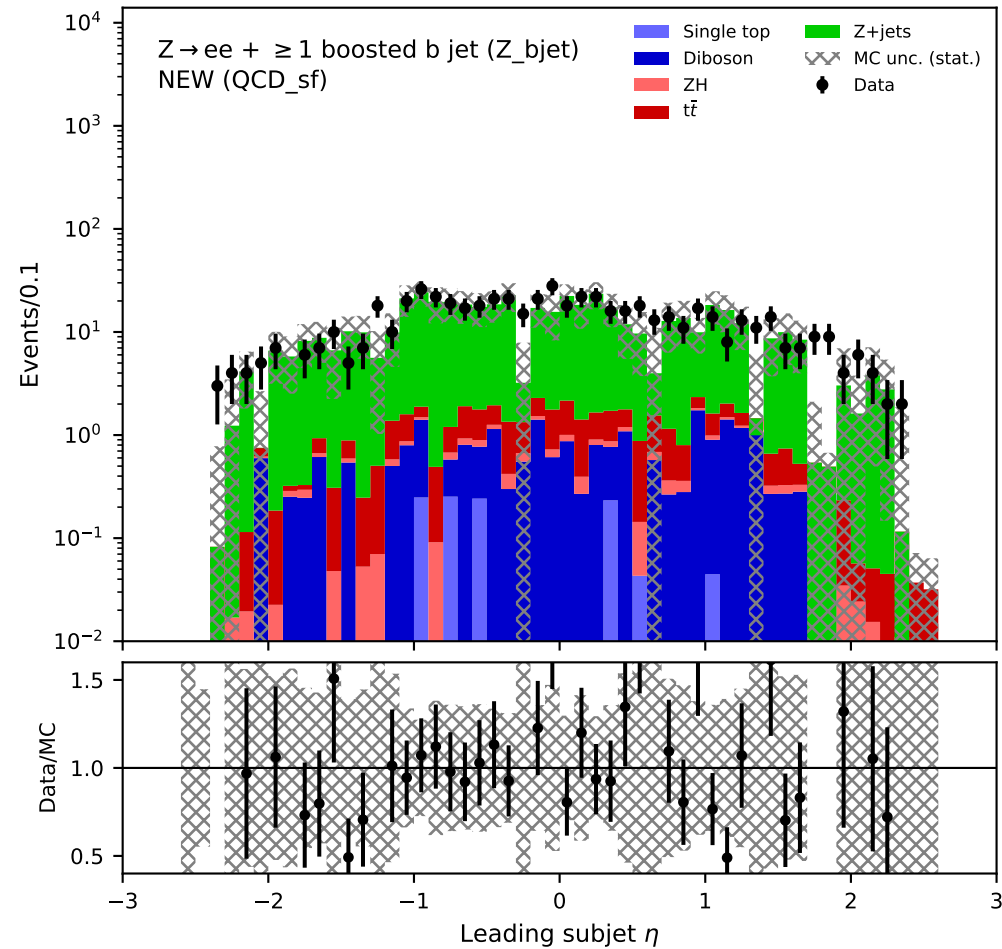
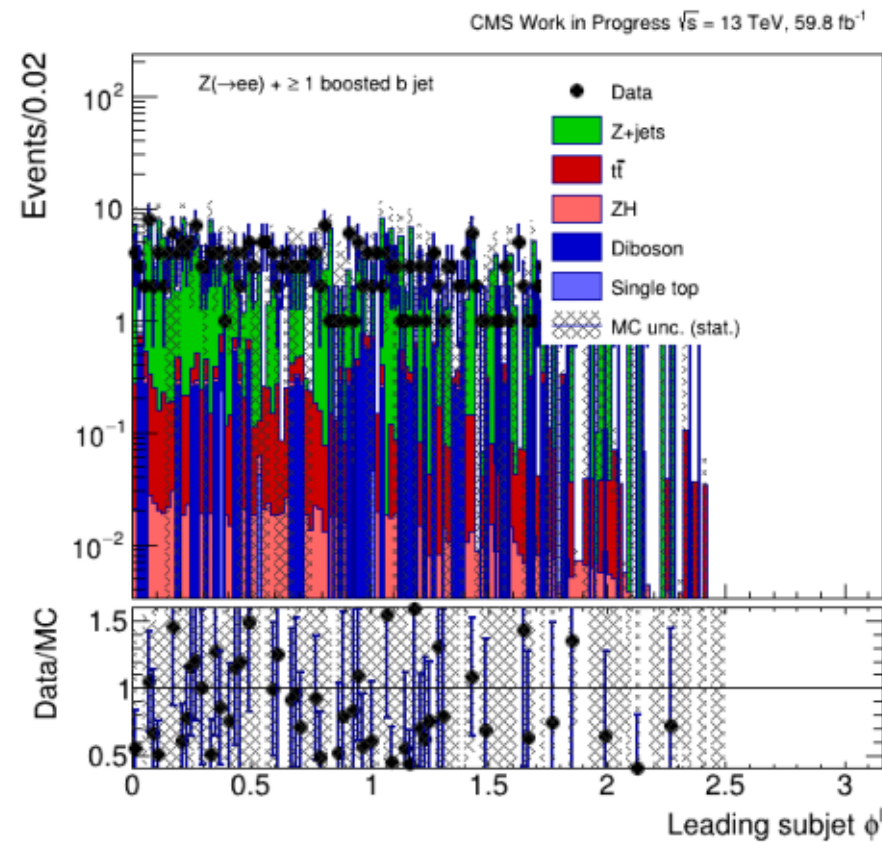
OLD: phi_sub0_Z_jet_Zmm



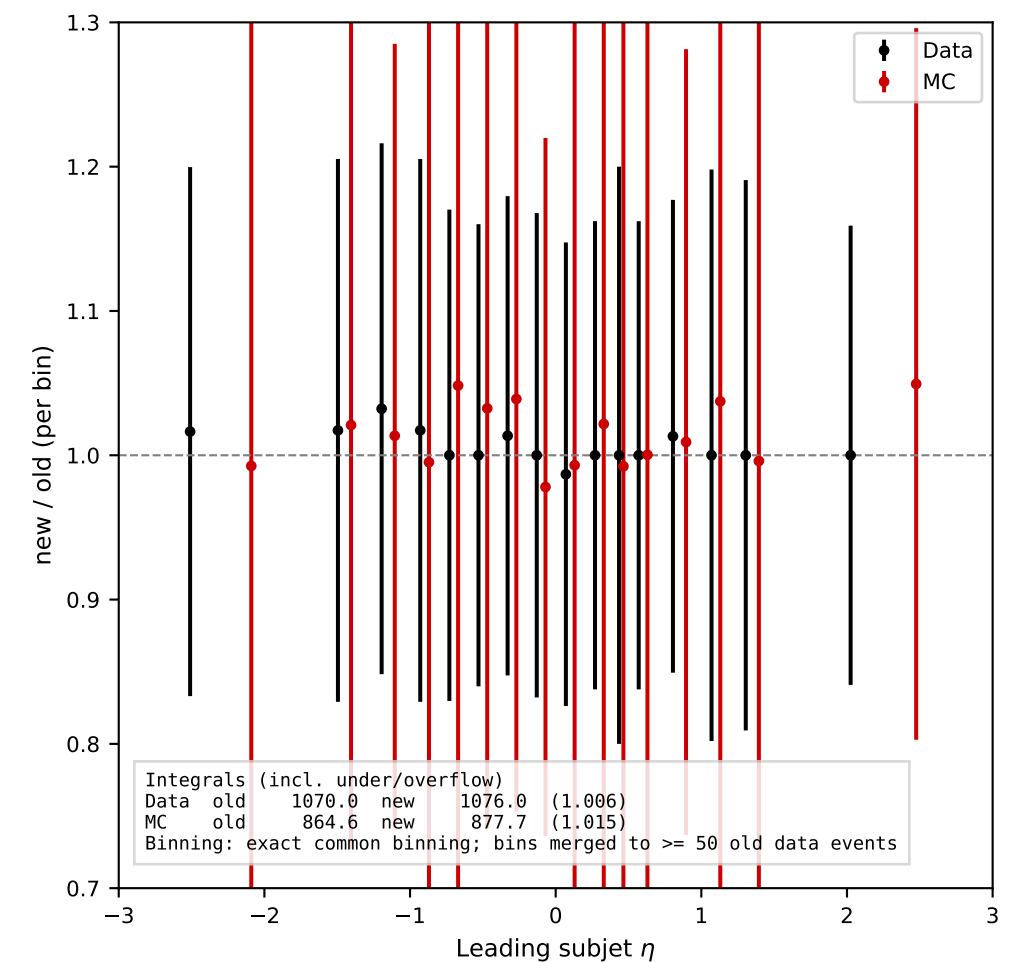
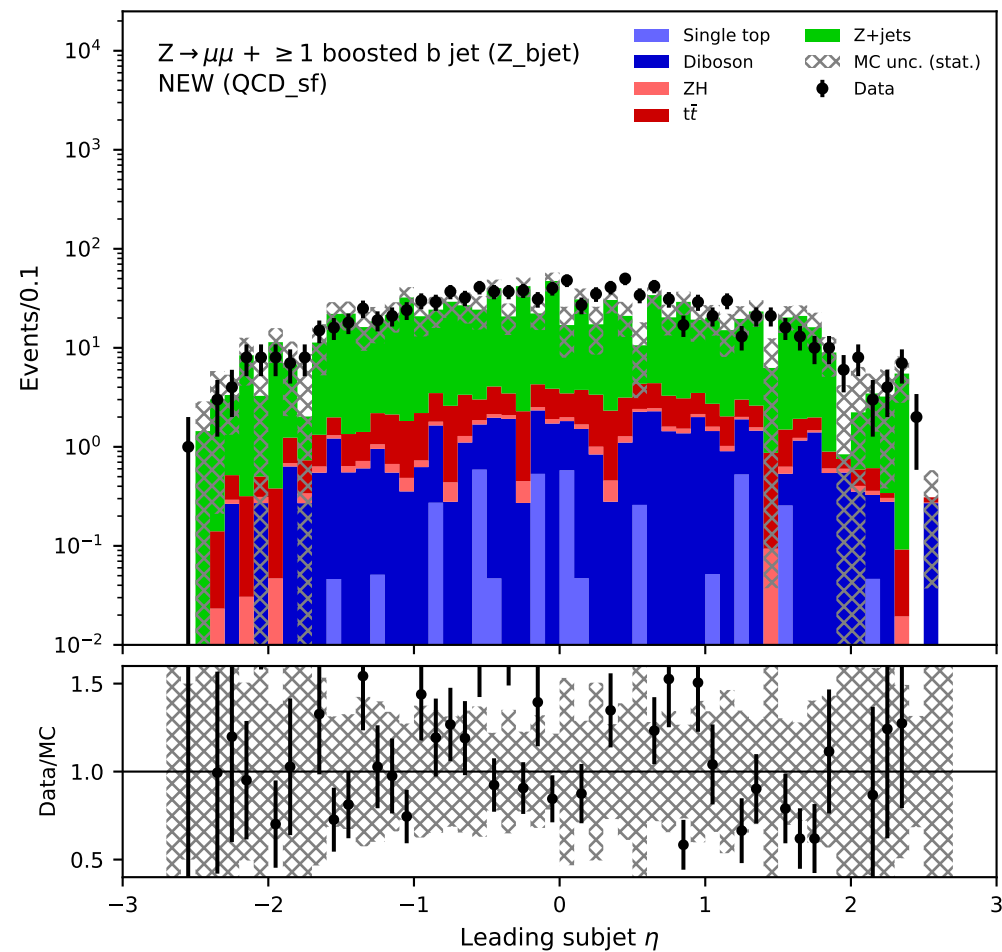
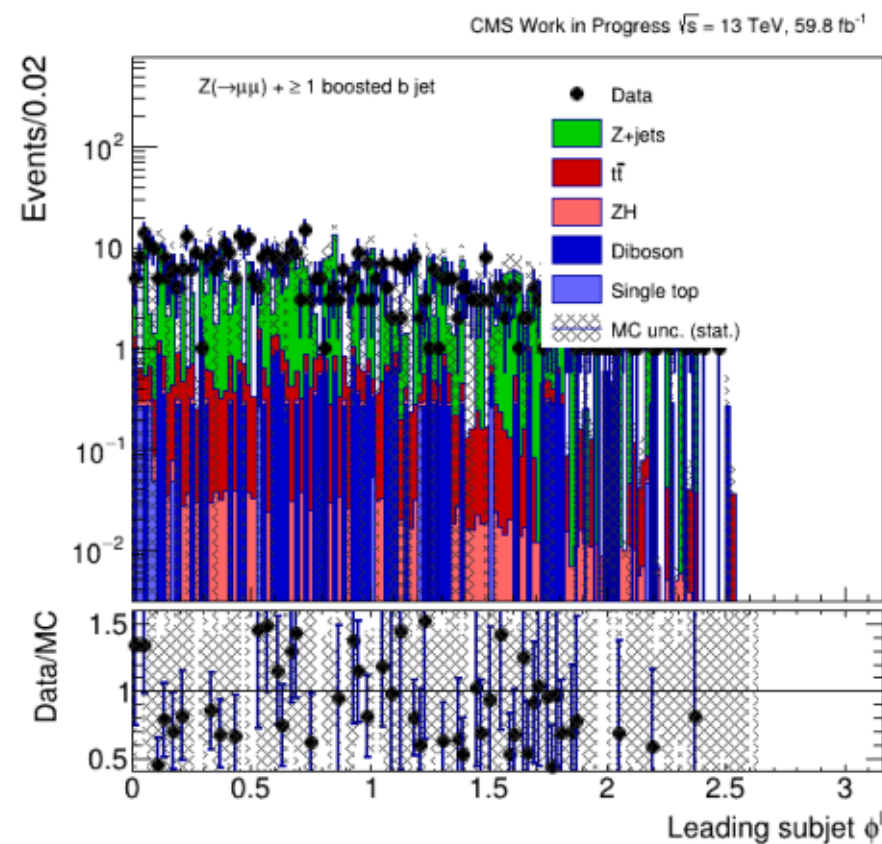
eta_sub0 | Z_bjet | 2018

Old plot shown is its 'phi_sub0', which holds subjet eta (old eta/phi swap). Z_bjet: the old framework fills subjet histograms from the leading jet's subjets, ours from the tagged jet's subjets; differences are expected.

OLD: phi_sub0_Z_bjet_Zee



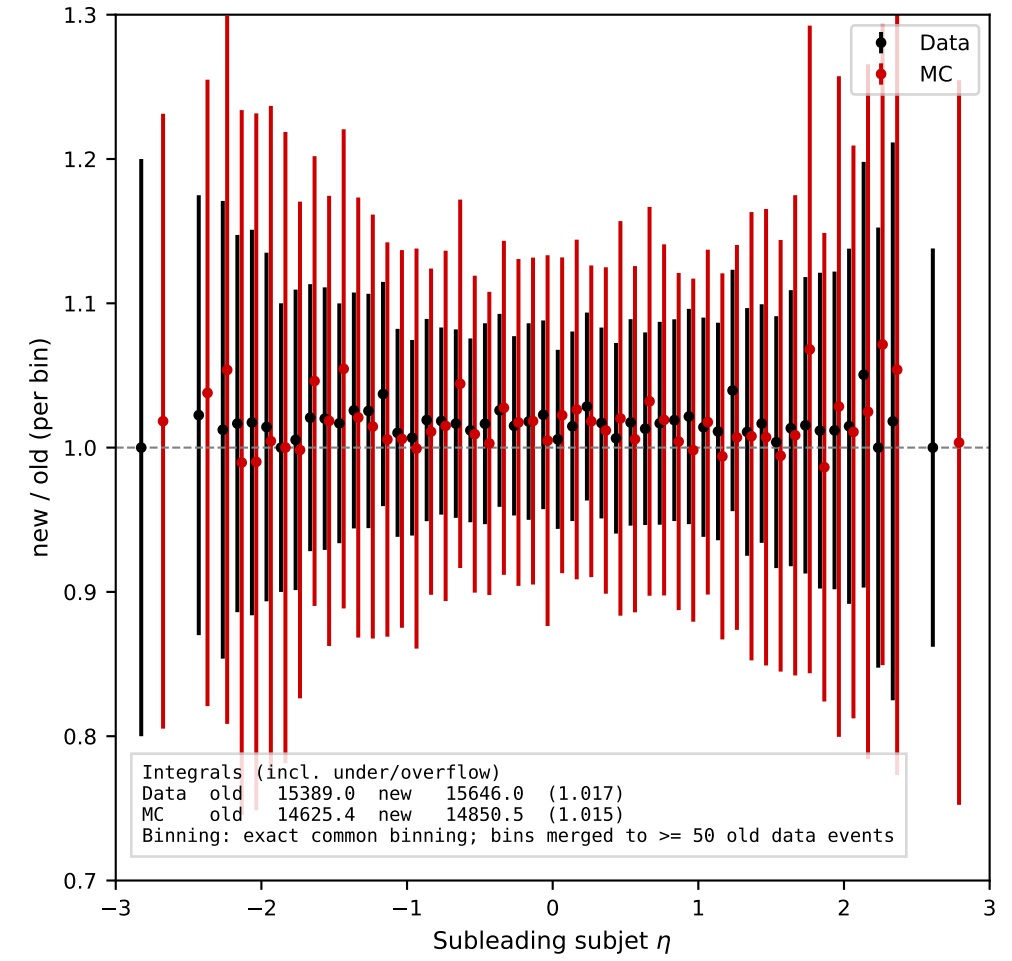
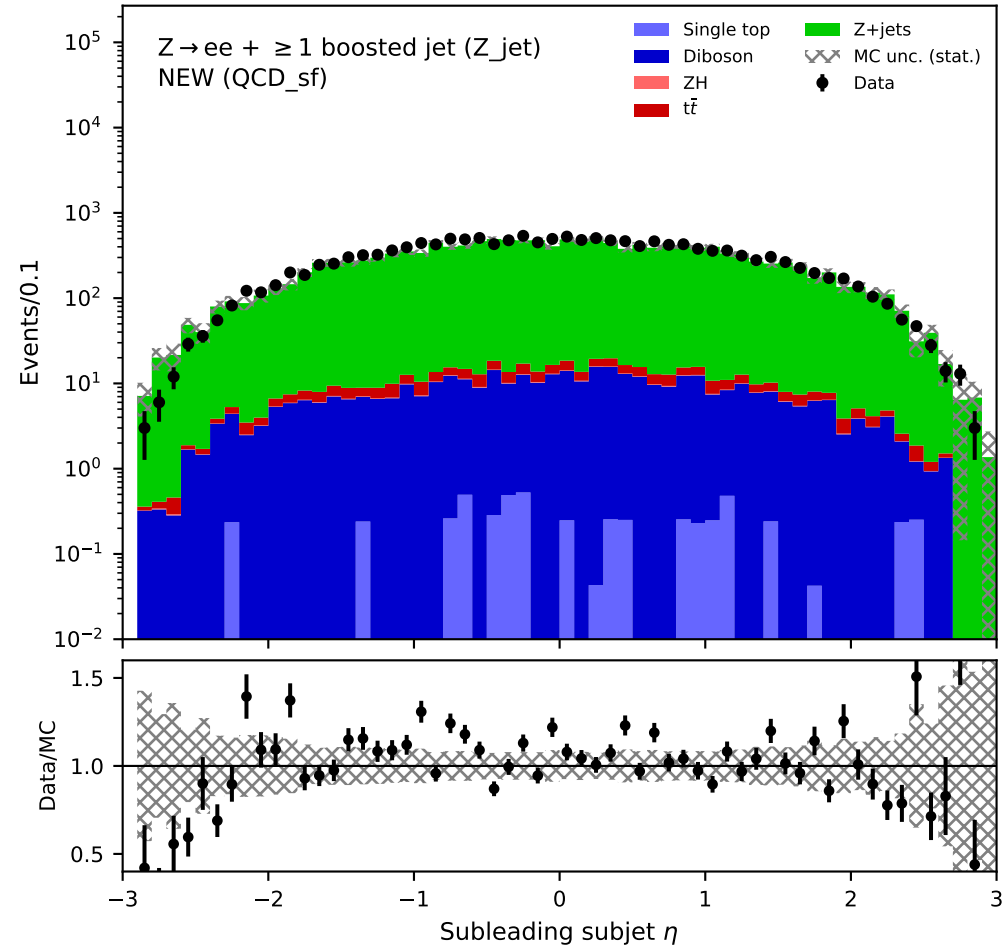
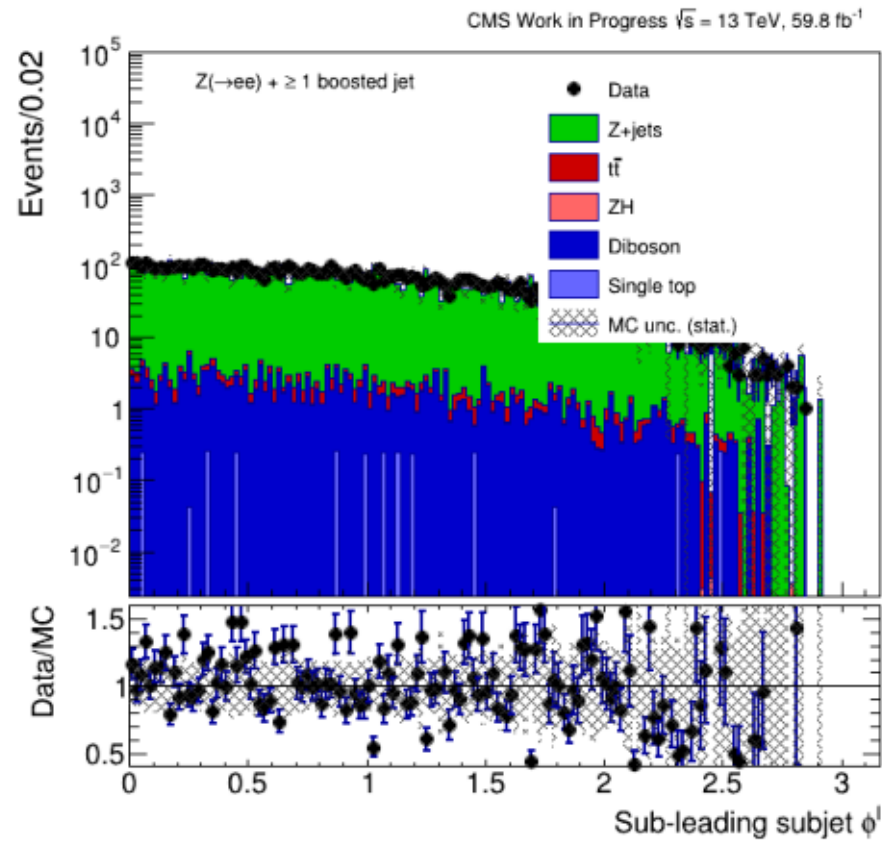
OLD: phi_sub0_Z_bjet_Zmm



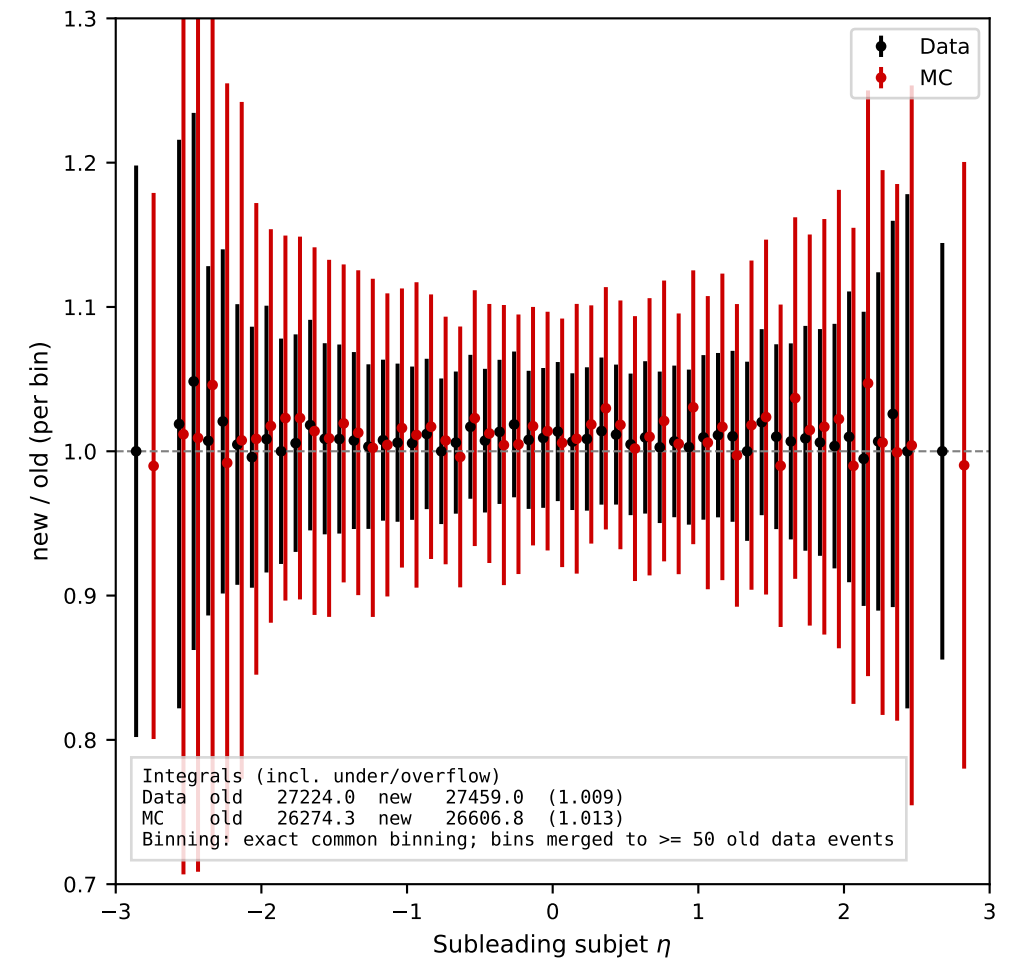
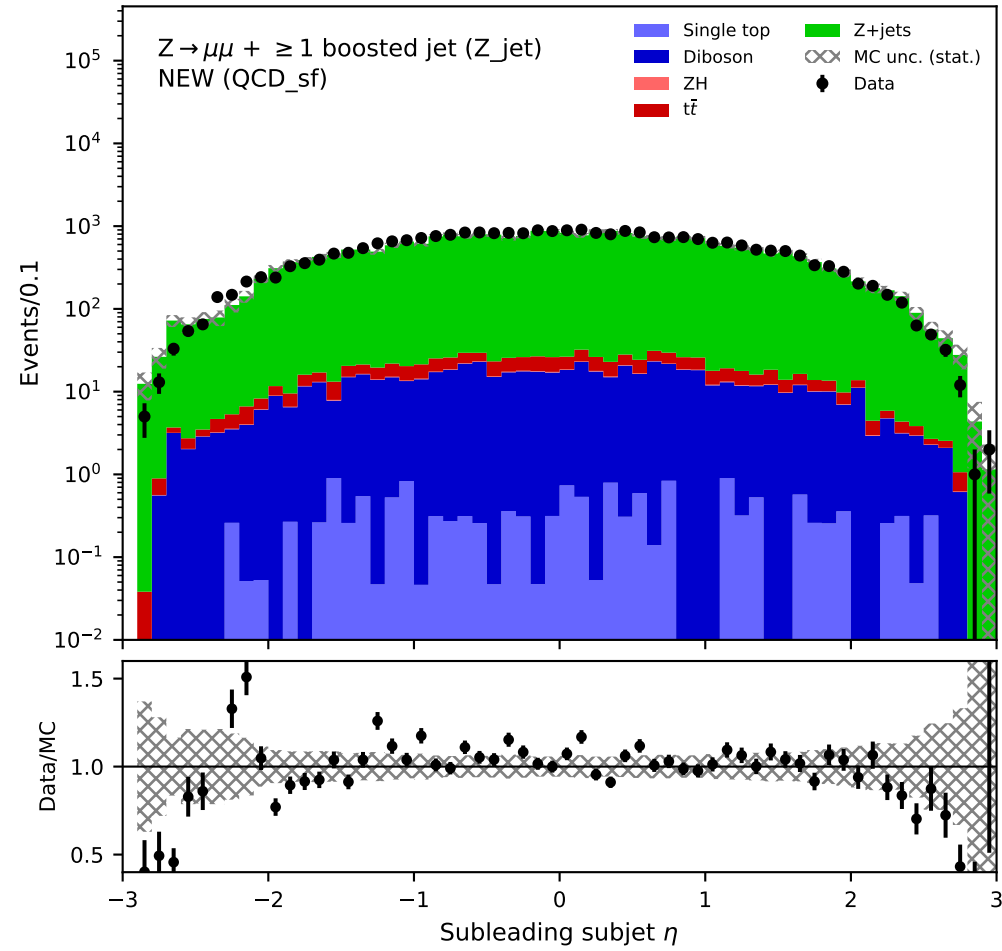
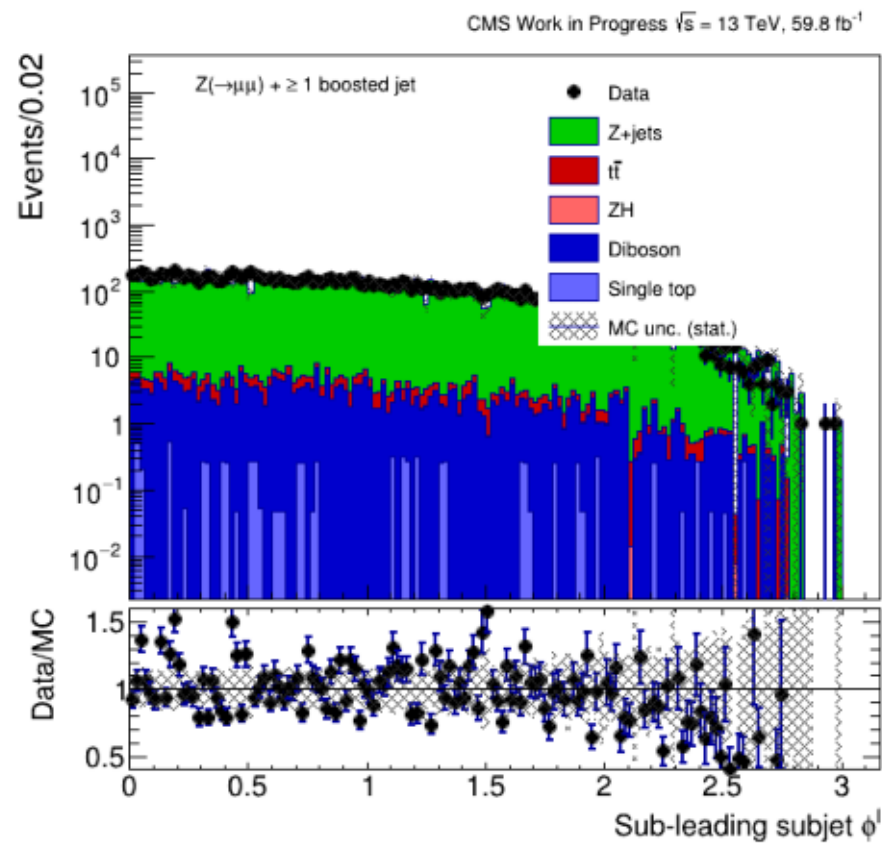
eta_sub1 | Z_jet | 2018

Old plot shown is its 'phi_sub1', which holds subjet eta (old eta/phi swap).

OLD: phi_sub1_Z_jet_Zee



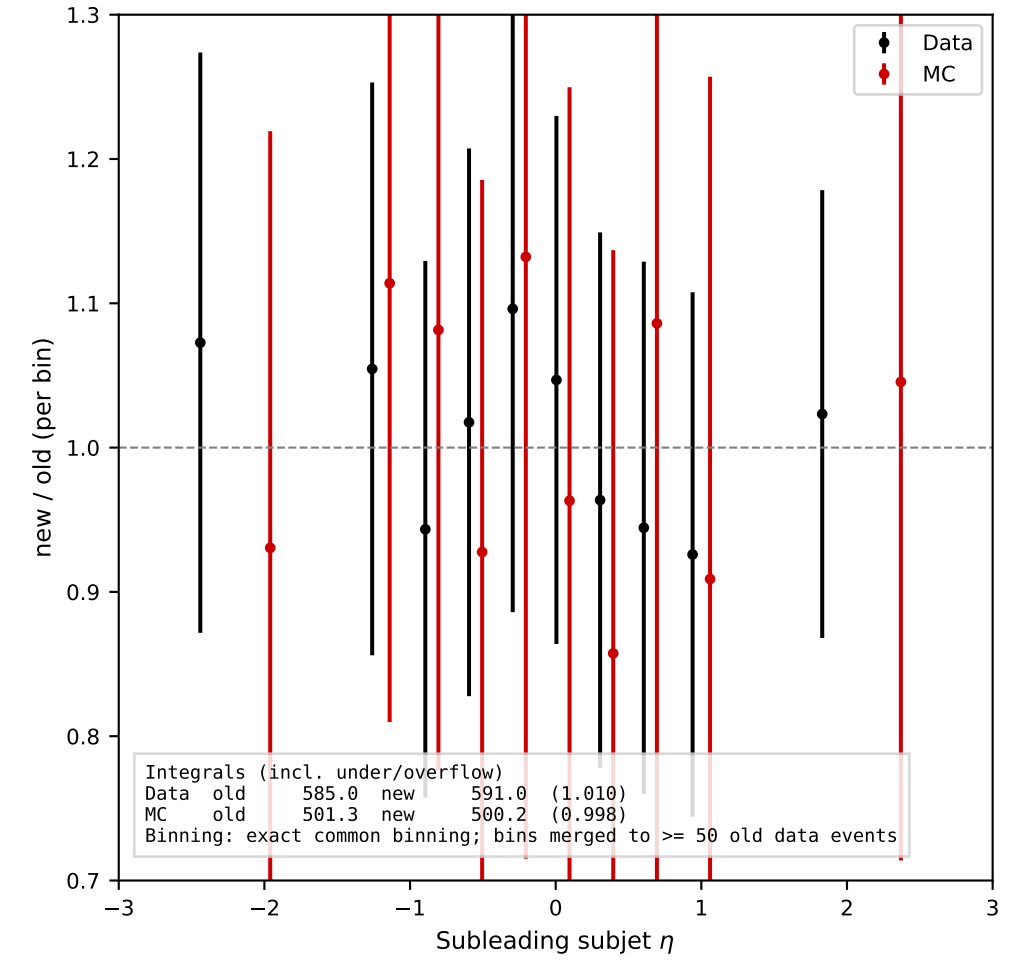
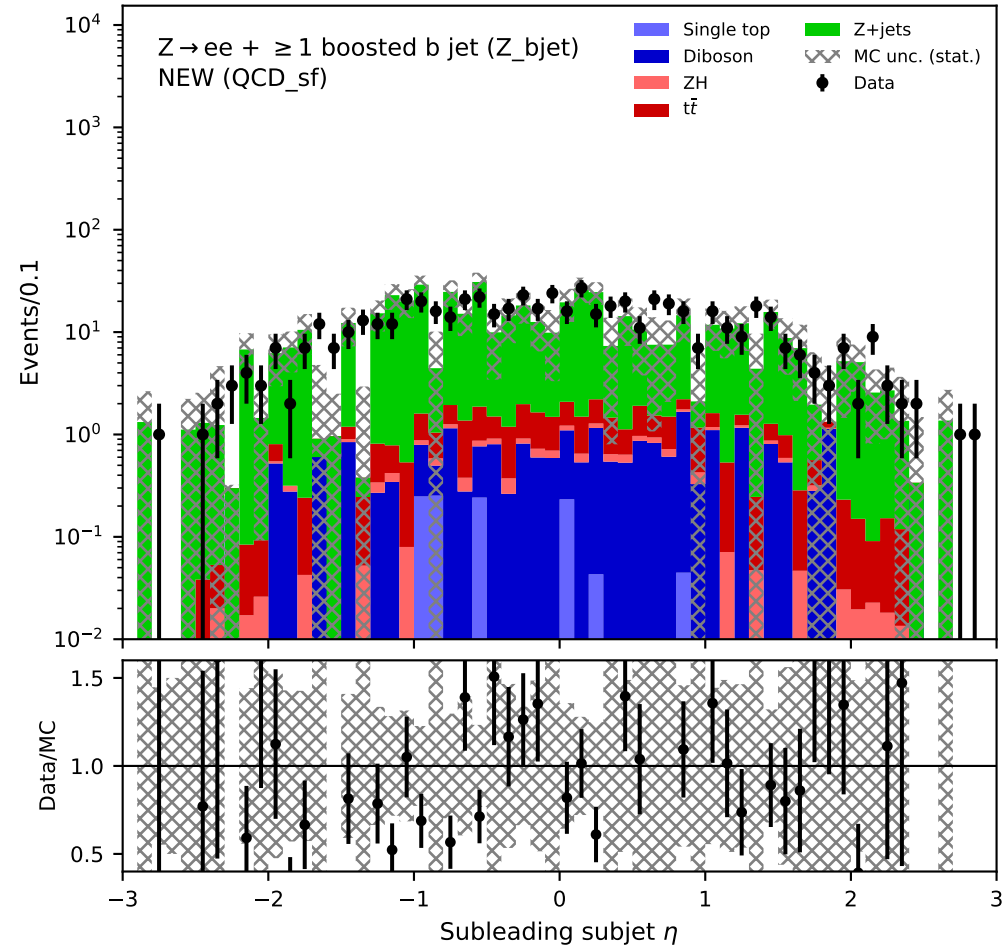
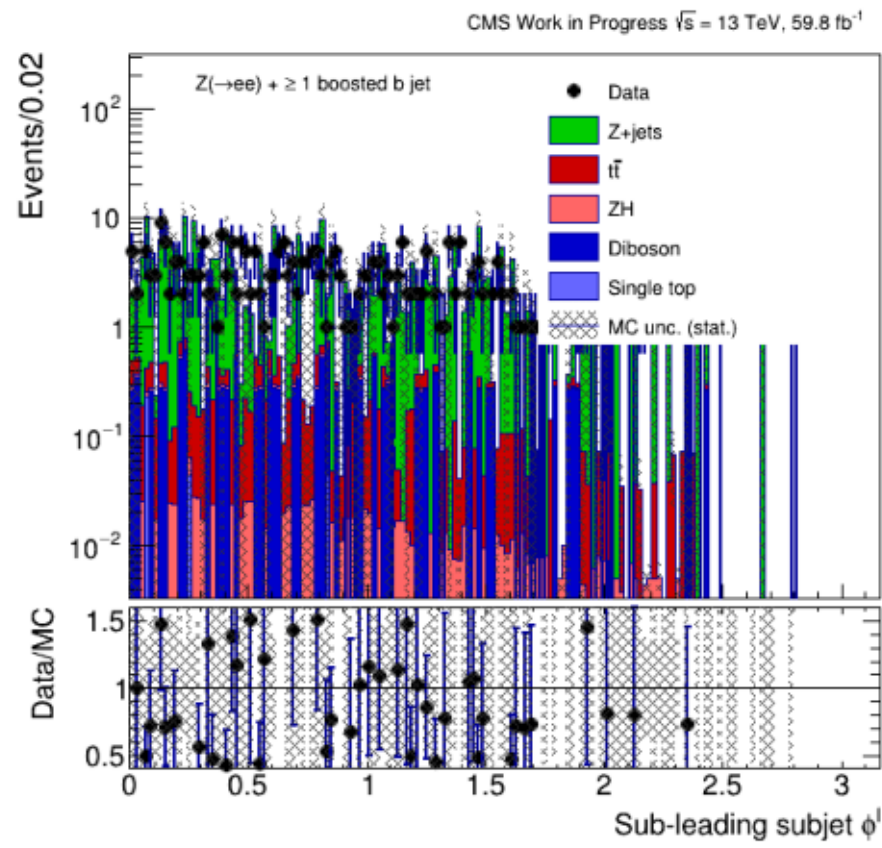
OLD: phi_sub1_Z_jet_Zmm



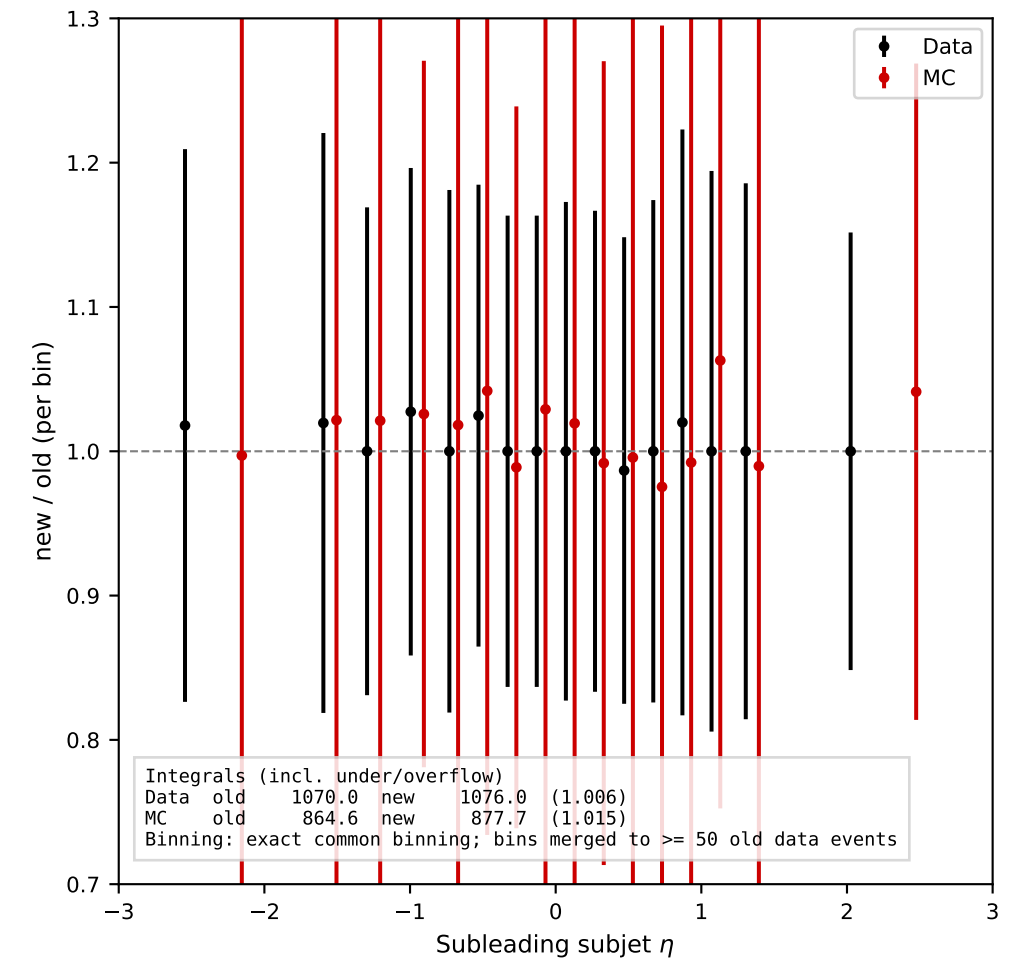
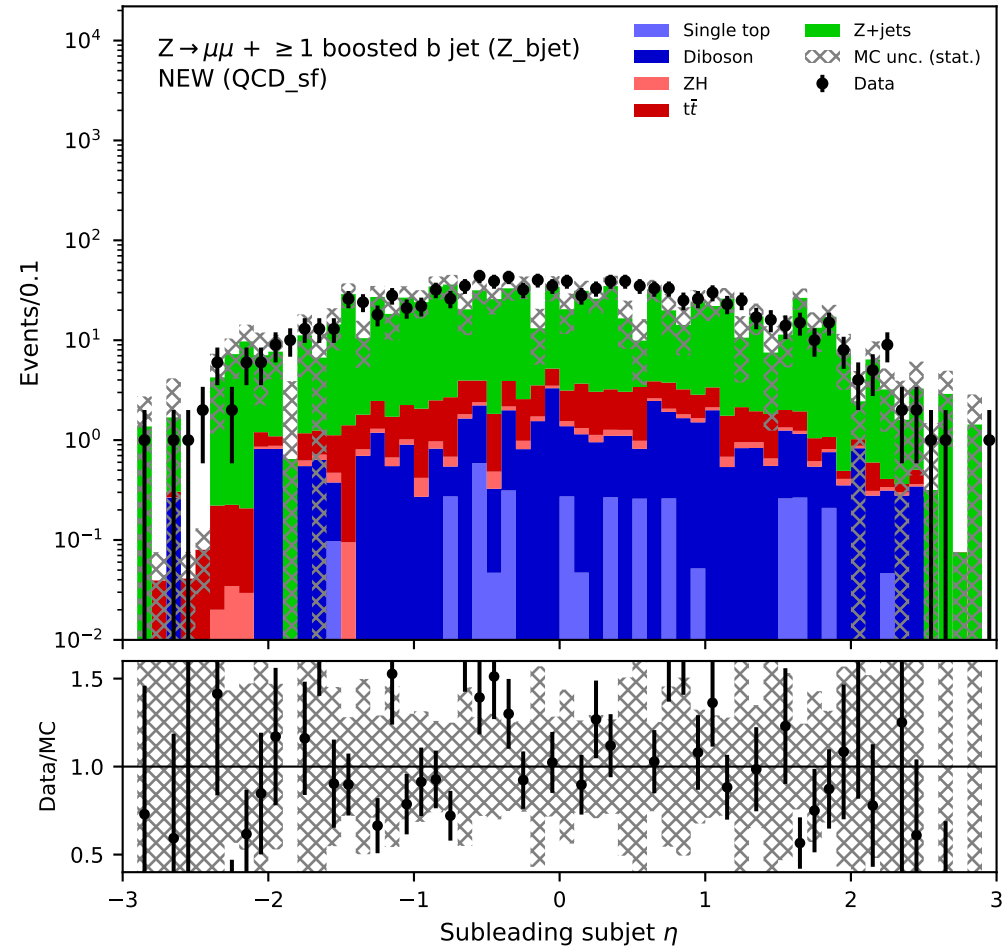
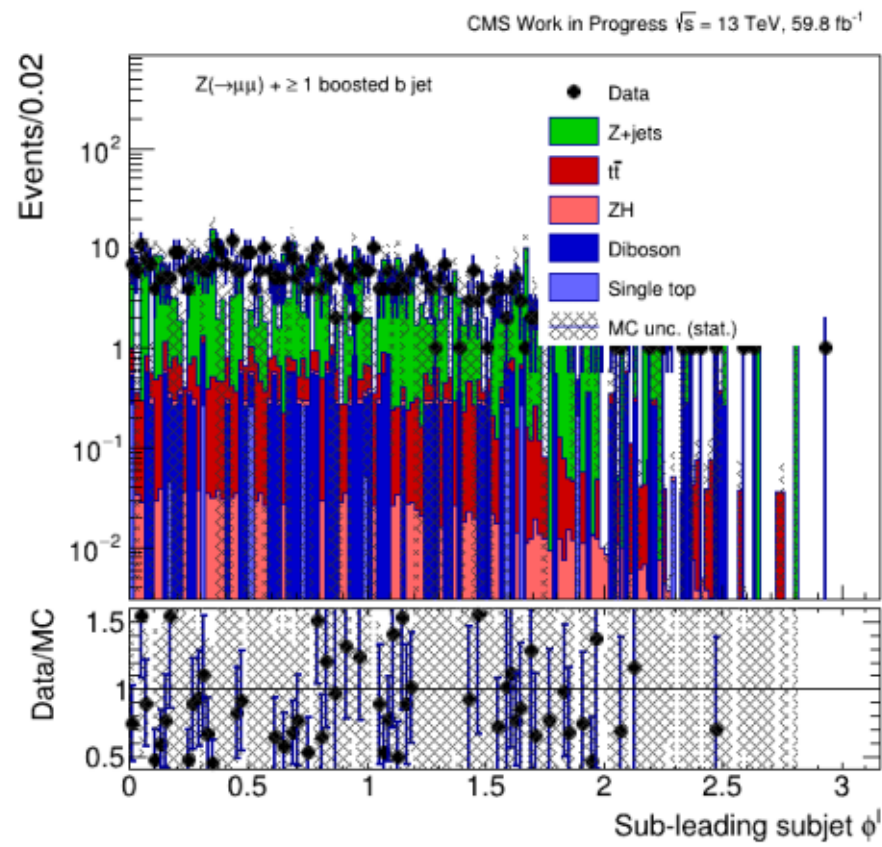
eta_sub1 | Z_bjet | 2018

Old plot shown is its 'phi_sub1', which holds subjet eta (old eta/phi swap). Z_bjet: the old framework fills subjet histograms from the leading jet's subjets, ours from the tagged jet's subjets; differences are expected.

OLD: phi_sub1_Z_bjet_Zee



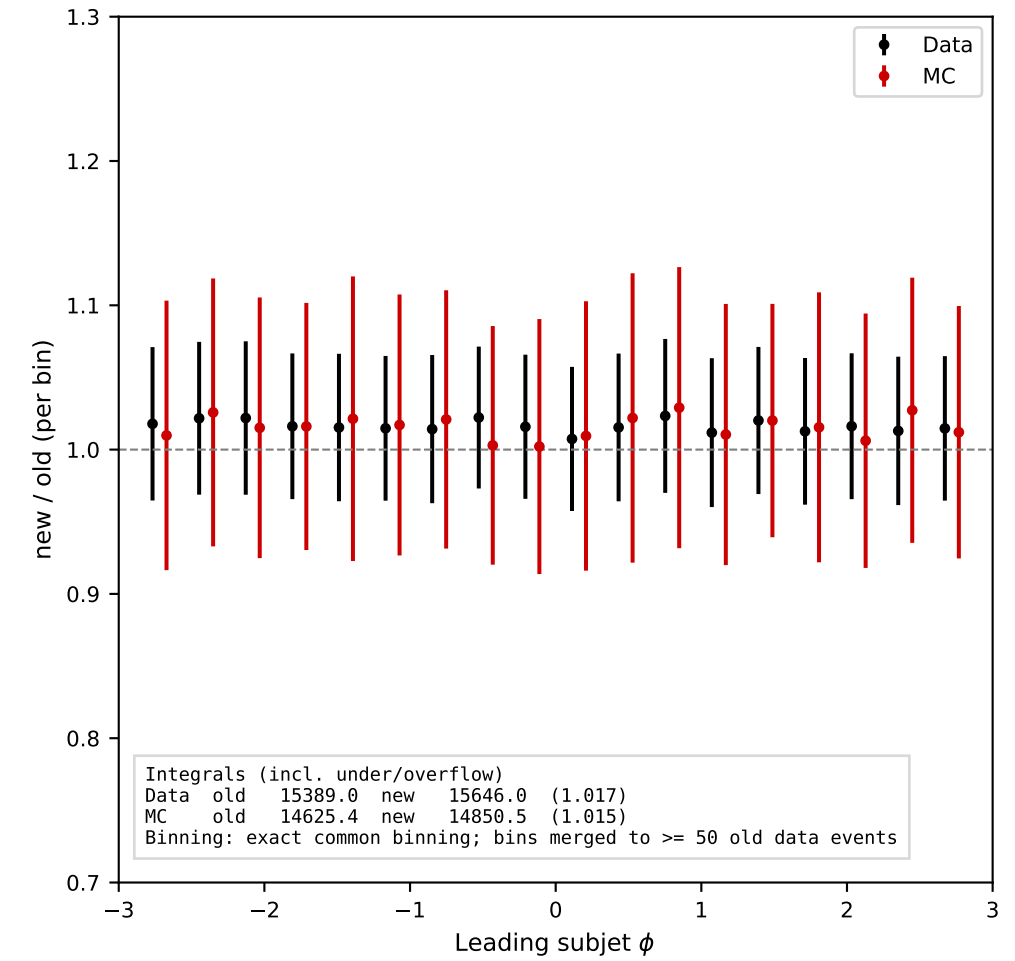
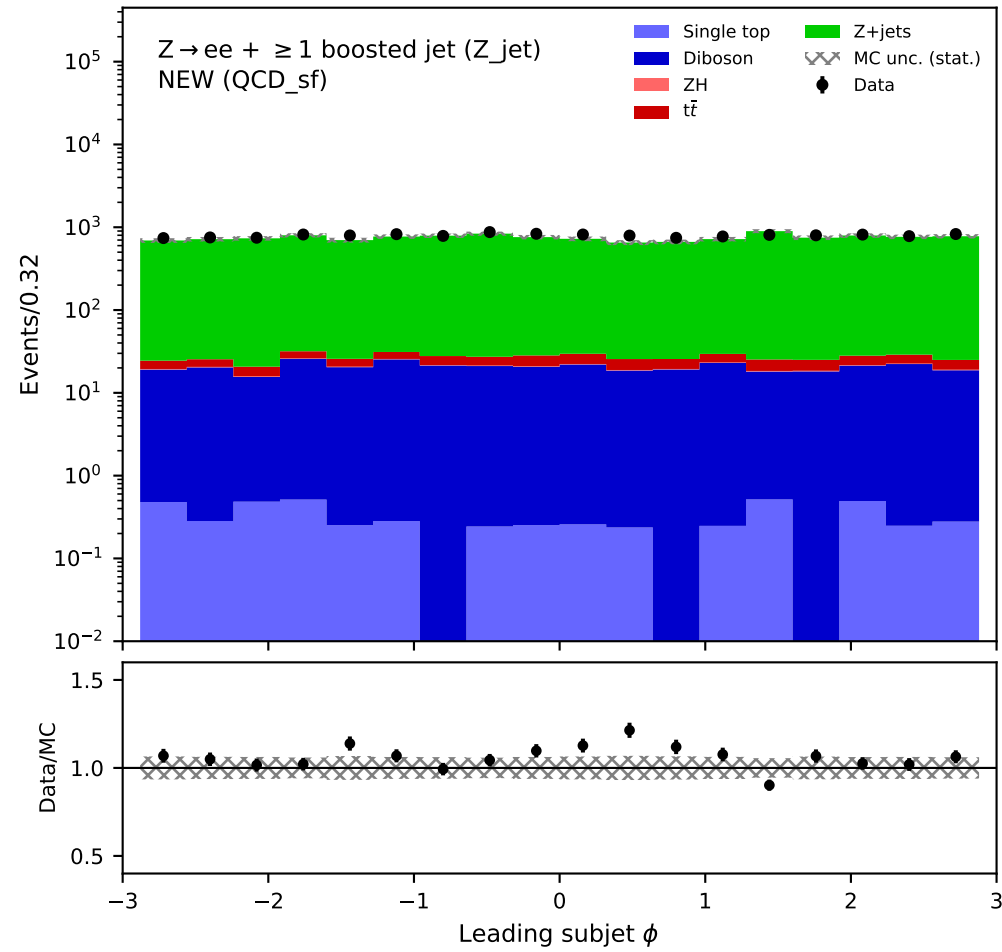
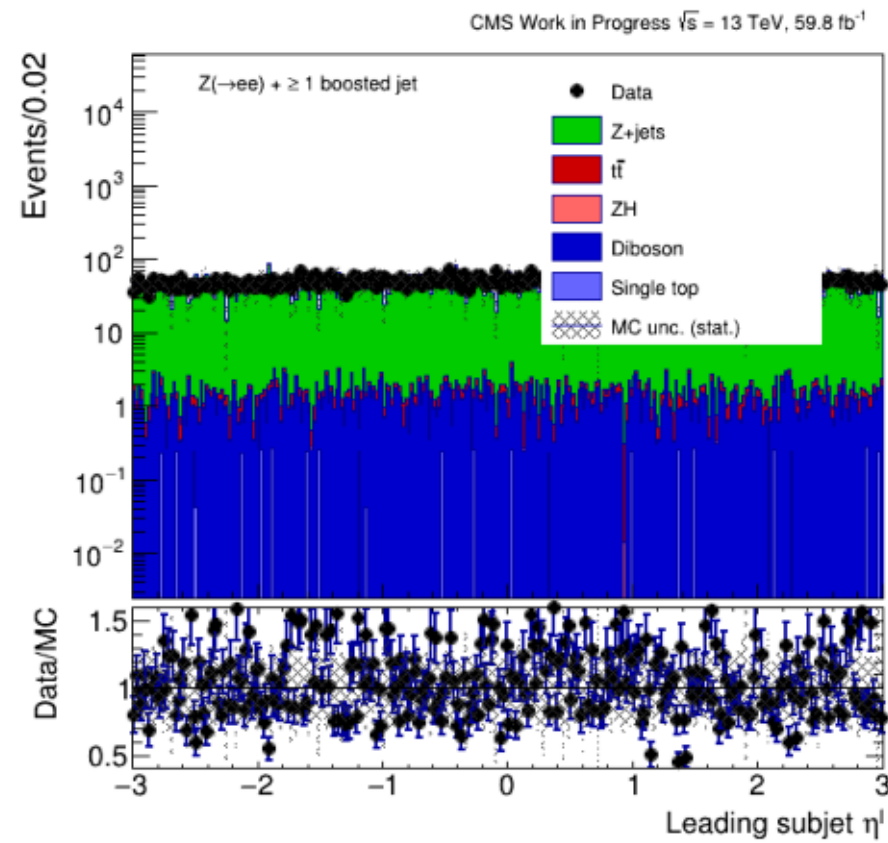
OLD: phi_sub1_Z_bjet_Zmm



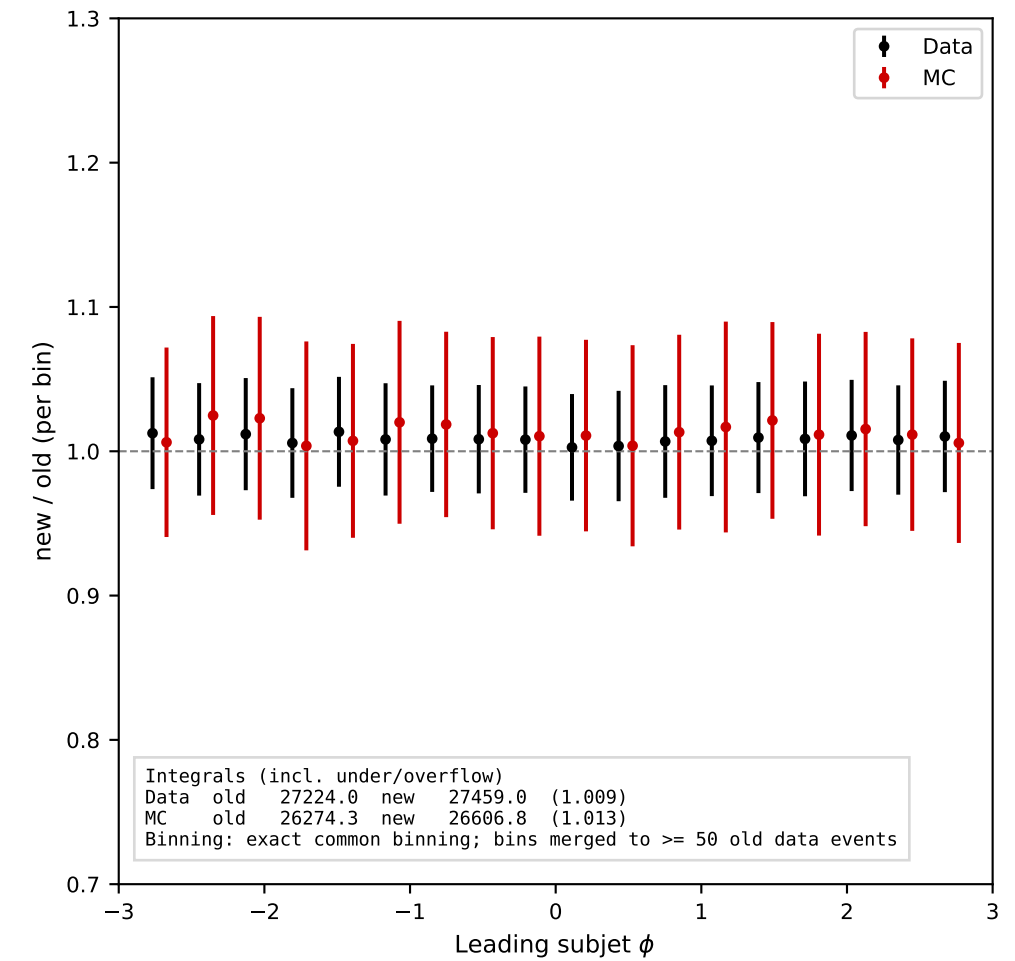
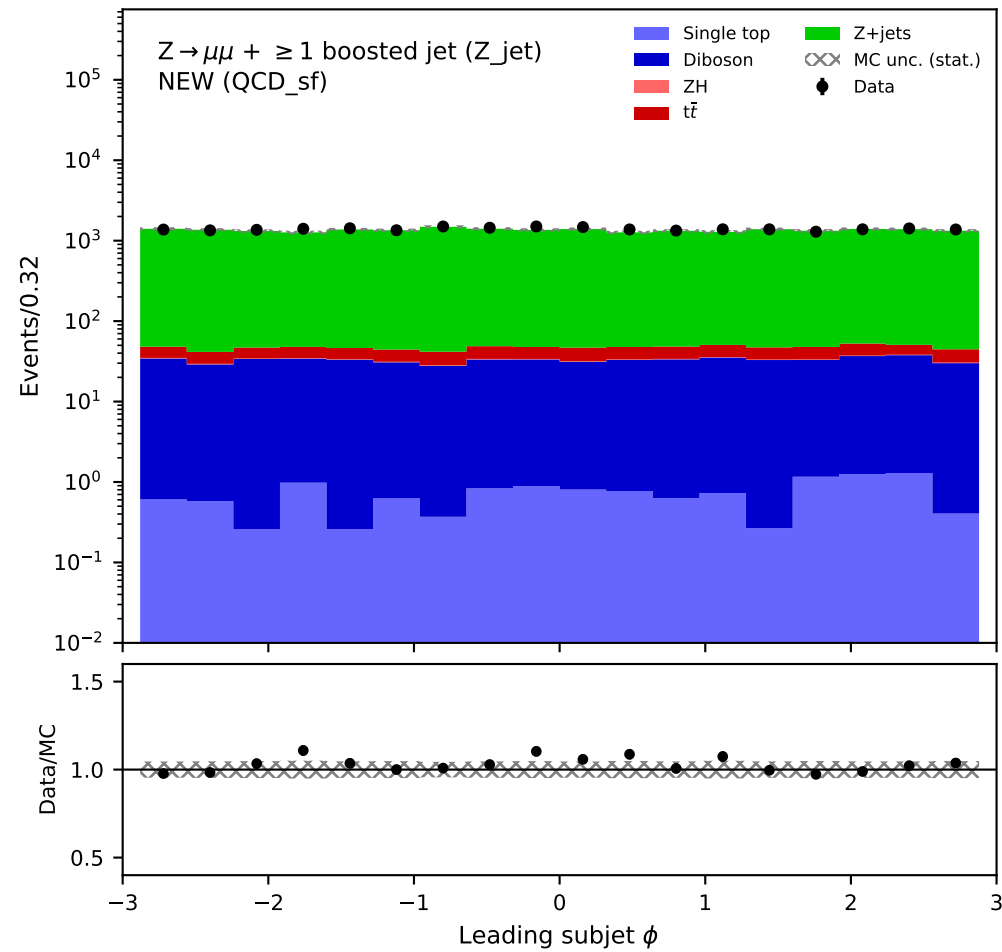
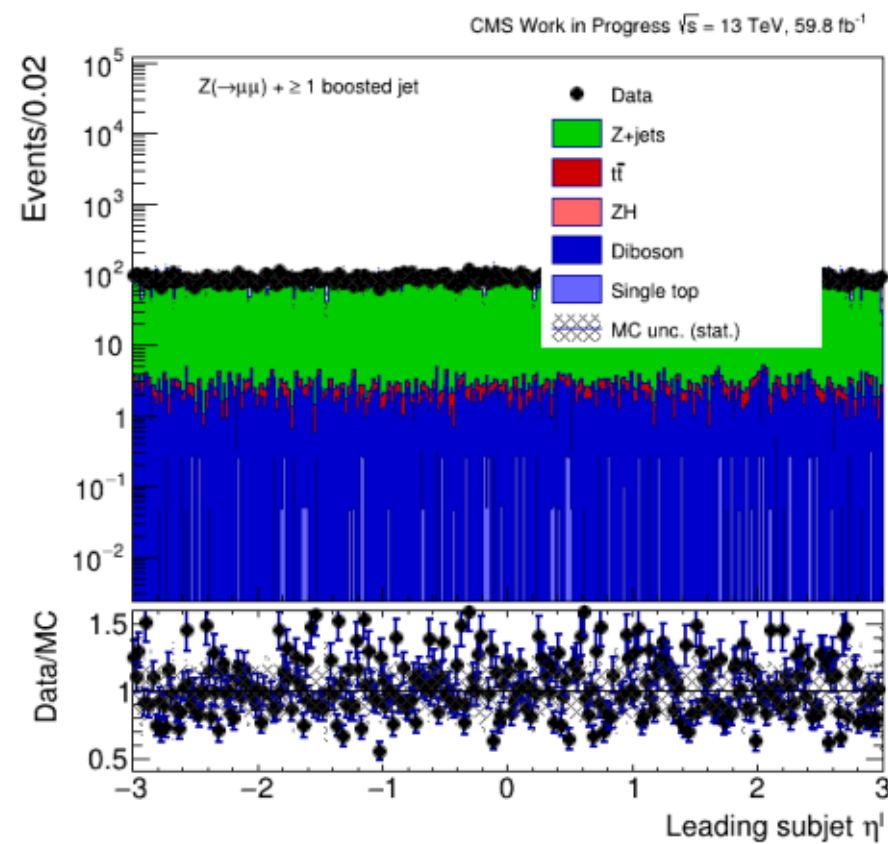
phi_sub0 | Z_jet | 2018

Old plot shown is its 'eta_sub0', which holds subjet phi (old eta/phi swap); it overflows beyond $|\phi| > 3$.

OLD: eta_sub0_Z_jet_Zee



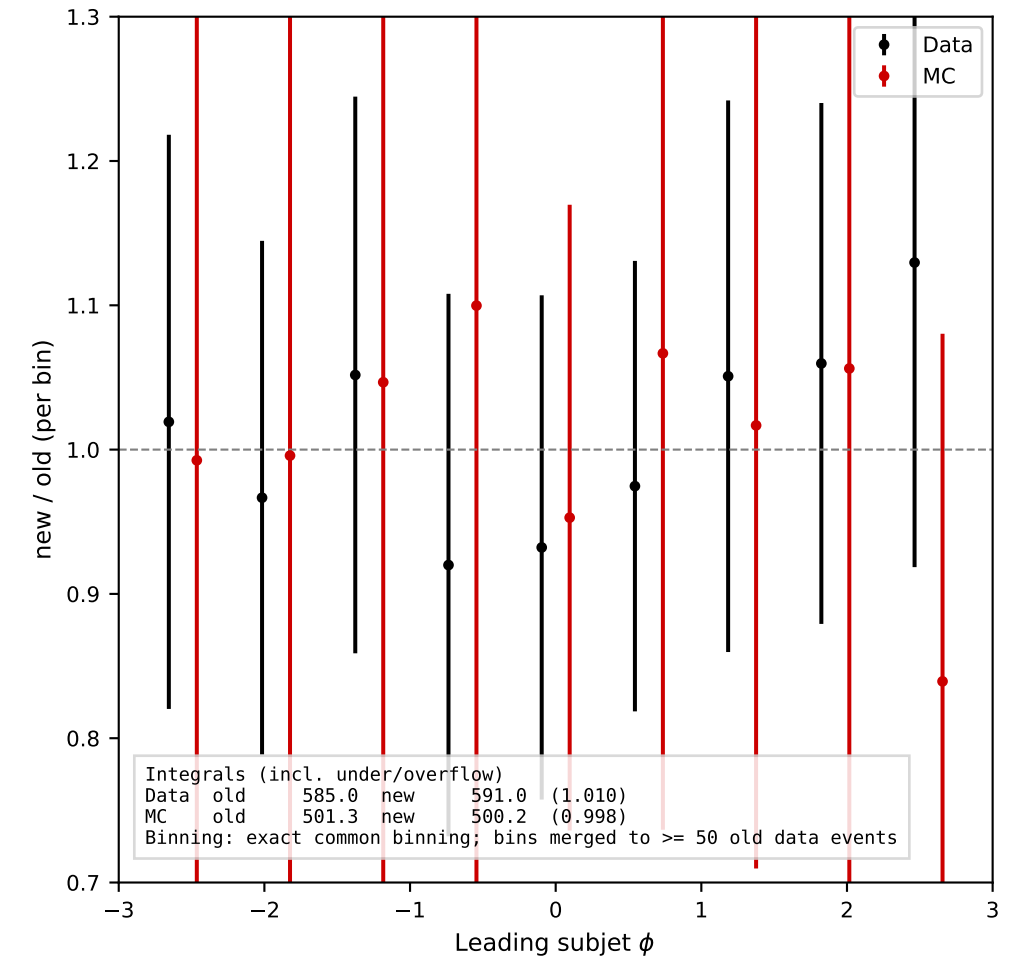
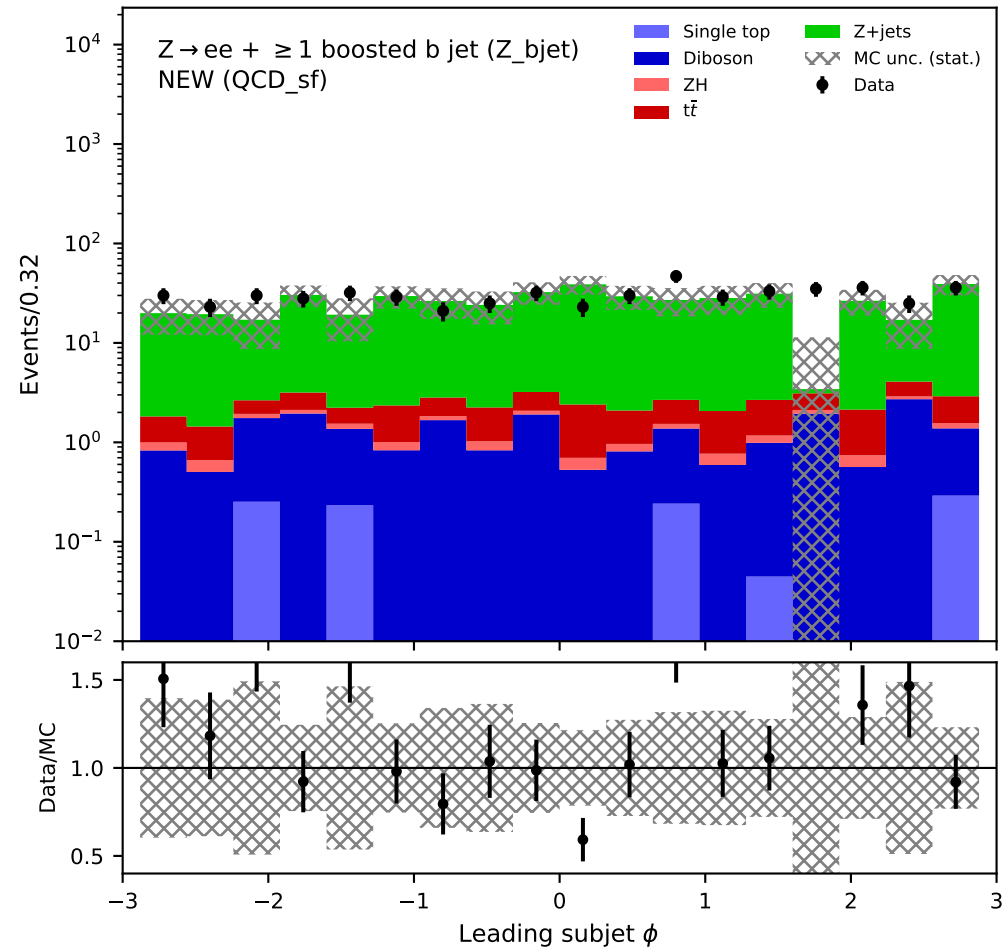
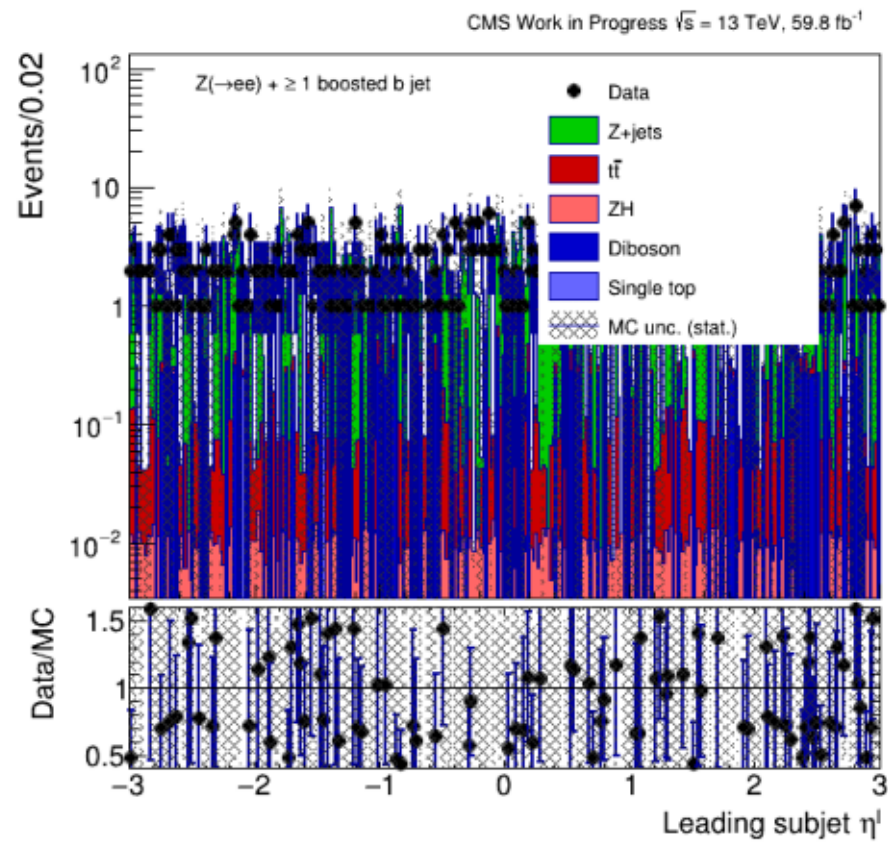
OLD: eta_sub0_Z_jet_Zmm



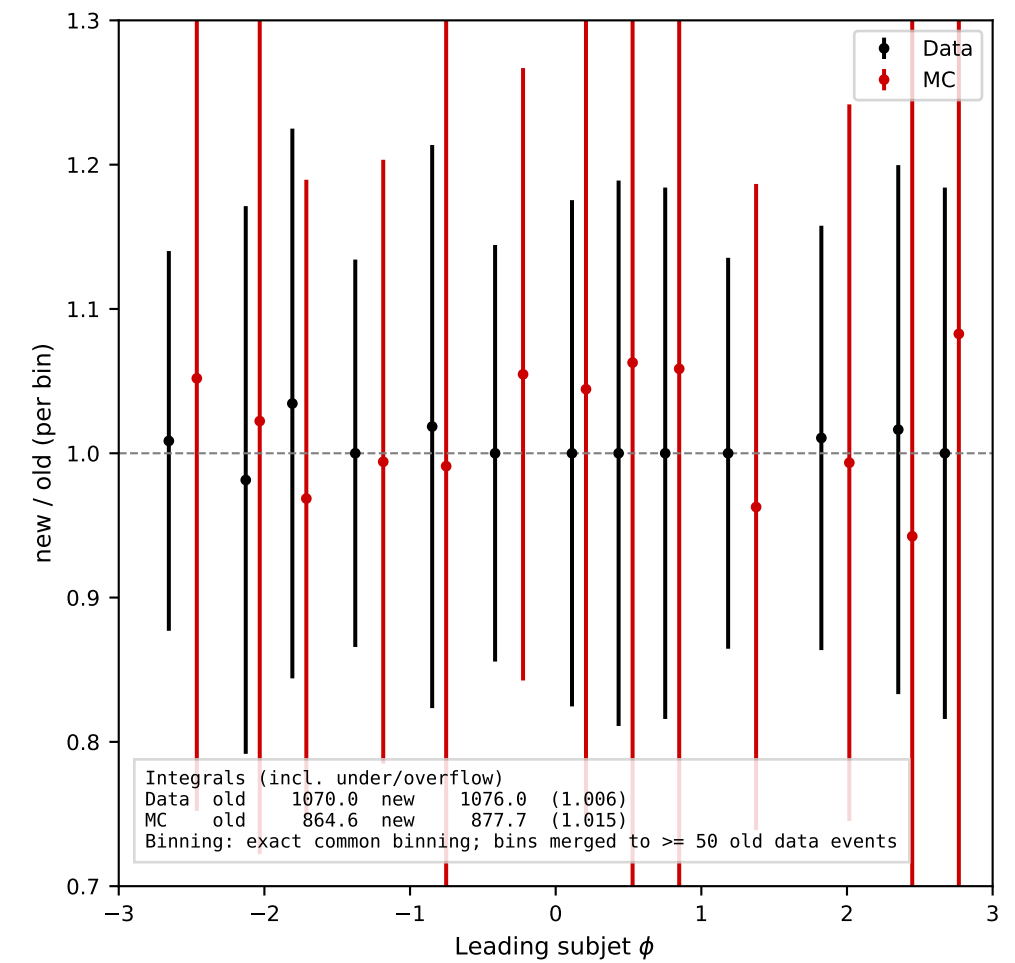
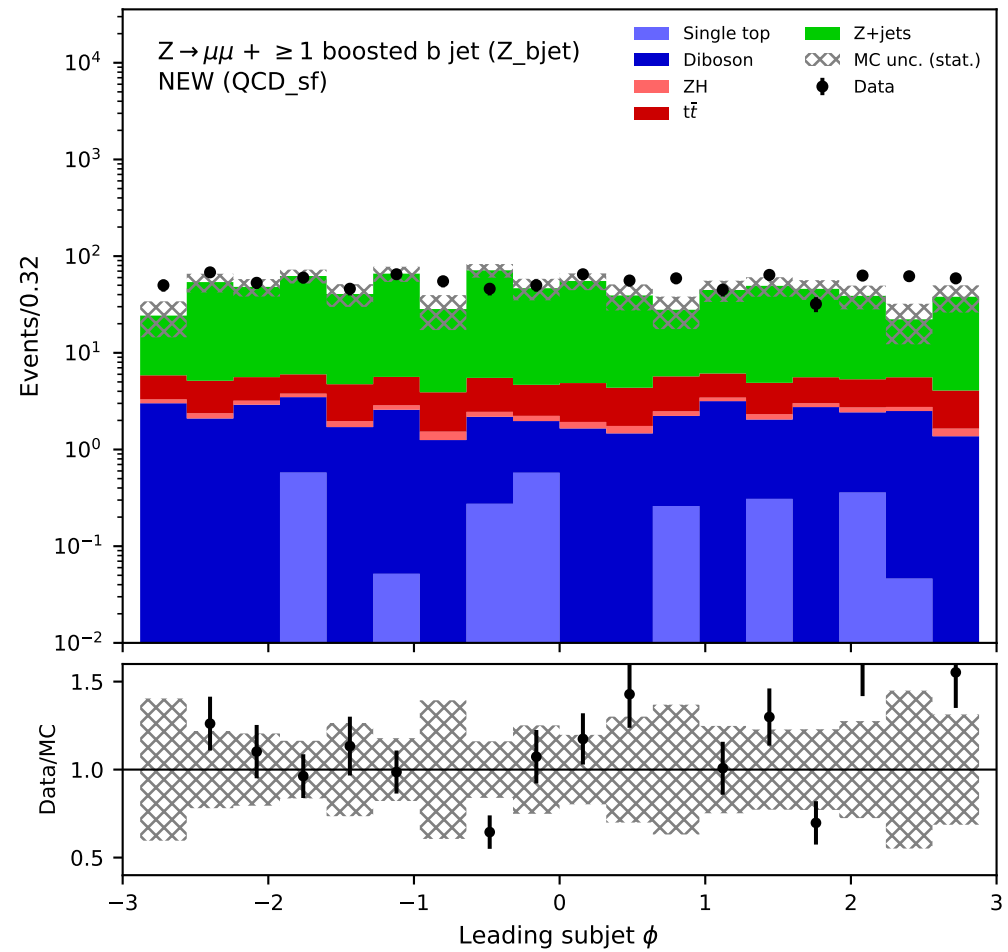
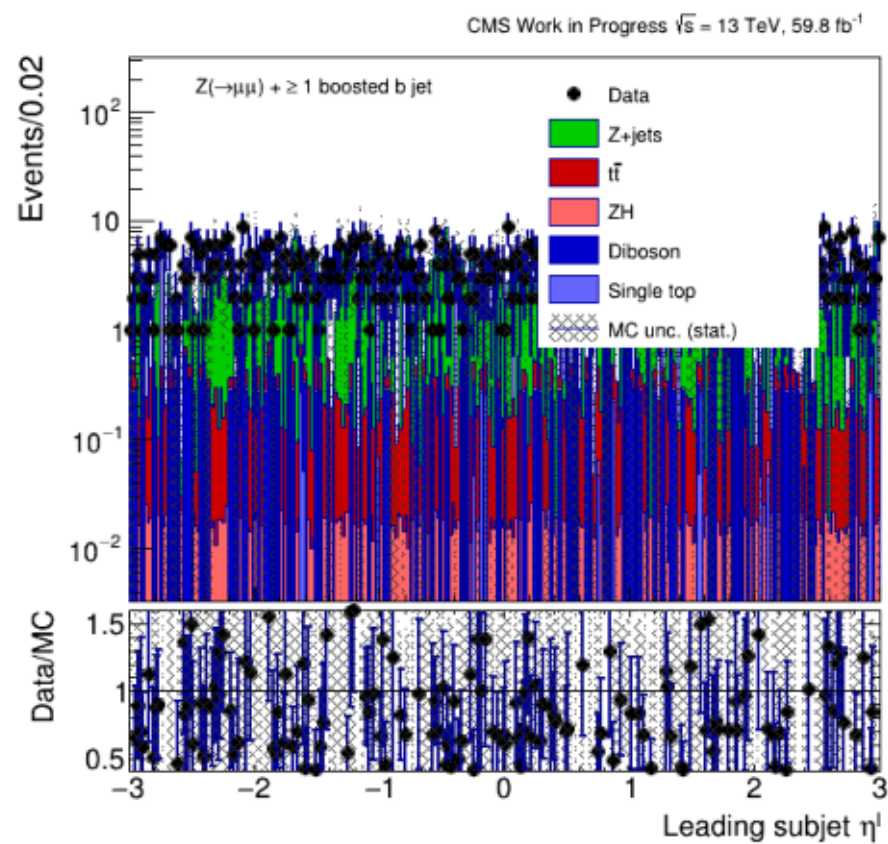
phi_sub0 | Z_bjet | 2018

Old plot shown is its 'eta_sub0', which holds subjet phi (old eta/phi swap); it overflows beyond $|\phi| > 3$. Z_bjet: the old framework fills subjet histograms from the leading jet's subjets, ours from the tagged jet's subjets; differences are expected.

OLD: eta_sub0_Z_bjet_Zee



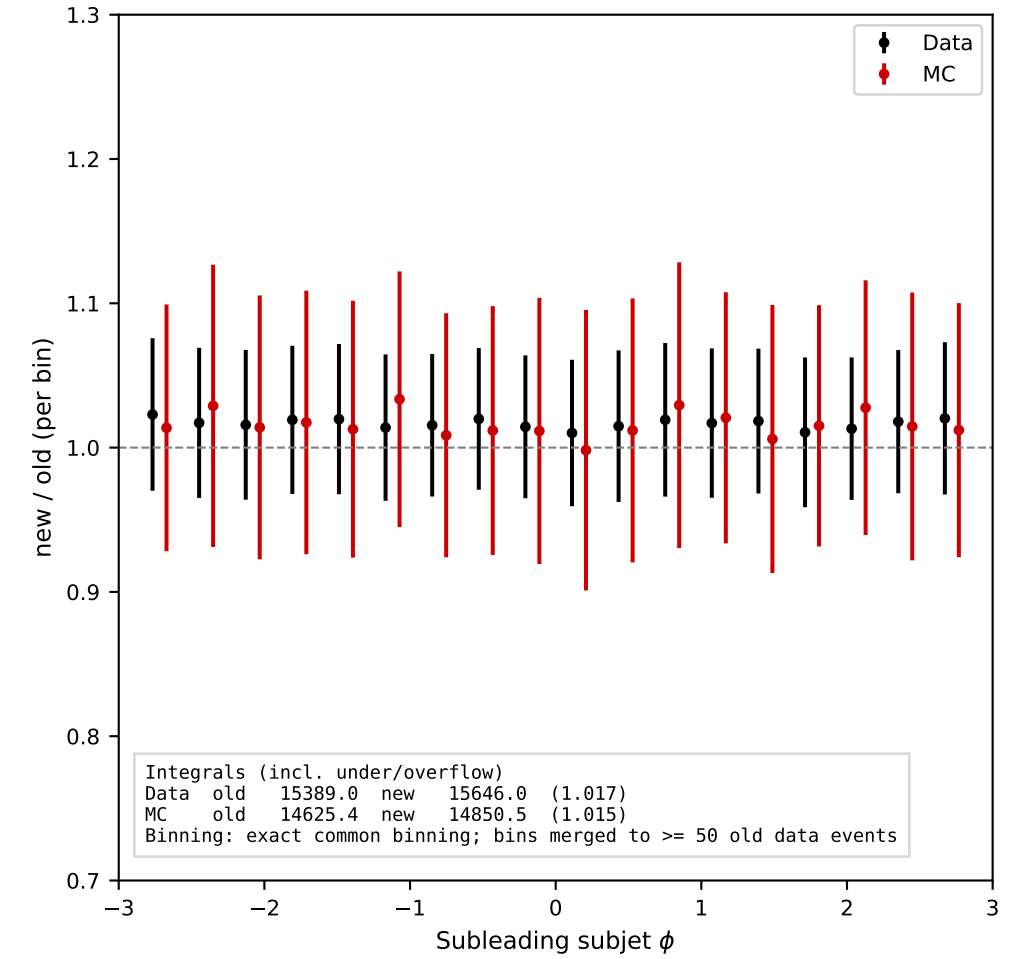
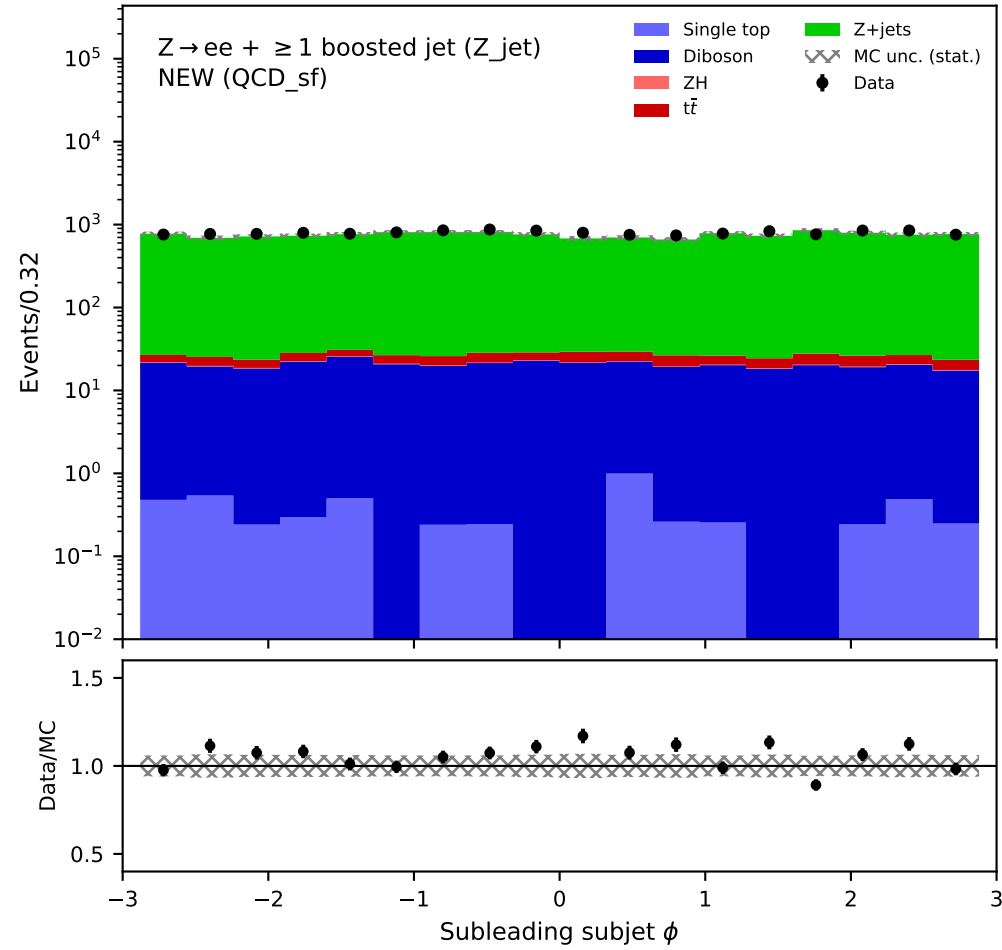
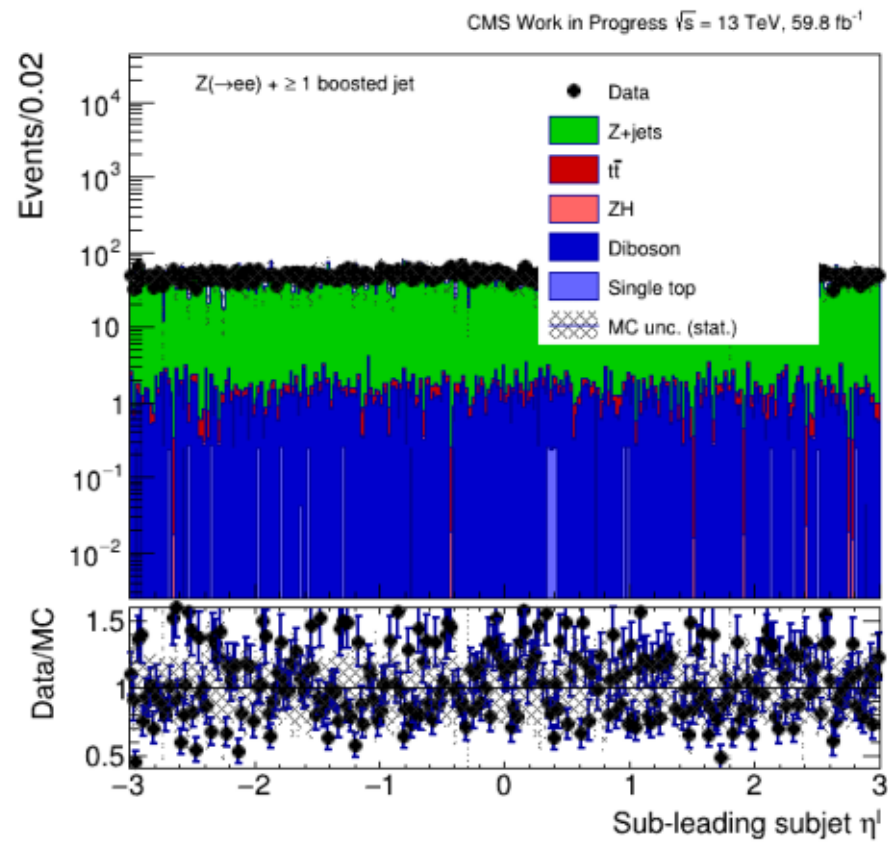
OLD: eta_sub0_Z_bjet_Zmm



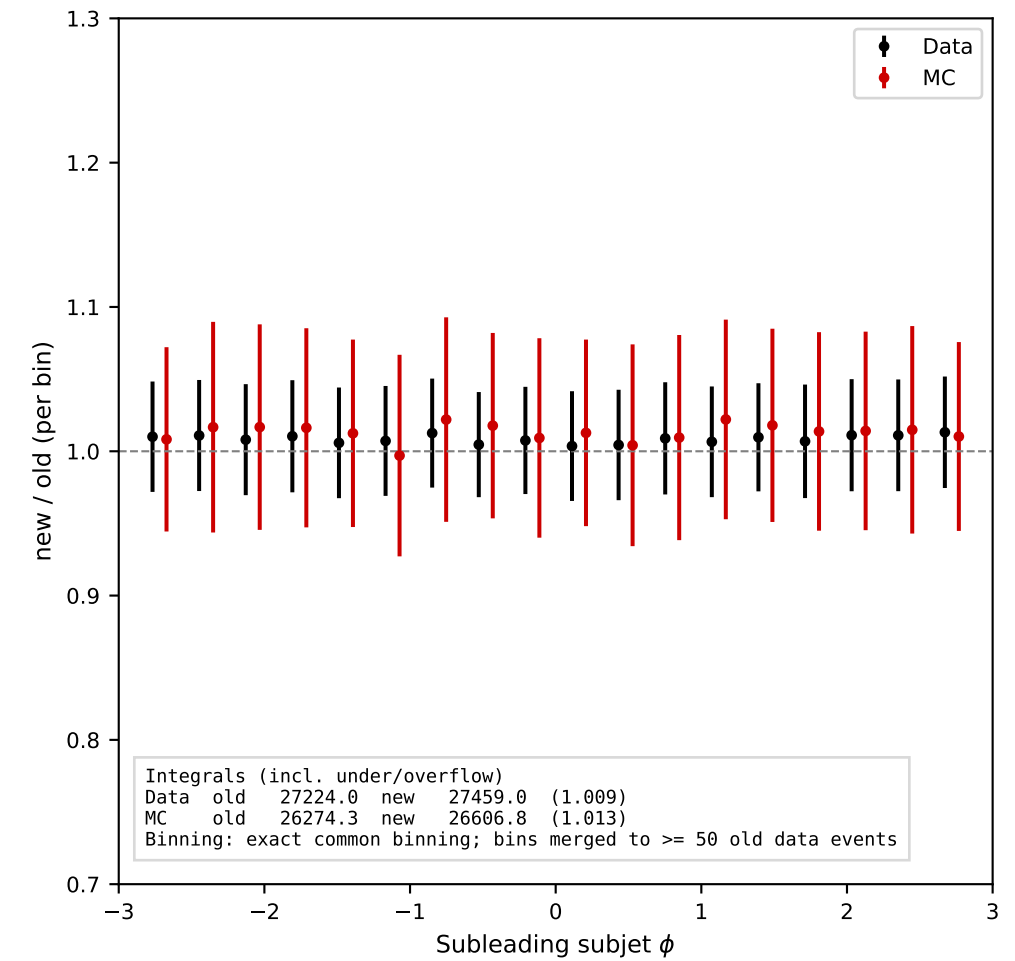
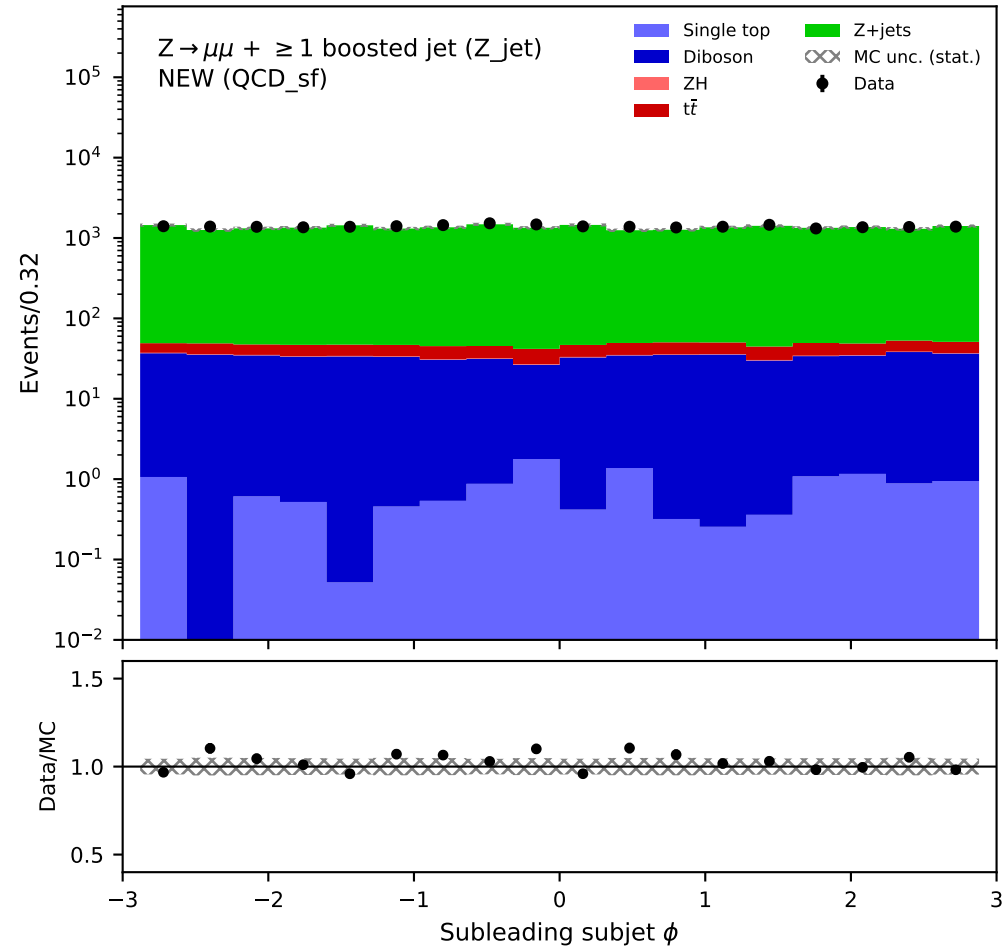
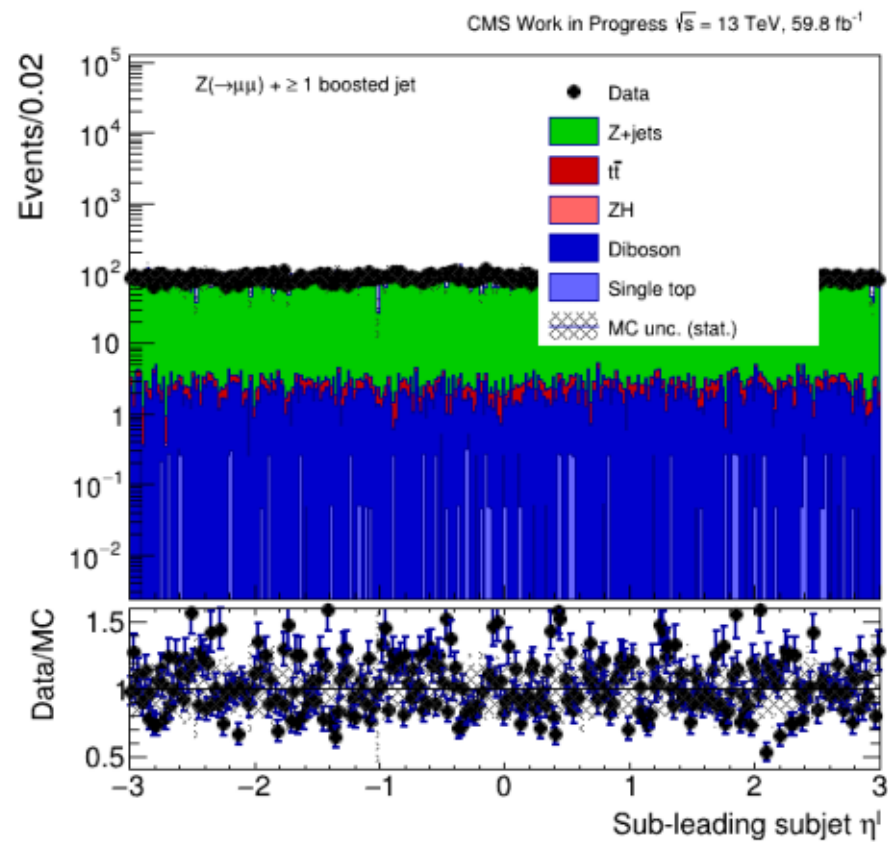
phi_sub1 | Z_jet | 2018

Old plot shown is its 'eta_sub1', which holds subjet phi (old eta/phi swap); it overflows beyond $|\phi| > 3$.

OLD: eta_sub1_Z_jet_Zee



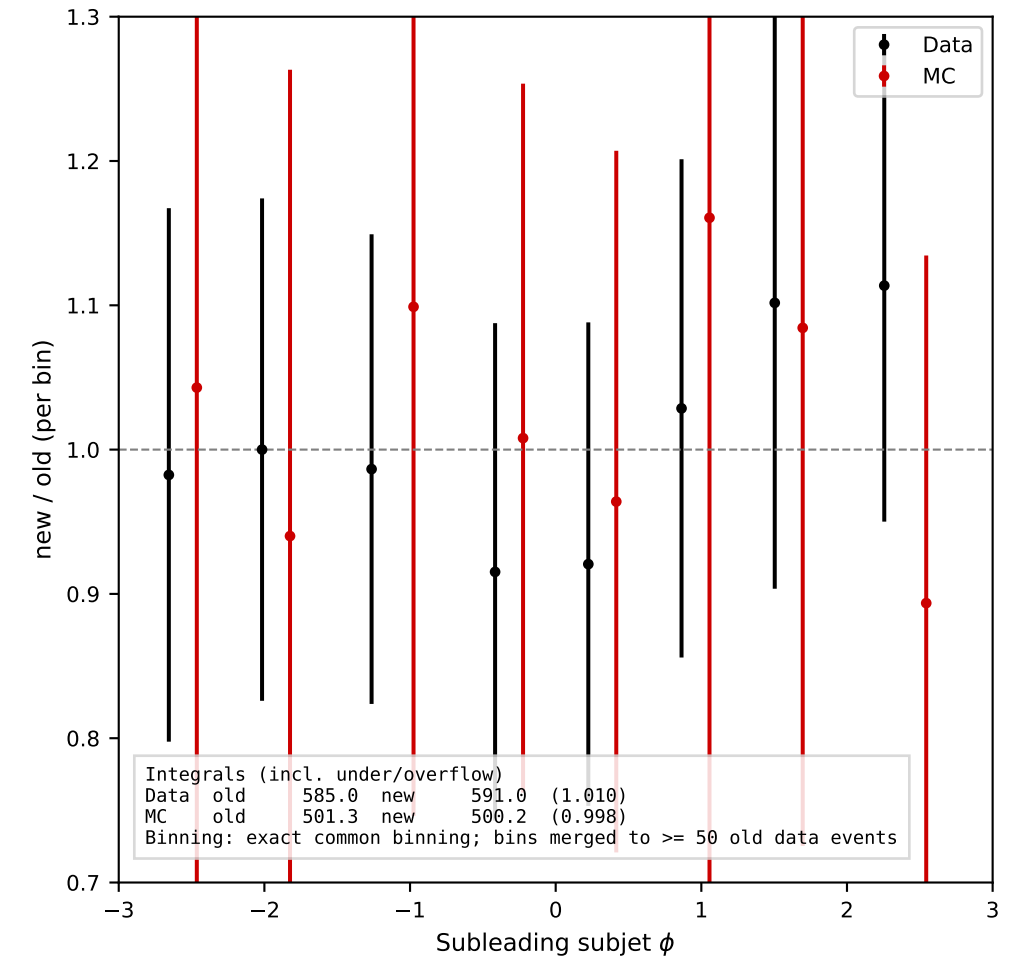
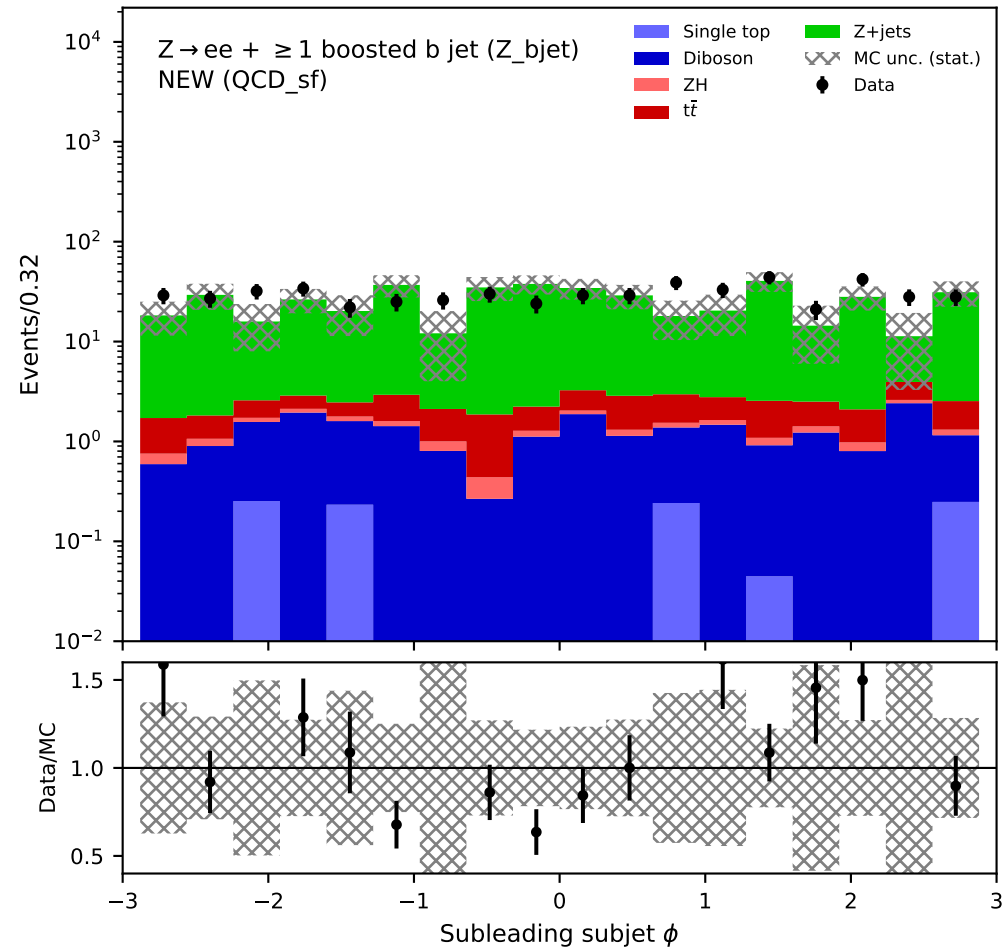
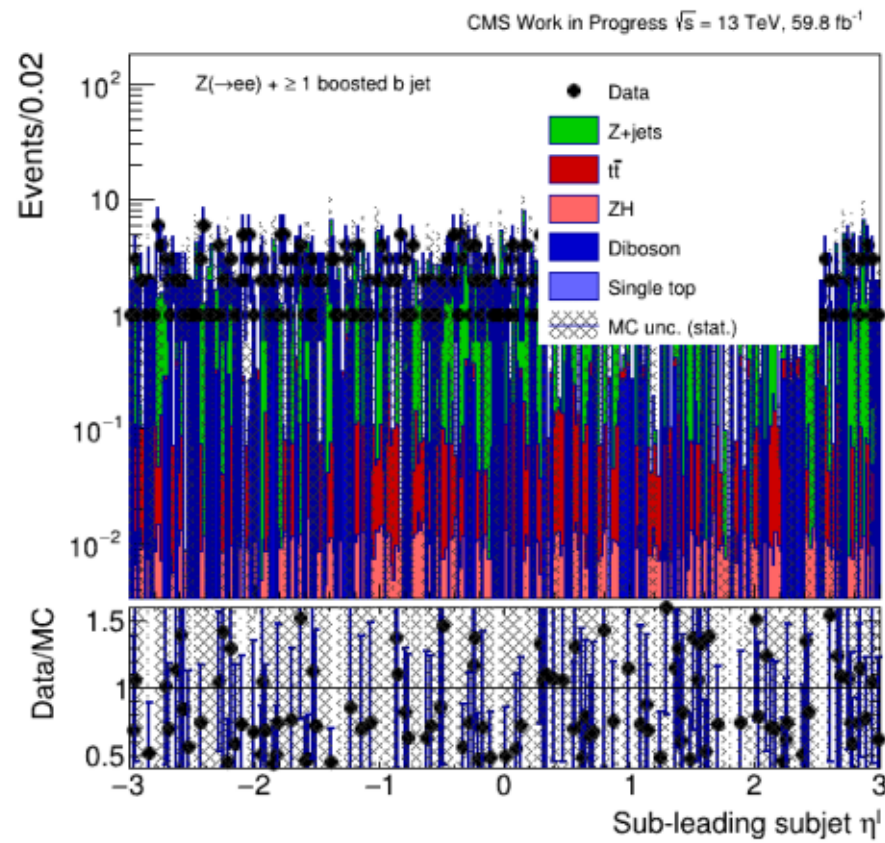
OLD: eta_sub1_Z_jet_Zmm



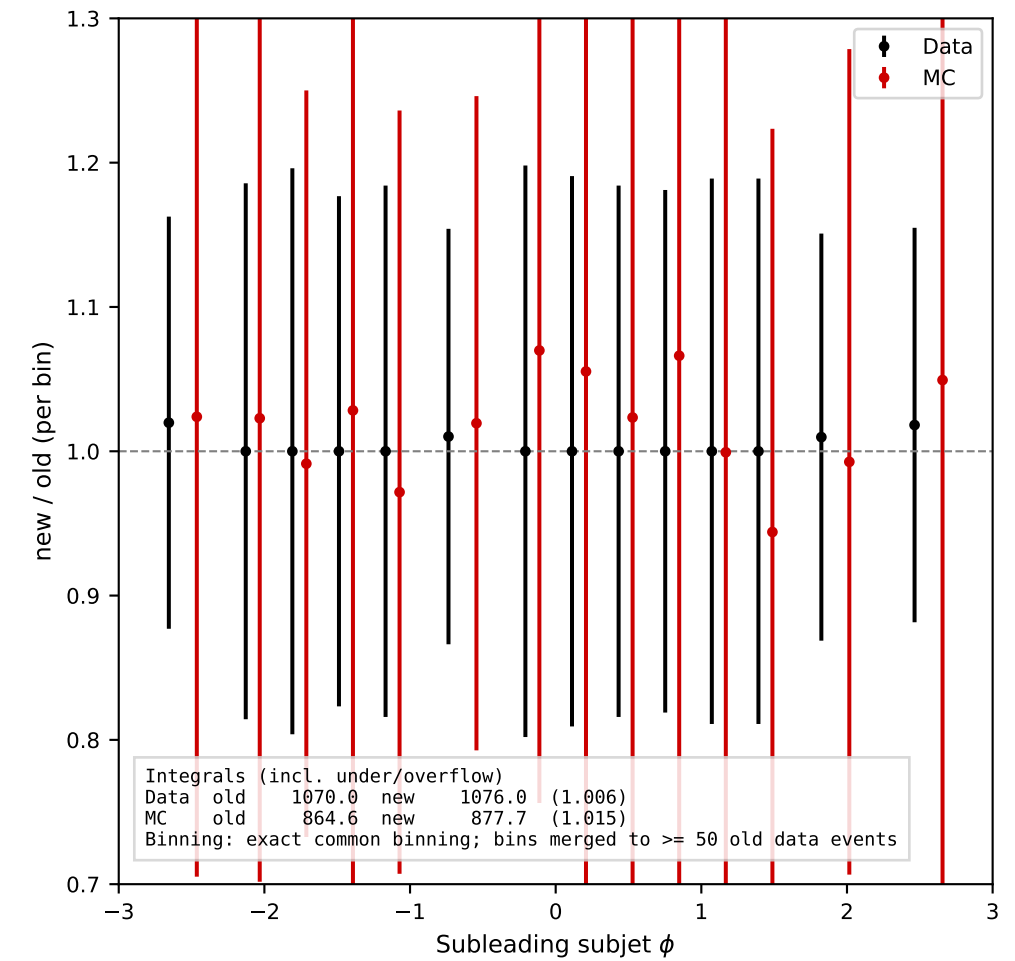
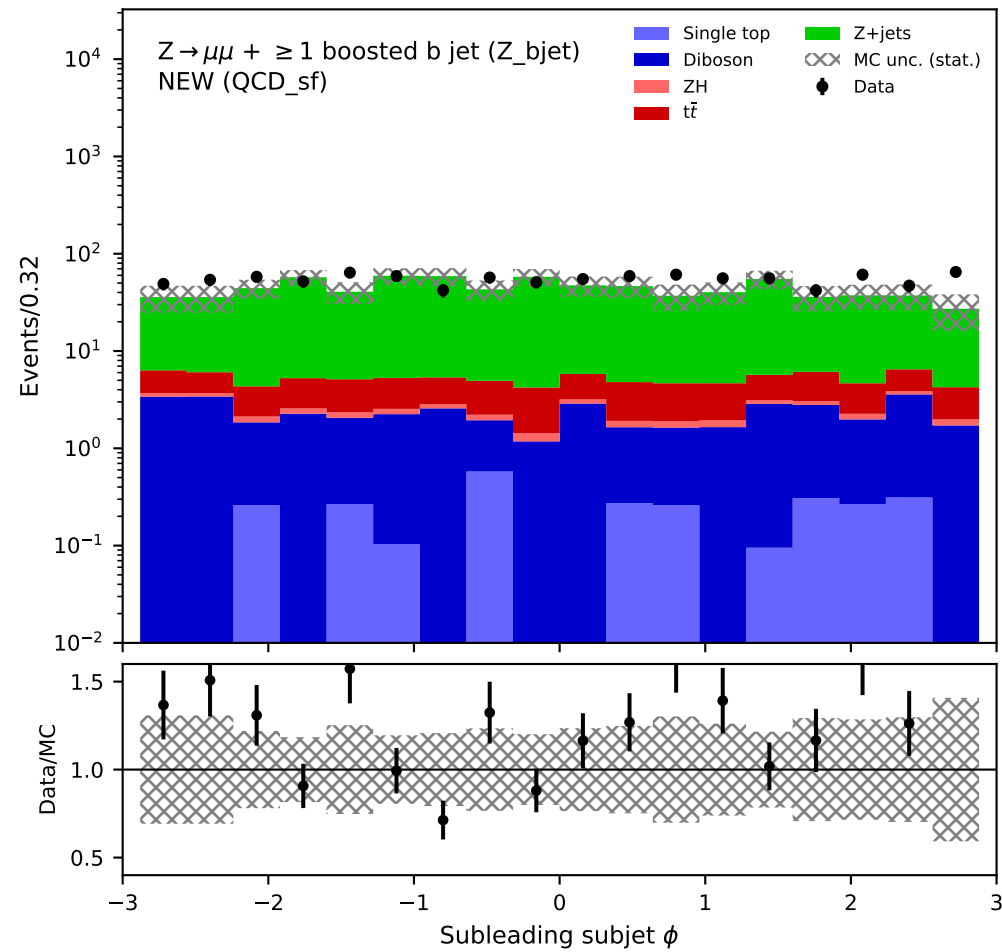
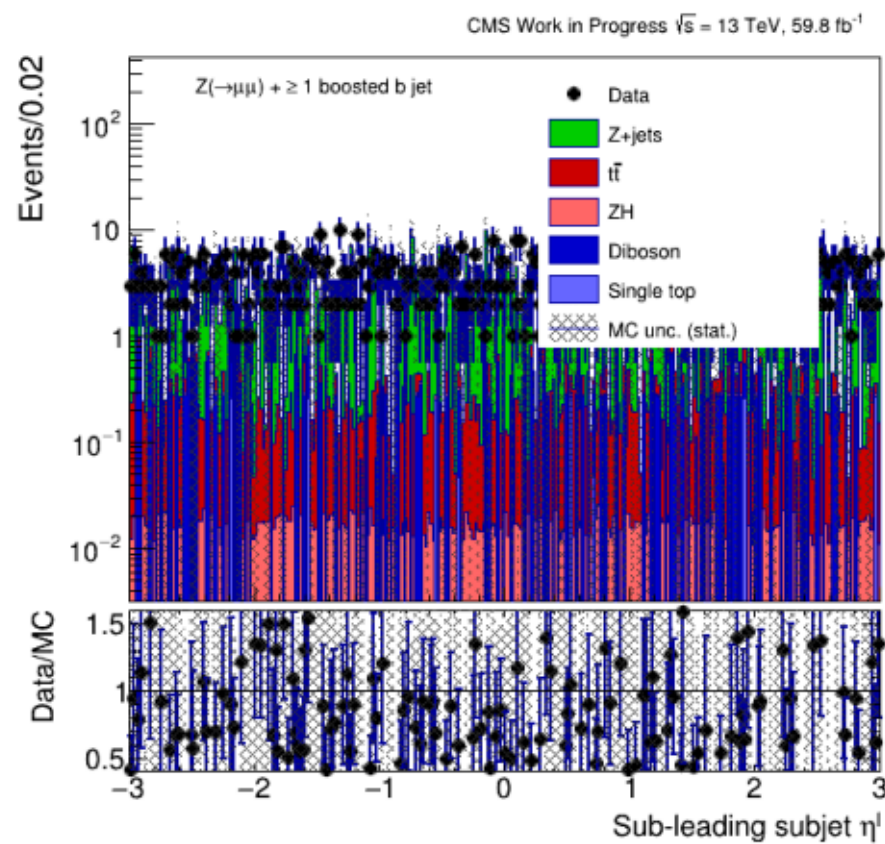
phi_sub1 | Z_bjet | 2018

Old plot shown is its 'eta_sub1', which holds subjet phi (old eta/phi swap); it overflows beyond $|\phi| > 3$. Z_bjet: the old framework fills subjet histograms from the leading jet's subjets, ours from the tagged jet's subjets; differences are expected.

OLD: eta_sub1_Z_bjet_Zee

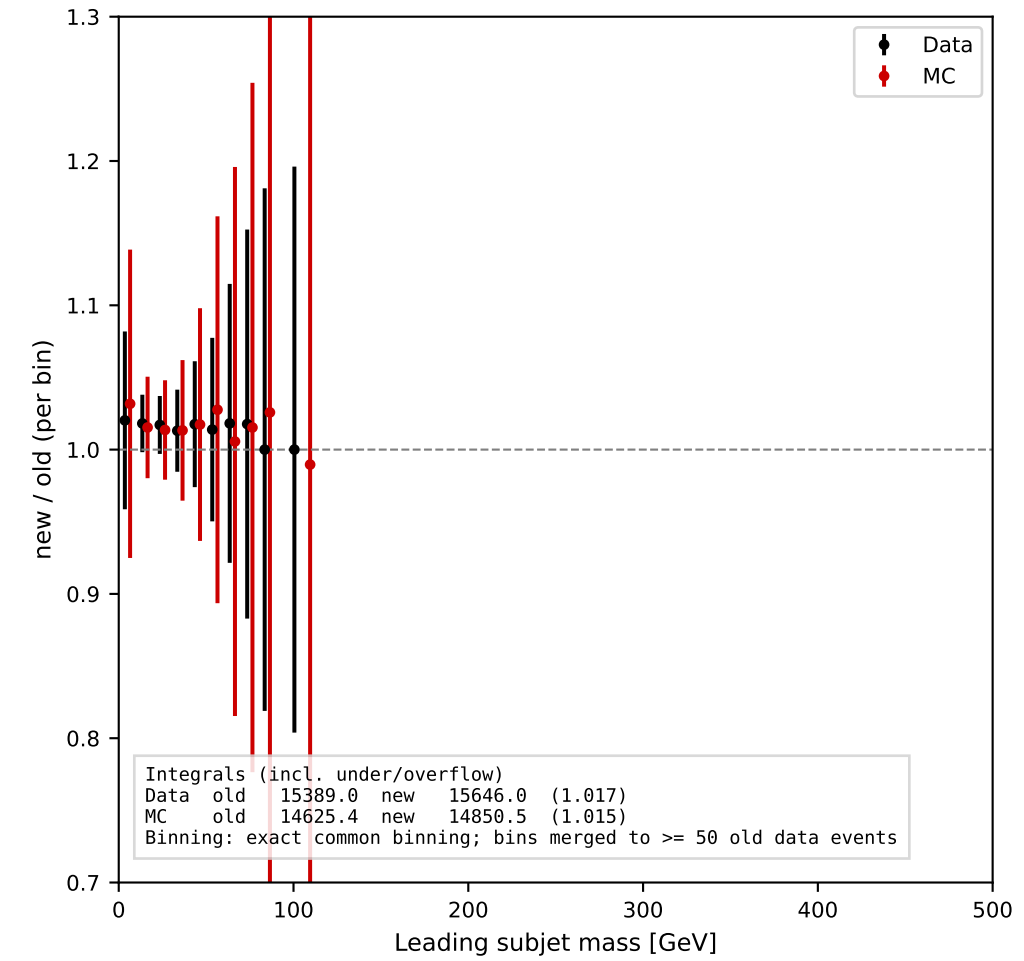
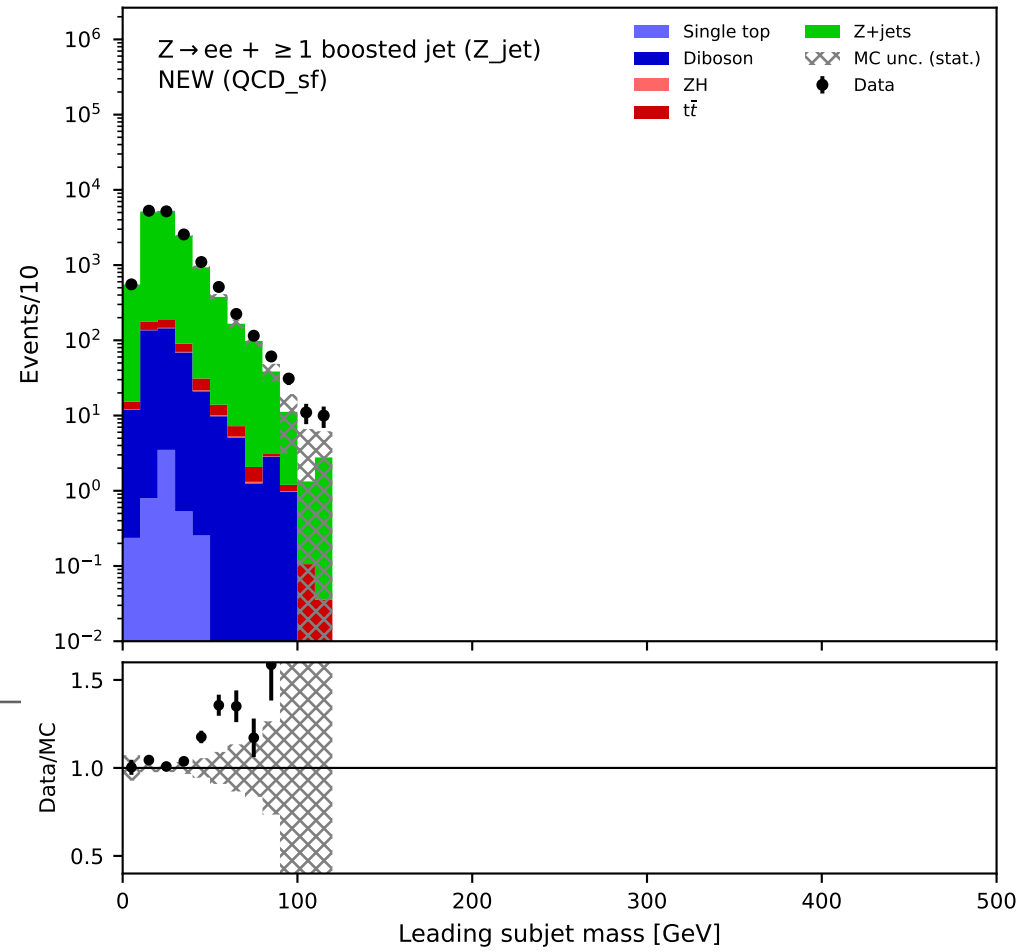
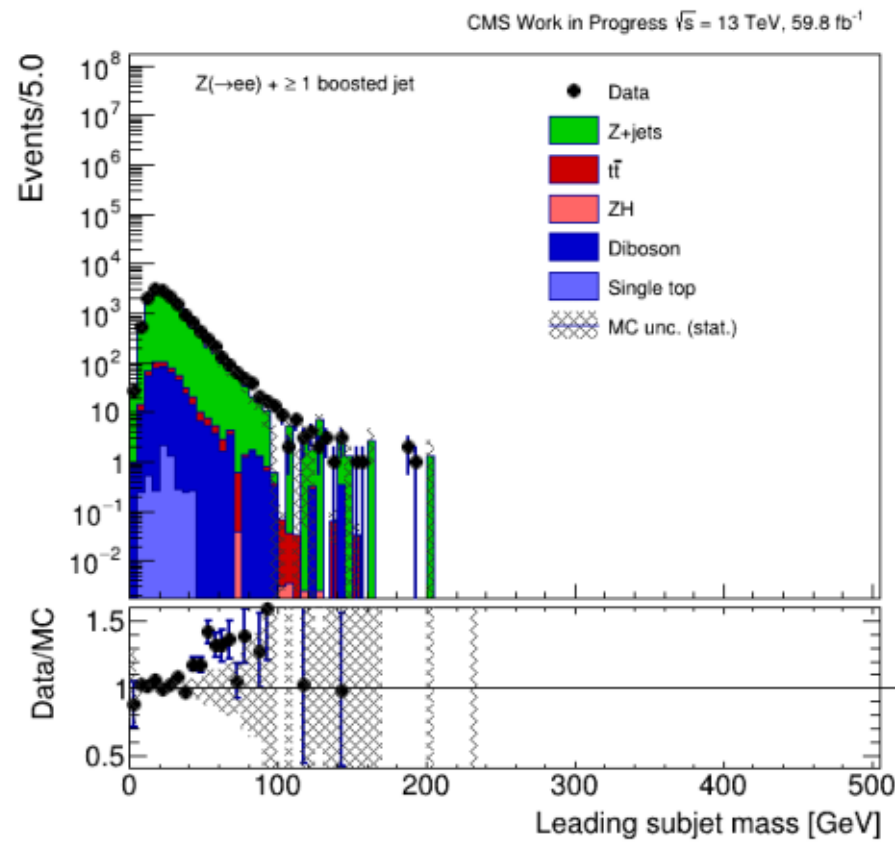


OLD: eta_sub1_Z_bjet_Zmm

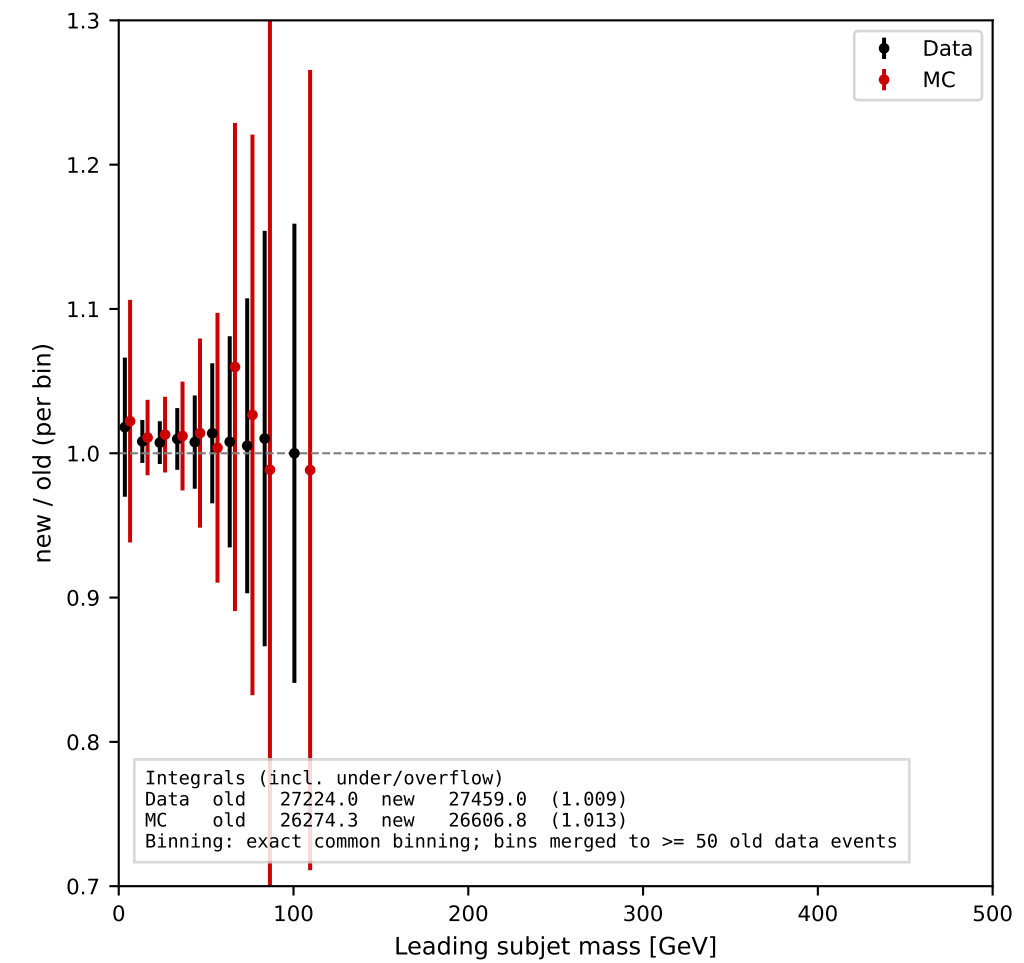
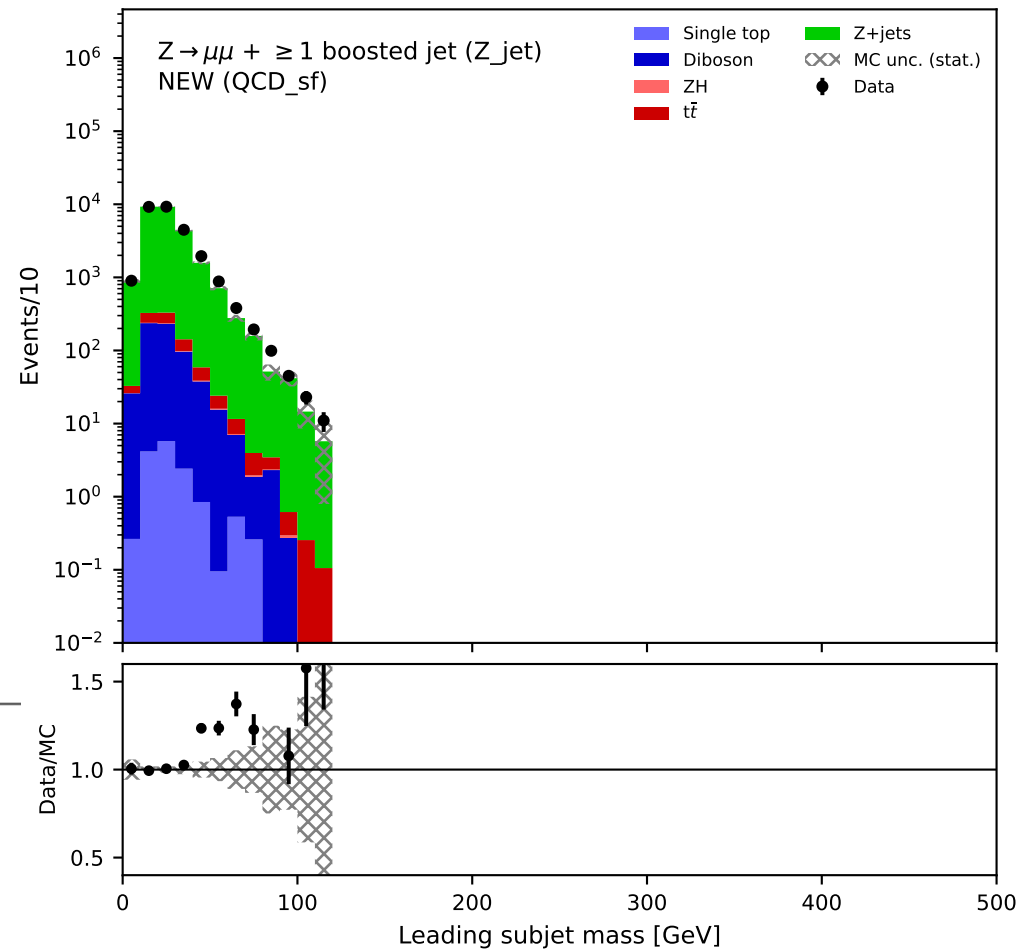
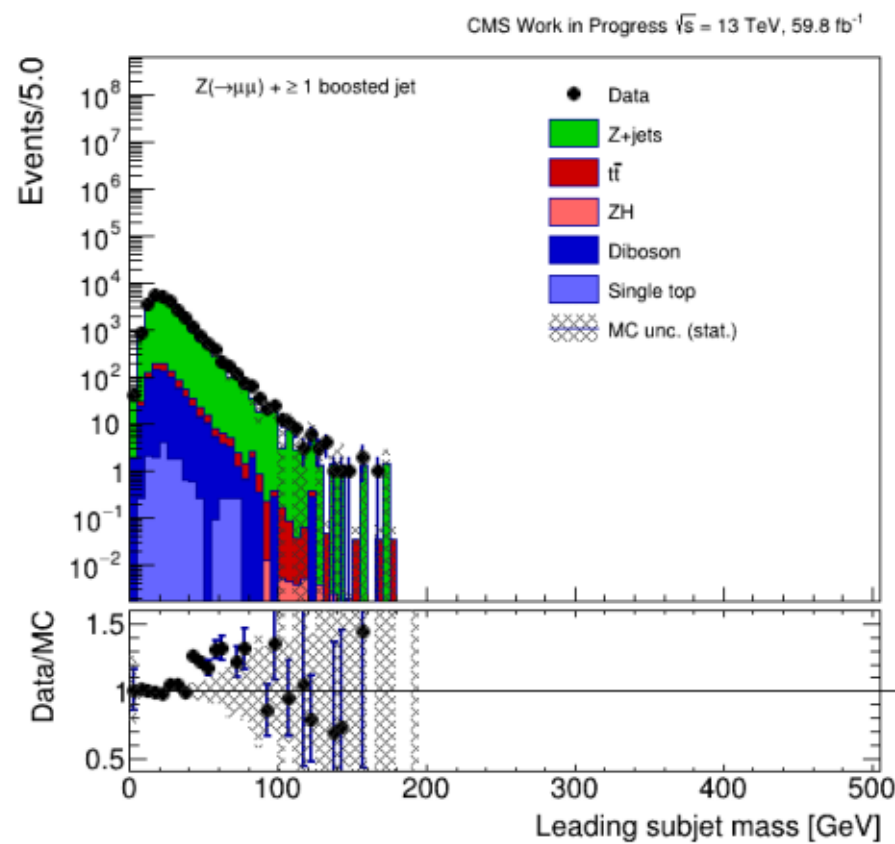


mass_sub0 | Z_jet | 2018

OLD: mass_sub0_Z_jet_Zee



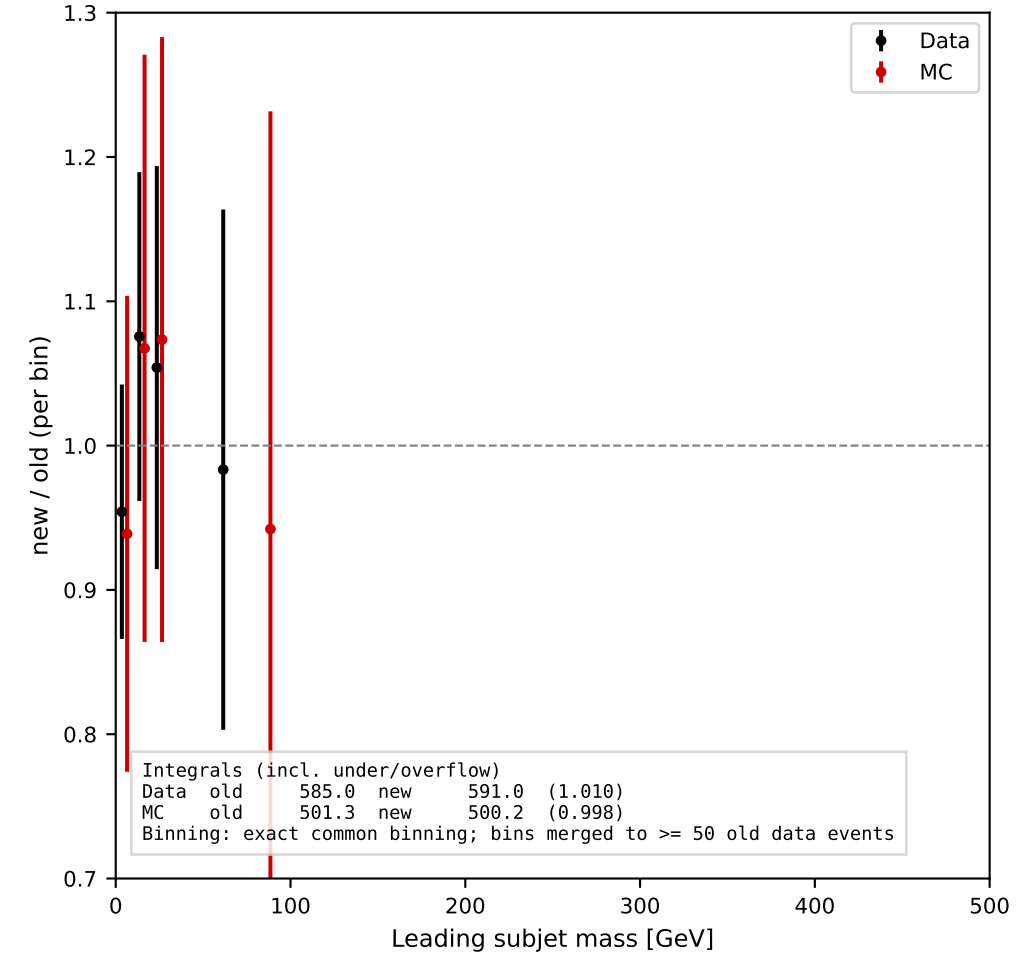
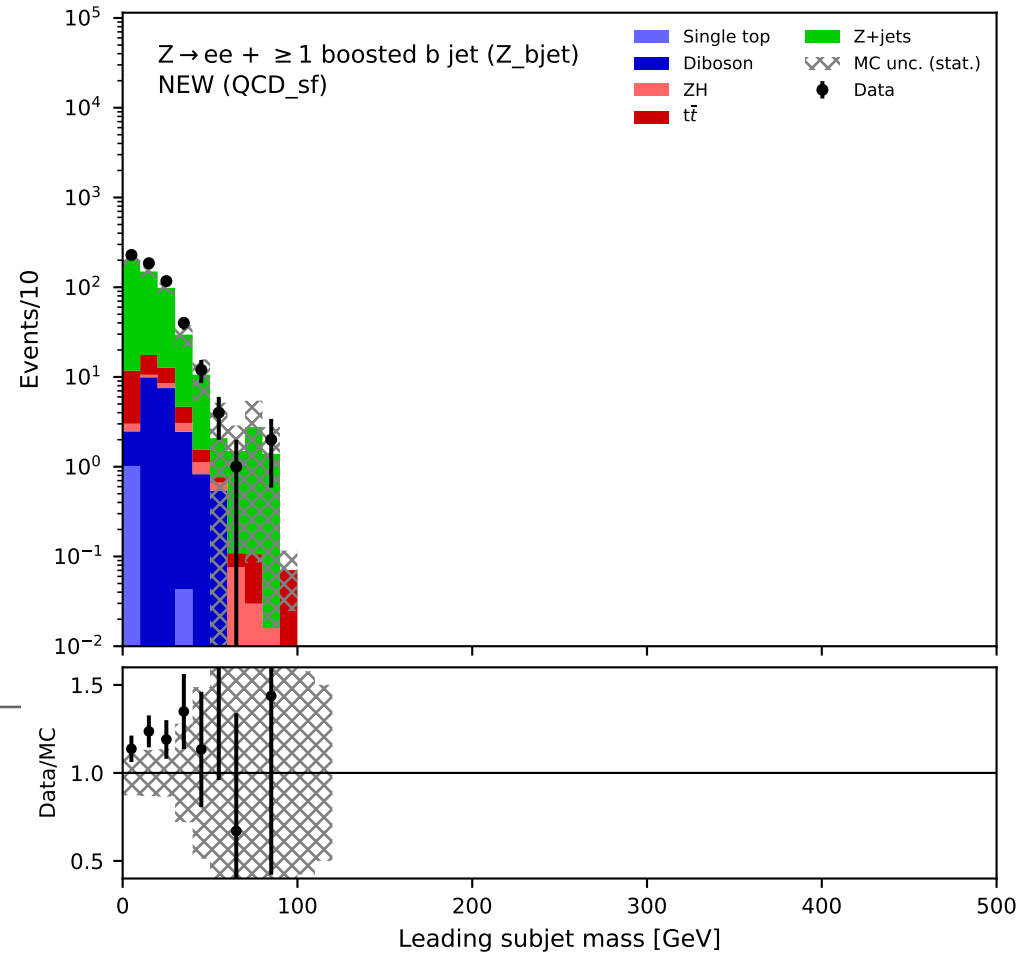
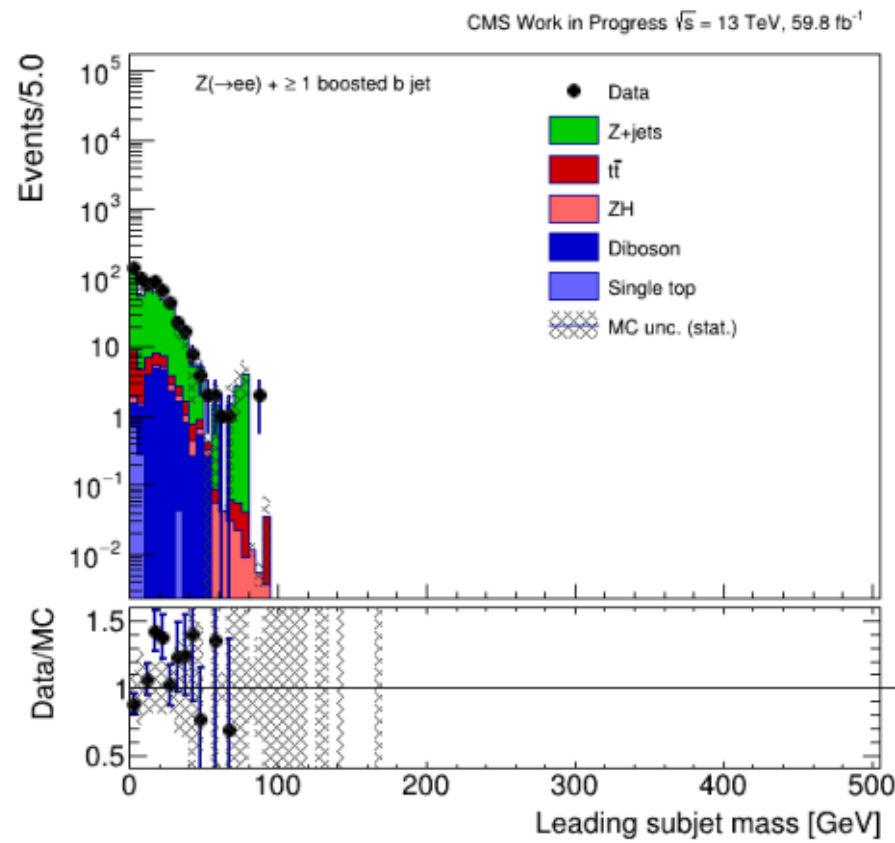
OLD: mass_sub0_Z_jet_Zmm



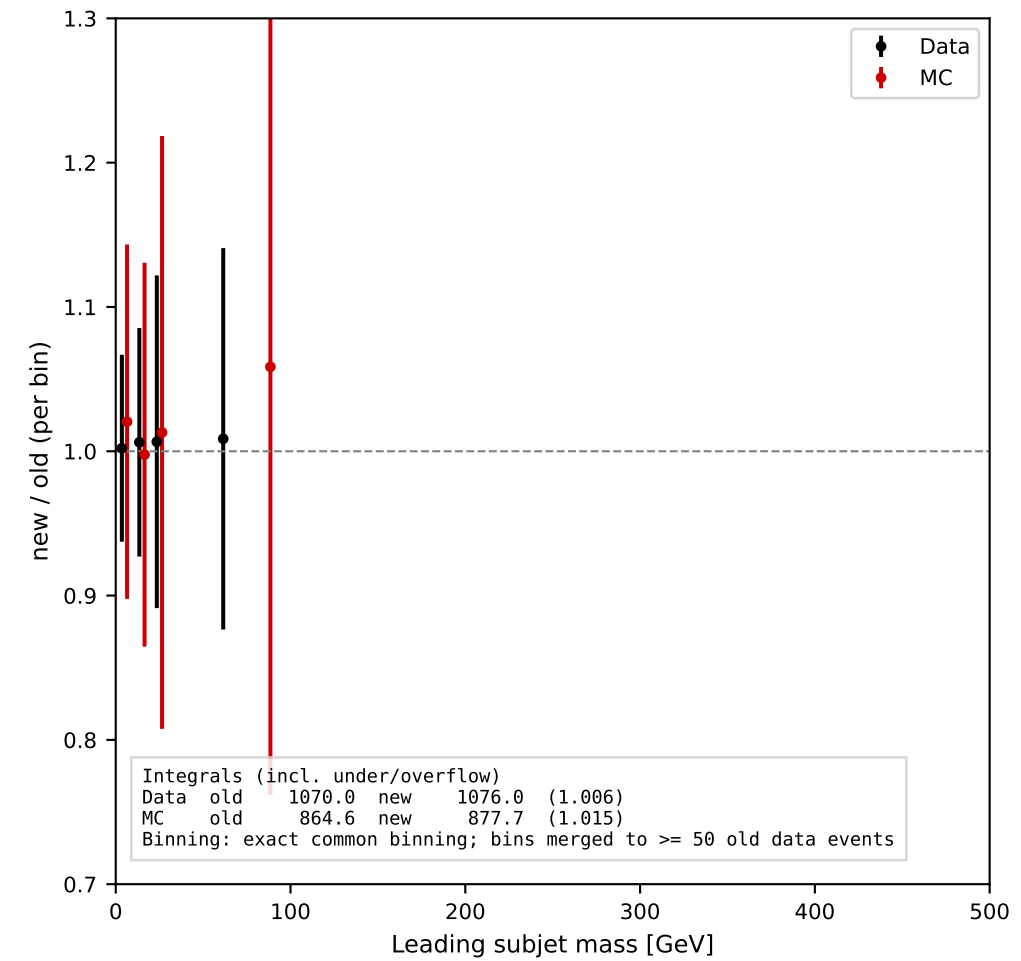
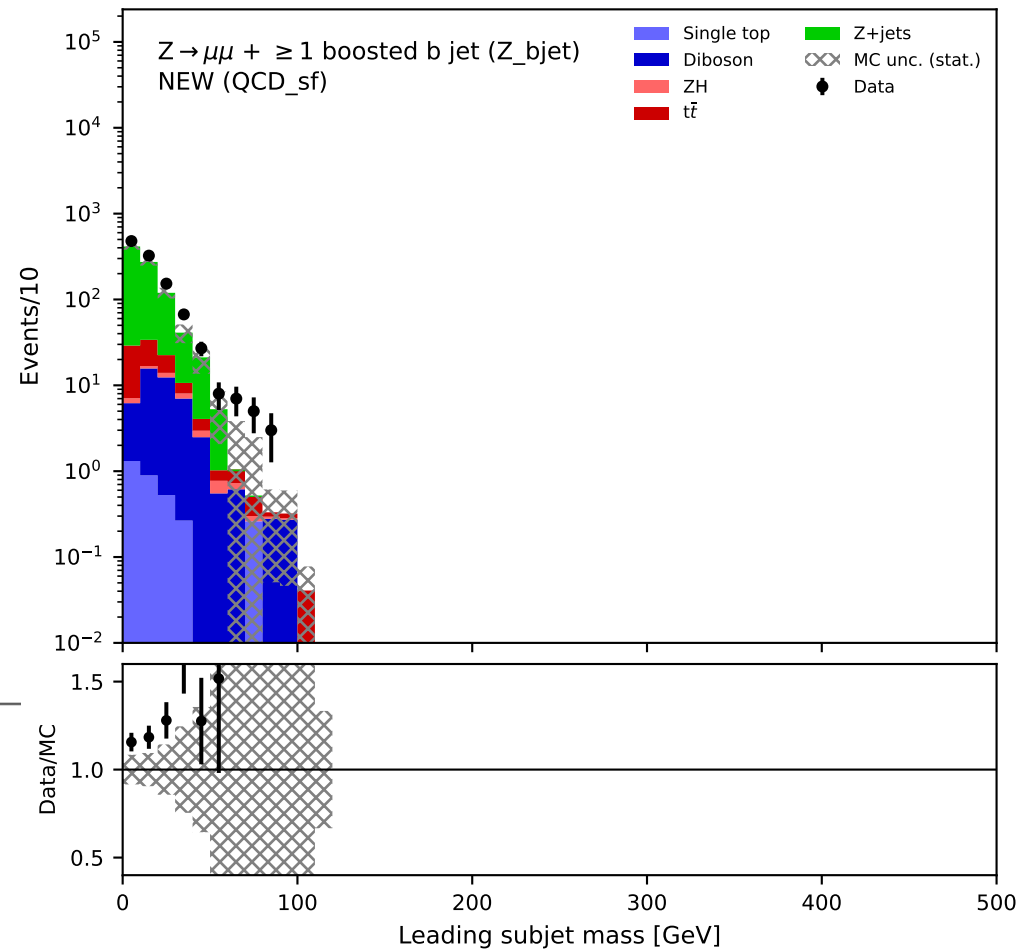
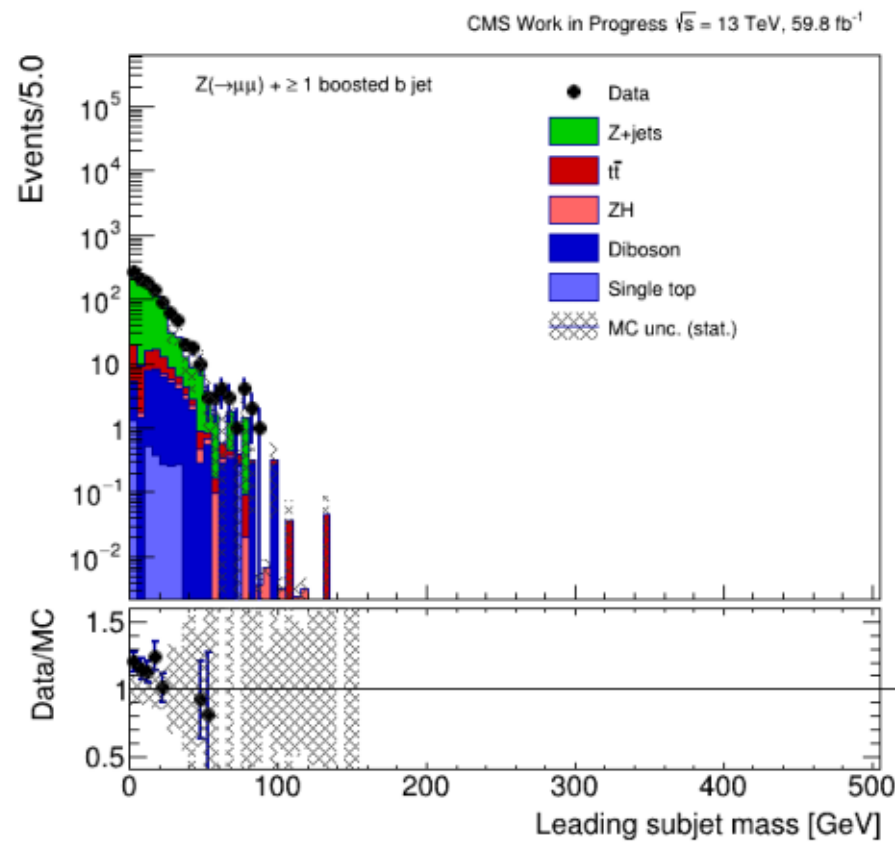
mass_sub0 | Z_bjet | 2018

Z_bjet: the old framework fills subjet histograms from the leading jet's subjets, ours from the tagged jet's subjets; differences are expected.

OLD: mass_sub0_Z_bjet_Zee

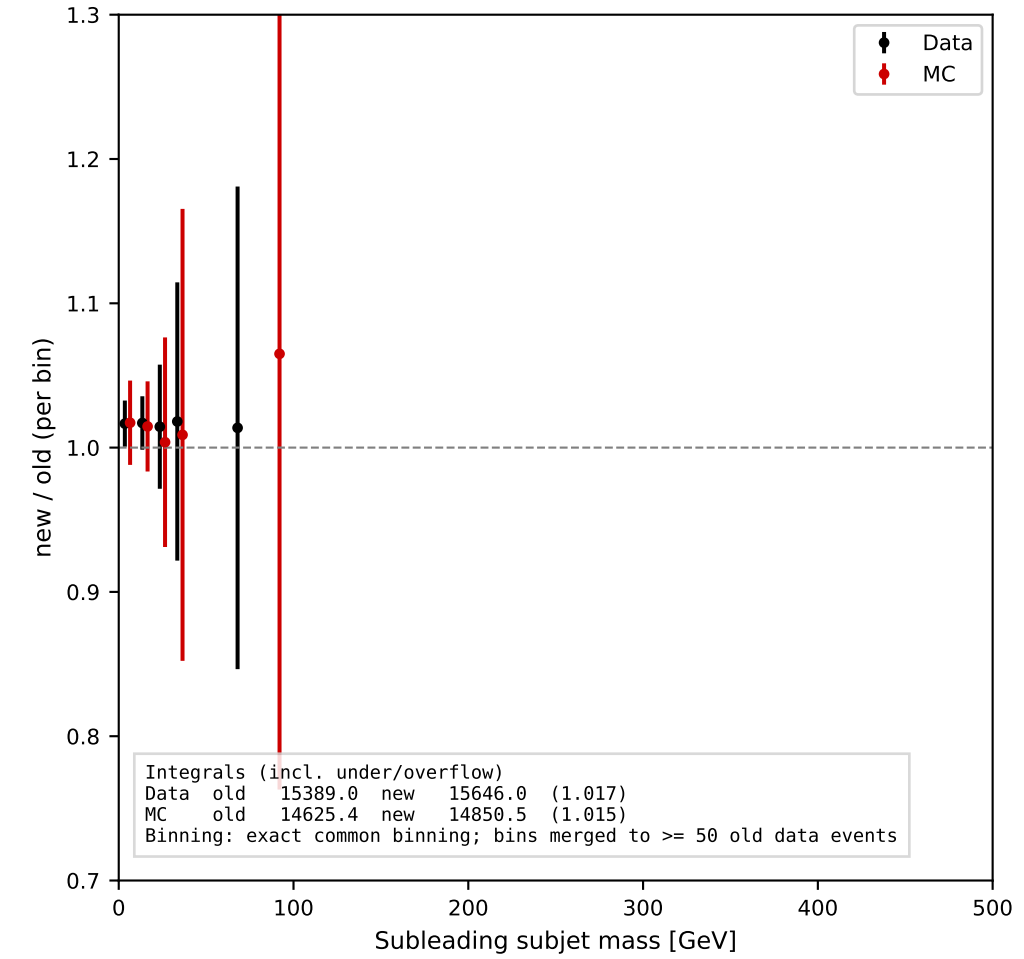
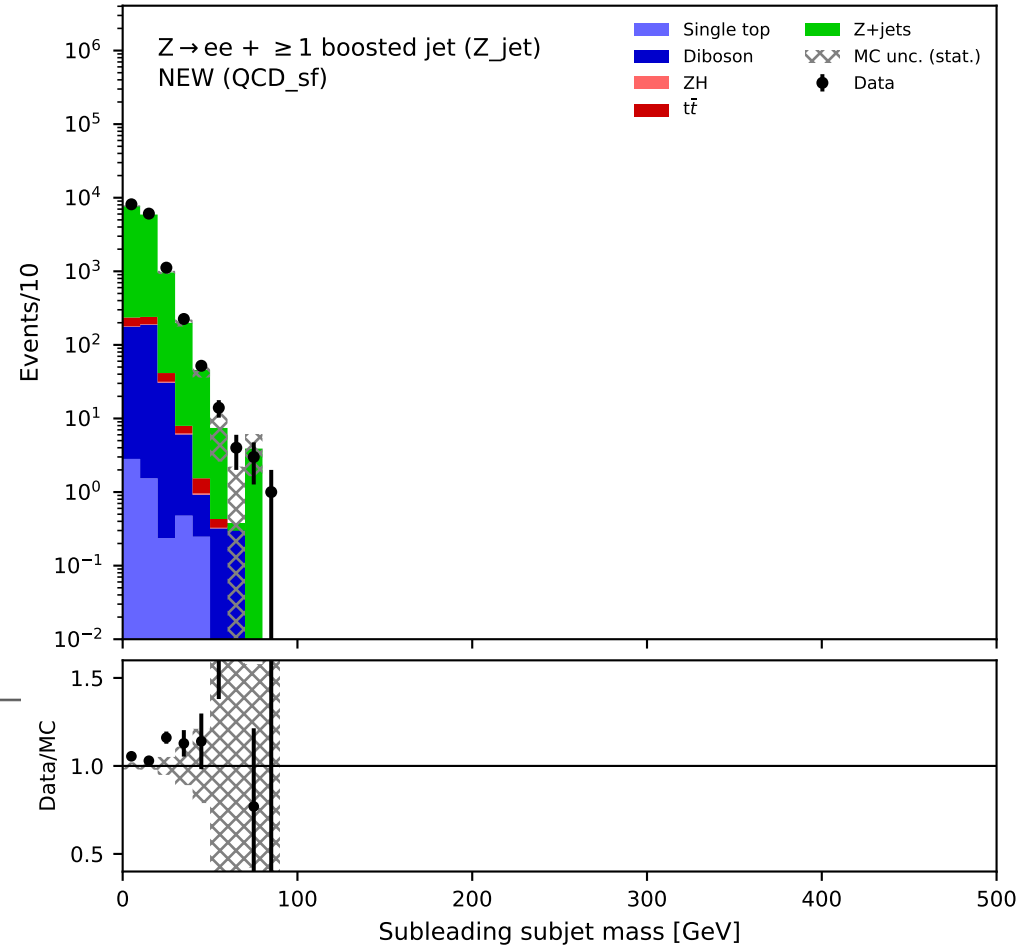
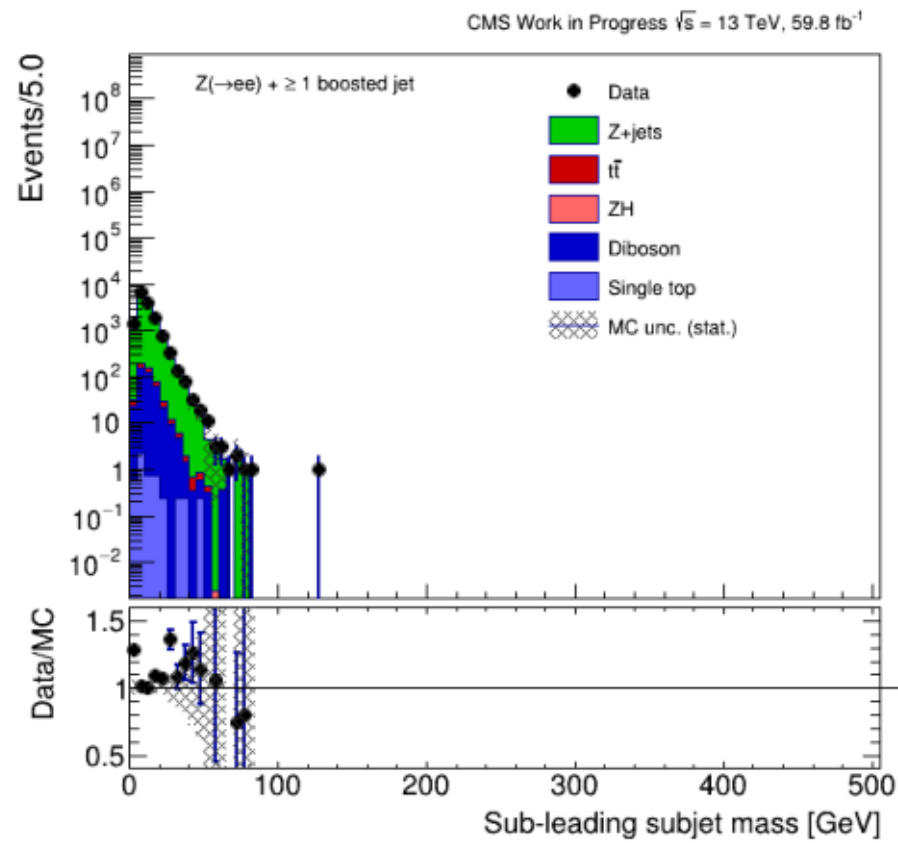


OLD: mass_sub0_Z_bjet_Zmm

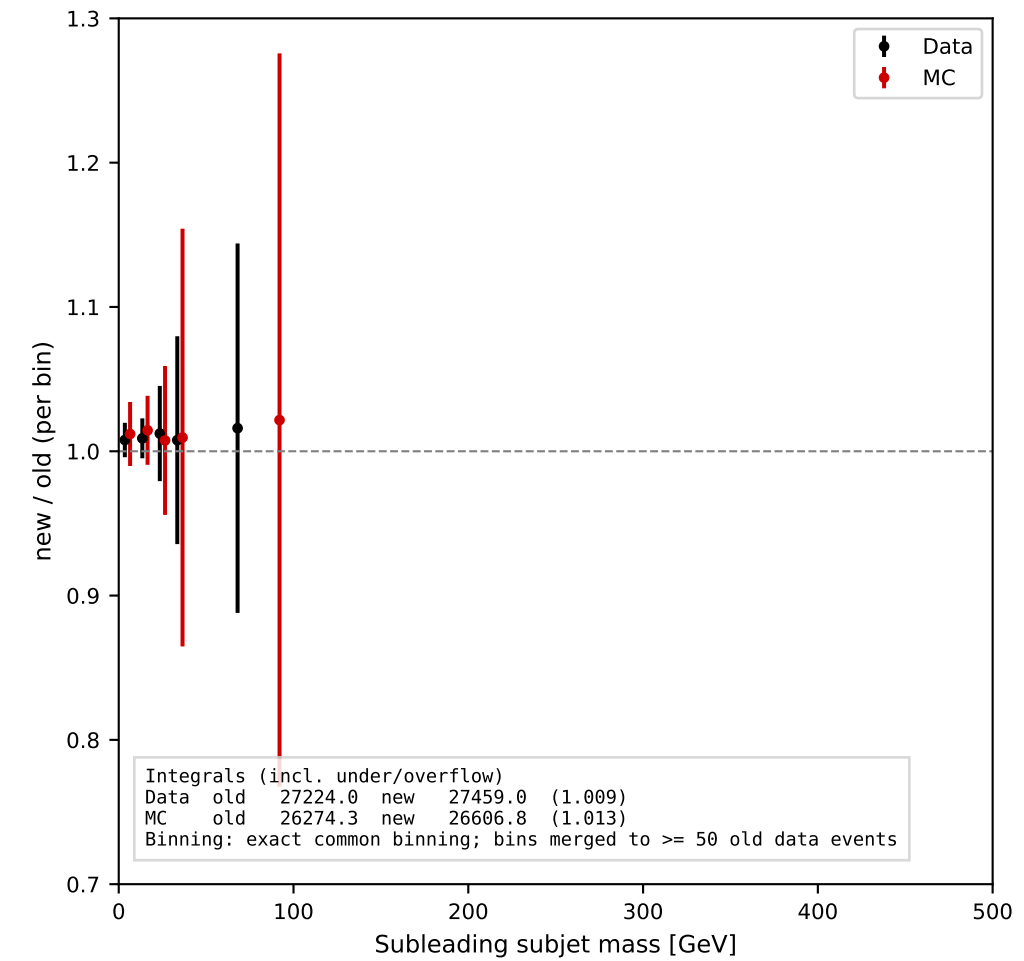
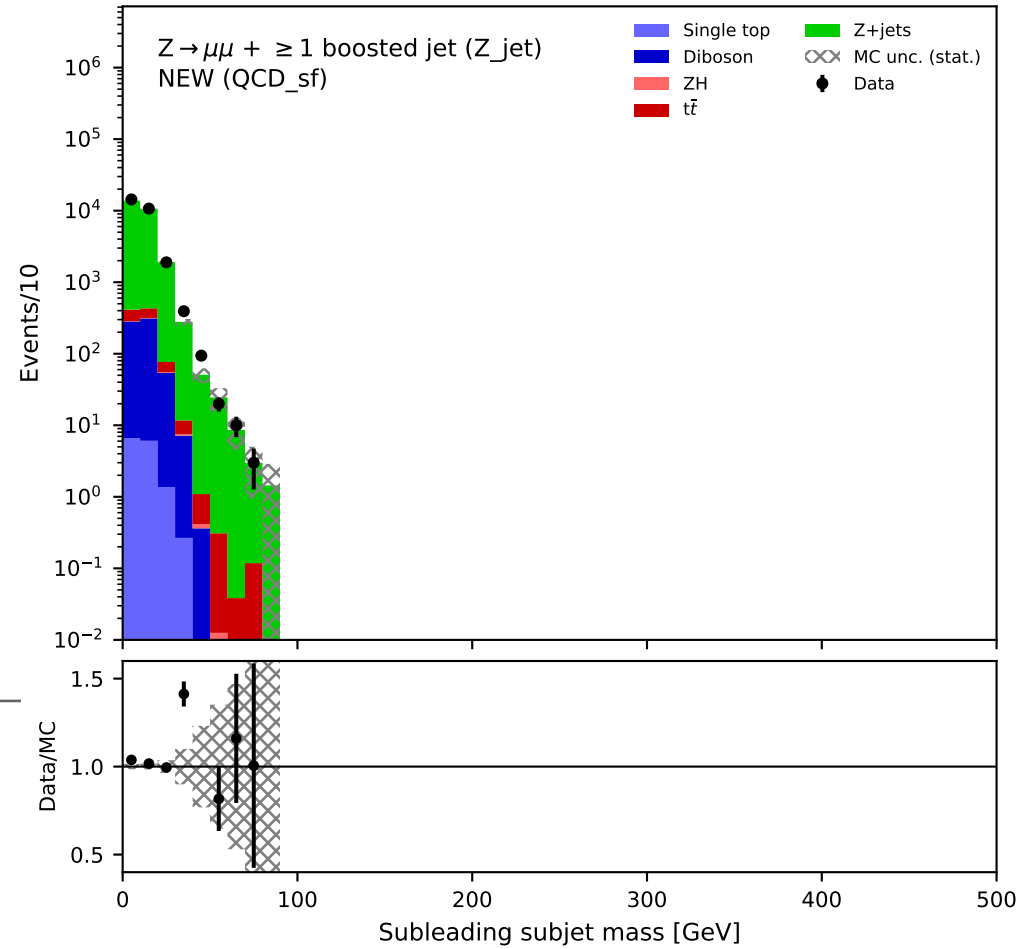
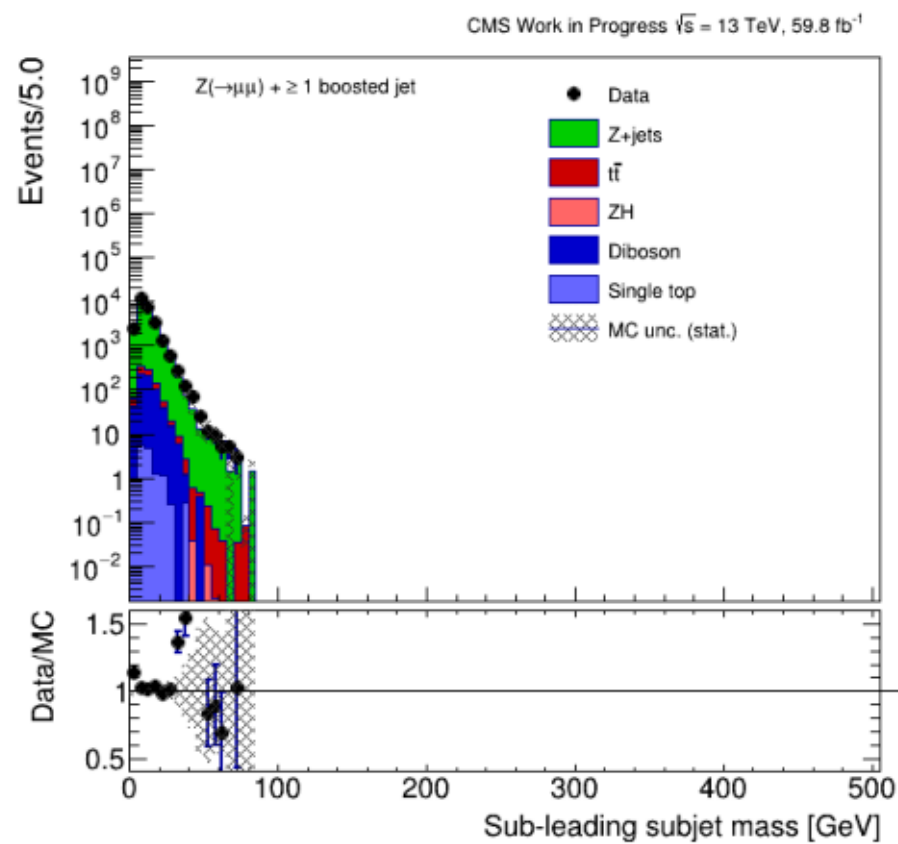


mass_sub1 | Z_jet | 2018

OLD: mass_sub1_Z_jet_Zee



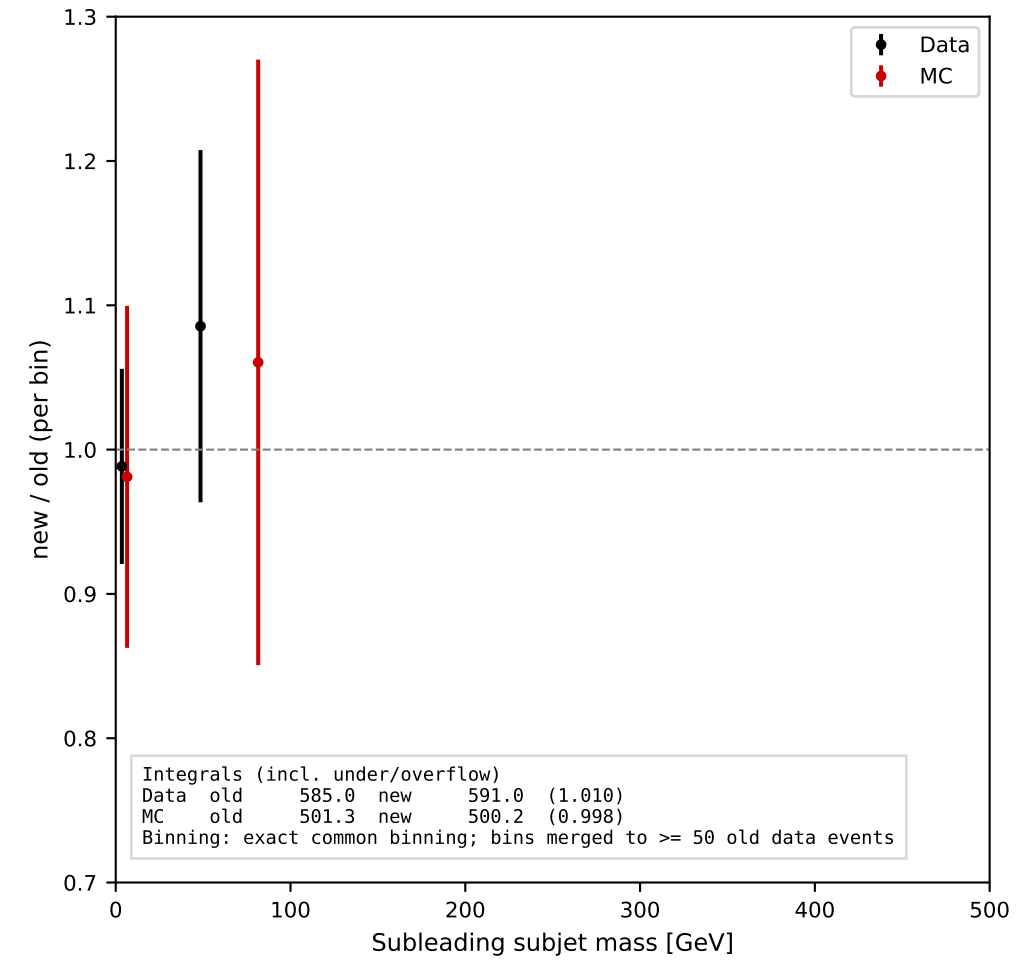
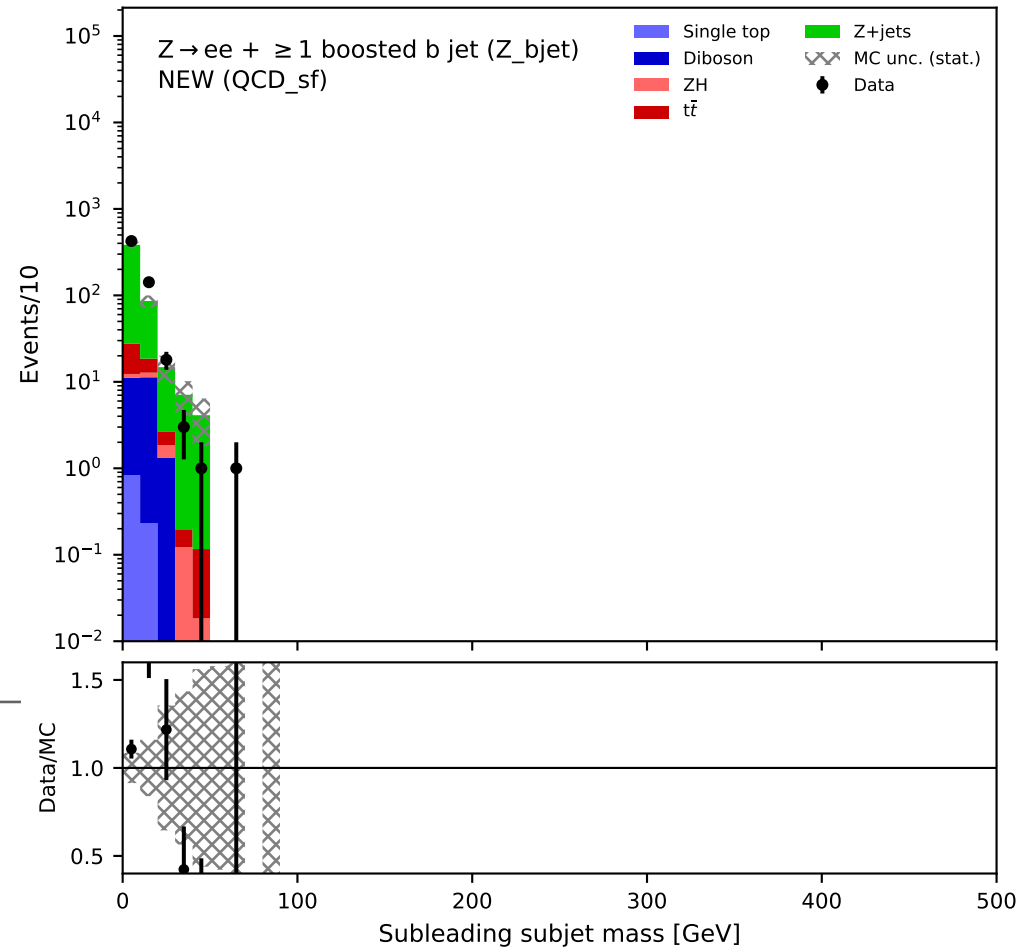
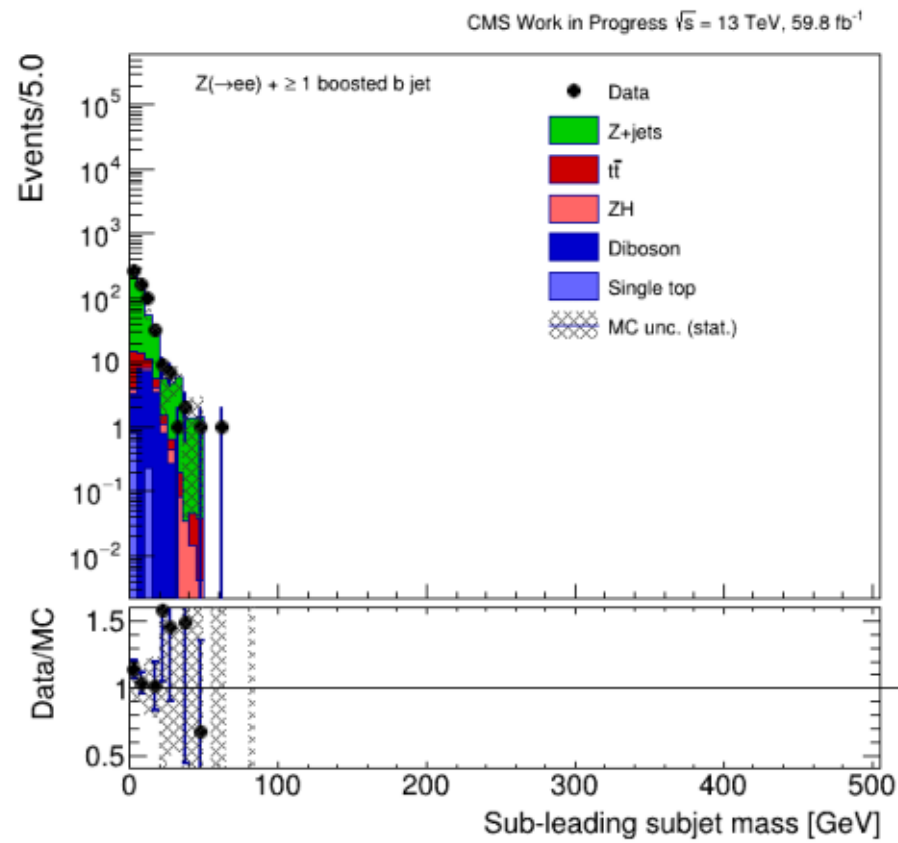
OLD: mass_sub1_Z_jet_Zmm



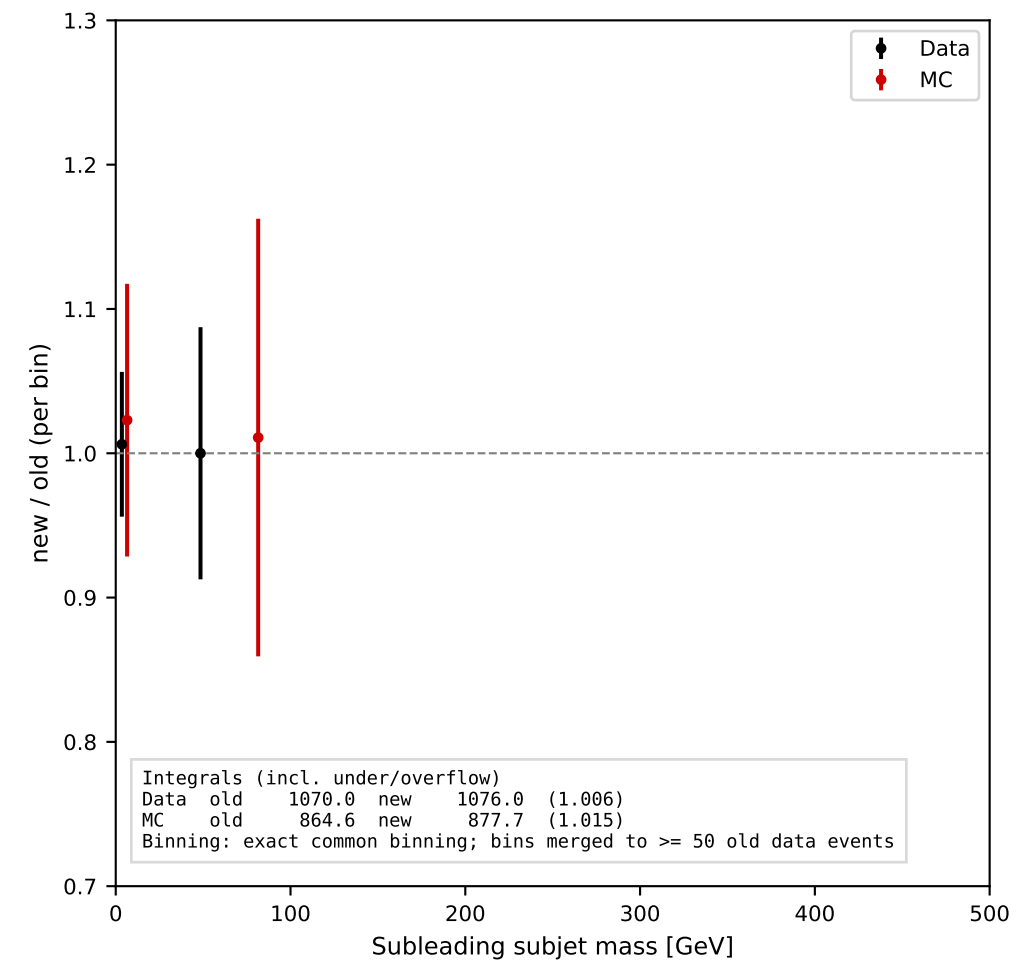
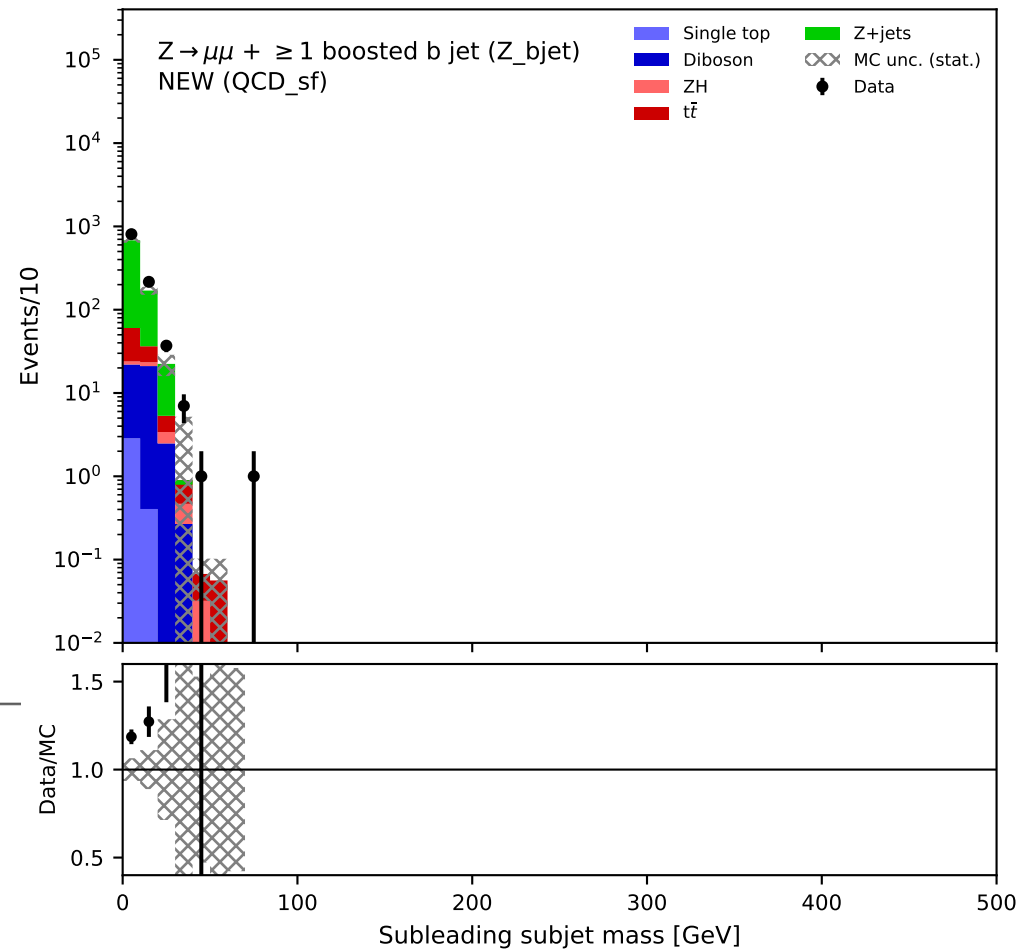
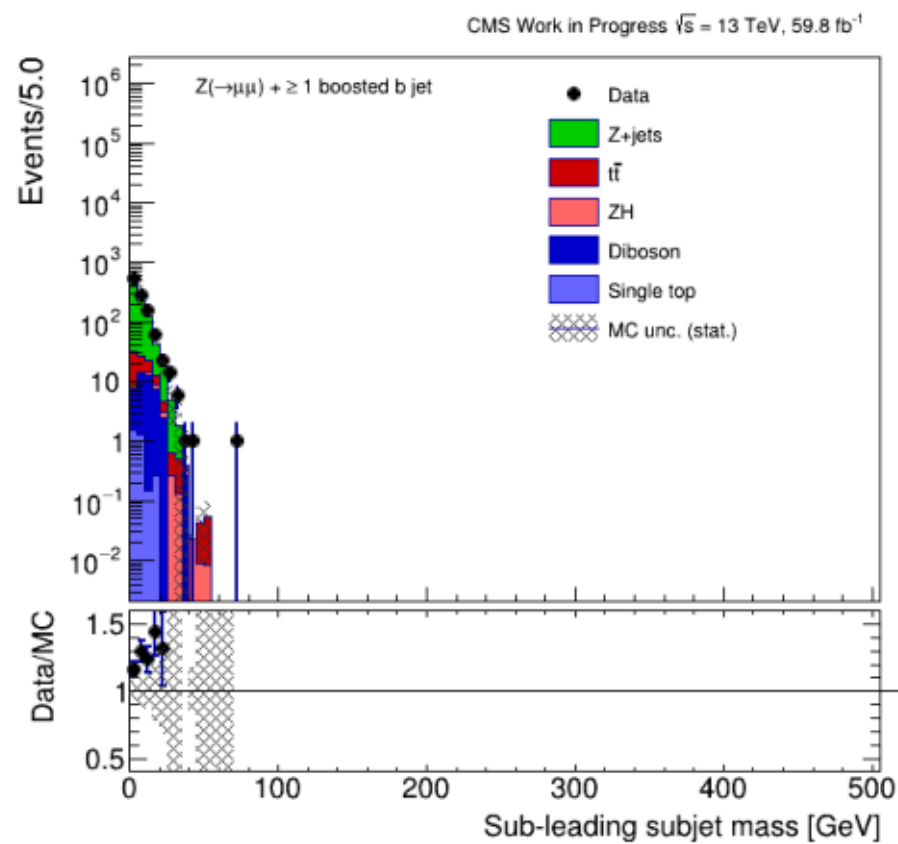
mass_sub1 | Z_bjet | 2018

Z_bjet: the old framework fills subjet histograms from the leading jet's subjets, ours from the tagged jet's subjets; differences are expected.

OLD: mass_sub1_Z_bjet_Zee

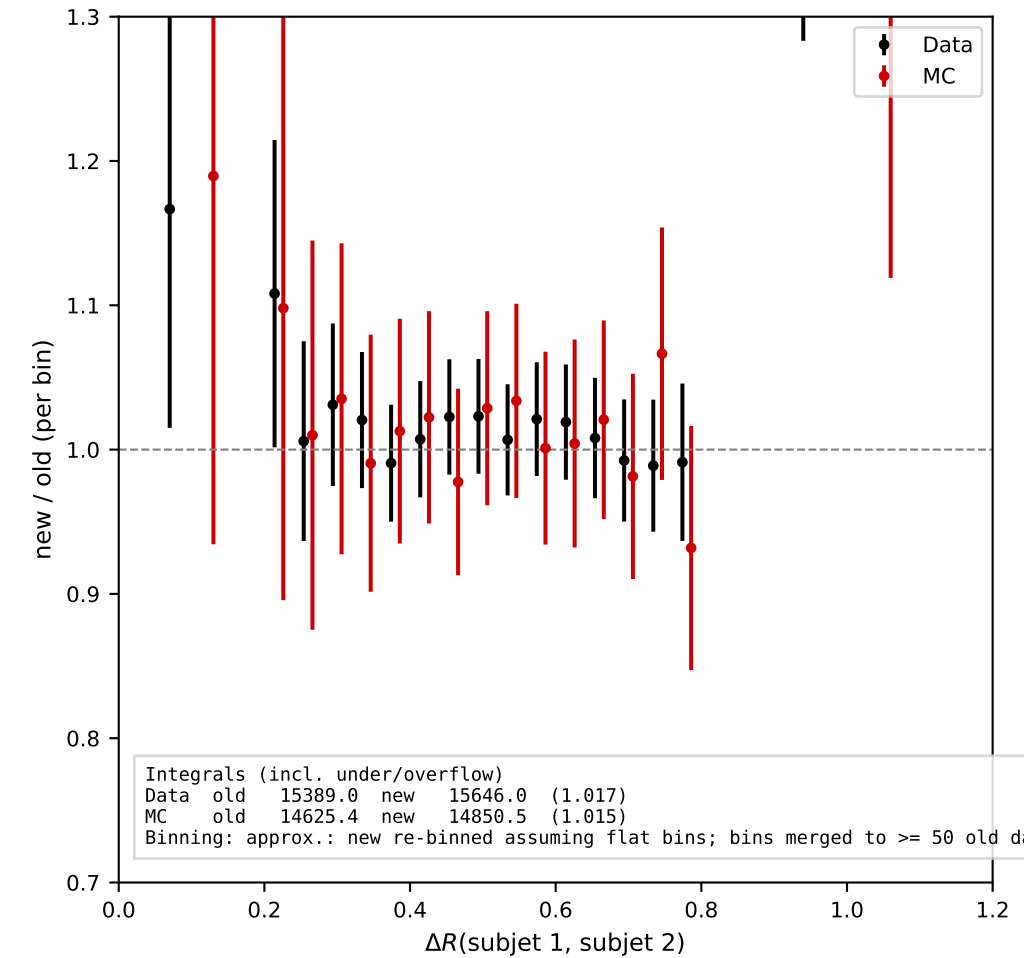
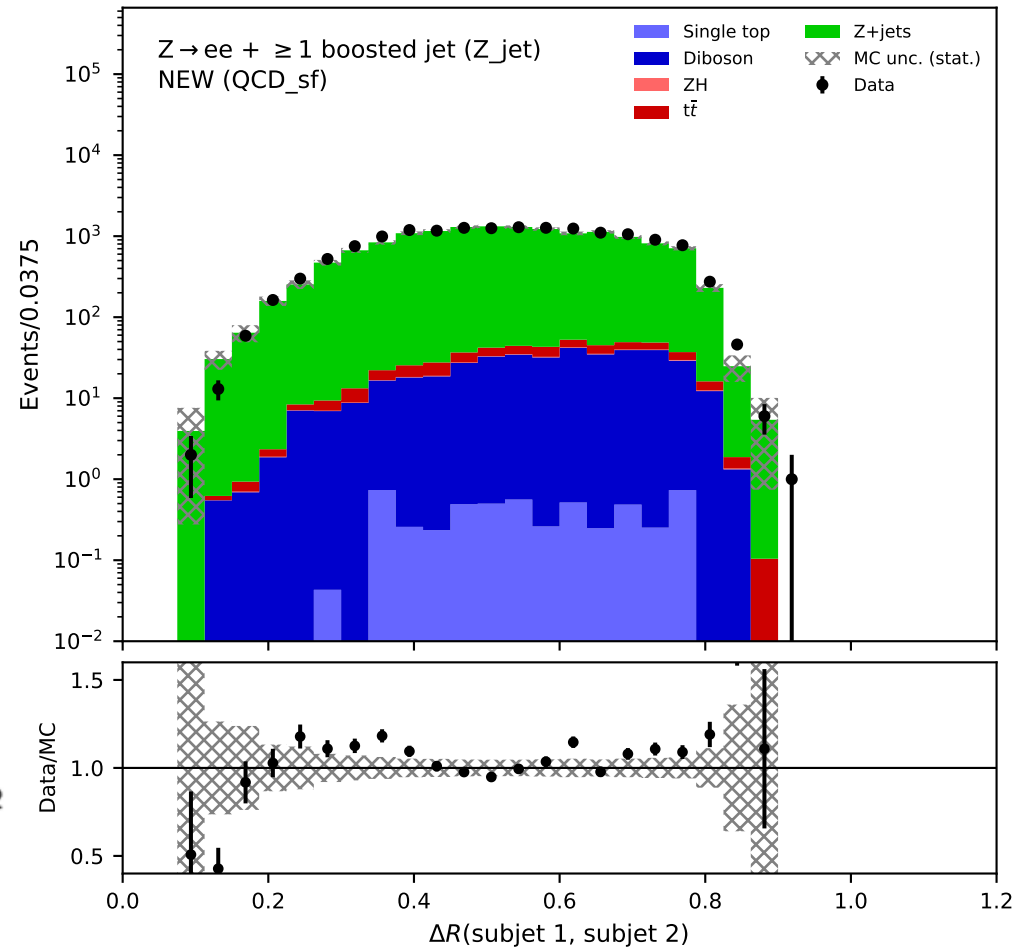
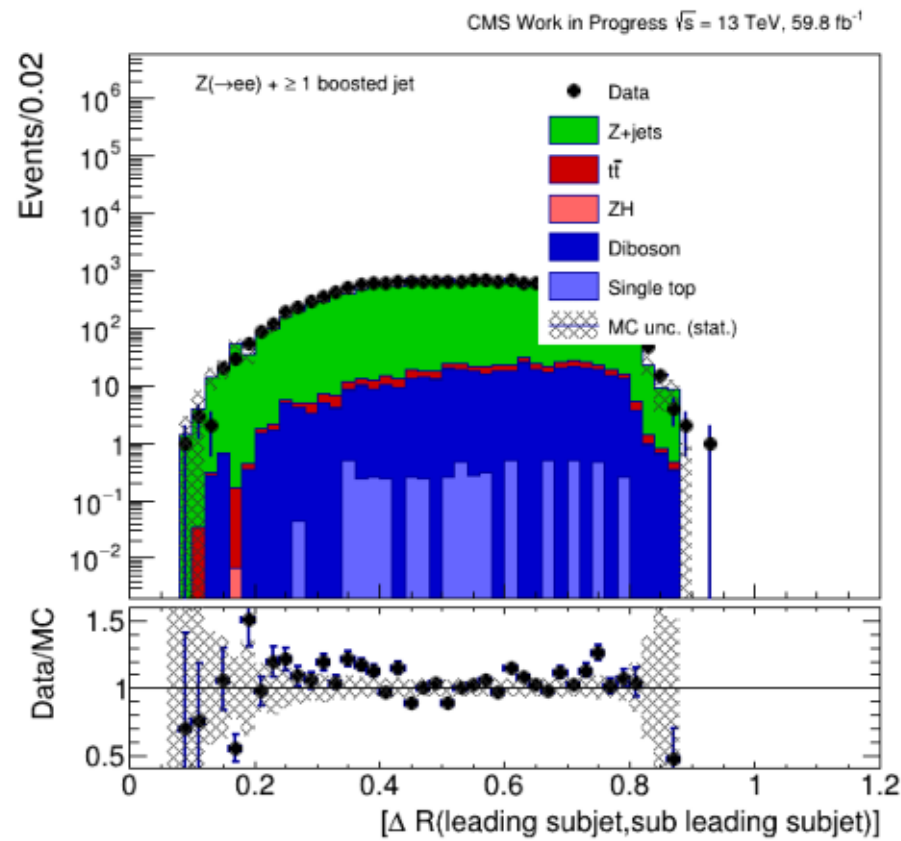


OLD: mass_sub1_Z_bjet_Zmm

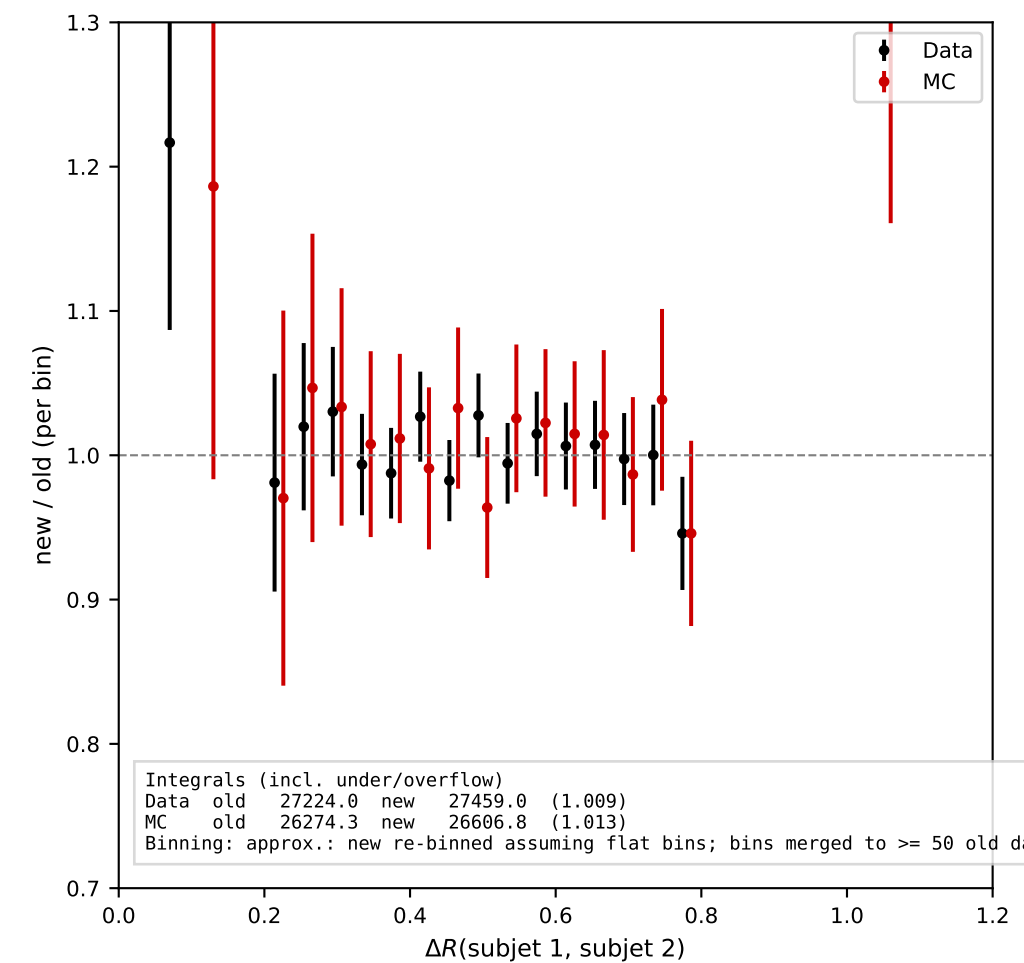
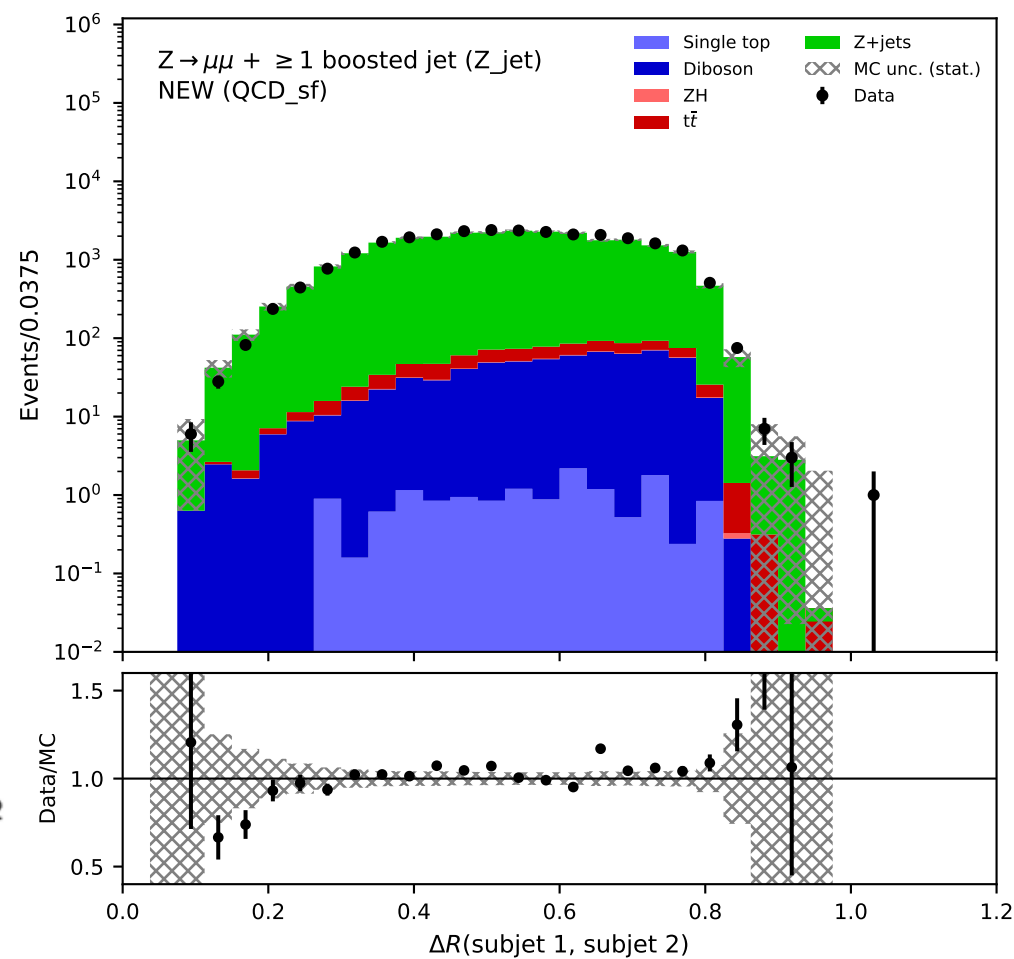
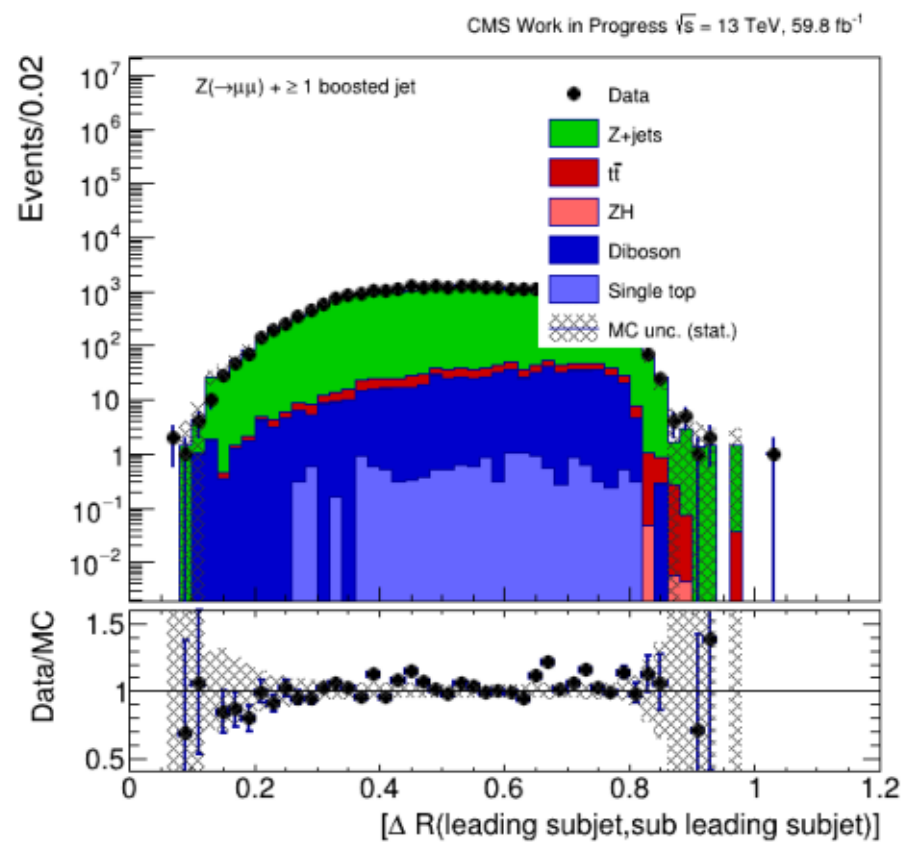


deltaR_subjets | Z_jet | 2018

OLD: deltaR_subjets_Z_jet_Zee



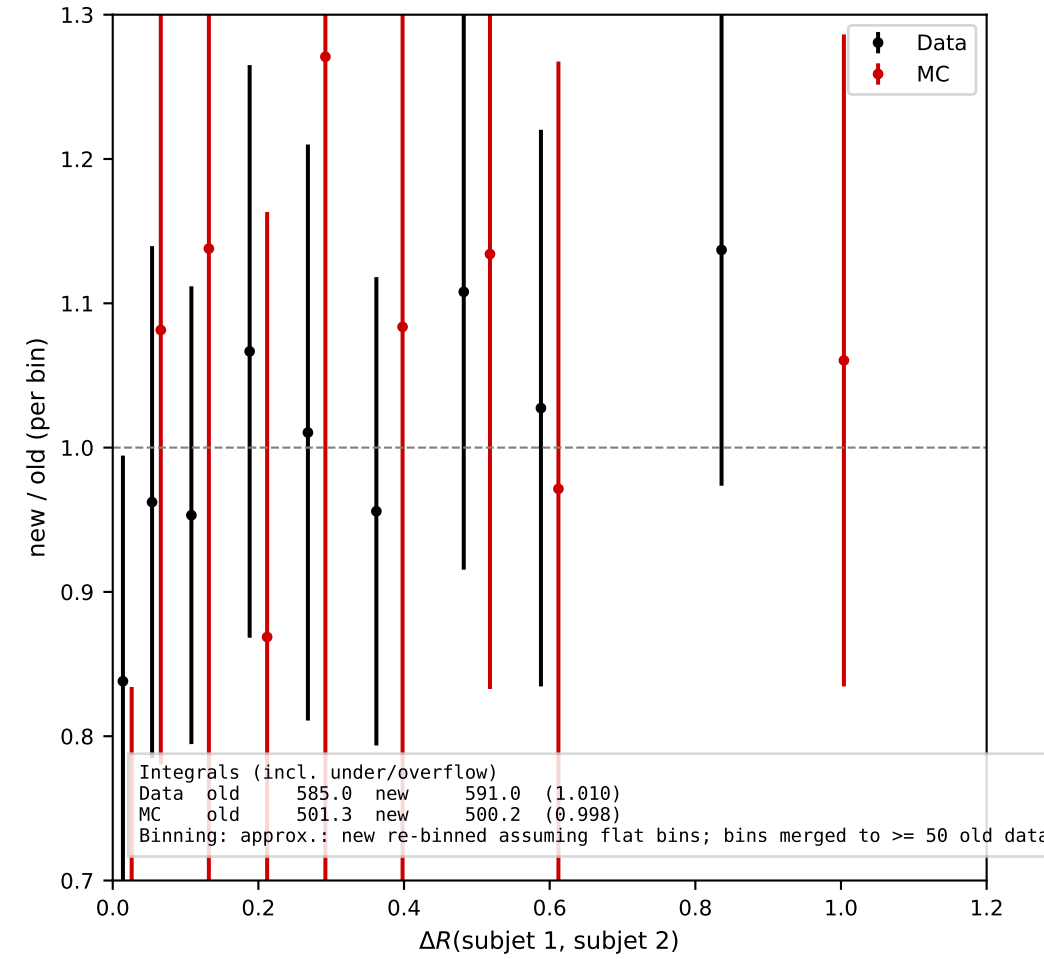
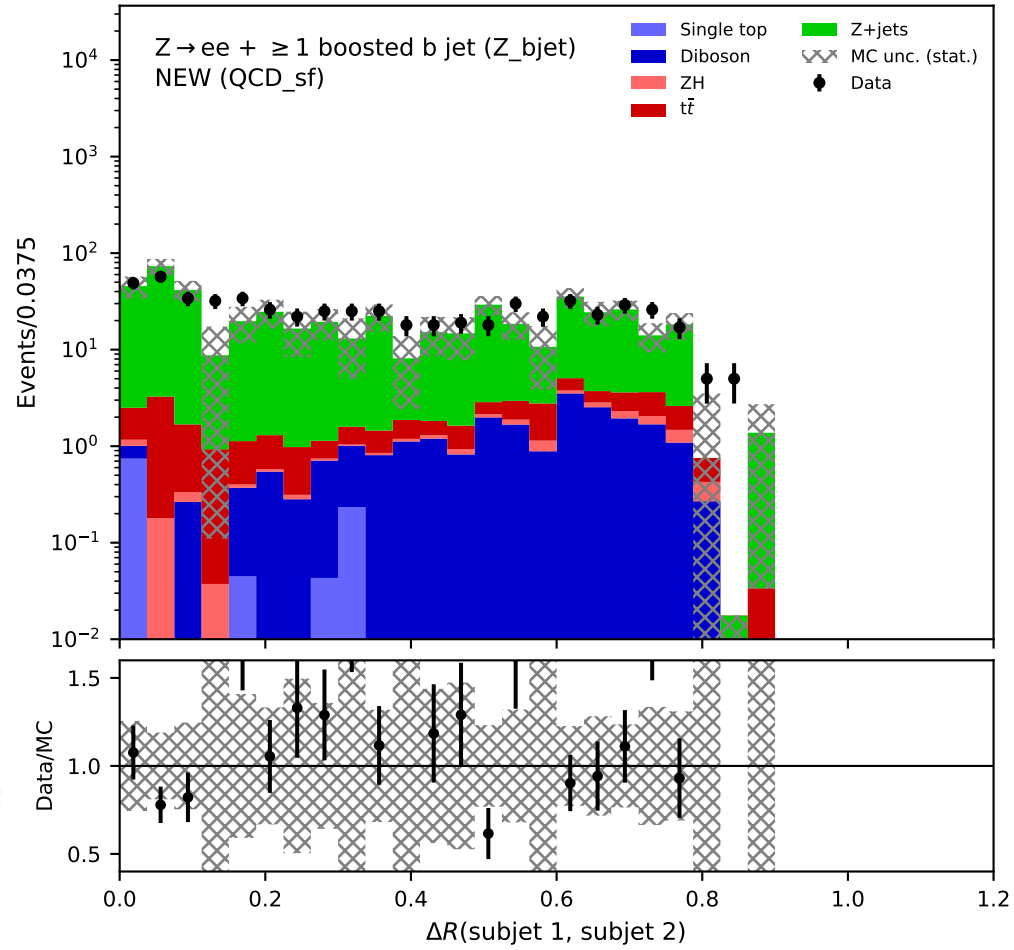
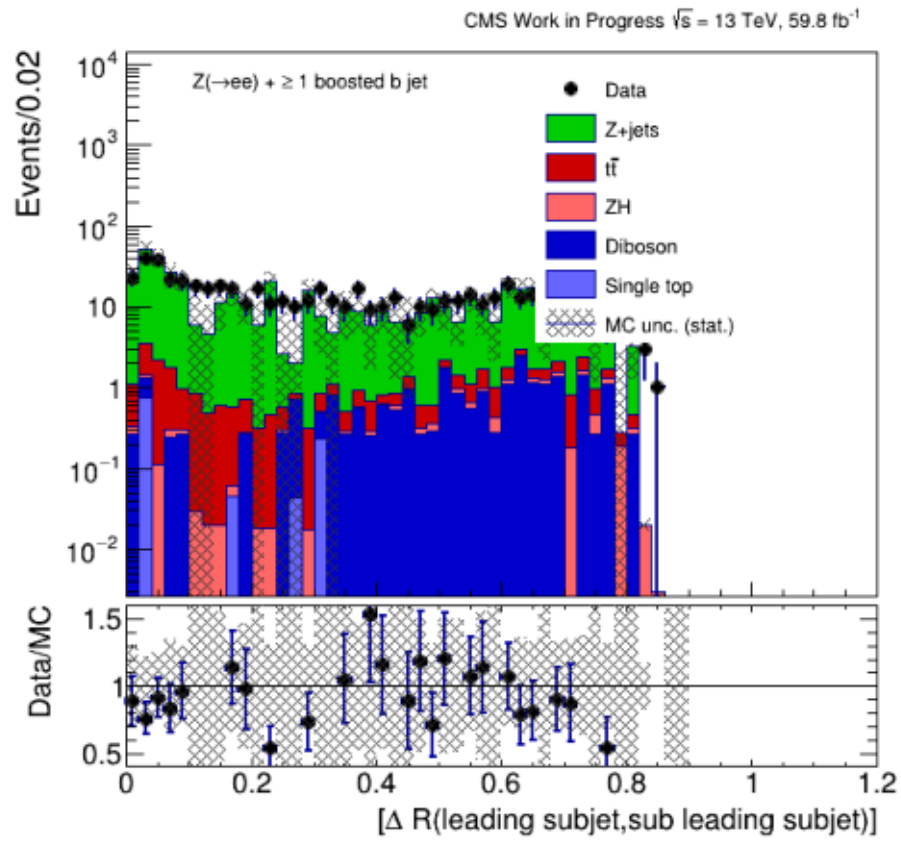
OLD: deltaR_subjets_Z_jet_Zmm



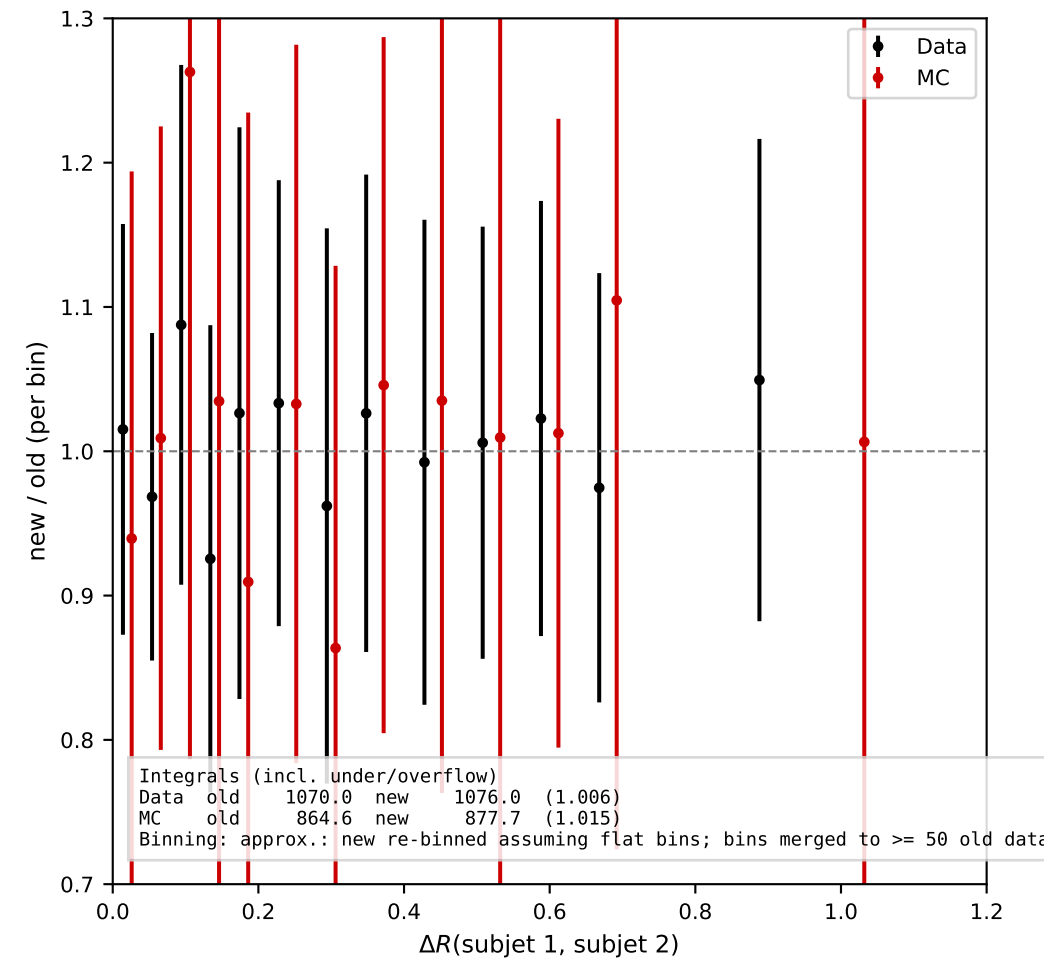
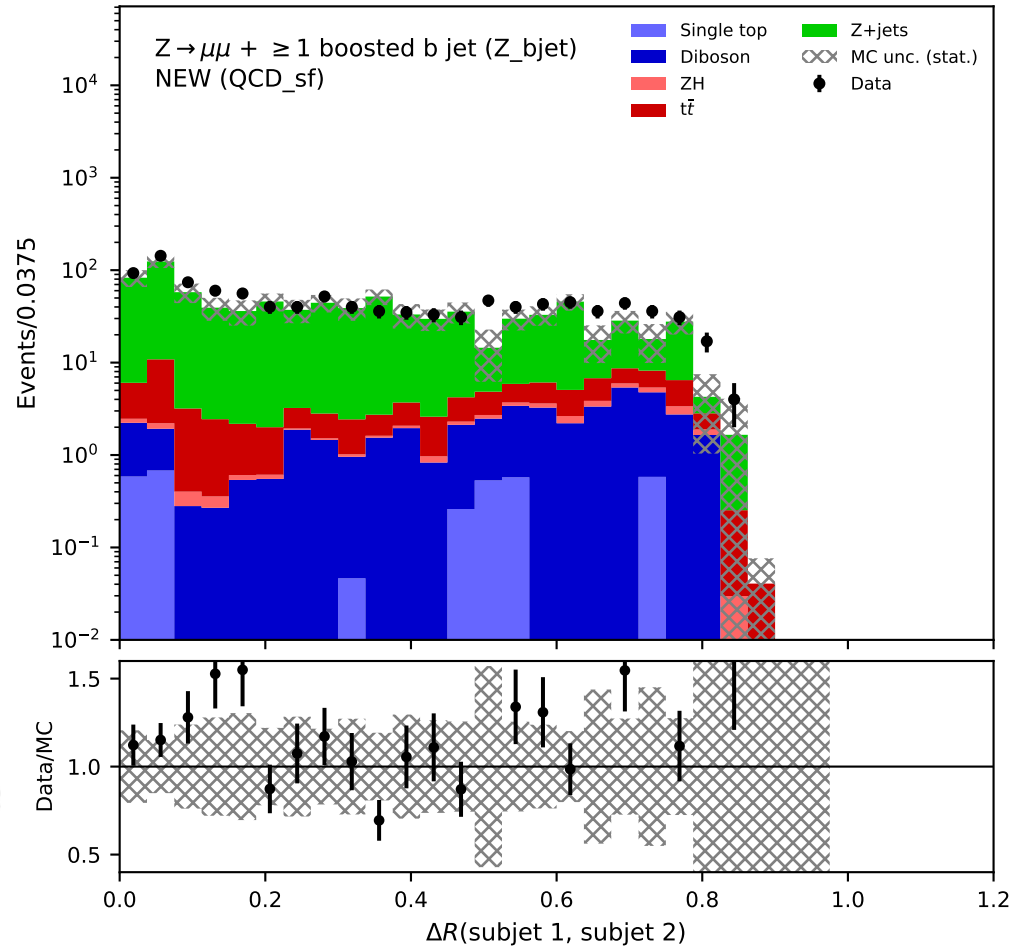
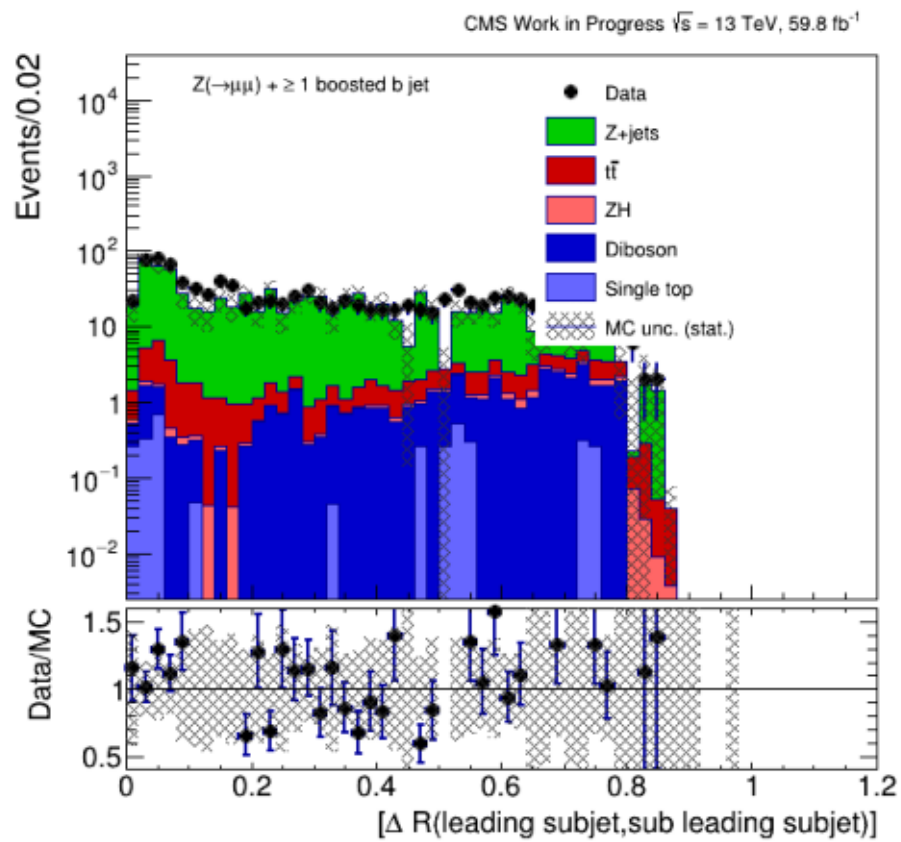
deltaR_subjets | Z_bjet | 2018

Z_bjet: the old framework fills subjet histograms from the leading jet's subjets, ours from the tagged jet's subjets; differences are expected.

OLD: deltaR_subjets_Z_bjet_Zee

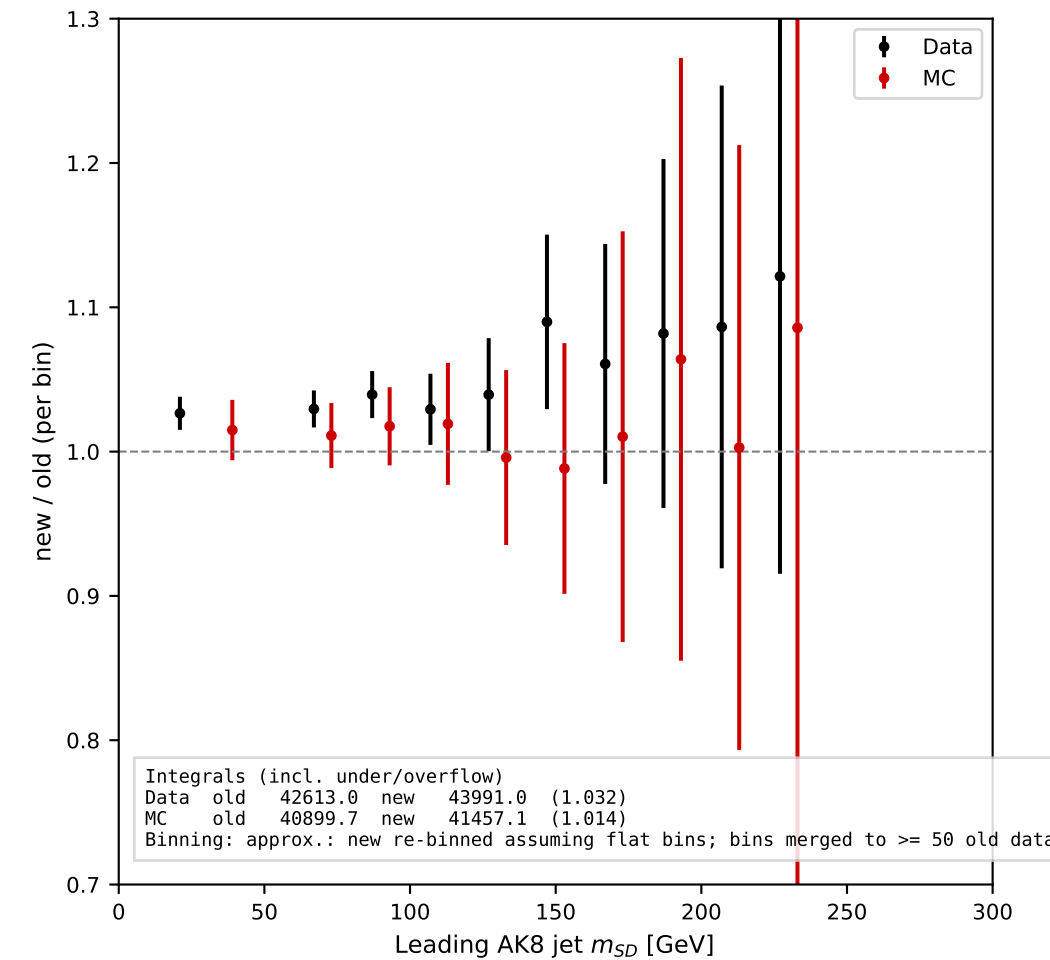
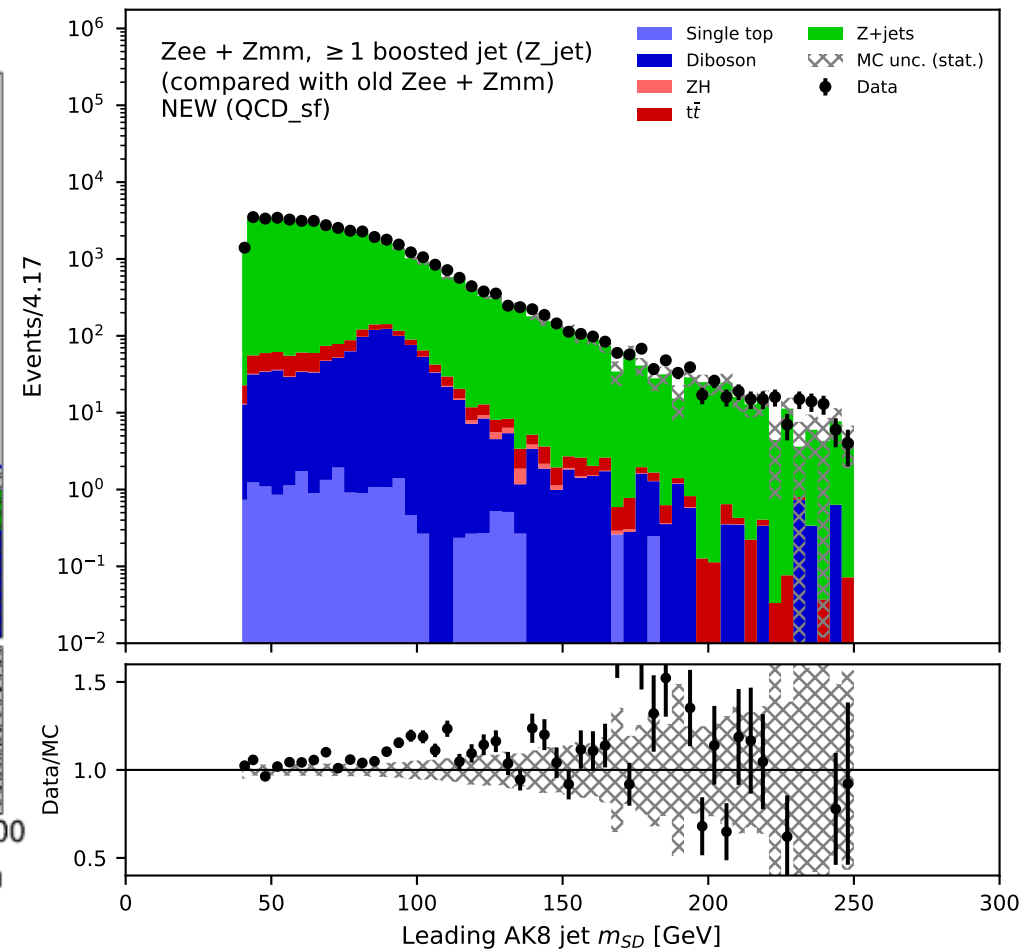
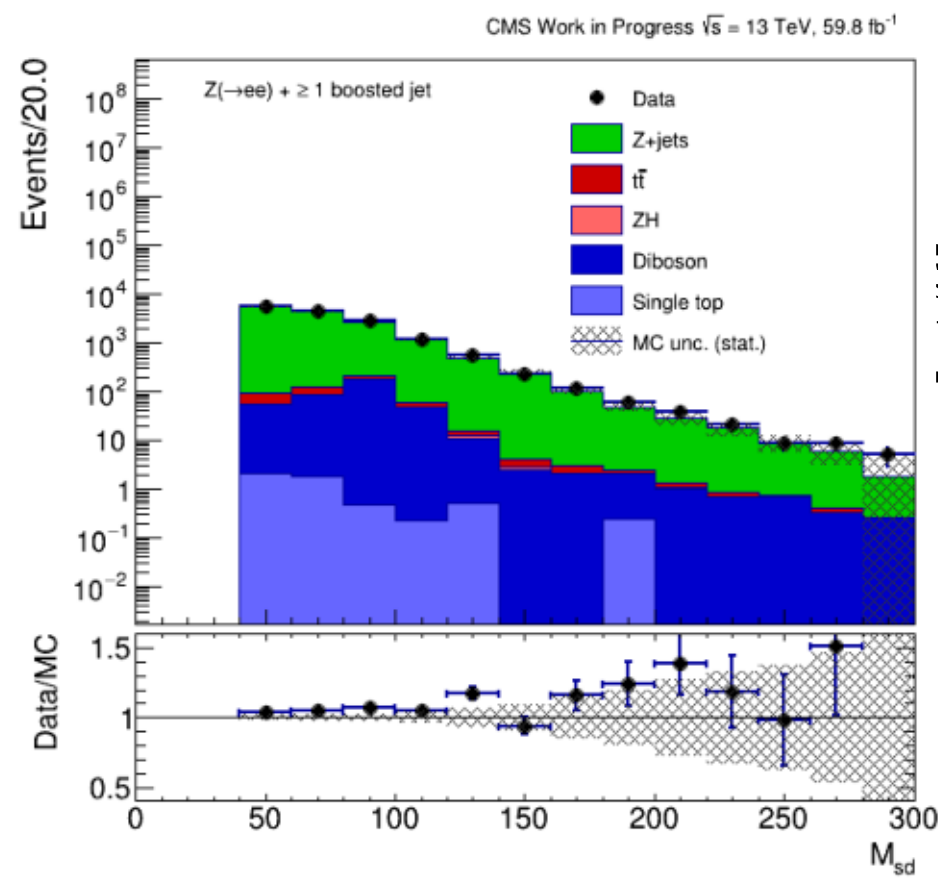


OLD: deltaR_subjets_Z_bjet_Zmm

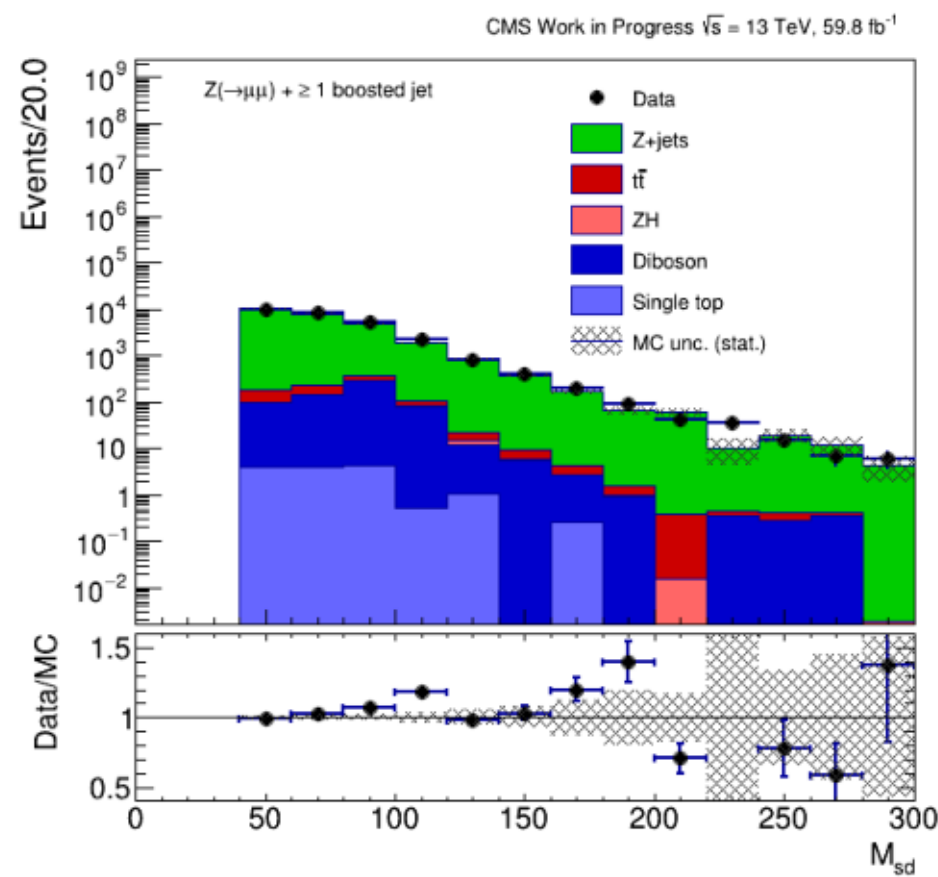


mSD | Z_jet | 2018

OLD: mSD_Z_jet_Zee

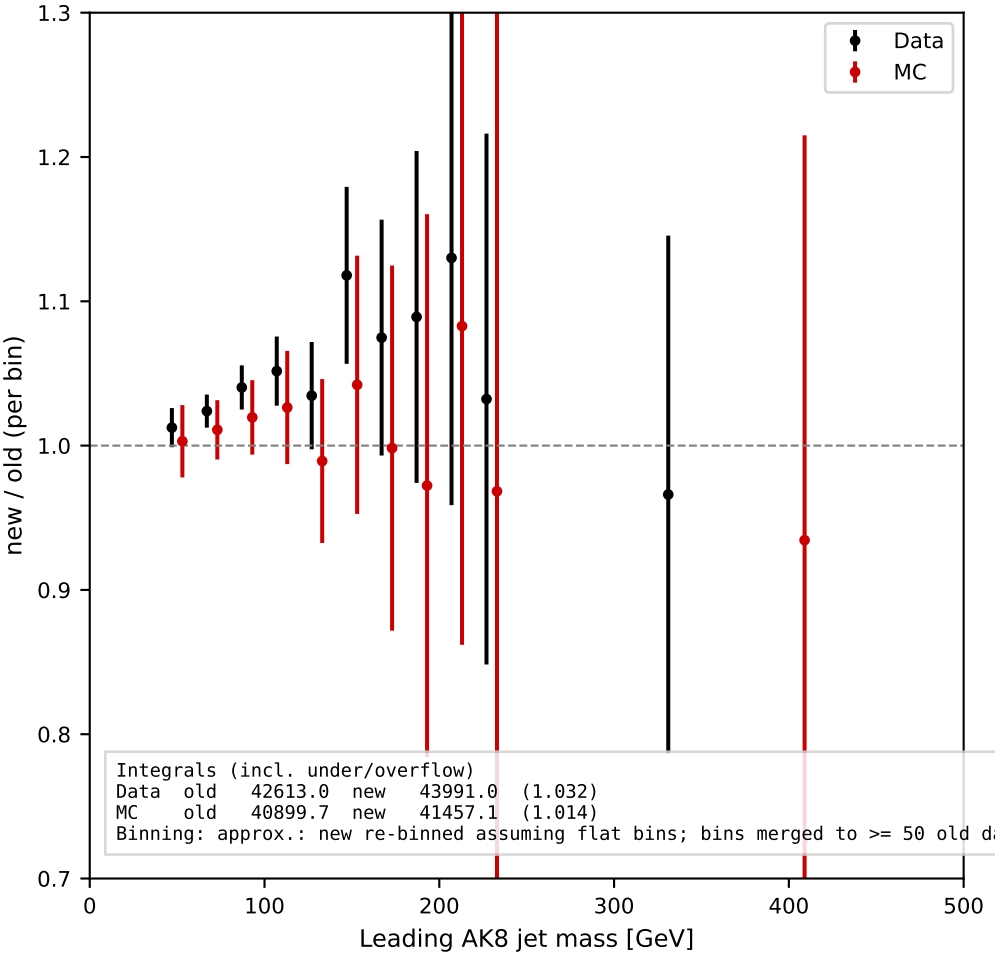
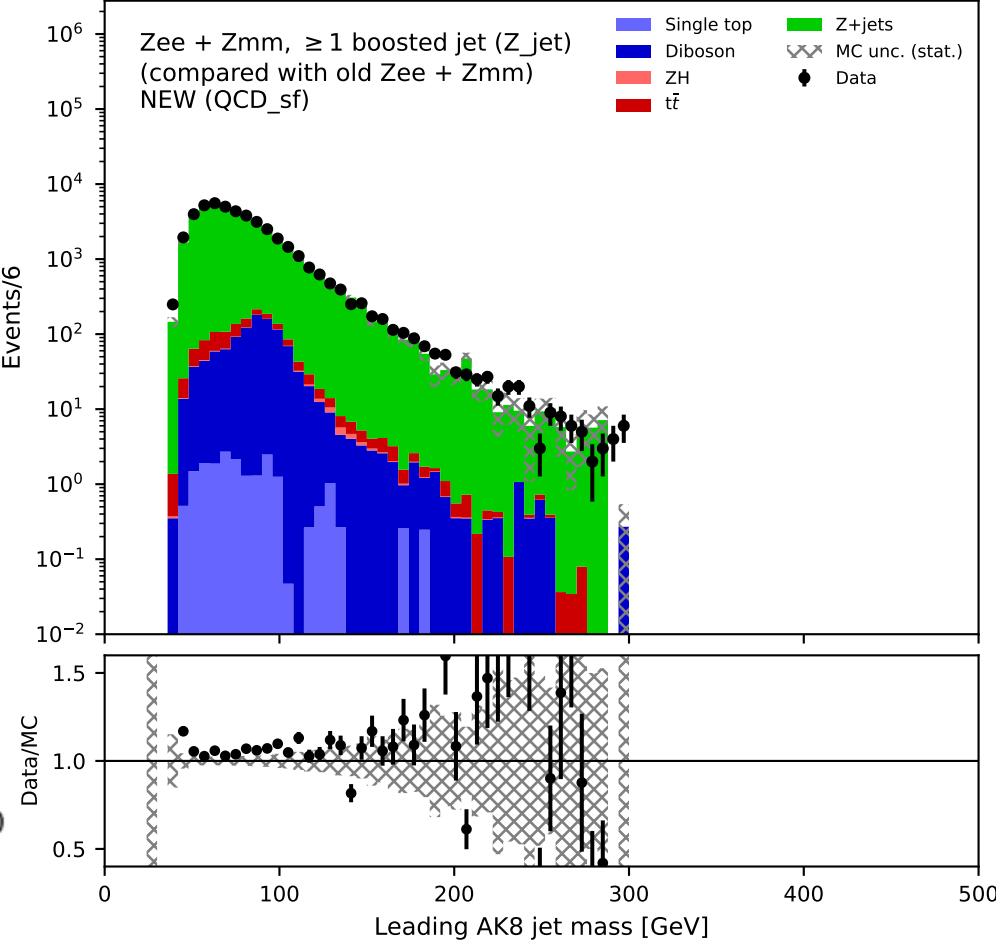
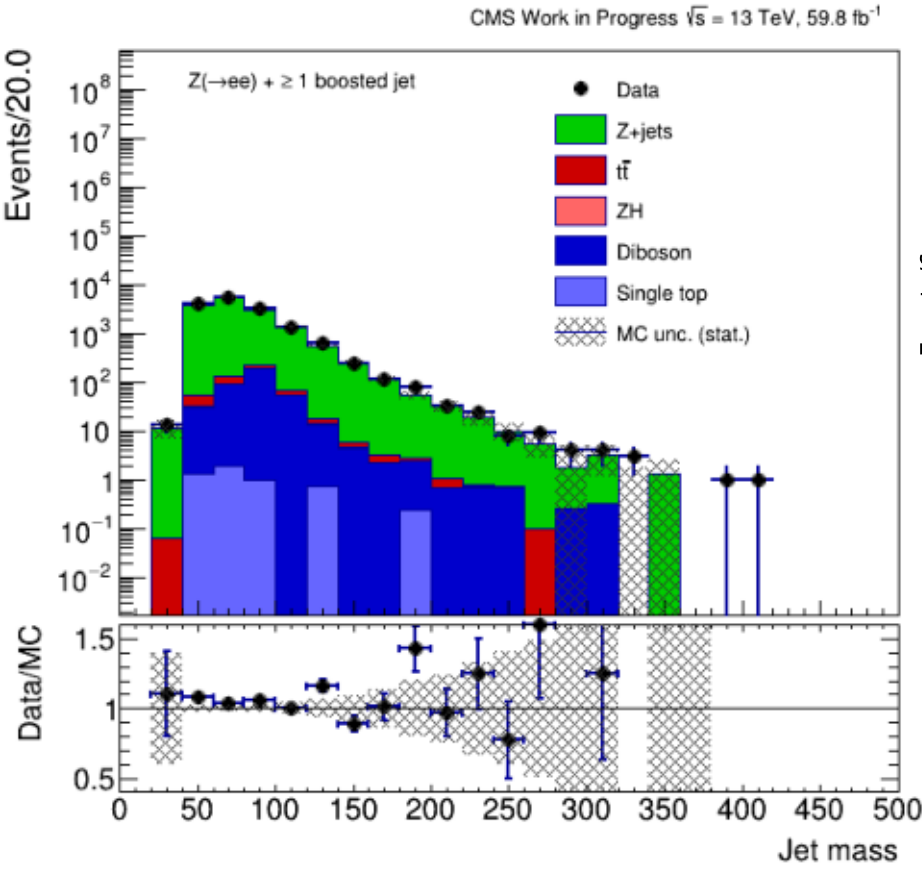


OLD: mSD_Z_jet_Zmm

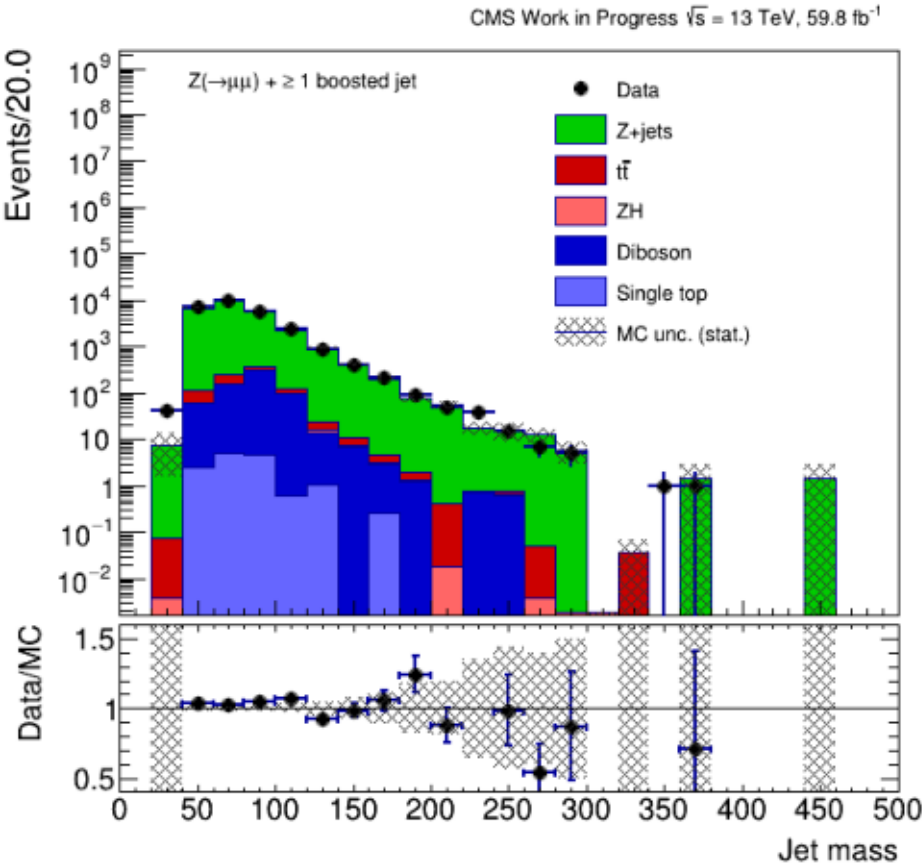


mass_jet | Z_jet | 2018

OLD: mass_jet_Z_jet_Zee

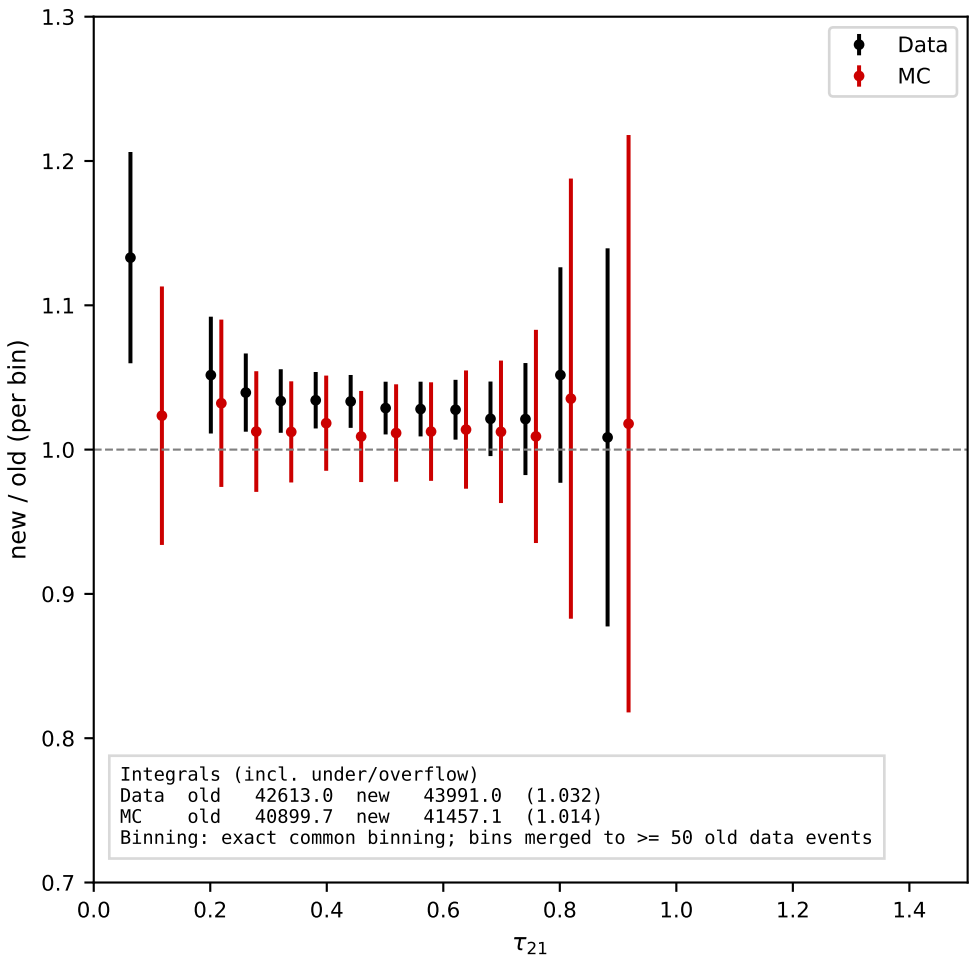
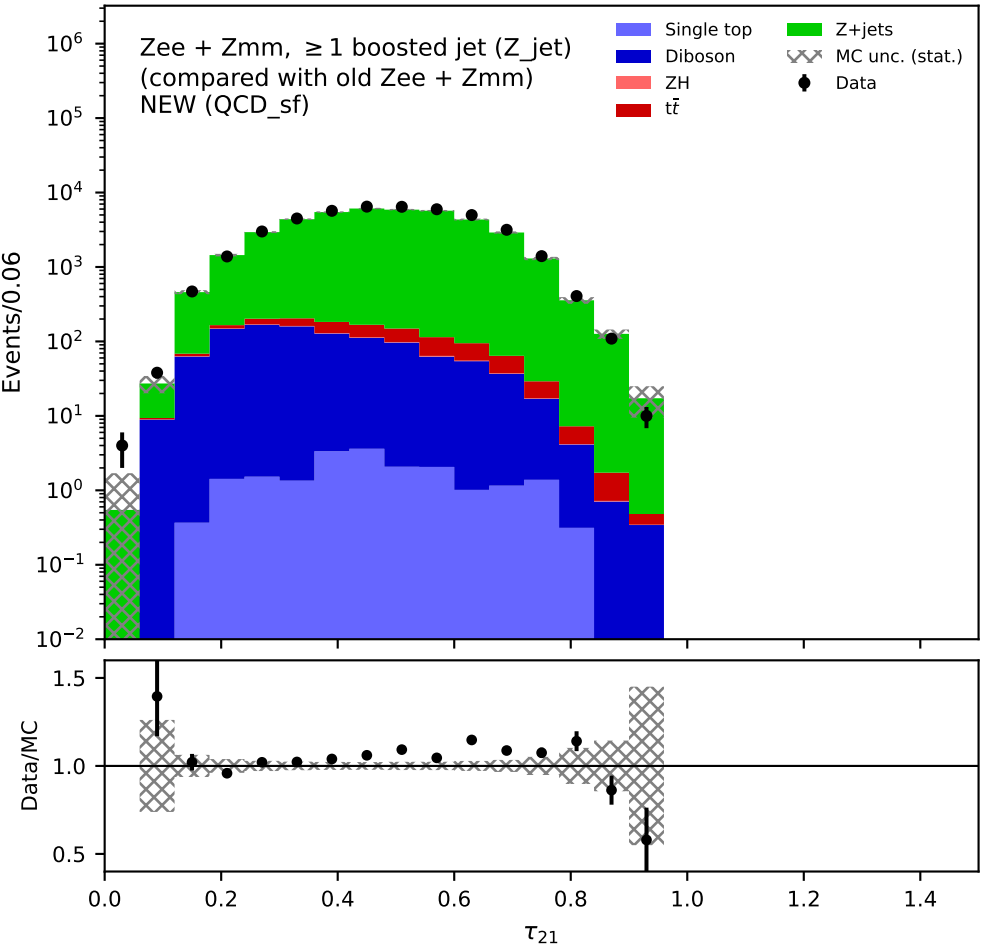
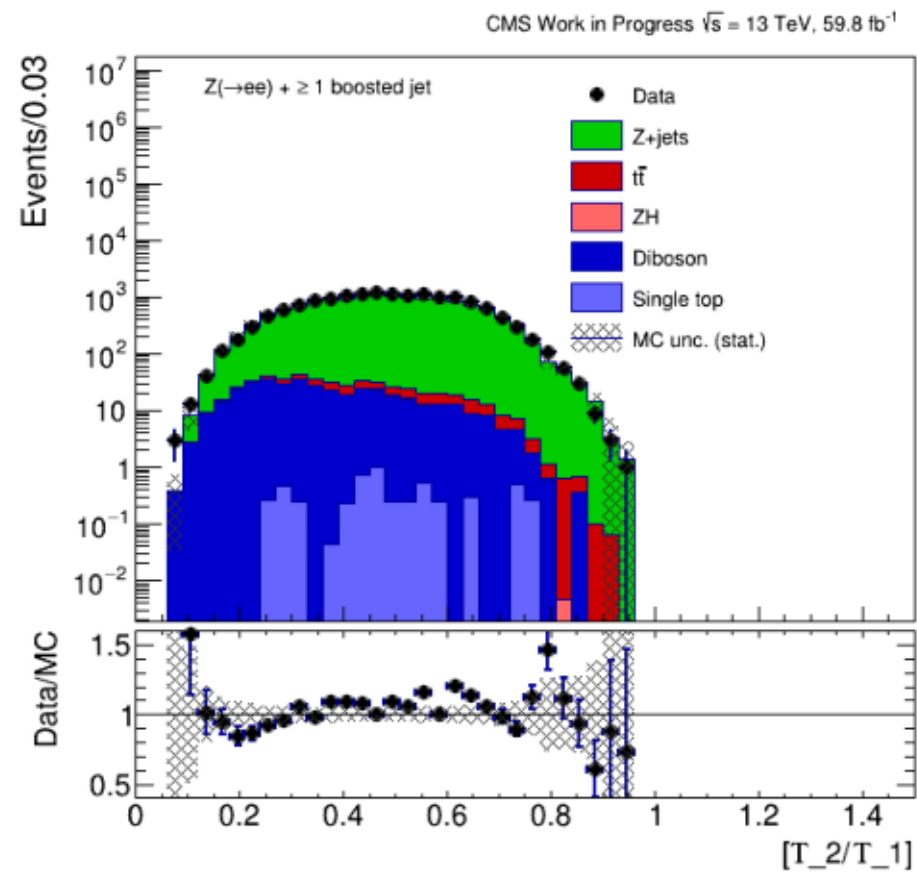


OLD: mass_jet_Z_jet_Zmm

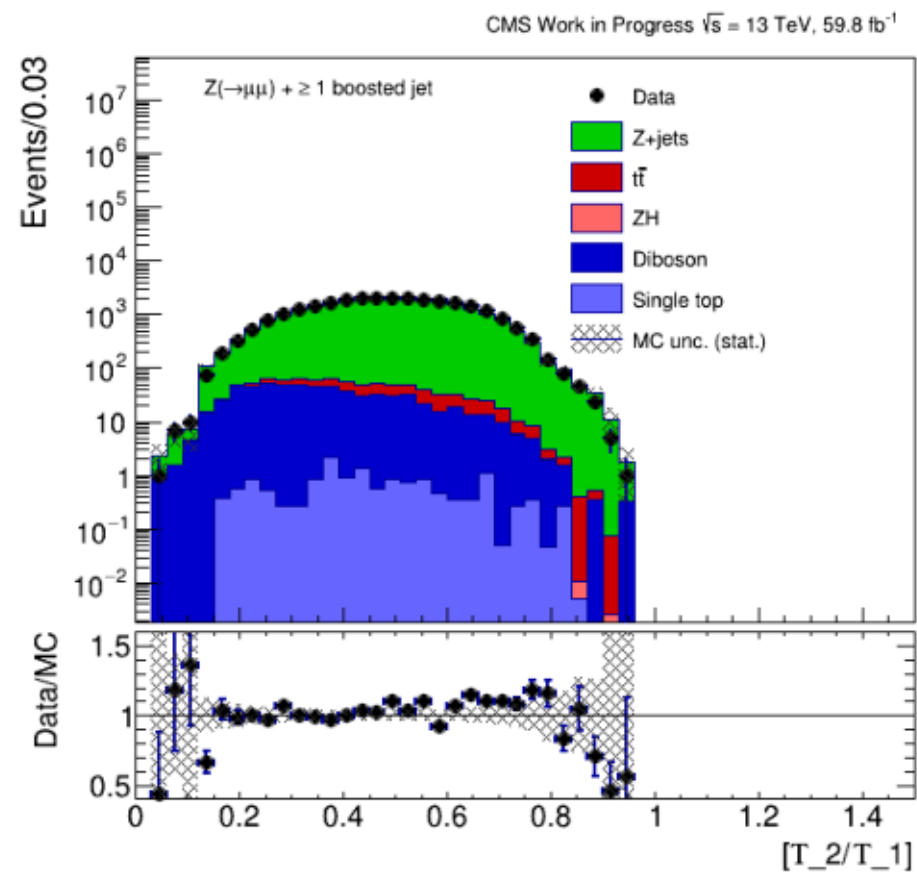


tau21_jet | Z_jet | 2018

OLD: tau21_jet_Z_jet_Zee

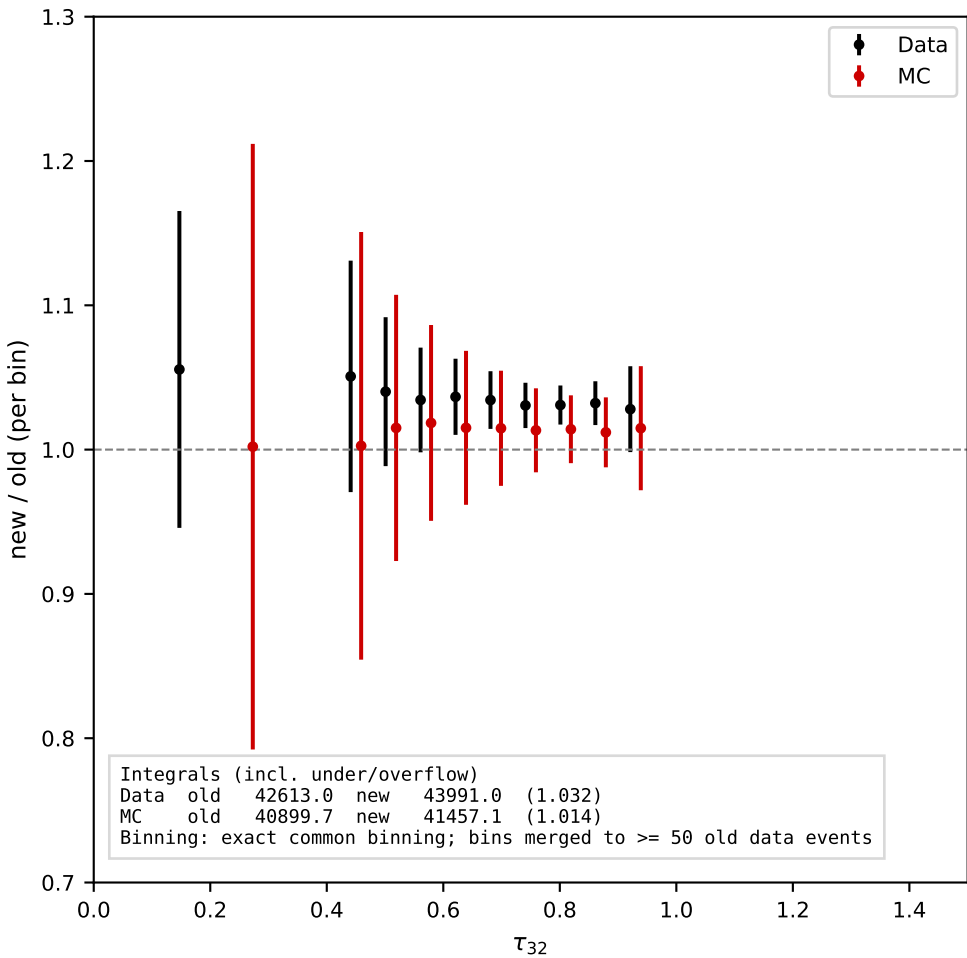
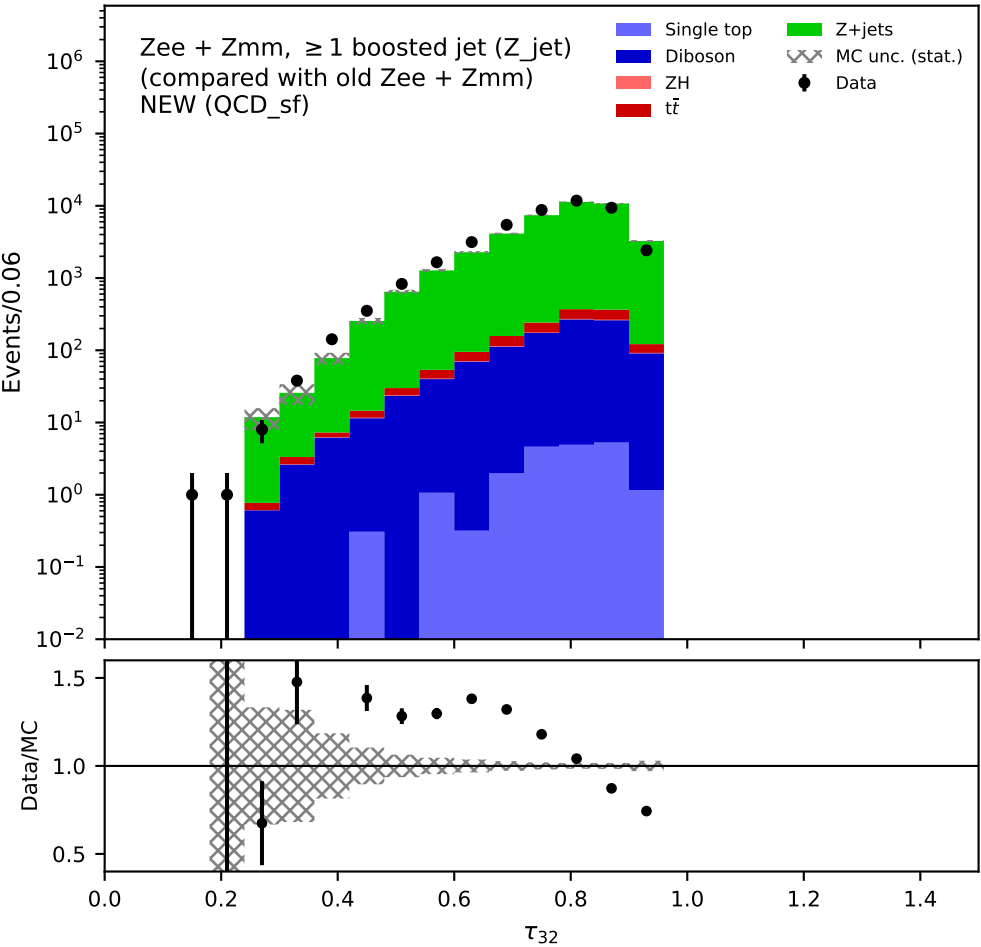
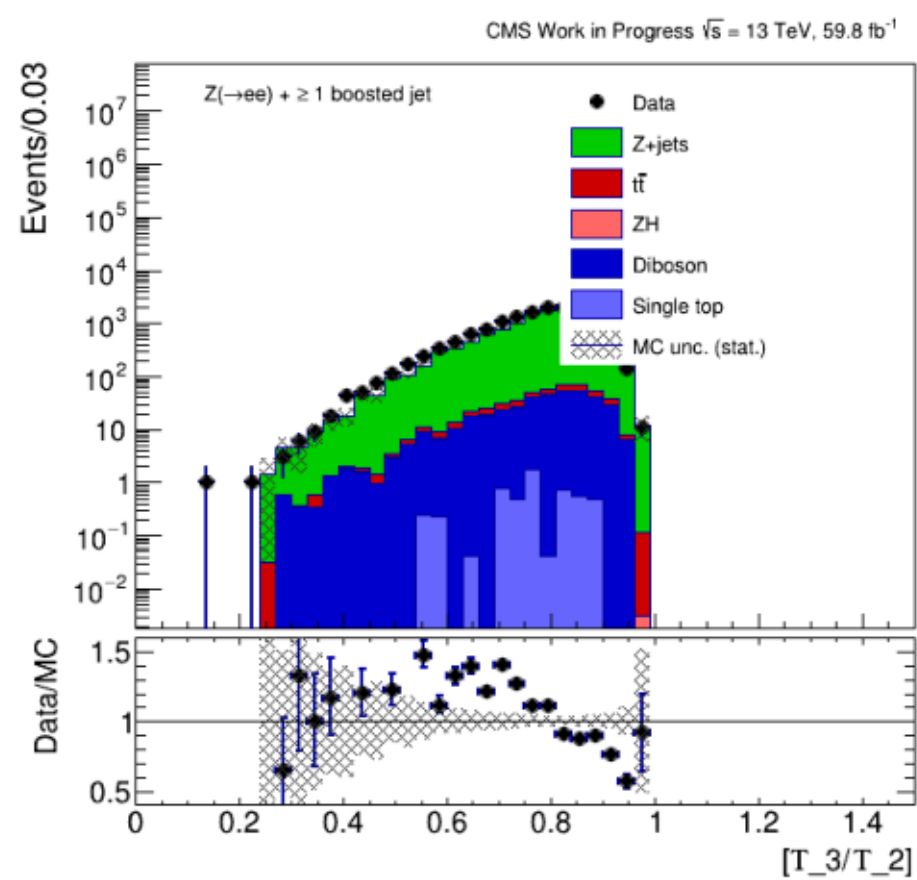


OLD: tau21_jet_Z_jet_Zmm

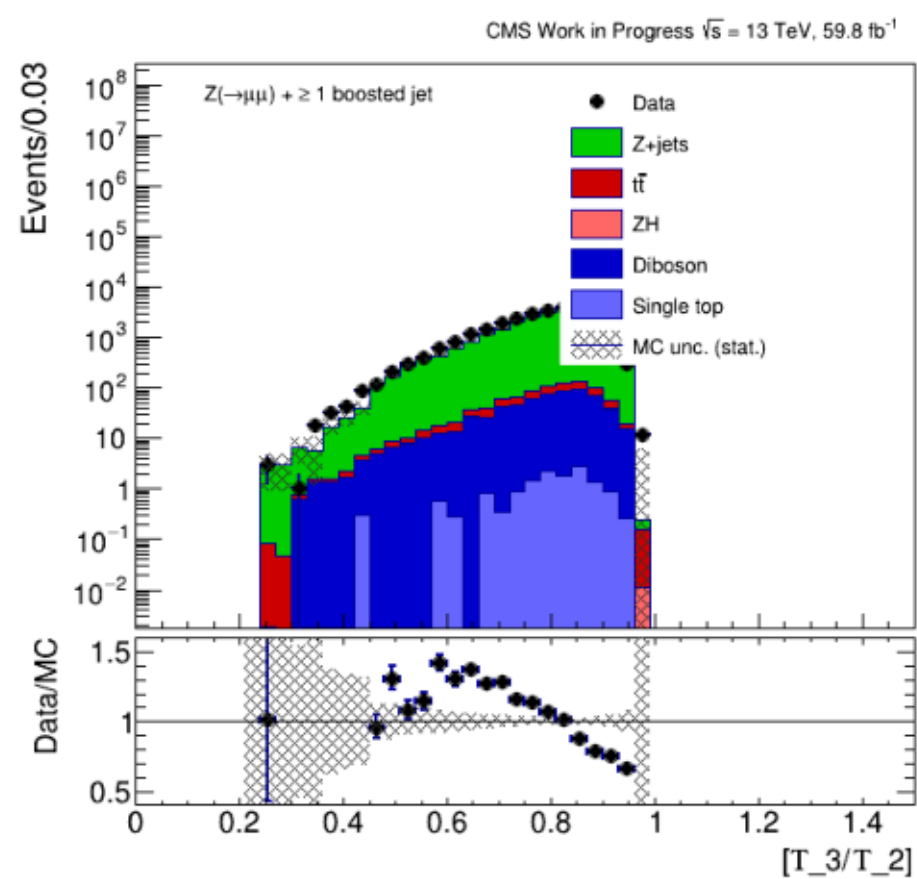


tau32_jet | Z_jet | 2018

OLD: tau32_jet_Z_jet_Zee



OLD: tau32_jet_Z_jet_Zmm



Old plots without a new counterpart

These variables are not filled by the new QCD_sf workflow yet:

- dPhi_Z_jet_Z_jet (Zee, Zmm)
- deltaR_lep_lep_Z_jet (Zee, Zmm)
- deltaR_lep_jet_Z_jet (Zee, Zmm)
- deltaR_Zjet_Z_jet (Zee, Zmm)
- deltaR_lep_Z_Z_jet (Zee, Zmm)